

**ACADEMIC REGULATIONS
AND
SYLLABUS**
CHOICE BASED CREDIT SYSTEM
MLR18
MECHANICAL ENGINEERING

Bachelor of Technology (B.Tech)

B. Tech. - Regular Four Year Degree Program
(For batches admitted from the academic year 2018 - 19)
&
B. Tech. - Lateral Entry Scheme
(For batches admitted from the academic year 2019 - 20)



MLR Institute of Technology

(Autonomous)

Laxman Reddy Avenue, Dundigal

Hyderabad – 500043, Telangana State

www.mlrit.ac.in, Email: director@mlrinstitutions.ac.in

FOREWORD

The autonomy is conferred on MLR Institute of Technology by UGC, based on its performance as well as future commitment and competency to impart quality education. It is a mark of its ability to function independently in accordance with the set norms of the monitoring bodies like UGC and AICTE. It reflects the confidence of the UGC in the autonomous institution to uphold and maintain standards it expects to deliver on its own behalf and thus awards degrees on behalf of the college. Thus, an autonomous institution is given the freedom to have its own **curriculum, examination system and monitoring mechanism**, independent of the affiliating University but under its observance.

MLR Institute of Technology is proud to win the credence of all the above bodies monitoring the quality in education and has gladly accepted the responsibility of sustaining, if not improving upon the standards and ethics for which it has been striving for more than a decade in reaching its present standing in the arena of contemporary technical education. As a follow up, statutory bodies like Academic Council and Boards of Studies are constituted with the guidance of the Governing Body of the College and recommendations of the JNTU Hyderabad to frame the regulations, course structure and syllabi under autonomous status.

The autonomous regulations, course structure and syllabi have been prepared after prolonged and detailed interaction with several expertise solicited from academics, industry and research, in accordance with the vision and mission of the college in order to produce quality engineering graduates to the society.

All the faculty, parents and students are requested to go through all the rules and regulations carefully. Any clarifications, if needed, are to be sought, at appropriate time with principal of the college, without presumptions, to avoid unwanted subsequent inconveniences and embarrassments. The Cooperation of all the stake holders is sought for the successful implementation of the autonomous system in the larger interests of the college and brighter prospects of engineering graduates.

PRINCIPAL

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B. Tech. - Regular Four Year Degree Programme
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&
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For pursuing four year under graduate Bachelor Degree Programme of study in Engineering (B.Tech) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified herein approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2018-19 onwards. Any reference to "Institute" or "College" in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

1. ADMISSION

1.1. Admission into first year of four year B. Tech. degree Programmes of study in Engineering

1.1.1. Eligibility:

A candidate seeking admission into the first year of four year B. Tech. degree Programmes should have:

(i) Passed either Intermediate Public Examination (I.P.E) conducted by the Board of Intermediate Education, Telangana, with Mathematics, Physics and Chemistry as optional subjects or any equivalent examination recognized by Board of Intermediate Education, Telangana or a Diploma in Engineering in the relevant branch conducted by the Board of Technical Education, Telangana or equivalent Diploma recognized by Board of Technical Education for admission as per guidelines defined by the Regulatory bodies of Telangana State Council for Higher Education (TSCHE) and AICTE.

(ii) Secured a rank in the EAMCET examination conducted by TSCHE for allotment of a seat by the Convener, EAMCET, for admission.

1.1.2. Admission Procedure:

Admissions are made into the first year of four year B. Tech. Degree Programmes as per the stipulations of the TSCHE.

(a) Category A seats are filled by the Convener, TSEAMCET.

(b) Category B seats are filled by the Management.

1.2. Admission into the second year of four year B. Tech. degree Program in Engineering

1.2.1. Eligibility:

A candidate seeking admission under lateral entry into the II year I Semester B. Tech. degree Programmes should have passed the qualifying exam (B.Sc. Mathematics or Diploma in concerned course) and based on the rank secured by the candidate at Engineering Common Entrance Test ECET (FDH) in accordance with the instructions received from the Convener, ECET and Government of Telangana.

1.2.2 Admission Procedure:

Admissions are made into the II year of four year B. Tech. degree Programmes through Convener, ECET (FDH) against the sanctioned strength in each Programmes of study as lateral entry students.

2. PROGRAMMES OFFERED

MLR Institute of Technology, an autonomous college affiliated to JNTUH, offers the following B. Tech. Programmes of study leading to the award of B. Tech. degree under the autonomous scheme.

- 1) B.Tech. - Aeronautical Engineering
- 2) B.Tech. - Computer Science and Engineering
- 3) B.Tech. - Electronics and Communication Engineering
- 4) B.Tech. - Electrical & Electronics Engineering
- 5) B.Tech. - Information Technology
- 6) B.Tech. - Mechanical Engineering

3. DURATION OF THE PROGRAMMES

3.1 Normal Duration

- 3.1.1 B. Tech. degree programme extends over a period of four academic years leading to the Degree of Bachelor of Technology (B.Tech.) of the Jawaharlal Nehru Technological University Hyderabad.
- 3.1.2 For students admitted under lateral entry scheme, B. Tech. degree programme extends over a period of three academic years leading to the Degree of Bachelor of Technology (B. Tech.) of the Jawaharlal Nehru Technological University Hyderabad.

3.2 Maximum Duration

- 3.2.1 The maximum period within which a student must complete a full-time academic programme is 8 years for B. Tech. If a student fails to complete the academic programme within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.2 For students admitted under lateral entry scheme in B. Tech. degree programme, the maximum period within which a student must complete a full-time academic programme is 6 years. If a student fails to complete the academic programme within the maximum duration as specified above, he shall forfeit the seat in B.Tech and his admission shall stand cancelled.
- 3.2.3 The period is reckoned from the academic year in which the student is admitted first time into the degree Programme.

4. AWARD OF B.Tech. DEGREE

A student will be declared eligible for the award of the B.Tech. degree if he/she fulfils the following academic regulations:

- 4.1 The candidate shall pursue a course of study for not less than four academic years and not more than eight years.
- 4.2 The candidate shall register for 160 credits and secure 160 credits.
- 4.3 The degree will be conferred and awarded by Jawaharlal Nehru Technological University Hyderabad on the recommendations of the Chairman, Academic Council.

5. PROGRAMME STRUCTURE

- 5.1 UGC/AICTE specified Definitions/Descriptions are adopted appropriately for various terms and abbreviations used in these Academic Regulations/ Norms, which are listed below.

Semester Scheme:

Each UGP is of 4 Academic Years (8 Semesters), each year divided into two Semesters of 22 weeks (≥ 90 working days), each Semester having - 'Continuous Internal Evaluation (CIE)' and 'Semester End Examination (SEE)'. Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) as denoted by UGC, and Curriculum/Course Structure as suggested by AICTE are followed.

5.1.2 The B.Tech. Programme of MLR Institute of Technology are of Semester pattern, with 8 Semesters constituting 4 Academic Years, each Academic Year having TWO Semesters (First/Odd and Second/Even Semesters). Each Semester shall be of 22 Weeks duration (inclusive of Examinations), with a minimum of 90 Instructional Days per Semester.

5.1.3 Credit Courses:

a) All Courses are to be registered by a student in a Semester to earn Credits. Credits shall be assigned to each Subject/ Course in a L: T: P: C (Lecture Periods: Tutorial Periods: Practical Periods: Credits) Structure, based on the following general pattern ..

- One Credit - for One hour/Week/Semester for Theory/Lecture(L)/Tutorial(T) Courses; and
- One Credit - for Two hours/Week/Semester for Laboratory/Practical (P) Courses, Mini Project...
- Mandatory Courses will not carry any Credits.

5.1.4 **Course Classification:**

All Courses offered for the UGP are broadly classified as:

- Basic Science Courses (BSC) include Mathematics, Physics, Chemistry, Biology etc
- **Engineering Science Courses (ESC)** courses include Materials, Workshop, Basics of Electrical/Electronics/ Mechanical/Computer Science & Engineering, Engineering Graphics, Instrumentation, Engineering Mechanics, Instrumentation etc.
- **Humanities and Social Science including Management Courses (HSMC)** courses include English, Communication skills, Management etc.
- Professional Core Courses (PCC) are core courses relevant to the chosen specialization/branch.
- **Professional Elective Courses (PEC)** are courses relevant to the chosen specialization/ branch offered as electives.
- **Open Elective Courses (OEC)** courses from other technical and/or emerging subject areas offered in the College by the Departments of Engineering, Science and Humanities.
- **Mandatory Course:** Course work on peripheral subjects in a programme, wherein familiarity considered mandatory. To be included as non-Credit, Mandatory Courses, with only a pass in each required to qualify for the award of degree from the concerned institution.
- **Project Work** and/or internship in industry or elsewhere, seminar.

5.1.5 **Course Nomenclature:**

The Curriculum Nomenclature or Course-Structure Grouping for the each of the UGP E&T (B.Tech. Degree Programme), is as listed below (along with AICTE specified Range of Total Credits).

S. No.	Broad Course Classification	Course Group/ Category	Course Description	Range of Credits
1)	BSC,ESC & HSMC	BSC – Basic Sciences Courses	Includes - Mathematics, Physics and Chemistry Subjects	25*
2)		ESC - Engineering Sciences Courses	Includes fundamental engineering subjects.	24*

3)		HSMC – Humanities and Social Sciences including Management	Includes subjects related to Humanities, Social Sciences and Management.	12*
4)	PCC	PCC – Professional Core Courses	Includes core subjects related to the Parent Discipline/ Department/ Branch of Engg.	48*
5)	PEC	PEC– Professional Elective Courses	Includes Elective subjects related to the Parent Discipline / Department / Branch of Engg.	18*
6)	OEC	OEC – Open Elective Courses	Elective subjects which include inter-disciplinary subjects or subjects in an area outside the Parent Discipline/ Department / Branch of Engg.	18*
7)	PWC	Project Work	Major Project.	15*
8)		Industrial Training/ Mini-Project	Industrial Training/ Internship/ Mini-Project.	
9)		Seminar	Seminar / Colloquium based on core contents related to Parent Discipline/ Department/ Branch of Engg.	
10)	MC	Mandatory Courses	Mandatory Courses (non-credit)	-
Total Credits for UGP (B. Tech.)Programme				160 (100%)

- Minor variations as per AICTE guidelines

6. COURSE REGISTRATION

- 6.1 A 'Faculty Advisor or Counsellor' shall be assigned to each student, who advises him/her about the UGP, its Course Structure and Curriculum, Choice/Option for Subjects/Courses, based on his/her competence, progress, pre-requisites and interest.
- 6.2 Academic Section of the College invites 'Registration Forms' from students prior (before the beginning of the Semester), ensuring 'DATE and TIME Stamping'. The Registration Requests for any 'CURRENT SEMESTER' shall be completed BEFORE the commencement of SEEs (Semester End Examinations) of the 'PRECEDING SEMESTER'
- A Student can apply for Registration, which includes approval from his faculty advisor, and then should be submitted to the College Academic Section through the Head of Department (a copy of the same being retained with Head of Department, Faculty Advisor and the Student).
- 6.4 A student may be permitted to register for his/her course of CHOICE with a Total of prescribed credits per Semester (permitted deviation being $\pm 12\%$), based on his PROGRESS and SGPA/CGPA, and completion of the 'PRE-REQUISITES' as indicated for various courses in the Department Course Structure and Syllabus contents.
- 6.5 Choice for 'additional Courses' must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Counsellor.
- 6.6 If the Student submits ambiguous choices or multiple options or erroneous (incorrect) entries during Registration for the Course(s) under a given/specified Course Group/ Category as listed in the Course Structure, only the first mentioned Course in that Category will be taken into consideration.

- 6.7 Dropping of Courses or changing of options may be permitted, ONLY AFTER obtaining prior approval from the Faculty Advisor, 'within 15 Days of Time' from the commencement of that Semester. Course Options exercised through Registration are final and CAN NOT be changed, and CAN NOT be inter-changed; further, alternate choices will also not be considered. However, if the Course that has already been listed for Registration (by the Head of Department) in a Semester could not be offered due to any unforeseen or unexpected reasons, then the Student shall be allowed to have alternate choice - either for a new Subject (subject to offering of such a Subject), or for another existing Subject (subject to availability of seats), which may be considered. Such alternate arrangements will be made by Head of the Department, with due notification and time-framed schedule, within the FIRST WEEK from the commencement of Class-work for that Semester.

7. COURSES TO BE OFFERED

- 7.1 A typical section (or class) strength for each Semester shall be 60.
- 7.2 courses may be offered to the Students, only if minimum of 20 students ($1/3^{\text{rd}}$ of the section strength) opt for it.
- 7.2 More than ONE TEACHER may offer the SAME SUBJECT (Lab/Practical's may be included with the corresponding Theory Subject in the same Semester) in any Semester. However, selection choice for students will be based on - 'CGPA Basis Criterion' (i.e., the first focus shall be on early Registration in that Semester, and the second focus, if needed, will be on CGPA of the student).
- 7.3 If more entries for Registration of a Subject come into picture, then the concerned Head of the Department shall take necessary decision, whether to offer such a Subject/Course for TWO (or multiple) SECTIONS or NOT.
- 7.4 OPEN ELECTIVES will be offered by a department to the students of other departments.

8. ATTENDANCE REQUIREMENTS

- A student will be eligible to appear for the End Semester Examinations, if he acquires a minimum of 75% of attendance in aggregate of all the Subjects/Courses (excluding Mandatory or Non-Credit Courses) for that Semester.
- Condoning of shortage of attendance in aggregate up to 10% (65% and above, and below 75%) in each Semester may be granted by the College Academic Committee on genuine and valid grounds, based on the student's representation with supporting evidence by following the govt. rules in vogue.
- A stipulated fee shall be payable towards condoning of shortage of attendance.
- Shortage of Attendance below 65% in aggregate shall in No case be condoned.
- A student shall not be promoted to the next Semester unless he/she satisfies the attendance requirements of the current Semester. The student may seek readmission for the Semester when offered next. He / She shall not be allowed to register for the subjects of the Semester while he/she is in detention. A student detained due to shortage of attendance, will have to repeat that Semester when offered next.
- Students whose attendance is less than 75% are not entitled to get the scholarship / fee reimbursement in any case as per the TS Govt. Rules in force.

9. ACADEMIC REQUIREMENTS FOR PROMOTION / COMPLETION OF REGULAR B.TECH PROGRAMME COURSE STUDY.

- 9.1 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to each Course, if he secures not less than 35% marks in the End Semester Examination, and a minimum of 40% of marks in the sum Total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together; in terms of Letter Grades, this implies securing P Grade or above in that Course.

- 9.2 A student shall be deemed to have satisfied the Academic Requirements and earned the Credits allotted to – Internship, if he/she secures not less than 40% of the Total marks to be awarded for each. The student would be treated as failed, if he/she - (i) does not submit a report on his/her Internship, or does not make a presentation of the same before the Evaluation Committee as per the schedule, or (ii) does not present the Seminar as required in the IV year I/II Semester, or (iii) secures less than 40% of marks in Internship evaluations. He/She may reappear once for each of the above evaluations, when they are scheduled again; if he/she fails in such 'one reappearance' evaluation also, he/she has to reappear for the same in the next subsequent Semester, as per the schedule.
- 9.3 A Student will not be promoted from I Year to II Year, unless he/she fulfils the Attendance requirements
- 9.4 A Student will not be promoted from II Year to III Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total 50% of Credits up to II Year II Semester from all the relevant regular and supplementary examinations.
- 9.5 A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total 50% of Credits up to III Year II Semester, from all the regular and supplementary examinations.
- 9.6 After securing the necessary 160 Credits as specified for the successful completion of the entire UGP, resulting in 160 Credits for UGP performance evaluation, i.e., the performance of the Student in these 160 Credits shall alone be taken into account for the calculation of the final CGPA.
If a Student registers for some more 'extra courses' (in the parent Department or other Departments/Branches of Engg.) other than those listed courses Totalling to 160 Credits as specified in the Course Structure of his/her Department, the performances in those 'extra courses' (although evaluated and graded using the same procedure as that of the required 160 Credits) will not be taken into account while calculating the SGPA and CGPA. For such 'extra courses' registered, % marks and Letter Grade alone will be indicated in the Grade Card, as a performance measure, subject to completion of the Attendance and Academic Requirements as stated in items 8 and 9.1-9.6.
- 9.7 Students who fail to earn minimum of 160 Credits as per the Course Structure, and as indicated above, within 8 Academic Years from the Date of Commencement of their I Year shall forfeit their seats in B.Tech Programme and their admissions shall stand cancelled.
When a Student is detained due to shortage of attendance/lack of credits in any Semester, he may be re-admitted into that Semester, as and when offered. However the regulations at the time of admissions hold good.

10. EVALUATION - DISTRIBUTION AND WEIGHTAGE OF MARKS

- 10.1 The performance of a student in each Semester shall be evaluated Course-wise (irrespective of Credits assigned) with a maximum of 100 marks for Theory. The B.Tech Project Work (Major Project) will be evaluated for 100 marks in Phase-I and 200 Marks in Phase-II.
- 10.2 For all Theory Courses as mentioned above, the distribution shall be 30 marks for CIE, and 70 marks for the SEE.
- 10.3 a) For Theory Subjects (inclusive of Minor Courses), during the Semester, there shall be 2 mid-term examinations for 30 marks each. Each mid-term examination consists of one subjective paper for 25 marks, and assignment for 5 marks for each subject.

Question paper contains 2 Parts (Part-A and Part-B.) The distribution of marks for PART-A and PART-B will be 10 marks & 15 marks respectively for UG programme.

Pattern of the question paper is as follows:

PART-A

Consists of **one compulsory question** with five sub questions each carrying two mark. For the I-Mid examinations the sub question would be from first 2 ½ units and for the II-Mid examination the sub question would be from the remaining 2 ½ units.

PART-B

Consists of five questions (out of which students have to answer three questions) carrying five marks each. Each question there will be an “either” “or” choice (that means there will be two questions from each unit and the student should answer any one question). The questions can consist of sub questions also.

- b) The first mid-term examination shall be conducted for the first 50% of the syllabus, and the second mid-term examination shall be conducted for the remaining 50% of the syllabus.
 - c) First Assignment should be submitted before the commencement of the first mid-term examinations, and the Second Assignment should be submitted before the commencement of the second mid-term examinations. The assignments shall be specified/given by the concerned subject teacher.
 - d) If any candidate is absent for the MID term examination or those who want to improve their internal marks in any subject can opt for Computer Based Test (CBT) as and when offered. The CBT is a 45 minutes duration ONLINE exam consisting of 30 objective questions from the entire syllabus of the subject. Best marks is consider as a final marks from the average of two mid examinations or CBT examination marks. The CBT can be taken after the payment of prescribed fee. There is no Internal Improvement for the courses Machine Drawing, Production Drawing, Engineering Drawing, Engineering Graphics and practical courses.
- 10.4 For Practical Courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned for Lab/Practical End Semester Examination (SEE). Out of the 30 marks for internals, day-to-day work in the laboratory shall be evaluated for 15 marks; and for the remaining 15 marks - two internal practical tests (each of 15 marks) shall be conducted by the concerned laboratory teacher and the average of the two tests is taken into account. The SEE for Practical's shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.
- 10.5 For the Subjects having Design and/or Drawing, (such as Engineering Graphics, Engineering Drawing, Machine Drawing, Production Drawing Practice, and Estimation), the distribution shall be 30 marks for CIE (10 marks for day-to-day work and 20 marks for internal tests) and 70 marks for SEE. There shall be two internal tests in a semester and the average of the two shall be considered for the award of marks for internal tests.
- 10.6 **Open Elective Course:** Students can choose One Open Elective Course (OEC-I) during V Semester, one (OEC-II) during VI Semester, one (OEC-III) in VII Semester, and one (OEC-IV) in VIII Semester from the list of Open Elective Courses given. However, Students cannot opt for an Open Elective Courses offered by their own (parent) Department, if it is already listed under any category of the Subjects offered by parent Department in any Semester.
- 10.7 There shall be an internship to be taken up during the vacation after II year II Semester (IV Semester) examination. However, the internship and its report shall be evaluated in III Year I Semester (VI Semester) SEE. The internship shall be submitted in a report form and presented before the committee. It shall be evaluated for 100 marks. The committee consists of head of the department, and a senior faculty member(s)(minimum Two) of the department. There shall be no internal marks for internship.
- 10.8 There shall be a Comprehensive Viva in VI Semester and will be conducted SEE by through a test or a committee consisting of One External Examiner, Head of the Department and two Senior faculty members of the Department. The comprehensive viva is intended to assess the student's understanding of the subjects he/she studied during the B.Tech course of study. The Comprehensive Viva is evaluated for 100 marks by the test or committee. There shall be no CIE for Comprehensive Viva.
- 10.9 Each Student shall start the Project Work Phase-I during the IV B.Tech I Semester(VII Semester), as per the instructions of the Project Guide/Project Supervisor assigned by the Head of Department. Total 100 marks allotted for the Project Work Stage-I. 40% of marks shall be evaluated Project Guide/Project supervisor CIE (Continuous Internal Evaluation) based on the reports submitted and conduct presentations. Remaining 60% of marks shall be evaluated by committee comprising of the

Head of the Department, project supervisor and senior faculty member from concerned department based on Viva/Seminar Presentation. He/She must secure the 40% of the marks from CIE. For Project work Phase-II in IV Year II Sem. There is an internal marks of 50, the evaluation should be done by the supervisor. There is an external marks of 150 and the same evaluated by the external examiner appointed by the Chief Controller of Examinations and he secures a minimum of 35% of marks in the Semester End Examination and a minimum aggregate of 40% of the total marks in the Semester End Examination and Continuous Internal Evaluation taken together.

10.10. End Semester examination:

- a) Question paper contains 2 Parts (Part-A and Part-B) having the questions distributed equally among all units.
- b) The distribution of marks for PART-A and PART-B will be 20 marks & 50 marks respectively for UG programs. Pattern of the question paper is as follows:

PART-A

Consists of one question which are compulsory. The first question consists of ten sub-questions two from each unit and carry 2 marks each.

PART-B

Consists of 5 questions carrying 10 marks each. Each of these questions is from one unit and may contain sub questions. Each question there will be an "either" "or" choice (that means there will be two questions from each unit and the student should answer any one question).

- 10.11 For Mandatory Non-Credit Courses offered in a Semester, after securing $\geq 65\%$ attendance and has secured not less than 35% marks in the SEE, and a minimum of 40% of marks in the sum Total of the CIE and SEE taken together in such a course, then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be allotted for these courses/activities. However, for non credit courses '**Satisfactory**' or '**Unsatisfactory**' shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.
- 10.12. For social innovation and Engineering Exploration courses, there shall be a Continuous Internal Evaluation (CIE) during the Semester for 30 internal marks, and 70 marks are assigned End Semester Examination (SEE). Internal evaluations done through presentations in CIE-1 & CIE-2. The SEE for shall be conducted at the end of the Semester by Two Examiners appointed by the Chief Controller of Examinations in consultation with the Head of the Department.

11. AWARD OF DEGREE

After a student has satisfied the requirement prescribed for the completion of the Program and is eligible for the award of B. Tech. Degree he shall be placed in one of the following four classes Shown in Table.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

Class Awarded	Grade to be Secured
First Class with Distinction	CGPA ≥ 8.00
First Class	≥ 6.50 to < 8.00 CGPA
Second Class	≥ 5.50 to < 6.50 CGPA
Pass Class	≥ 5.00 to < 5.50 CGPA
FAIL	CGPA < 5

a) Improvement of Grades and Completion of the Course

- i) Candidates who have passed in a theory course in a Semester are allowed to appear for improvement of Grade in the next immediate supplementary examination for a maximum of

three subjects only. Candidates will not be allowed to improve grade in the Laboratory, Seminars, Internships and Project Work.

- ii) Improved grade will not be counted for the award of prizes/medals and Rank. However the previous grade will be considered for the award of prizes/medals and rank in case of toppers.
- iii) If the candidate does not show improvement in the grade, his/her previous grade will be taken into consideration.

12. LETTER GRADE AND GRADE POINT

12.1 Marks will be awarded to indicate the performance of each student in each Theory Subject, or Lab/Practical's, or Seminar, or Project, or Internship*/Mini-Project, Minor Course etc., based on the %marks obtained in CIE+SEE (Continuous Internal Evaluation + Semester End Examination, both taken together), and a corresponding Letter Grade shall be given.

* For internship only SEE will be taken into consideration. There is no CIE for Internship.

12.2 As a measure of the student's performance, a 10-point Absolute Grading System using the following Letter Grades (UGC Guidelines) and corresponding percentage of marks shall be followed...

% of Marks Secured (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
90% and above ($\geq 90\%$, $\leq 100\%$)	O (Outstanding)	10
Below 90% but not less than 80% ($\geq 80\%$, $< 90\%$)	A ⁺ (Excellent)	9
Below 80% but not less than 70% ($\geq 70\%$, $< 80\%$)	A (Very Good)	8
Below 70% but not less than 60% ($\geq 60\%$, $< 70\%$)	B ⁺ (Good)	7
Below 60% but not less than 50% ($\geq 50\%$, $< 60\%$)	B (above Average)	6
Below 50% but not less than 40% ($\geq 40\%$, $< 50\%$)	C (Average)	5
Below 40% ($< 40\%$)	F (FAIL)	0
Absent	AB	0

12.3 A student obtaining F Grade in any Subject shall be considered 'failed' and will be required to reappear as 'Supplementary Candidate' in the End Semester Examination (SEE), as and when offered. In such cases, his Internal Marks (CIE Marks) in those Subject(s) will remain same as those he obtained earlier.

12.4 A Letter Grade does not imply any specific % of Marks.

12.5 In general, a student shall not be permitted to repeat any Subject/Course (s) only for the sake of 'Grade Improvement' or 'SGPA/CGPA Improvement'. However, he has to repeat all the Subjects/Courses pertaining to that Semester, when he is detained.

12.6 A student earns Grade Point (GP) in each Subject/Course, on the basis of the Letter Grade obtained by him in that Subject/Course (excluding Mandatory non-credit Courses). Then the corresponding 'Credit Points' (CP) are computed by multiplying the Grade Point with Credits for that particular Subject/Course.

Credit Points (CP) = Grade Point (GP) x Credits For a Course

12.7 The Student passes the Subject/Course only when he gets $GP \geq 4$ (P Grade or above).

12.8 The Semester Grade Point Average (SGPA) is calculated by dividing the Sum of Credit Points (ΣCP) secured from ALL Subjects/Courses registered in a Semester, by the Total Number of Credits registered during that Semester. SGPA is rounded off to TWO Decimal Places. SGPA is thus computed as

$SGPA = \{\sum_{i=1}^N C_i G_i\} / \{\sum_{i=1}^N C_i\}$ For each Semester,

where 'i' is the Subject indicator index (takes into account all Subjects in a Semester), 'N' is the no. of Subjects 'REGISTERED' for the Semester (as specifically required and listed under the Course

Structure of the parent Department), C_i is the no. of Credits allotted to that i th Subject, and G_i represents the Grade Points (GP) corresponding to the Letter Grade awarded for that i th Subject.

Illustration of Computation of SGPA Computation

Course	Credit	Grade Letter	Grade Point	Credit Point (Credit x Grade)
Course1	3	A	8	$3 \times 8 = 24$
Course2	4	B+	7	$4 \times 7 = 28$
Course3	3	B	6	$3 \times 6 = 18$
Course4	3	O	10	$3 \times 10 = 30$
Course5	3	C	5	$3 \times 5 = 15$
Course6	4	B	6	$4 \times 6 = 24$

Thus, **SGPA = $139/20 = 6.95$**

- 12.9 The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student over all Semesters considered for registration. The CGPA is the ratio of the Total Credit Points secured by a student in ALL registered Courses in ALL Semesters, and the Total Number of Credits registered in ALL the Semesters. CGPA is rounded off to TWO Decimal Places. CGPA is thus computed from the I Year Second Semester onwards, at the end of each Semester, as per the formula

$$\text{CGPA} = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all } S \text{ Semesters registered}$$

(i.e., up to and inclusive of S Semesters, $S \geq 2$),

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the j th Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that j th Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

For CGPA Computation

Semester 1	Semester 2	Semester 3	Semester 4	Semester 5	Semester 6
Credits : 20 SGPA : 6.9	Credits : 22 SGPA : 7.8	Credits : 25 SGPA : 5.6	Credits : 26 SGPA : 6.0	Credits : 26 SGPA : 6.3	Credits : 25 SGPA : 8.0

$$\text{Thus, CGPA} = \frac{20 \times 6.9 + 22 \times 7.8 + 25 \times 5.6 + 26 \times 6.0 + 26 \times 6.3 + 25 \times 8.0}{144} = 6.73$$

144

- 12.10 For Merit Ranking or Comparison Purposes or any other listing, ONLY the 'ROUNDED OFF' values of the CGPAs will be used.
- 12.11 For Calculations listed in Item 12.6–12.10, performance in failed Subjects/Courses (securing F Grade) will also be taken into account, and the Credits of such Subjects/Courses will also be included in the multiplications and summations. However, Mandatory Courses will not be taken into consideration.
- 12.12 Conversion formula for the conversion of GPA into indicative percentage is
 $[\text{CGPA Earned} - 0.50] \times 10 = \% \text{ of marks scored}$

13. DECLARATION OF RESULTS

Computation of SGPA and CGPA are done using the procedure listed in 12.6– 2.10.
 No SGPA/CGPA is declared, if a candidate is failed in any one of the courses of a given Semester.

14. WITH HOLDING OF RESULTS

If the student has not paid fees to College at any stage, or has pending dues against his name due to any reason what so ever, or if any case of indiscipline is pending against him, the result of the student may be withheld, and he will not be allowed to go into the next higher Semester. The Award or issue of the Degree may also be withheld in such cases.

15. REVALUATION

Students shall be permitted for revaluation after the declaration of end Semester examination results

within due dates by paying prescribed fee. After revaluation if there is any betterment in the grade, then improved grade will be considered. Otherwise old grade shall be retained.

16. SUPPLEMENTARY EXAMINATIONS

16.1 Supplementary examinations for the odd Semester shall be conducted with the regular examinations of even Semester and vice versa, for those who appeared and failed or absent in regular examinations. Such candidates writing supplementary examinations may have to write more than one examination per day.

16.2 ADVANCED SUPPLEMENTARY EXAMINATION

Advanced supplementary examinations will be conducted for IV year II Semester after announcement of regular results.

17. TRANSCRIPTS

After successful completion of prerequisite credits for the award of degree a Transcript containing performance of all academic years will be issued as a final record. Duplicate transcripts will also be issued if required after the payment of requisite fee and also as per norms in vogue.

18. RULES OF DISCIPLINE

- 18.1 Any attempt by any student to influence the teachers, Examiners, faculty and staff of controller of Examination for undue favours in the exams, and bribing them either for marks or attendance will be treated as malpractice cases and the student can be debarred from the college.
- 18.2 When the student absents himself, he is treated as to have appeared and obtained zero marks in that course(s) and grading is done accordingly.
- 18.3 When the performance of the student in any subject(s) is cancelled as a punishment for indiscipline, he is awarded zero marks in that subject(s).
- 18.4 When the student's answer book is confiscated for any kind of attempted or suspected malpractice the decision of the Examiner is final.

19. MALPRACTICE PREVENTION COMMITTEE

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convenor
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

20. TRANSITORY REGULATIONS

Student who has discontinued for any reason, or has been detained for want of attendance or lack of required credits as specified, or who has failed after having undergone the Degree Programme, may

be considered eligible for readmission to the same Subjects/Courses (or equivalent Subjects/Courses, as the case may be), and same Professional Electives/Open Electives (or from set/category of Electives or equivalents suggested, as the case may be) as and when they are offered (within the time-frame of 8 years from the Date of Commencement of his I Year I Semester).

21. AMENDMENTS TO REGULATIONS

The Academic Council of MLR Institute of Technology reserves the right to revise, amend, or change the regulations, scheme of examinations, and / or syllabi or any other policy relevant to the needs of the society or industrial requirements etc., without prior notice.

22. STUDENT TRANSFERS

There shall be no Branch transfers after the completion of Admission Process. Transfer of students from other colleges or universities are permitted subjected to the rules and regulations of TSCHE (TE Department) and JNTUH in vogue.

23. GRADUATION DAY

The College shall have its own Annual Graduation Day for the award of Degrees issued by the College/University.

24. AWARD OF MEDALS

Institute will award Medals to the outstanding students who complete the entire course in the first attempt within the stipulated time.

25. SCOPE

- i) Where the words "he", "him", "his", occur in the write-up of regulations, they include "she", "her".
- ii) Where the words "Subject" or "Subjects", occur in these regulations, they also imply "Course" or "Courses".
- iii) The Academic Regulations should be read as a whole, for the purpose of any interpretation.
- iv) In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of the Academic Council is final.

Academic Regulations for B. Tech. (Lateral Entry Scheme)

(Effective for the students getting admitted into II year from the Academic Year 2018-2019 onwards)

1. The Students have to acquire 120 credits from II to IV year of B.Tech Programme (Regular) for the award of the degree.
2. Students, who fail to fulfil the requirement for the award of the degree in 6 consecutive academic years from the year of admission, shall forfeit their seat.
3. The same attendance regulations are to be adopted as that of B. Tech. (Regular)
4. **Promotion Rule:**
5. A Student will not be promoted from III Year to IV Year, unless he/she fulfils the Attendance and Academic Requirements and (i) secures a Total of 50% Credits up to III Year II Semester, from all the regular and supplementary examinations.
6. **Award of Class:**
After the student has satisfied the requirements prescribed for the completion of the program and is eligible for the award of B. Tech. Degree, he/she shall be placed in one of the following four classes: The marks obtained for 120 credits will be considered for the calculation of CGPA and award of class shall be shown separately.

Table: **Declaration of Class based on CGPA (Cumulative Grade Point Average)**

Class Awarded	Grade to be Secured
First Class with Distinction	CGPA $\geq$ 8.00
First Class	$\geq$ 6.50 to $<$ 8.00 CGPA
Second Class	$\geq$ 5.50 to $<$ 6.50 CGPA
Pass Class	$\geq$ 5.00 to $<$ 5.50 CGPA
FAIL	CGPA $<$ 5

7. All other regulations as applicable for B. Tech. Four-year degree course (Regular) will hold good for B.Tech (Lateral Entry Scheme).

MALPRACTICES RULES - DISCIPLINARY ACTION FOR IMPROPER CONDUCT IN EXAMINATIONS

S. No	Nature of Malpractices / Improper Conduct	Punishment
1 (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, Cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which he is appearing but has not made use of (material shall include any marks on the body of the candidate which can be used as an aid in the subject of the examination)	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other candidate orally or by any other body language methods or communicates through cell phones with any candidate or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the candidates involved. In case of an outsider, he will be handed over to the police and a case is registered against him.

2	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the candidate is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The Hall Ticket of the candidate is to be cancelled and sent to the Principal.
3	Impersonates any other candidate in connection with the examination.	The candidate who has impersonated shall be expelled from examination hall. The candidate is also debarred and forfeits the seat. The performance of the original candidate who has been impersonated, shall be cancelled in all the subjects of the examination (including practical's and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.
4	Smuggles in the Answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
5	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.

6	Refuses to obey the orders of the Addl. Controller of examinations / any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the addl. Controller of examinations or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the addl. Controller of examinations, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which result in damage to or destruction of property in the examination hall or any part of the College campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the candidate(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that Semester/year. The candidates also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.
7	Leaves the exam hall taking away answer script or intentionally tears of the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred for two consecutive Semesters from class work and all examinations. The continuation of the course by the candidate is subject to the academic regulations in connection with forfeiture of seat.
8	Possess any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred and forfeits the seat.

9	If student of the college, who is not a candidate for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Student of the colleges expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year. The candidate is also debarred and forfeits the seat. Person(s) who do not belong to the College will be handed over to police and, a police case will be registered against them.
10	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the candidate has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that Semester/year.
11	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the candidate has appeared including practical examinations and project work of that Semester/year examinations.
12	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the principal for further action to award suitable punishment.	

COURSE STRUCTURE

I B.Tech.- I SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS02	Calculus and Applications	BSC	3	1	0	4	30	70	100
A4BS11	Chemistry	BSC	4	0	0	4	30	70	100
A4CS01	Programming for Problem Solving	ESC	3	0	0	3	30	70	100
A4HS01	English	HSMC	2	0	0	2	30	70	100
A4BS10	Programming for Problem Solving Lab	ESC	0	0	4	2	30	70	100
A4BS12	Chemistry Lab	BSC	0	0	3	1.5	30	70	100
A4HS02	English Language and Communication Skills Lab	HSMC	0	0	2	1	30	70	100
A4HS03	Social Innovation	HSMC	0	0	2	1	30	70	100
TOTAL			12	1	11	18.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC04	Technical Seminar-I	MC	0	0	2	0	30	70	100

I YEAR II SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS03	Integral Calculus and Transforms	BSC	3	1	0	4	30	70	100
A4BS09	Engineering Physics	BSC	3	1	0	4	30	70	100
A4ES01	Engineering Mechanics	ESC	3	1	0	4	30	70	100
A4ES02	Engineering Graphics and Design	ESC	1	0	4	3	30	70	100
A4BS13	Engineering Physics Lab	BSC	0	0	3	1.5	30	70	100
A4AE64	Workshop and Manufacturing Practices	ESC	1	0	4	3	30	70	100
A4HS04	Engineering Exploration	HSMC	0	0	2	1	30	70	100
TOTAL			11	3	13	20.5	210	490	700
Mandatory Course (Non-Credit)									
A4MC05	Technical Seminar-II	MC	0	0	2	0	30	70	100

II B.Tech.- I SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4BS06	Numerical Methods, Probability and Statistics	BSC	3	0	0	3	30	70	100
A4ME01	Mechanics of Solids	PCC	3	0	0	3	30	70	100
A4ME02	Metallurgy and Material science	PCC	3	0	0	3	30	70	100
A4ME03	Production Technology	PCC	3	0	0	3	30	70	100
A4ME04	Thermodynamics	PCC	3	0	0	3	30	70	100
A4ME05	Production Technology Lab	PCC	0	0	3	1.5	30	70	100
A4ME06	Material science and Metallurgy lab	PCC	0	0	3	1.5	30	70	100
A4ME07	Machine Drawing	PCC	0	0	3	1.5	30	70	100
TOTAL			15	0	9	19.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC02	Gender Sensitization	MC	2	0	0	0	30	70	100

II B.Tech.- II SEMESTER

Code	Course	Course Area	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4EE64	Basic Electrical and Electronics Engineering	ESC	3	0	0	3	30	70	100
A4ME08	Kinematics of Machinery	PCC	4	0	0	4	30	70	100
A4ME09	Thermal Engineering-I	PCC	3	0	0	3	30	70	100
A4ME10	Fluid mechanics and Hydraulic Machines	PCC	3	0	0	3	30	70	100
A4ME11	Design of machine members-1	PCC	3	0	0	3	30	70	100
A4ME12	Mechanics of Solids Lab	PCC	0	0	3	1.5	30	70	100
A4ME13	Fluid mechanics and Hydraulic Machines Lab	PCC	0	0	3	1.5	30	70	100
A4EE65	Basic Electrical and Electronics Engineering Lab	ESC	0	0	3	1.5	30	70	100
TOTAL			16	0	9	20.5	240	560	800
Mandatory Course (Non-Credit)									
A4MC01	Environmental Studies	MC	2	0	0	0	30	70	100

III B.Tech.- I SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4ME14	Dynamics of Machinery	PCC	3	0	0	3	30	70	100
A4ME15	Design of Machine Members-II	PCC	3	0	0	3	30	70	100
A4ME16	Metrology and Machine tools	PCC	3	0	0	3	30	70	100
A4HS06	Open Elective-I	HSMC	3	0	0	3	30	70	100
A4ME17	Thermal Engineering-II	PCC	3	0	0	3	30	70	100
A4ME18	Metrology and Machine Tools Lab	PCC	0	0	3	1.5	30	70	100
A4ME19	Thermal Engineering Lab	PCC	0	0	3	1.5	30	70	100
A4ME20	Theory of Machines Lab	PCC	0	0	3	1.5	30	70	100
A4ME21	Summer Internship	PWC	0	0	0	2	-	100	100
TOTAL			15	0	9	21.5	240	660	900
Mandatory Course (Non-Credit)									
A4MC06	Essence of Indian Traditional knowledge	MC	2	0	0	0	30	70	100

III B.Tech.- II SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal	External	L
	Professional Elective-I	PEC	3	0	0	3	30	70	100
A4ME22	Heat Transfer	PCC	3	0	0	3	30	70	100
A4ME23	CAD/ CAM	PCC	3	0	0	3	30	70	100
	Open Elective-II	OEC	3	0	0	3	30	70	100
	Professional Elective-II	PEC	3	0	0	3	30	70	100
A4ME24	Heat Transfer Lab	PCC	0	0	3	1.5	30	70	100
A4ME25	Production Drawing and CAD/ CAM Lab	PCC	0	0	3	1.5	30	70	100
A4ME33	Comprehensive viva	PWC	0	0	0	1	-	100	100
A4HS07	Advanced English communication skills Lab	HSMC	0	0	3	1.5	30	70	100
TOTAL			15	0	9	20.5	240	660	900
Mandatory Course (Non-Credit)									
A4MC03	Constitution of India	MC	2	0	0	0	30	70	100

IV YEAR I SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A4ME26	Composite Materials	PCC	3	0	0	3	30	70	100
A4ME27	Instrumentation and Control Systems	PCC	3	0	0	3	30	70	100
	Professional Elective-III	PEC	3	0	0	3	30	70	100
	Professional Elective-IV	PEC	3	0	0	3	30	70	100
	Open Elective-III	OEC	3	0	0	3	30	70	100
A4ME28	Composite Materials Lab	PCC	0	0	3	1.5	30	70	100
A4ME29	Instrumentation Lab	PCC	0	0	3	1.5	30	70	100
A4ME30	Project Phase -I	PWC	0	0	8	4	100	0	100
TOTAL			15	0	14	22	310	490	800

IV YEAR II SEMESTER

Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
	Professional Elective-V	PEC	3	0	0	3	30	70	100
	Professional Elective-VI	PEC	3	0	0	3	30	70	100
	Open Elective-IV	OEC	3	0	0	3	30	70	100
A4ME32	Project Phase-II	PWC	0	0	16	8	50	150	200
TOTAL			9	0	16	17	140	360	500

PROFESSIONAL ELECTIVES

	<u>Professional Elective-I</u>
A4ME35	FINITE ELEMENT METHODS
A4ME36	NANO TECHNOLOGY
A4ME37	COMPUTATIONAL FLUID DYNAMICS
A4ME38	PRODUCT LIFE CYCLE MANAGEMENT
	<u>Professional Elective-II</u>
A4ME39	INDUSRTIAL ENGINEERING
A4ME40	FRACTURE MECHANICS
A4ME41	REFRIGERATION AND AIR CONDITIONING
A4ME42	DESIGN FOR MANUFACTURING
	<u>Professional Elective-III</u>
A4ME43	INDUSTRIAL ROBOTICS
A4ME44	TOTAL QUALITY MANAGEMENT
A4ME45	AUTOMOBILE ENGINEERING
A4ME46	ADDITIVE MANUFACTURING
	<u>Professional Elective-IV</u>
A4ME47	OPERATIONS RESEARCH
A4ME48	FLEXIBLE MANUFACTURING SYSTEMS
A4ME49	RENEWABLE ENERGY SOURCES
A4ME50	MECHANICAL VIBRATIONS
	<u>Professional Elective-V</u>
A4ME51	POWER PLANT ENGINEERING
A4ME52	MACHINE TOOL DESIGN
A4ME53	AUTOMATION IN MANUFACTURING
A4ME54	VALUE ENGINEERING
	<u>Professional Elective-VI</u>
A4ME55	PRODUCT DESIGN AND PROCESS PLANNING
A4ME56	TRIBOLOGY
A4ME57	MECHATRONICS
A4ME58	UNCONVENTIONAL MACHINING PROCESSES

OPEN ELECTIVE COURSES

OPEN ELECTIVE-I			
S. No.	Course Code	Course Name	Offering Department
1	A4AE17	Fabrication Processes	Aeronautical Engineering
2	A4AE18	Fundamentals of Avionics	
3	A4AE19	Introduction to jets and rockets	
4	A4CS43	Fundamentals of Databases	Computer Science and Engineering
5	A4CS44	Introduction to C++	
6	A4CS45	Core Java Programming	
7	A4EC59	Logic Design	Electronics & Communication Engineering
8	A4EC60	Principles of Communications	
9	A4EC61	Measurements and Instruments	

10	A4EE52	Electrical Wiring and Safety Measures	Electrical & Electronics Engineering
11	A4EE53	Electrical Materials	
12	A4EE54	New trends Electrical Energy	
13	A4IT11	Fundamentals of Data Structures	Information Technology
14	A4IT12	Software Engineering Principles	
15	A4IT13	Computer Organization and Operating Systems	
16	A4ME60	Elements Of Mechanical Engineering	Mechanical Engineering
17	A4ME61	Fundamentals Of Engineering Materials	
18	A4ME62	Fundamentals Of Theory Of Machines	
19	A4HS10	Business Economics and Financial Analysis	HS
20	A4HS11	Basics of Entrepreneurship	
21	A4HS12	Human Values and Professional Ethics	

OPEN ELECTIVE-II			
S. No.	Course Code	Course Name	Offering Department
1	A4AE25	Introduction to aircraft Industry	Aeronautical Engineering
2	A4AE26	Non destructive testing Methods	
3	A4AE27	Fundamentals of Finite element methods	
4	A4CS46	Introduction to Computer Networks	Computer Science and Engineering
5	A4CS47	Introduction to Linux Programming	
6	A4CS48	Fundamentals of Artificial Intelligence	
7	A4EC62	Fundamentals of Integrated Circuits	Electronics & Communication Engineering
8	A4EC63	Signal Transmission through linear systems	
9	A4EC64	Fundamentals of VLSI Design	
10	A4EE55	Power Plant Engineering	Electrical & Electronics Engineering
11	A4EE56	Analysis of Linear Systems	
12	A4EE57	Neural Networks and Fuzzy Logic	
13	A4IT14	Basics of Python Programming	Information Technology
14	A4IT15	Human Computer Interaction	
15	A4IT16	Software Testing Fundamentals	
16	A4ME63	Fundamentals of Mechatronics	Mechanical Engineering
17	A4ME64	Basics Of Thermodynamics	
18	A4ME65	Fundamentals of Smart Materials	
19	A4HS13	Advanced Entrepreneurship	HS

OPEN ELECTIVE -III			
S. No.	Course Code	Course Name	Offering Department
1	A4AE40	Guidance and control of aerospace vehicles	Aeronautical Engineering
2	A4AE41	Wind tunnel Techniques	
3	A4AE42	Introduction to Aerospace Technology	
4	A4CS49	Soft Computing	Computer Science and Engineering
5	A4CS50	Introduction to Cloud Computing	
6	A4CS51	Web Programming	
7	A4EC65	Introduction of Microprocessors and Microcontrollers	Electronics &

8	A4EC66	Fundamentals of Image processing	Communication Engineering
9	A4EC67	TV Engineering	
10	A4EE58	Electrical Engineering Materials	Electrical & Electronics Engineering
11	A4EE59	Non-Conventional Power Generation	
12	A4EE60	Solar Energy and Applications	
13	A4IT17	E- Commerce	Information Technology
14	A4IT18	Cyber Forensics	
15	A4IT19	Discrete Structures	
16	A4ME66	Fundamentals of Operation Research	Mechanical Engineering
17	A4ME67	Fundamentals of Energy Sources and Applications	
18	A4ME68	Basics of Robotics	
19	A4HS14	Indian Ethos & Business Ethics	HS

OPEN ELECTIVE-IV			
S. No.	Course Code	Course Name	Offering Department
1	A4HS15	Management Science	HSM
2	A4HS16	Intellectual Property Rights	
3	A4HS17	Number Theory	
4	A4HS18	Physics and Technology of Thin films	
5	A4HS19	Polymer chemistry	
7	A4CS52	Introduction to Network Security	Computer Science and Engineering
8	A4CS53	Data Analytics	
9	A4CS54	.NET & C# Programming	
10	A4EC68	Introduction to signal Processing	Electronics & Communication Engineering
11	A4EC69	Introduction to mobile communication	
12	A4EC70	Basic Embedded systems Design	
13	A4EE61	Instrumentation and Control	Electrical & Electronics Engineering
14	A4EE62	Energy Audit and Management Systems	
15	A4EE63	Energy Storage Systems	
16	A4IT20	Introduction to Mobile Application Development	Information Technology
17	A4IT21	Distributed Databases	
18	A4IT22	Big Data	
19	A4ME69	Introduction to Material Handling	Mechanical Engineering
20	A4ME70	Non Conventional Energy Sources	
21	A4ME71	Aspects of Heat Transfer	

I B.TECH I SEMESTER SYLLABUS

CALCULUS AND APPLICATIONS

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A4BS02	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OVERVIEW :

This course is intended to enable students to explore on concepts of theory of matrices, ordinary differential equations, sequences and series, definite integrals etc.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Solve the system of linear equations using rank of the matrices.
2. Identify the different types of differential equations and solve them using appropriate methods.
3. Apply the appropriate test to find the convergence and divergence of the given series.
4. Evaluate the improper integrals using beta and gamma functions.
5. Find the Maxima and Minima of several variable functions.

UNIT-I THEORY OF MATRICES

Finding rank of a matrix by reducing to Echelon form, Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix, Eigen values and Eigen vectors and its properties (without proof), Cayley-Hamilton theorem (Statement and verification)-Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalisation of matrices.

UNIT-II ORDINARY DIFFERENTIAL EQUATIONS

Introduction- Exact and reducible to Exact differential equations -Equations not of first degree: Equations solvable for p-equations, solvable for y- equations and solvable for x and Clairaut's type. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\cos ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters.

UNIT-III SEQUENCES AND SERIES

Basic definitions of Sequences and series – Convergence and divergence –Comparison Test- Ratio Test – Raabe's Test-Integral Test – Cauchy's n^{th} root Test –Absolute and Conditional convergence – Power Series.

UNIT-IV BETA GAMMA FUNCTIONS AND APPLICATIONS OF DEFINITE INTEGRALS

Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.

Application of definite integrals: Lengths, evaluate surface areas and volumes of revolution of curves (only in Cartesian co-ordinates).

UNIT-V	CALCULUS OF SEVERAL VARIABLES
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative – Chain rule, Jacobians -functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange's method (with constraints)	
Text Books:	
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.	
Reference Books:	
1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.	
Web references:	
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/	
E -Text Books:	
1.https://www.e-booksdirectory.com/details.php?ebook=10166 2. https://www.e-booksdirectory.com/details.php?ebook=10166	
MOOCS Course:	
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/	

CHEMISTRY								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS11	BSC	L	T	P	C	CIE	SEE	Total
		4	0	0	4	30	70	100
COURSE OVERVIEW : The primary objective of this course is to introduce the students to the concepts and applications of chemistry in Engineering. It should cultivate in them an ability to identify chemistry in each piece of finely engineered products used in households and industry. It aims to strengthen the fundamental concepts of chemistry and then builds an interface with their industrial applications. It deals with applied and industrially useful topics such as Molecular structure, Theory of bonding, Water Technology, Electro chemistry, Sterio chemistry, Spectroscopic techniques etc.								
COURSE OUTCOMES: At the end of the course the students should gain: 1. The knowledge of atomic, molecular and electronic changes, band theory related to conductivity. 2. The required principles and concepts of electrochemistry, corrosion and in understanding the problem of water and its treatments. 3. The knowledge of sterio chemistry, reaction mechanism and synthesis of drug molecules. 4. The required skills to get clear concepts on basic spectroscopy Skill on spectroscopy application to medical and other fields. 5. The knowledge of configurationally and conformational analysis of molecules and reaction mechanisms.								
UNIT-I	MOLECULAR STRUCTURE AND THEORIES OF BONDING							
Atomic and Molecular orbital's. Linear Combination of Atomic Orbital's (LCAO), molecular orbital's of diatomic molecules, molecular orbital energy level diagrams of N2, O2 and F2 molecules. π molecular orbital's of butadiene and benzene. Crystal Field Theory (CFT): Salient Features of CFT – Crystal Field Splitting of transition metal ion d- orbital's in Tetrahedral, Octahedral and square planar geometries. Band structure of solids and effect of doping on conductance.								
UNIT-II	WATER AND ITS TREATMENT							
Introduction – hardness of water – Causes of hardness - Types of hardness: temporary and permanent – expression and units of hardness – Estimation of hardness of water by complexometric method. Potable water and its specifications. Steps involved in treatment of water – Disinfection of water by chlorination and ozonization. Boiler feed water and its treatment – Calgon conditioning, Phosphate conditioning and Colloidal conditioning. External treatment of water – Ion exchange process. Desalination of water – Reverse osmosis. Numerical problems.								

UNIT-III	ELECTROCHEMISTRY AND CORROSION
Electro chemical cells – electrode potential, standard electrode potential, types of electrodes – calomel, Quinhydrone and glass electrode. Nernst equation Determination of pH of a solution by using quinhydrone and glass electrode. Electrochemical series and its applications. Numerical problems. Potentiometric titrations. Batteries – Primary (Lithium cell) and secondary batteries (Lead – acid storage battery). Corrosion: Causes and effects of corrosion – theories of chemical and electrochemical corrosion – mechanism of electrochemical corrosion, Types of corrosion: Galvanic, water-line and pitting corrosion. Factors affecting rate of corrosion, Corrosion control methods- Cathodic protection– Sacrificial anode and impressed current cathodic methods. Surface coatings – metallic coatings –Hot dipping , metal cladding and electro plating(copper plating)..	
UNIT-IV	STEREOCHEMISTRY, REACTION MECHANISM AND SYNTHESIS OF DRUG MOLECULES
Introduction to representation of 3-dimensional structures, Structural and stereoisomers, configurations, symmetry and chirality. Enantiomers, diastereomers, optical activity and Absolute configuration. Conformation analysis of n- butane. Substitution reactions: Nucleophilic substitution reactions: Mechanism of SN₁, SN₂ reactions. Electrophilic and nucleophilic addition reactions: Addition of HBr to propene. Markownikoff and anti Markownikoff's additions. Grignard additions on carbonyl compounds. Elimination reactions: Dehydro halogenation of alkylhalides. Saytzeff rule. Oxidation reactions: Oxidation of alcohols using KMnO₄ and chromic acid. Reduction reactions: reduction of carbonyl compounds using LiAlH₄ & NaBH₄. Hydroboration of olefins. Structure, synthesis and pharmaceutical applications of Aspirin.	
UNIT-V	SPECTROSCOPIC TECHNIQUES AND APPLICATIONS
Principles of spectroscopy, selection rules and applications of electronic spectroscopy, vibrational and rotational spectroscopy. Basic concepts of Nuclear magnetic resonance Spectroscopy, chemical shift. Introduction to Magnetic resonance imaging. Fluorescence and its applications in medicine.	
Text Books:	
1.Engineering Chemistry 14th edition by P.C.Jain & M.Jain; Dhanpat Rai Publishing Company (P) Ltd., New Delhi. 2.Fundamentals of Molecular Spectroscopy 5th edition by C.N. Banwell Mc.Graw-Hills book company.	
Reference Books:	
1. Organic Chemistry 7th Edition, Robert Thornton Morrison, Robert Neilson Boyd, Saibal Kanti Bhattacharjee Pearson Education Singapore Pvt.Ltd. 2. Engineering Chemistry (NPTEL Web-book), 11th edition by B.L. Tembe, Kamaluddin and M.S. Krishnan. 3. Physical Chemistry 11th edition by P.W. Atkins OUP Oxford.	
Web References:	
1. https://www.scribd.com/document/23180395/Engineering-Chemistry-Unit-I-Water-Treatment 2. https://chem.libretexts.org/Core/Inorganic_Chemistry/Descriptive_Chemistry/Periodic_Trends_of_Elemental_Properties/Periodic_Properties_of_the_Elements 3. https://www.khanacademy.org/science/biology/chemistry--of-life/chemical-bonds-and-reactions/v/intermolecular-forces-and-molecular-bonds 4. https://study.com/academy/lesson/the-relationship-between-free-energy-and-the-equilibrium-constant.html	

E-Text Books:

1. <http://www.freebookcentre.net/Chemistry/Chemistry-Books-Online.html>
2. https://www.sdu.dk/en/om_sdu/institutter_centre/fysik_kemi_og_farmaci

MOOC Course:

1. <http://nptel.ac.in/courses/122101001/34>
2. <https://ocw.mit.edu/courses/chemistry/>

PROGRAMMING FOR PROBLEM SOLVING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS01	ESC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100
COURSE OVERVIEW : This course is intended to enable students to acquire knowledge on various concepts of programming for problem solving such as writing algorithms for arithmetic and logical problems, execute programs, synthesizing of complete program etc.								
COURSE OUTCOMES: At the end of the course, student will be able to 1. Formulate simple algorithms for arithmetic and logical problems. 2. Test and execute the programs and correct syntax and logical errors. 3. Implement conditional branching, iteration and recursion. 4. Decompose a problem into functions and synthesize a complete program; Use arrays, pointers, strings and structures to formulate algorithms and programs. 5. Apply programming to solve problems related to matrices, Searching and sorting; Use files to perform read and write operations.								
UNIT-I	INTRODUCTION							
Introduction to Programming: Computer system, components of a computer system, computing environments, computer languages, creating and running programs, Algorithms, flowcharts. Introduction to C language: History of C, basic structure of C programs, process of compiling and running a C program, C tokens, keywords, identifiers, constants, strings, special symbols, variables, data types, I/O statements.								
UNIT-II	OPERATORS, EXPRESSIONS AND CONTROL STRUCTURES							
Operators and expressions: Operators, arithmetic, relational and logical, assignment operators, increment and decrement operators, bitwise and conditional operators, special operators, operator precedence and associativity, evaluation of expressions, type conversions in expressions. Control structures: Decision statements; if and switch statement; Loop control statements: while, for and do while loops, jump statements, break, continue, goto statements.								
UNIT-III	ARRAYS AND FUNCTIONS							
Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays, Basic Algorithms: Searching, Basic Sorting Algorithms- Bubble sort, Insertion sort and Selection sort. Functions: User defined and built-in Functions, storage classes, Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference, Recursion, as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc, Quick sort or Merge sort.								
UNIT-IV	STRINGS AND POINTERS							
Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, string handling functions. Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, functions returning pointers, Dynamic memory allocation.								

UNIT-V	STRUCTURES, UNIONS AND FILE HANDLING
Structures and unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, structures and functions, self referential structures, unions, typedef, enumerations. File handling: command line arguments, File modes, basic file operations read, write and append, example programs	
Text Books:	
1. Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017. 2. E. Balagurusamy, "Programming in ANSI C", McGraw Hill Education, 6th Edition, 2012.	
Reference Books:	
1. W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988. 2. Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003. 3. Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014. 4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012. 5. Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006. 6. Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.	
Web References:	
1. https://www.bfoit.org/itp/Programming.html 2. https://www.khanacademy.org/computing/computer-programming 3. https://www.edx.org/course/programming-basics-iitbombavx-cs101-1x-0 4. https://www.edx.org/course/introduction-computer-science-harvardx-cs50x	
E-Text Books:	
1. http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm 2. http://www.imada.sdu.dk/~svalle/courses/dm14-2005/mirror/c/ 3. http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf	
MOOC Course:	
1. https://onlinecourses.nptel.ac.in/noc18_cs33/preview 2. https://www.alison.com/courses/Introduction-to-Programming-in-c 3. http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.htm	

ENGLISH								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS01	HSMC	L	T	P	C	CIE	SEE	Total
		2	0	0	2	30	70	100
COURSE OVERVIEW : In view of the growing importance of English as a tool for global communication and the consequent emphasis on training students to acquire language skills, the syllabus of English has been designed to develop linguistic, communicative and critical thinking competencies of Engineering students. The students should be encouraged to read the texts leading to reading comprehension and different passages may be given for practice in the class. The time should be utilized for working out the exercises given after each excerpt, and also for supplementing the exercises with authentic materials of a similar kind, for example, newspaper articles, advertisements, promotional material etc. The focus in this syllabus is on skill development, fostering ideas and practice of language skills in various contexts and cultures.								
COURSE OUTCOMES: The course should enable the students to: 1. Improve language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills. 2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically. 3. Analyze a variety of texts and interpret them to demonstrate in writing or speech. 4. Write clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject. 5. Develop language components to communicate effectively in formal and informal situations.								
UNIT-I	THE RAMAN EFFECT							
Chapter entitled ‘The Raman Effect’ from the prescribed textbook, ‘English for Engineers’ published by Cambridge University Press. V- The concept of Word Formation, Root Words and their use in English G- Types of Sentences, Parts of Speech, Nouns, Pronouns, Adjectives R- Reading and its importance W- Sentence Structures, Punctuation, Writing Introductions and Conclusions								
UNIT-II	LETTER WRITING							
Letter Writing. V- Synonyms and Antonyms, Standard Abbreviations G- Verbs, Modal Auxiliaries, Adverbs, Prepositions, Conjunctions R- Improving Comprehension Skills, Reading and Comparing Two Articles W- Writing Paragraphs, Letter Writing- Letters of Request, Apology and Complaint- Letter of Application with Resume.								

UNIT-III	BLUE JEANS
Chapter entitled 'Blue Jeans' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- Prefixes and Suffixes, Idioms and Phrasal verbs G- Articles, Tenses and its forms R- Sub skills of Reading- Skimming and Scanning W- Essay writing and Describing Objects, Places and Events	
UNIT-IV	WHAT SHOULD YOU BE EATING
Chapter entitled 'What Should You Be Eating' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- One word Substitutes, Words often confused G- Active and passive Voice Subject Verb Agreement (Concord) R- Reading Comprehension- Intensive and Extensive W- Technical Report Writing, E-mail writing, Picture Essay	
UNIT-V	HOW A CHINESE BILLIONAIRE BUILT HER FORTUNE
Chapter entitled 'How a Chinese Billionaire Built Her Fortune' from the prescribed textbook, 'English for Engineers' published by Cambridge University Press. V- Technical Vocabulary and their Usage, Misplaced Modifiers, Redundancies G- Direct and Indirect Speech, Degrees of Comparison, Common Errors in English R- Effective Reading and Exercises for Practice W- Memo, Précis and Resume Writing	
Text Books:	
1. Sudarshan, N. P. and Savitha, C. (2018). English for Engineers. Cambridge University Press. 2. Wren & Martin. (2017). High School English Grammar and Composition Book. S Chand Publishing.	
Reference Books:	
1. Murphy, R. (2015). Essential Grammar in Use. Cambridge University Press. 2. Wood, F.T. (2007). Remedial English Grammar. Macmillan. 3. Swan, M. (2016). Practical English Usage. Oxford University Press. 4. Kumar, S and Lata, P. (2018). Communication Skills. Oxford University Press. 5. Zinsser, William. (2001). On Writing Well. Harper Resource Book. 6. Hamp-Lyons, L. (2006). Study Writing. Cambridge University Press. 7. Exercises in Spoken English. Parts I –III. CIEFL, Hyderabad. Oxford University Press.	
Web References:	
1. https://www.oxfordlineenglish.com 2. https://www.bbclearningenglish.com 3. https://www.learnenglish.britishcouncil.org 4. https://www.fluentu.com/english	

E-Text Books:

1. https://www.uop.edu.io/download/research/members/oxford_guide_to_english_grammar.pdf
2. <http://www.espressoenglish.net/wp-content/uploads/2012/07/Free-Grammar-Ebook-Level-2.pdf>
3. <https://update24hour.com/2018/03/wren-and-martin-english-grammar-pdf>

MOOC Course:

1. <http://www.cambridgeenglish.org/learning-english/free-resources/mooc/>
2. <https://www.britishcouncil.org.tr/en/english/mooc>
3. <https://mooc.com>

PROGRAMMING FOR PROBLEM SOLVING LAB

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS02	ESC	L	T	P	C	CIE	SEE	Total
		0	0	4	2	30	70	100

COURSE OVERVIEW:

This course is intended to enable students to acquire knowledge on various concepts of programming for problem solving such as writing algorithms for simple problems, execute programs, Identifying and correcting syntax errors etc.

COURSE OUTCOMES:

At the end of the course, student will be able to

1. Formulate the algorithms for simple problems and translate given algorithms to a working and correct program
2. Correct syntax errors as reported by the compilers and Identify and correct logical errors encountered at run time
3. Write iterative as well as recursive programs and represent data in arrays, strings and structures and manipulate them through a program
4. Declare pointers of different types and use them in defining self-referential structures.
5. Create, read and write to and from simple text files.

LIST OF EXPERIMENTS**Week-1 INTRODUCTION TO LINUX COMMANDS**

- a. Basic Linux commands
- b. Write a C program to use printf() and scanf() functions
- c. Write C programs to implement basic arithmetic operations – sum, average, product, difference, quotient and remainder of given numbers etc.

Week-2 OPERATORS AND EVALUATION OF EXPRESSIONS

- a. Write a C program to check whether a number is even or odd using ternary operator.
- b. Write a C program to perform the addition of two numbers without using +operator.
- c. Write a C program to evaluate the arithmetic expression $((a + b / c * d - e) * (f - g))$. Read the values a, b, c, d, e, f, g from the standard input device.
- d. Write a C program to find the sum of individual digits of a 3 digit number.
- e. Write a C program to read the values of x and y and print the results of the following expressions in one line:
 - i. $(x + y) / (x - y)$
 - ii. $(x + y)(x - y)$

Week-3	CONDITIONAL STATEMENTS
a. Write a C program to find largest and smallest of given numbers. b. Write a C program to find roots of a quadratic equation. c. Write a C program which takes two integer operands and one operator from the user(+,-,*,/,% use switch)	
Week-4	LOOPING STATEMENTS
a. Write a C program to find Sum of individual digits of given integer b. Write a C program to generate first n terms of Fibonacci series c. Write a C program to generate prime numbers between 1 and n	
Week-5	LOOPING STATEMENTS
a. Write a C Program to find the Sum of Series $SUM=1-x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$ b. Write a C program to generate Pascal's triangle. c. Write a C program to generate pyramid of numbers. <pre> 1 1 3 1 1 3 5 3 1 </pre>	
Week-6	ARRAYS
a. Write a C Program to implement following sorting methods i. Bubble sort ii. Selection sort iii. Insertion sort b. Write a C program to find largest and smallest number in a list of integers	
Week-7	ARRAYS
a. Write a C program i. To add two matrices ii. To multiply two matrices b. Write a C program to find Transpose of a given matrix	
Week-8	FUNCTIONS
a. Write a C program to find the factorial of a given integer using functions b. Write a C program to find GCD of given integers using functions c. Write a C Program to find the power of a given number using functions	

Week-9	RECURSION
a. Write a C Program to find binary equivalent of a given decimal number using recursive functions. b. Write a C Program to print Fibonacci sequence using recursive functions. c. Write a C Program to find LCM of 3 given numbers using recursive functions	
Week-10	STRINGS
a. Write a C program using functions to a. Insert a sub string into a given main string from a given position b. Delete n characters from a given position in a string b. Write a C program to determine if given string is palindrome or not	
Week-11	POINTERS AND STRUCTURES
a. Write a C program to print 2-D array using pointers b. Write a C program to allocate memory dynamically using memory allocation functions (malloc, calloc, realloc, free) c. Write a C Program using functions to a. Reading a complex number b. Writing a complex number c. Add two complex numbers d. Multiply two complex numbers Note: represent complex number using structure.	
Week-12	FILES
a. Write a C program to copy one file to other b. Write a C program to copy one file to other c. Write a C Program to merge two files into a third file	
Text Books:	
Yashavant Kanetkar, "Let Us C", BPB Publications, New Delhi, 13 th Edition, 2012. Oualline Steve, "Practical C Programming", O'Reilly Media, 3 rd Edition, 1997.	
Reference Books:	
1. King KN, "C Programming: A Modern Approach", Atlantic Publishers, 2 nd Edition, 2015. Kochan Stephen G, "Programming in C: A Complete Introduction to the C Programming Language", Sam's Publishers, 3 rd Edition, 2004. 3. Linden Peter V, "Expert C Programming: Deep C Secrets", Pearson India, 1 st Edition, 1994.	

Web References:

1. <http://www.sanfoundry.com/c-programming-examples>
2. <http://www.geeksforgeeks.org/c>
3. <http://www.cprogramming.com/tutorial/c>
4. <http://www.cs.princeton.edu>

CHEMISTRY LAB								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS12	BSC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100
COURSE OVERVIEW : The course consists of experiments related to the principles of chemistry required for engineering student. COURSE OUTCOMES: The course should enable the students to: 1. Determine parameters like hardness and chloride content in water. 2. Estimate rate constant of a reaction from concentration – time relationships. 3. Determine physical properties like adsorption and viscosity. 4. Calculate R_f values of some organic molecules by TLC technique 5. Synthesise Thiokol Rubber.								
LIST OF EXPERIMENTS								
Week-1	DETERMINATION OF TOTAL HARDNESS OF WATER BY COMPLEXOMETRIC METHOD USING EDTA							
Week-2	ESTIMATION OF AN HCL BY CONDUCTOMETRIC TITRATIONS							
Week-3	ESTIMATION OF ACETIC ACID BY CONDUCTOMETRIC TITRATIONS							
Week-4	ESTIMATION OF HCL BY POTENTIOMETRIC TITRATIONS							
Week-5	ESTIMATION OF ACETIC ACID BY POTENTIOMETRIC TITRATIONS							
Week-6	DETERMINATION OF RATE CONSTANT OF ACID CATALYSED HYDROLYSIS OF METHYL ACETATE							
Week-7	SYNTHESIS OF ASPIRIN							
Week-8	THIN LAYER CHROMATOGRAPHY CALCULATION OF R_f VALUES. EG ORTHO AND PARA NITRO PHENOLS							
Week-9	VERIFICATION OF FREUNDLICH ADSORPTION ISOTHERM-ADSORPTION OF ACETIC ACID ON CHARCOAL							
Week-10	DETERMINATION OF VISCOSITY OF CASTOR OIL AND GROUND NUT OIL BY USING OSTWALD'S VISCOMETER							
Week-11	DETERMINATION OF SURFACE TENSION OF A GIVE LIQUID USING STALAGMOMETER							
Week-12	SYNTHESIS OF THIOKOL RUBBER							

Week-13	DETERMINATION OF CHLORIDE CONTENT OF WATER USING ARGENTOMETRIC METHOD
Week-14	DETERMINATION OF RATE CONSTANT OF ACID CATALYSED HYDROLYSIS OF METHYL ACETATE
Reference Books:	
1. Senior practical physical chemistry, B.D. Khosla, A. Gulati and V. Garg (R. Chand & Co., Delhi). 2. An introduction to practical chemistry, K.K. Sharma and D. S. Sharma (Vikas publishing, N. Delhi). 3. Vogel's text book of practical organic chemistry 5th edition. 4. Text book on Experiments and calculations in Engineering chemistry – S.S. Dara.	
Web References:	
1. http://amrita.olabs.edu.in/?sub=73&brch=8&sim=153&cnt=2 2. https://chem.libretexts.org/Textbook_Maps/Analytical_Chemistry_Textbook_Maps/Map%3A_Analytical_Chemistry_2.0_(Harvey)/11_Electrochemical_Methods/11.2%3A_Potentiometric_Methods 3. http://fch.upol.cz/skripta/fcc_and_zvem_english/FCH/Adsorption%20of%20oxalic%20acid%20on%20activated%20charcoal.htm 4. https://www.askiitians.com/iit-jee-chemistry/physical-chemistry/electrolytic-conductance-molar-conductance-and-specific-conductance.aspx 5. https://owlcation.com/stem/tlc-thin-layer-chromatography-Principle-Procedure	

ENGLISH LANGUAGE AND COMMUNICATION SKILLS LAB

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS02	HSMC	L	T	P	C	CIE	SEE	Total
		0	0	2	1	30	70	100

COURSE OVERVIEW :

In this competitive world, the students need to prepare themselves for their career which requires them to listen to, read, speak and write in English both for their professional and interpersonal communication. The main purpose of this course is to enhance communication skills of students of Engineering and make them employable.

COURSE OUTCOMES:

At the end of course , students will have ability to:

1. Acquire better perception of nuances of English language through audio- visual experience
2. Neutralize accent for intelligibility
3. Take part in group activities
4. Have Speaking skills with clarity and confidence
5. Enhance their employability

LIST OF ACTIVITIES**Week-1 GENERAL INTRODUCTIONS AND FORMAL GREETINGS**

- a. Introductions and greetings in formal and informal situations
- b. Worksheets to extract information
- c. Questionnaires to enquire about the expressions used during formal introductions
- d. Ice Breaking activity by preparing and asking five questions each
- e. Creation of dialogues using greetings, leave- taking and introductions

Week-2 JAM- JUST A MINUTE

- a. Strengthen the ability to analyze a topic and logical organization of thoughts.
- b. Logically starting with introductory sentence, points of discussion and closing sentence.
- c. Practicing to speak within one minute
- d. Activity based on JAM on a familiar topic
- e. Planning and executing JAM considering the parameters

Week-3 PHONETICS

- a. Speech sounds and their prominence in pronunciation
- b. Understanding and practicing word stress
- c. Neutralizing the accent and practicing the right intonation
- d. Knowing the differences between different accents
- e. Increase fluency with the help of Phonetics

Week-4	LISTENING SKILLS
a. Developing good listening skills for effective communication b. Enhancing listening skills through audio tracks and oral conversation c. Empathizing others point of view while they speak d. Incorporating non verbal communications while listening e. Improving overall performance listening to the audio tracks	
Week-5	SITUATIONAL DIALOGUES AND GIVING DIRECTIONS
a. Creating dialogues in any given situations b. Framing and choosing appropriate words to frame the dialogues in any situation c. Guiding and giving directions using appropriate expressions d. Activities on how to make polite requests, offers, rejections etc e. Practicing to speak confidently in different situations	
Week-6	ROLE PLAY
a. Understanding a Role play and its procedure b. Planning and Executing a Role Play accordingly c. Practicing to get into the role and perform within stipulated time d. Activities based on Role Play with different situations e. Performing a Role Play considering the parameters	
Week-7	GROUP DISCUSSIONS
a. Understanding a Group Discussion (GD) and its procedure b. Following the rules of a GD c. Planning and Executing a GD within the stipulated time d. Activities based on GD e. Performing a GD considering the parameters	
Week-8	DEBATE
a. Understanding the procedure of a Debate b. Planning and executing a Debate following its rules c. Strengthen the ability to analyze a topic and logical organization of thoughts. d. Logically arranging the arguments e. Performing a Debate considering the parameters	
Week-9	TELEPHONIC ETIQUETTES
a. Understanding basic Telephonic Etiquettes b. The approach one needs to follow while making and answering a call c. Making a formal telephonic conversation d. Activities based on modulating voice and tone e. Interpersonal skills required to overcome rude and hostile behavior f. Write a C program to compare two strings using pointers. g. Write a C program to copy a string from source to destination using pointers. h. Write a C program to reverse a string using pointers.	

Week-10	PRESENTATION SKILLS
a. Planning a Presentation b. Enhancing skills required for making effective presentations c. Usage of different tools that help us to give effective presentations d. Executing a presentation effectively e. Activities based on presentations	
Week-11	ORAL PRESETATIONS AND EXTEMPORE
a. Planning an oral presentation or an Extempore b. Preparing good PPT c. Using appropriate body language in public speaking domain d. Planning and Executing oral presentation e. Activities based on oral presentations and extempore	
Week-12	INTERVIEW SKILLS
a. Preparing to succeed in Interviews b. Preparing a strong Resume for interviews c. Practicing different techniques to overcome nervousness in interviews d. Using appropriate body language in interviews e. Activities based on Interviews skills	
Week-13	INFORMATION TRANSFER
a. Extracting Information Transfer from different kinds of representation b. Reading and decoding the information given in various types c. Representing the information in charts or graphs in a written document d. Developing writing skills from these aspects e. Activity on transferring given data into graphs or charts for presentation skills	
Reference Books:	
1. E. Suresh Kumar. A Handbook for English Language Laboratories (with CD) Revised Edition 2. Normal Whitby. Business Benchmarch. Cambridge University Press(with CD) 2 nd Edition 3. Liz Hamp-Lyons and Ben Heasly. Study Writing.Cambridge University Press. 2006. 4. Sanjay Kumar and Pushp Lata.Communication Skills. Oxford University Press. 2011. 5. Exercises in Spoken English. Parts. I-III. CIEFL, Hyderabad. Oxford University Press 6. Raman Sharma, Technical Communications, Oxford Publication, London, 2004. 7. Dale Jungk, Applied Writing for Technicians, McGraw Hill, New York, 2004. 8. T. Balasubramanian, A Text book of English Phonetics for Indian Students, Macmillan, 2008. 9. Edgar Thorpe, Winning at Interviews, Pearson Education, 2006. 10.J. Sethi et al, A Practical Course in English Pronunciation (with CD), Prentice Hall of India, 2005. 11.Hari Mohan Prasad, How to Prepare for Group Discussions and Interviews, Tata McGraw Hill, 2006	

Websites:

1. <https://www.britishcouncil.org>
2. <https://www.bbc.co.uk>
3. <https://www.grammarly.com>
4. <https://www.fluentu.com>
5. <https://www.cambridgeenglish.org/exams-and-tests/business-preliminary>
6. <https://www.cambridgeenglish.org/exams-and-tests/business-vantage>

SOCIAL INNOVATION

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS03	HSMC	L	T	P	C	CIE	SEE	Total
		2	-	-	1	30	70	100

COURSE OVERVIEW:

Social innovation is the process of developing and deploying effective solutions to challenging and often systemic social and environmental issues in support of social progress. Social innovation is not the prerogative or privilege of any organizational form or legal structure. Solutions often require the active collaboration of constituents across government, business, and the nonprofit world. This course will enable students to understand concept of social innovation, interact and engage with society and present different case studies with a focus on impact and vision on society.

COURSE OUTCOMES:

The course should enable the students to

1. Demonstrate social innovation concepts and approaches.
2. Discuss the community problems, social and economical change.
3. Identify new and unaddressed social needs.
4. Analyze social innovation disclosures in different sectors.
5. Design innovative solutions with Social impact through application of new models of leadership, collective intelligence and creativity techniques.

UNIT-I**INTRODUCTION TO SOCIAL INNOVATION**

Core definitions, core elements and common features of social innovation, a topology of social innovations, history of social innovation, social and economic change, Swachh Bharat, Unnat Bharat Abhiyan, National Service Scheme (NSS).

UNIT-II**INTERACTION AND ENGAGEMENT WITH SOCIETY**

Engage with community, interact with them to understand the community problems, Understanding social and economical change – individuals, organizations and movements.

UNIT-III**PROCESS OF SOCIAL INNOVATION**

Understanding the pain/need, description and problem definition, social and economic constraints for affordable and appropriate technology.

UNIT-IV**SOCIAL INNOVATION ACROSS FOUR SECTORS IN INDIA AND GLOBAL SCENARIO**

The four sectors – the non-profit sector, public sector, the private sector, the informal sector, links between and cross sectors.

UNIT-V**SOCIAL INNOVATION – CASE STUDIES**

Designing and implementing social innovations, report writing and documentation, presentation of the case studies with a focus on impact and vision on society.

Text Books:

1. The Power of Social Innovation: How Civic Entrepreneurs Ignite Community Networks for Good 1st Edition by Stephen Goldsmith, Michael R. Bloomberg, Gigi Georges, Tim Glynn Burke.
2. The Open Book of Social Innovation: Ways to Design, Develop and Grow Social Innovation Paperback – March, 2010 by Robin Murray, Julia Caulier-Grice, Geoff Mulgan.

Reference Books:

1. Social innovator series: ways to design, develop and grow social innovation, the open book of social innovation by robin murrayjuliecaulier-gricegeoffmulgan.
2. The International Handbook on Social Innovation: Collective Action, Social Learning and Transdisciplinary Research Paperback by Frank Moulaert , Diana MacCallum.
3. Guide to Social Innovation by Johannes HAHN and Laszlo ANDOR

Web References:

1. [http://s3platform.irc.ec.europa.eu/documents/20182/84453/Guide to Social Innovation.pdf](http://s3platform.irc.ec.europa.eu/documents/20182/84453/Guide%20to%20Social%20Innovation.pdf)
2. <https://www.si-drive.eu/wp-content/uploads/2016/12/SI-DRIVE-CA-short-2016-11-30-Druckversion.pdf>

E-Text Books:

1. <https://epdf.tips/the-power-of-social-innovation-how-civic-entrepreneurs-ignite-community-networks.html>
2. <https://youngfoundation.org/wp-content/uploads/2012/10/The-Open-Book-of-Social-Innovationg.pdf>
3. <http://www.idmais.org/desislab/wp-content/media/social.pdf>

MOOC Course:

1. <https://iversity.org/en/courses/social-innovation-101-en>
2. <https://www.class-central.com/tag/social%20innovation>
3. <https://www.mooc-list.com/tags/social-innovation>

I B.TECH II SEMESTER SYLLABUS

INTEGRAL CALCULUS AND TRANSFORMS**II Semester: ECE/EEE/MECH/AERO**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS03	BSC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	30	70	100

COURSE OVERVIEW :

Integral calculus deals with total size or value, such as lengths, areas, and volumes. An integral transform is useful if it allows one to turn a complicated problem into a simpler one. This part of the course introduces two extremely powerful methods to solving differential equations.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Evaluate multiple integrals.
2. Verify vector integral theorems.
3. Solve the differential equations using Laplace transform techniques.
4. Find the Fourier series of the given functions.
5. Find the Fourier transforms of the given functions.

UNIT-I MULTIPLE INTEGRALS

Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double, centre of mass and gravity.

UNIT-II VECTOR CALCULUS

Scalar and vector point functions - Gradient, divergence, curl and their related properties - Solenoidal and irrotational vector point functions - Scalar potential function - Laplacian operator - Line integral - work done - surface integrals - volume integral - Vector integral theorems - Green's theorem in a plane - Stoke's theorem - Gauss divergence theorem (all theorem statements and their verification).

UNIT-III LAPLACE TRANSFORMS

Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n - Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem – Applications of Laplace transforms to ordinary differential equations.

UNIT-IV	FOURIER SERIES
Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Even Odd periodic continuation-Half range Fourier sine and cosine expansions.	
UNIT-V	FOURIER TRANSFORMS
Fourier integral theorem (statement)-Fourier sine and cosine integrals –Fourier transforms –Fourier sine and cosine transforms-properties- Inverse transforms-Finite Fourier transforms.	
Text Books:	
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.	
Reference Books:	
1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.	
Web references:	
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/	
E -Text Books:	
1. https://www.e-booksdirectory.com/details.php?ebook=10166 2. https://www.e-booksdirectory.com/details.php?ebook=10166	
MOOCS Course:	
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/	

ENGINEERING PHYSICS								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS09	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	30	70	100
COURSE OVERVIEW : The course aims at making students to understand the basic concepts of principles of Physics in a broader sense with a view to lay foundation for the various engineering courses .								
COURSE OUTCOMES: The student will able to 1. Analyze the Optical phenomenon of light and estimate their applications in different situations 2. Evaluate the mobility of charge carrier concentration of a given semiconductor material. 3. Justify how the graded index optical fiber is more efficient than step index optical fiber in fiber optic communication system. 4. Classify various magnetic, dielectric materials and can apply knowledge in various fields 5. Describe the behaviour of waves at interfaces (impedance) and their behaviour in dissipative media (damping)								
UNIT-I	WAVE OPTICS							
Huygen's principle, Superposition of waves and interference of light by wave front splitting and amplitude splitting, Young's double slit experiment, Newton's rings, Michelson's interferometer, Mach-Zehnder interferometer, Fraunhofer diffraction from a single slit and circular aperture, Diffraction grating: Grating spectrum and resolving power.								
UNIT-II	QUANTUM PHYSICS							
Black body radiation, Planck's law, photoelectric effect Compton effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer's Experiment, Heisenberg's Uncertainty Principle, Schrodinger's Time Independent Wave Equation-Physical Significance of the wave Function, Particle in One Dimensional Potential Box.								
UNIT-III	LASER & FIBER OPTIC COMMUNICATION							
Laser: Characteristics of Laser beams, Energy levels in atoms, Einstein's theory of matter radiation interaction and A and B coefficients; amplification of light by population inversion, different types of lasers: gas lasers -He-Ne, solid-state lasers -ruby, Neodymium, Applications of Lasers.								
Fiber Optic Communication: Structure of Optical fibers, Basic principle of fibers, Acceptance angle and Numerical aperture, Types of Optical Fibers-Step Index and Graded Index fibers; Modes of fibers-SMSI, MMSI, MMGI., Optical fiber Communication System with block diagram. Applications of fibers, fiber optic sensors – Basic principle, Intrinsic, Extrinsic sensors. Working of Pressure and Temperature Sensors.								

UNIT-IV	DIELECTRICS AND MAGNETISM
Dielectrics: Introduction, Types of polarizations (Electronic, Ionic and Orientation Polarizations) and calculation of their polarizabilities, Internal fields in a solid, Clausius-Mossotti relation. Magnetism: Introduction, Bohr magneton, classification of dia, para and ferro magnetic materials on the basis of magnetic moment, Hysteresis curve based on domain theory, Soft and hard magnetic materials, Properties of anti-ferro and ferri magnetic materials.	
UNIT-V	INTRODUCTION TO OSCILLATIONS
Harmonic oscillators: Mechanical and electrical simple harmonic oscillators, Damped harmonic oscillator – heavy, critical and light damping, energy decay in a damped harmonic oscillator, quality factor. Non-dispersive transverse and longitudinal waves in one dimension Transverse wave on a string, the wave equation on a string, Harmonic waves, reflection and transmission of waves at a boundary, impedance matching, standing waves and their eigen frequencies, longitudinal waves and the wave equation for them, acoustics waves and speed of sound, standing sound waves.	
Text Books:	
1. A textbook of Engineering Physics, Dr. M.N. Avadhanulu, Dr. P.G.Kshirsagar – S.Chand 2. Haliday and Resnick, Physics-Wiley 3. P K Palanisamy, Engineering Physics, Sitech Publications, 2008. 4. D. J. Griffiths, “Quantum mechanics”, Pearson Education, 2014.	
Reference Books:	
1. Richard Robinett, Quantum mechanics. 2. O.Svelto, Principles of lasers. 3. Bottaccini M.R, E.E. Merill, Instruments and Measurements, Bell and Howell 4. I. G. Main, “Vibrations and waves in physics”, Cambridge University Press, 1993. 5. Introduction to Solid State Physics by Charles Kittel, Wiley student edition 6. Ajoy Ghatak, “Optics”, Mc Graw-Hill Education, 2012	
Web References:	
1. https://www.edx.org/course?search_query=semiconductor+physics 2. https://www.edx.org/course/nanotechnology-fundamentals-purdue-nano530x 3. https://www.edx.org/course/physics-electronic-polymers-pep-purdue-nano600	
E-Text Books:	
1. https://www.researchgate.net/publication/274876688_Vibrations_and_Waves_in_Physics. 2. http://www.phys.sinica.edu.tw/TIGP-NANO/Course/2010_Fall/classnotes/NanoB_week14.pdf 3. https://subodhtrpathi.files.wordpress.com/2012/01/optical-fiber-communications-by-gerd-keiser_2.pdf	
MOOC Course:	
1. http://nptel.ac.in/noc/individual_course.php?id=noc16-ch06 2. http://nptel.ac.in/courses/104104085/2	

ENGINEERING MECHANICS								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ES01	ESC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	30	70	100
COURSE OVERVIEW: Engineering mechanics is the study of forces that act on bodies and the resultant motion that those bodies experience. This course is designed to introduce the basic principles of engineering mechanics with emphasis on analysis of rigid bodies in rest and their application to solve real-time engineering problems.								
COURSE OUTCOMES : The course should enable the students to: 1. Resolve various force systems and calculate reactions of supports on rigid bodies in equilibrium 2. Analyze frames and frictional forces of rigid bodies on rough horizontal and inclined planes. 3. Evaluate Geometric properties of composite areas and solids. 4. Analyze rigid bodies in motion, work-energy problems and their relationship to engineering applications 5. Explain basic concepts of mechanical vibrations.								
UNIT-I		INTRODUCTION & EQUILIBRIUM OF PLANAR SYSTEM OF FORCES						
Introduction: Basic concepts, Classification of a force system, Parallelogram law, Resultant of coplanar concurrent force system, Moment of a force, its applications, Varignon's theorem, Resultant of coplanar non-concurrent force systems, Couples. Equilibrium of Planar System of forces: Equilibrium, Conditions of equilibrium, Lami's theorem, free body diagrams, Types of supports, their reactions, Analysis of beams.								
UNIT-II		ANALYSIS OF PERFECT FRAMES & FRICTION						
Analysis of perfect frames: Types of frames, Method of joints and Method of Sections Friction: Introduction, types of friction, laws of friction, coefficient of friction, angle of friction, angle of repose, analysis of bodies on rough horizontal, inclined planes, wedge friction, ladder friction and screw friction.								
UNIT-III		CENTROID, CENTRE OF GRAVITY & MOMENT OF INERTIA						
Centroid & Centre of Gravity: Introduction, centroids & centre of gravity of simple figures (from first principles), centroid of composite sections, centre of gravity of composite solids, theorems of pappus & guldinus. Moment of Inertia: Area moment of inertia of plane sections (from first principles), Parallel axis theorem, Perpendicular axis theorem, Moment of inertia of standard sections and composite sections. Mass moment of inertia of circular disc, cylinder and sphere.								
UNIT-IV		KINEMATICS & KINETICS						
Kinematics of a particle: Rectilinear motion, curvilinear motion, direct and oblique impact. Kinetics of rigid bodies: Analysis of connecting bodies, kinetics of rigid body rotation, D' Alembert's principle, work-energy theorem, impulse-momentum principle, their applications								
UNIT-V		VIRTUAL WORK & MECHANICAL VIBRATIONS						
Virtual work: Concept of virtual work, Principle of virtual work, its applications Mechanical Vibrations: Basic terminology, free and forced vibrations, resonance, simple harmonic motion, simple, compound and torsional pendulums.								

Text Books:

1. Reddy Vijay Kumar and K. Suresh Kumar, Singer's Engineering Mechanics
2. Engineering Mechanics/Timoshenko and D.H. Young, Mc Graw Hill Book Company

Reference Books:

1. Engineering Mechanics/Irving Shames/Prentice Hall
2. Engineering Mechanics/N.H Dubey/Tata Mc Graw Hill
3. Tayal A.K, Engineering Mechanics, Umesh Publications

Web References:

1. <http://nptel.ac.in/courses/112103109/>
2. <http://www.iitq.ac.in/rkbc/me101/Presentation/L01-03.pdf>
3. <https://www.udemy.com/myengineeringmechanics/>

E-Text Books:

1. <https://www.pdfdrive.net/engineering-mechanics-e15629836.html>
2. <https://www.pdfdrive.net/engineering-mechanics-dynamics-si-version-e4362556.html>
3. <https://www.pdfdrive.net/engineering-mechanics-statics-3rd-ed-e4229691.html>

MOOC Course:

1. <https://www.coursera.org/learn/engineering-mechanics-statics>
2. <https://www.edx.org/course/engineering-mechanics>

ENGINEERING GRAPHICS AND DESIGN

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ES02	ESC	L	T	P	C	CIE	SEE	Total
		1	0	4	3	30	70	100

COURSE OVERVIEW :

This course is introduced to provide the basic understanding of the fundamentals of Engineering graphics and design, mainly principles of engineering graphics, their significance, conventions and standards related to engineering drawings, concepts of orthographic and isometric projections, computer aided drafting and modeling.

COURSE OUTCOMES:

The course should enable the students to

1. Illustrate various concepts of dimensioning, conventions and standards related to engineering drawings. Also learn to sketch basic engineering curves.
2. Compare different angles of projections and solve projection of lines and planes.
3. Prepare drawings of projection of solids and also develop new shapes in sheet metal applications using AutoCAD software.
4. Analyze the given drawings and convert isometric projections to orthographic and vice-versa using AutoCAD software.
5. Develop three-dimensional models of engineering components.

UNIT-I**INTRODUCTION**

Principles of Engineering Graphics and their significance, usage of drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloid, Hypocycloid and Involute.

UNIT-II**DRAWING OF PROJECTIONS OR VIEWS: ORTHOGRAPHIC PROJECTION IN FIRST ANGLE PROJECTION ONLY**

Principles of orthographic projections – conventions – first and third angle projections. Projections of points- Projection of lines inclined to both the planes.

PROJECTIONS OF PLANES: Projections of regular planes, inclined to both planes.

UNIT-III**INTRODUCTION TO COMPUTER AIDED DRAFTING**

INTRODUCTION TO COMPUTER AIDED DRAFTING: Generation of points, lines, curves, polygons, simple solids, dimensioning. LAYERS: Concept of layers, working with layers, creating, display, locking, unlocking, and delete commands, Coordinate system in AutoCAD, UCS, WCS, MCS

PROJECTION OF SOLIDS-Solids inclined to both planes.

DEVELOPMENT OF SURFACES OF SOLIDS: Theory of development, development of lateral surface along with base

UNIT-IV	ISOMETRIC DRAWINGS
Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections; Drawing Isometric circles, Dimensioning Isometric Objects; Conversion of Isometric view to Orthographic views and Orthographic to isometric views	
UNIT-V	3D MODELING
Types of 3D models, 3D Coordinate Systems, basic commands in 3D, PEDIT command. CREATING SOLID MODELS: creating pre-defined Solid Primitives, Dynamic UCS, methods of creating solids by - Extrude, Revolve, Swept, Loft, & Presspull, in 3Dcreating solid models, Dynamic UCS. MODIFYING 3D OBJECTS: Fillet, Chamfer, Rotate, Mirror, Array, Slicing solid Models. EDITING 3D OBJECTS: SOLVIEW, SOLDRAW, SOLPROF	
Text Books:	
1. Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar Publishing House 2. Shah, M.B. & Rana B.C. (2008), Engineering Drawing and Computer Graphics, Pearson Education 3. Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMH Publication 4. Narayana, K.L. & P Kanniah (2008), Text book on Engineering Drawing, Scitech Publishers . 5. D.M. Kulkarni,A.P.Rastogi,A.K. Sarka "Engineering Graphics with AutoCAD" PHI publications, 2013	
Reference Books:	
1. Johle (2009), Engineering Drawing, Tata Mc Graw Hill, New Delhi, India. 2. Trymbaka Murthy (2007), Computer Aided Engineering Drawing, I.K. International Publishers, New Delhi. 3. Sham Tickoo, D. saravanan, "AutoCAD 2010 for engineers and designers" Dreamtech Press, 2010 2. 4. Sham Tickoo " AutoCAD 2011: A Problem solving approach" Autodesk Press, USA.	
Web References:	
1. nptel.ac.in/courses/112103019/ 2. web.iitd.ac.in/~achawla/public_html/201/lectures/sp46.pdf	
E-Text Books:	
1. https://books.google.co.in/books?id=_hdOU8kRb2AC&printsec=frontcover&source=gbg_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://www.pdfdrive.com/engineering-drawing-by-nd-bhatt-e43754521.html	
MOOC Course:	
1. https://www.udemy.com/topic/autocad/ 2. https://nptel.ac.in/courses/112104172/1	

ENGINEERING PHYSICS LAB								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS13	BSC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100
COURSE OVERVIEW : The basic focus in the Engineering Physics Laboratory is on various experiments to realize the theoretical concepts of Physics. It aims to develop scientific temper and encourage students for innovative engineering inventions through Physics experiments.								
COURSE OUTCOMES: At the end of the course student will be able 1. To find dispersive power, angular rotation, dispersion using minimum deviation method for a given prism and V-I, P-V and P-I characteristics of a solar cell 2. To find the numerical aperture and acceptance angle of an optical fiber. 3. To determine the radius of curvature of a given convex lens by using Newton's Rings apparatus 4. To determine energy gap of a semi conductor and to Study the magnetic field along the axis of a current carrying coil 5. To analyze LCR and RC circuits malus law, torsional oscillator, laser-diffraction and melde's experiment								
LIST OF EXPERIMENTS								
Week-1	DISPERSIVE POWER OF PRISM-SPECTROMETER							
a. Find the dispersive power of a material of prism using Spectrometer for angle of minimum deviation. b. Determine the angular rotation for a given wavelength for a given prism. What is your analysis? c. Using minimum deviation method how do you find the dispersion of a given prism. Analyze the results. d. Let the angle of the prism is 50 degree and refractive index of the prism is 1.33. Find the angle of minimum deviation. e. Define refractive index and minimum deviation. Evaluate the dispersive power of prism for blue and green colour.								
Week-2	SOLAR CELL							
a. Study the V-I characteristics of a solar cell for the voltage interval of 0.4 V when the given electric bulb is at a distance of 10cm. Will there be any change if the distance is doubled? b. Study the P-V characteristics of a solar cell for the voltage interval of 0.3 V for four different distances of the given electric bulb. Discuss your observations. c. Study the P-I characteristics of a solar cell for the given electric bulb for the distances 30 cm. What is expected if the distance is doubled?								

Week-3	OPTICAL FIBER
To determine the numerical aperture and acceptance angle of an optical fiber.	
Week-4	NEWTON'S RINGS
To determine the radius of curvature of a given convex lens by using Newton's Rings apparatus.	
Week-5	ENERGY GAP OF PN JUNCTION DIODE
a. Using PN junction diode, determine the energy gap of a semiconductor by applying 2V of reverse bias by increasing the temperature from 30-70 degree centigrade. Analyze the obtained results. b. Evaluate the energy gap of Si PN junction diode by applying 1.5V of reverse bias for values of temperature 70,60,50,40,30 centigrade. c. Evaluate the energy gap of a semiconductor by applying 0.5V of reverse bias for temperatures at 75, 65, 55, 45 and 35 centigrade.	
Week-6	STEWART & GEE'S EXPERIMENT
a. Study the magnetic field along the axis of a current carrying coil of 500 turns for a constant current for the distances 0, 10, 15 cm respectively. Analyze the graph obtained. b. Study the magnetic field along the axis of a current carrying coil of 50 turns by applying current 100mA. for the distances 0,12,25 cm. Evaluate the result when the no. of turns are doubled.	
Week-7	LCR CIRCUIT
a. To determine the Resonance frequency and Quality factor of a LCR Circuit b. Study the Resonance frequency of an electrical circuit by varying values of inductance, Capacitance and resistors and analyze the results.	
Week-8	MALUS LAW
To study Polarization of light and verification of Maluslaw .	
Week-9	RC CIRCUIT
a. Analyze the time constant of R-C circuit by varying Resistance and Capacitance values in a electrical circuit. b. Determine the time constant of a given RC circuit by plotting a graph between Charging current versus time.	

Week-10	TORSIONAL OSCILLATOR
a. Making use of a torsional oscillator of 300gms circular disc, determine the rigidity modulus of given steel wire for 65 cm & 55 cm lengths. Analyze the results obtained. b. Determine the rigidity modulus of the given copper wire for 50, 40, 30 lengths by using torsional oscillator of 400gms circular disc. Analyze the results obtained. c. Give your analysis of $L \propto T^2$ behavior of a torsional oscillator. You may pick your own values for the analysis.	
Week-11	LASER - DIFFRACTION GRATING
a. Using a diffraction grating element of 2500 LPI determine the wavelength of LASER source for first and second order diffraction when the distance between the screen and grating is 50cm. What is your analysis? b. Determine the wavelength of a LASER source for first three orders of diffraction by maintaining a distance of 30 cms between grating material and the screen. Use diffraction grating element of 15000 LPI. What differences do you observe for the three orders.	
Week-12	MELDE'S EXPERIMENT
a. Determine the longitudinal frequency of tuning fork by using 100cm length of the thread by varying masses of 5gms, 10gms. Discuss your findings. b. Find out the transverse frequency of tuning fork by using 80cm length of the thread and by varying masses of 10gms and 40gms. Analyze the results obtained.	
Reference Books:	
1. "Optics, Principles and Applications" by K K Sharma. 2. "Principles of Optics" by M Born and E Wolf. 3. "Semiconductor Physics and Devices: Basic Principles" by Donald A Neamen. 4. "Oscillations and Waves" by Satya Prakash and Vinay Dua. 5. "Waves and Oscillations" by N Subrahmanyam and Brij Lal.	
Web References:	
1. http://www.arxiv.org/pdf/1510.00032 2. http://www.nptel.ac.in/courses/122103010/ 3. http://www.researchgate.net/.../276417736 Video Presentations in Engineering-Ph... 4. http://www.wileyindia.com/engineering-physics-theory-and-practical.html	

Note: Students can perform any 8 experiments.

WORKSHOP AND MANUFACTURING PRACTICES

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE64	ESC	L	T	P	C	CIE	SEE	Total
		1	0	4	3	30	70	100

COURSE OVERVIEW:

This basic focus of this course is to impart knowledge of various manufacturing processes, tools and equipment. It helps to develop and enhance relevant technical hand skills required to work in various engineering industries.

COURSE OUTCOMES:

At the end of course the student will be able to:

1. Prepare basic joints used in carpentry
2. Prepare edges for better joint for fitting
3. Perform basic house wiring connections
4. Prepare various shapes and objects by using Tin smithy Foundry and Black smithy
5. Perform machining operations.

VIDEO LECTURES

Week-1	Carpentry
Week-2	Fitting operations & Power Tools
Week-3	Casting
Week-4	Forming
Week-5	Machining
Week-6	Joining – Bolted & Riveted Joints
Week-7	Arc welding & Gas welding
Week-8	Brazing & Soldering
Week-9	Advanced Manufacturing methods
Week-10	CNC Machining
Week-11	Additive manufacturing
Week-12	Electrical & Electronics
Week-13	Plastic moulding, Glass cutting
Week-14	Metal casting

LIST OF EXPERIMENTS

WEEKS	BASIC TRADES	BASIC MANUFACTURING
	Fitting	Machine Shop
Week 1	Filing Four Sides of Work piece	Facing & Step Turning on Lathe
Week 2	L Fit	Milling and Drilling
	Carpentry	Black Smithy
Week 3	Half Lap Joint	Convert round rod to S-hook
Week 4	Dove Tail Joint	Convert round rod to Chisel

	Tin Smithy	Casting
Week 5	Tin Smithy- Prepare a Rectangular Tray	Preparation of Mould Cavity for Multi Piece Pattern
Week 6	Prepare A Square Tin	Casting of Simple pattern
	Electrical	Welding Shop
Week 7	House Wiring Parallel and Series Connection	Lap/Butt joint Using Arc Welding
Week 8	House Wiring Two Way Switch	Lap/Butt joint Using Gas Welding
	Electronics	Plastic Moulding & Glass Cutting
Week 9	Soldering Parallel Connection	Injection moulding of Simple Components
Week 10	Soldering Series Connection	Glass Cutting
Week 11	Revision/Practice	Revision/Practice
Text Books:		
1.Workshop Manual by P. Kannaiah and K. L. Narayana. 2.Rao P.N., "Manufacturing Technology", Tata McGraw Hill House, Vol. I and Vol. II.		
Reference Books:		
1. HajraChoudhury S.K., HajraChoudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010. 2. H. S. Bawa, "Workshop Practice", Tata McGraw-Hill Publishing Company Limited, New Delhi, (2007). 3. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4 th edition,2002.		
E-Text Books:		
1. https://blogpuneet.files.wordpress.com/2013/07/introduction-to-basic-manufacturing-processes-and-workshop-technology.pdf 2. https://soaneemrana.org/onewebmedia/Manufacturing%20Processes%20By%20H.N.%20Gupta.pdf		
MOOC Course:		
1. https://www.class-central.com/course/edx-fundamentals-of-manufacturing-processes-7224		

ENGINEERING EXPLORATION								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS04	HSMC	L	T	P	C	CIE	SEE	Total
		2	0	0	1	30	70	100
COURSE OVERVIEW : This course is introduced to explore different fields of engineering, understand design thinking, solve engineering problems. It also emphasizes to develop skills to write project reports and understand principles of project management.								
COURSE OUTCOMES: At the end of the course student will be able to 1. Explore multiple fields of engineering 2. Apply the engineering design process 3. Construct a prototype to solve an engineering problem 4. Formulate a plan for starting a college career in engineering. 5. Explain concepts of project management.								
UNIT-I	INTRODUCTION TO ENGINEERING AND ENGINEERING EXPLORATION							
Engineering Projects in Community Service, Design Thinking Process-Empathize, Define, Ideate, Prototype, Test. Engineering Ethics: Introduction to ethics, moral values, significance of professional ethics, code of conduct for engineers, identify ethical dilemmas in different tasks of engineering, applying moral theories and codes of conduct for resolution of ethical dilemmas.								
UNIT-II	PROBLEM IDENTIFICATION							
Authentic need in the community or society. Identify a real user or stakeholder, Interaction with Stakeholders, Viewpoints, Interviewing, Scenario.								
UNIT-III	SPECIFICATION DEVELOPMENT							
Clear and measurable requirements, criteria for success, Identifying relevant benchmarks, identifying the gap between the available and required products, requirements documentation.								
UNIT-IV	CONCEPTUAL DESIGN							
Ideation-generated multiple ideas, evaluation of ideas, systems model, Architectural Design, prototype development, testing – real/simulated users, feedback.								
UNIT-V	PROJECT MANAGEMENT							
Importance of team work, importance of project life cycle, project management, tools, various tools used in electronics documentation, importance of communication, usage of communication media.								
Text Books:								
1. Software Engineering: A Practitioner's Approach, Roger S. Pressman, 7 th Edition, Mc Graw Hill Education (India) Pvt. Ltd. 2. Software Engineering, Sommerville Ian, 7 th Edition, Pearson Education. 3. Examples of good practice in Special Needs Education & Community Based Programs, UNESCO PRESS. 4. Project Management, GRY R. Heerkens, McGraw-Hill								

Web References:

1. <http://www.purdue.edu/epics>
2. <http://epics.ieee.org/>
3. <https://www.uninettunouniversity.net/en/epics.aspx>

E-Text Books:

1. [http://www.uoitc.edu.iq/images/documents/informatics-institute/exam_materials/Software%20Engineering%20\(9th%20Edition\)%20by%20Ian%20Sommerville.pdf](http://www.uoitc.edu.iq/images/documents/informatics-institute/exam_materials/Software%20Engineering%20(9th%20Edition)%20by%20Ian%20Sommerville.pdf)
2. <https://engineering.purdue.edu/EPICS/k12/resources/1.6%20Teacher%20Toolbox%20EPICS%20High%20Design%20Process%20and%20Cycle.pdf>
3. https://launchschool.com/books/agile_planning/read/epics_and_stories
<http://www.enggnotebook.weebly.com/uploads/2/2/7/1/22718186/ge6151-notes.pdf>

MOOC Course

1. <https://www.mooc-list.com/tags/design-thinking>
2. <https://www.class-central.com/tag/design%20thinking>

II B.TECH I SEMESTER SYLLABUS

NUMERICAL METHODS, PROBABILITY AND STATISTICS

III Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4BS06	BSC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course is intended to introduce theory and applications of numerical methods, probability and statistics in order to solve various engineering problems.

COURSE OUTCOMES:

After learning the contents of this paper the student will be able to

1. Find Interpolating polynomial for the given tabular data.
2. Solve the first order ordinary differential equations using numerical techniques.
3. Evaluate the discrete and continuous random variables, mathematical expectation of mean and variance.
4. Apply the concepts of correlation and regression to find covariance and sampling distribution of mean and variance.
5. Evaluate the given data for appropriate test of hypothesis.

UNIT-I INTERPOLATION&CURVE FITTING

INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Differences of a polynomial – Missing terms - Newton's forward interpolation, Newton's backward interpolation, Gauss's forward and backward interpolation formulae and Stirling's formula. Interpolation with unequal intervals – Lagrange's interpolation.

CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y = a e^{bx}$, $y = a x^b$, $y = a b^x$ by the method of least squares.

UNIT-II NUMERICAL TECHNIQUES

ROOT FINDING TECHNIQUES : Bisection method-iteration method and Newton Raphson method.

NUMERICAL INTEGRATION : Newton-cotes quadrature formula - Trapezoidal rule - Simpson's one-third rule - Simpson's three-eighth rule.

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS: Taylor's series method – Picard's method - Euler's - modified Euler's Method – Runge-Kutta method

UNIT-III SINGLE RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

Basic definitions of probability, Random Variables – Discrete and Continuous. Probability distributions, mass function/ density function of a probability distribution- mathematical expectation, Mean, Variance. Binomial, Poisson & Normal distributions and their Properties.

UNIT-IV	CORRELATION , REGRESSION AND SAMPLING DISTRIBUTIONS
CORRELATION & REGRESSION: Coefficient of correlation, the rank correlation, Covariance of two random variables. Regression- Regression Coefficient, The lines of regression. SAMPLING: Definitions of population, sampling, statistic, parameter. Types of sampling, Expected values of Sample mean and variance, sampling distribution, Standard error, Sampling distribution of means and sampling distribution of variance. Parameter estimation- Point estimation and interval estimation.	
UNIT-V	TESTING OF HYPOTHESIS
TESTING OF HYPOTHESIS: Null hypothesis, Alternate hypothesis, type I & type II errors – critical region, confidence interval, Level of significance. One sided test, two sided test, LARGE SAMPLE TESTS:(i) Test of Equality of means of two samples equality of sample mean and population mean (cases of known variance & unknown variance, equal and unequal variances) (ii) Tests of significance of difference between sample S.D and population S.D. (iii) Tests of significance difference between sample proportion and population proportion & difference between two sample proportions. SMALL SAMPLE TESTS: Student t-distribution, its properties; Test of significance sample mean and population mean, difference between means of two small samples. Snedecor's F- distribution and its properties. Test of equality of two population variances. Chi-square distribution, it's properties, Chi-square test of goodness of fit.	
Text Books:	
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010. 3. Probability and Statistics for Engineers and Sciences by Jay L. Devore, Cengage Learning	
Reference Books:	
1. Fundamentals of Mathematical Statistics by S.C. Gupta & V.K. Kapoor, S. Chand 2. Introduction to Probability and Statistics for Engineers and Scientists by Sheldon M. Ross, Academic Press	
Web references:	
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/	

E -Text Books:

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>
2. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MECHANICS OF SOLIDS**III Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME01	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

Mechanics of solids is the branch of continuum mechanics which deals with behaviour of solids under different loading conditions. This course will help students to acquire knowledge on the concepts needed for the design of various machine elements.

COURSE OUTCOMES:

The course should enable the students to

1. Explain concepts of simple stresses and strains, calculate deformations, strains and stresses in axial bar elements subjected to different load conditions.
2. Draw and analyze shear force and bending moment diagrams of various beams.
3. Analyze the members subjected to bending and shear loads.
4. Explain principal stresses, principal strains and derive slope and deflection equations for various beams for different loading conditions.
5. Analyze shafts subjected to torsion and thin cylindrical and spherical shells.

UNIT-I SIMPLE STRESSES AND STRAINS

Elasticity and plasticity, Types of stresses and strains, Hooke's law stress, strain diagram for mild steel, Working stress, Factor of safety, Lateral strain, Poisson's ratio and volumetric strain, Elastic module and the relationship between them, Bars of varying section, composite bars, Temperature stresses. Strain energy, Resilience - Gradual, Sudden, Impact, and Shock loadings

UNIT-II SHEAR FORCE AND BENDING MOMENT

Definition of beam, Types of beams, Concept of shear force and bending moment, Relation between Shear Force and Bending Moment. and rate of loading at a section of a beam. Shear Force and Bending Moment diagrams for cantilever simply supported and overhanging beams subjected to point loads, U.D.L., uniformly varying loads and combination of these loads, Point of contra flexure.

UNIT-III FLEXURAL STRESSES & SHEAR STRESSES

FLEXURAL STRESSES: Theory of simple bending, Assumptions, Derivation of bending equation: $M/I = f/y = E/R$ Neutral axis, Determination bending stresses, section modulus of rectangular and circular sections (Solid and Hollow), I, T, Angle and Channel sections, Design of simple beam sections.

SHEAR STRESSES: Derivation of formula, Shear stress distribution across various beams sections like rectangular, circular, I, T, angle sections.

UNIT-IV	PRINCIPAL STRESSES AND STRAINS & DEFLECTION OF BEAMS
PRINCIPAL STRESSES AND STRAINS: Introduction - Stresses on an inclined section of a bar under axial loading - compound stresses - Normal and tangential stresses on an inclined plane for biaxial stresses - Two perpendicular normal stresses accompanied by a state of simple shear - Mohr's circle of stresses - Principle stresses and strains - Analytical and graphical solutions. DEFLECTION OF BEAMS: Bending into a circular arc slope, deflection and radius of curvature, Differential equation for the elastic line of a beam, Double integration and Macaulay's methods, Determination of slope and deflection for cantilever and simply supported beams subjected to point loads.	
UNIT-V	TORSION OF CIRCULAR SHAFTS & THIN CYLINDERS
TORSION OF CIRCULAR SHAFTS: Theory of pure torsion - derivation of Torsion equations: $T/J = \tau/R = G\theta/L$ - Assumptions made in the theory of pure torsion - Torsional moment of resistance - Polar section modulus - Power transmitted by shafts THIN CYLINDERS: Thin seamless cylindrical shells, Derivation of formula for longitudinal and circumferential stresses hoop, longitudinal and volumetric strains, changes in dia, and volume of thin cylinders, Riveted boiler shells, Thin spherical shells.	
Text Books:	
1. Ramamrutham S, Strength of materials, Danpath rai publishing company. 2. Timoshenko and Gere, Mechanics of Materials, CBS Publishers, 2011	
Reference Books:	
1. Dr.S.Sadhu Singh, Strength of Materials, Khanna Publishers 2. Engineering mechanics of solids by EGOR P. Popov, Pearson publishers 3. Strength of Materials by Andrew Pytel and Ferdinand L. Singer Longman	
Web References:	
1. https://nptel.ac.in/Aeronautical/Strength%20of%20Materials/course_strength%20of%20materials.pdf 2. https://lecturenotes.in/notes/10620-notes-for-strength-of-materials-som-by-md-wesh-karni	
E-Text Books:	
1. https://books.google.co.in/books?id=2IHEqp8dNWwC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=oXBLj7Jrr7YC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.classcentral.com/course/coursera-mechanics-of-materials-i-fundamentals-of-stress-strain-and-axial-loading-5031 2. https://ocw.mit.edu/courses/civil-and-environmental-engineering/1-050-solid-mechanics-fall-2004/	

METALLURGY AND MATERIAL SCIENCE**III Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME02	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

The subject deals with the materials and their properties, commonly used for the manufacturing various components in industry. It also focuses on study of various phase diagrams of alloy materials. The ferrous materials and Non Ferrous materials, their properties and applications in the industries are incorporated. The importance of polymers, ceramics and composite materials are included.

COURSE OUTCOMES:

At the end of course students are able to

- 1.Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
- 2.Demonstrate the concept of alloying and formation of different types of phases in alloys.
- 3.Differentiate ferrous and non ferrous alloys.
- 4.Explain various heat treatment processes.
- 5.Classify and explain polymers, ceramics and composites.

UNIT-I CRYSTAL STRUCTURE

Unit cells, crystal systems of metals, crystal planes and directions, slip systems, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms, determination of grain size, effect of grain size on the properties of alloys and Hall - Petch equation.

UNIT-II ALLOYS & PHASE DIAGRAMS

Alloys- substitutional and interstitial solid solutions.
Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.

UNIT-III FERROUS AND NON FERROUS ALLOYS

Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons-grey, white, malleable and spheroidal cast irons; copper and copper alloys- brass, bronze and cupro-nickel; Aluminium and Aluminium alloys.

UNIT-IV HEAT TREATMENT OF STEEL

Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening. Isothermal transformation diagrams for Fe-C alloys and microstructure development, Continuous cooling curves and interpretation of final microstructures and properties.

UNIT-V	POLYMERS, CERAMICS AND COMPOSITES
Classification, properties and applications of polymers, ceramics, composites and nano materials.	
Text Books:	
1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999. 2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011. 3. Sidney H. Avener (2007,) <i>Introduction to Physical Metallurgy</i>, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.	
Reference Books:	
1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India. 2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002. 3. V. D. Kodgire (2006), <i>Material Science and Metallurgy for engineers</i>, 1st Edition, Everest, Pune, India.	
Web References:	
1. https://nptel.ac.in/courses/113106032/ 2. http://web.pdx.edu/~pmoeck/phy381/lecture1.pdf	
E-Text Books:	
1. https://books.google.co.in/books?id=XE8BAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=kBM8BAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course	
1. https://ocw.mit.edu/courses/materials-science-and-engineering/ 2. https://www.mooc-list.com/tags/materials-science	

PRODUCTION TECHNOLOGY**III Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME03	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course is intended to introduce various manufacturing processes such as casting, joining, forming and their relevance in current manufacturing industry. Also focus to explore processing methods of plastics.

COURSE OUTCOMES:

The course should enable the students to

1. Explain casting, casting methods, preparation of patterns and castings.
2. Enumerate various welding & cutting techniques.
3. Explain hot working, cold working, extrusion, rolling and forging processes.
4. Select equipment for various deformation processes.
5. Demonstrate various methods used for producing plastic shapes.

UNIT-I**CASTINGS & METHODS OF MELTING**

CASTING: Steps involved in making a casting, Advantage of casting and its applications. Patterns and Pattern making, Types of patterns, Materials used for patterns, pattern allowances and their construction, Principles of Gating, Gating ratio and design of Gating systems Solidification of casting, Concept, Solidification of pure metal and alloys, short and long freezing range alloys. Risers Types function and design, special casting processes 1) Centrifugal 2) Die 3) Investment.

METHODS OF MELTING: Crucible melting and cupola operation, steel making processes

UNIT-II**WELDING & CUTTING OF METALS**

WELDING: Classification of welding process types of welds and welded joints and their characteristics, design of welded joints, Gas welding, ARC welding, Forge welding, resistance welding, Thermit welding and Plasma (Air and water) welding. TIG and MIG, welding, Friction welding, Induction welding, Explosive welding, Laser welding, Soldering and Brazing. Heat affected zones in welding; welding defects, causes and remedies, destructive, non-destructive testing of welds.

CUTTING OF METALS: Oxy Acetylene Gas cutting, water plasma, cutting of ferrous, non-ferrous metals.

UNIT-III**HOT AND COLD WORKING, EXTRUSION**

HOT WORKING AND COLD WORKING: Strain hardening, recovery, recrystallisation and grain growth, Comparison of properties of cold and hot worked parts.

EXTRUSION OF METALS: Basic extrusion process and its characteristics. Hot extrusion and cold extrusion, Forward extrusion and backward extrusion, Impact extrusion, Hydrostatic extrusion.

UNIT-IV	ROLLING
ROLLING: Fundamentals, theory of rolling, types of Rolling mills and Forces in rolling and power requirements. Stamping, forming and other cold working processes: Blanking and piercing, Bending and forming, Drawing and its types, wire drawing and Tube drawing, coining, Hot and cold spinning Forces and power requirement in the above operations.	
UNIT-V	FORGING & PROCESSING OF PLASTICS
FORGING: Principles of forging, Tools and dies, Types Forging, Smith forging, Drop Forging, Roll forging, Rotary forging, forging defects. PROCESSING OF PLASTICS: Types of Plastics, Properties, applications and their Processing methods and Equipment (blow and injection modelling).	
Text Books:	
1. P. N. Rao (2011), Manufacturing Technology, Vol -1, 3rd edition, Tata McGraw- Hill education (P) Ltd, New Delhi 2. S. Kalpakjain (2005), Manufacturing Engineering and Technology, 4th edition, Pearson Education, New Jersey.	
Reference Books:	
1. B.S. Raghuwanshi (2011), A course in workshop Technology, Vol - II, 3rd Edition, DhanpatRai& Co, New Delhi, India 2. Gosh (2004), Manufacturing science, Affiliated East-west press (p) Ltd, New Delhi, India	
Web References:	
1. http://nptel.ac.in/courses/112107145/ 2. http://nptel.ac.in/courses/Webcourse-contents/IIT-ROORKEE/MANUFACTURING-PROCESSES/index.htm 3. http://nptel.ac.in/courses/112107078/	
E-Text Books:	
1. http://nptel.ac.in/courses/112104204/45 2. http://nptel.ac.in/courses/112107144/13 3. http://nptel.ac.in/courses/112107144/27	
MOOC Course:	
1. https://onlinecourses.nptel.ac.in/noc18_me05/preview 2. https://www.edx.org/course/fundamentals-manufacturing-processes-mitx-2-008x-0 3. https://alison.com/topic/learn/37500/introduction-to-manufacturing-technologies	

THERMODYNAMICS**III Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME04	PC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course intended to introduce thermodynamic laws and their application to analyze boilers, heatpumps, heat engines etc., air standard cycles, properties of mixture of gases and pure substances.

COURSE OUTCOMES:

The course should enable the students to:

1. Demonstrate basic concepts of thermodynamics.
2. Apply the laws of thermodynamics to analyze boilers, heat pumps, heat engines, nozzles etc.
3. Analyze isochoric, isobaric, isothermal, isentropic and isenthalpic processes.
4. Analyze air standard cycles applied in prime movers.
5. Evaluate properties of mixture of gases and pure substances

UNIT-I INTRODUCTION

Macroscopic and Microscopic viewpoints, Quasi static Process, various flow and non-flow process, energy in State and in Transition, Work and Heat, Point and Path function. Zeroth Law of Thermodynamics, First law of Thermodynamics, PMM-I, Corollaries, First law applied to a Process, applied to a flow system.

UNIT-II LIMITATIONS OF FIRST LAW

Thermal Reservoir, Heat Engine, Heat pump Refrigerator. Parameters of performance, Second Law of Thermodynamics-Kelvin Planck and Clausius Statements and their Equivalence / Corollaries, PMM of Second kind, Carnot's principle, Carnot cycle and its specialties, Clausius Inequality, Entropy, Principle of Entropy Increase, Elementary Treatment of the Third Law of Thermodynamics.

UNIT-III PURE SUBSTANCES

P-V-T surfaces, T-S and h-s diagrams, Mollier Charts, Phase transformations, Triple point at critical state properties during change of phase, Dryness Fraction. Various thermodynamic processes and energy transfer, Steam calorimetry

UNIT-IV GAS LAWS & MIXTURE OF PERFECT GASES

Gas laws: Equation of State, specific and Universal Gas constants, Deviations from perfect Gas Model, Vander Waals Equation of State.

Mixtures of perfect gases: Mole fraction, Volume fraction, Mass fraction, Gravimetric and volumetric analysis, Dalton's law of partial pressure, Avogadro's law of additive volumes, Equivalent gas const and Molecular internal energy, enthalpy, Sp. heats and entropy of mixture of perfect gases and vapour.

UNIT-V	POWER CYCLES
Otto, Diesel, Dual Combustion cycles, Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis, comparison of Cycles.	
Text Books:	
1. Engineering Thermodynamics, P.K. Nag, 6th Edition, Mc Graw Hill Education. 2. Thermodynamics an engineering approach, Yunus A. Cengel & Michael A. Boles, 8th Edition, Mc Graw Hill Companies. 3. Fundamentals of Thermodynamics, Richard E. Sonntag Claus Borgnakke, 7th Edition, Wiley.	
Reference Books:	
1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2nd Edition, Prentice hall of India Pvt Ltd., 2006. 2. Thermodynamics, Arora.C.P, Tata Mc Graw Hill, New Delhi. 3. Applied Thermodynamics, Onkar Singh, 3rd Edition, New Age, India	
Web References:	
1. https://www.askitians.com/revision-notes/physics/thermodynamics/ 2. http://www.ncert.nic.in/ncerts/l/kech106.pdf	
E-Text Books:	
1. https://books.google.co.in/books?id=91MZ2ZOb3n8C&printsec=frontcover&source=gbs_ge_summary_r&ad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=w8GhW3J8RHIC&printsec=frontcover&source=gbs_ge_summary_r&ad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.coursera.org/learn/thermodynamics-intro 2. https://www.edx.org/course/thermodynamics-2	

PRODUCTION TECHNOLOGY LAB**III Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME05	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW :

This course is designed to impart hands-on practical exposure on manufacturing processes and equipment.

COURSE OUTCOMES:

The student will be able to:

1. Design and manufacture different patterns
2. Prepare various castings.
3. Operate arc welding, gas welding and resistance welding equipment
4. Perform operations such as blanking, piercing, deep drawing, extrusion, bending and other operations.
5. Use injection moulding and blow moulding equipment to produce plastic products such as water bottle, water bottle caps etc.

LIST OF EXPERIMENTS:**METAL CASTING**

1. Pattern Design and making - for one casting drawing.
2. Sand properties testing – 1 Exercise -for strength and permeability
3. Moulding Melting and Casting - 1 Exercise

II. WELDING

1. ARC Welding Lap & Butt Joint - 2 Exercises
2. Spot Welding - 1 Exercise
3. TIG Welding - 1 Exercise
4. Plasma welding and Brazing - 2 Exercises (Water Plasma Device)

III. MECHANICAL PRESS WORKING

1. Blanking & Piercing operation and study of simple, compound and progressive press tool.
2. Hydraulic Press: Deep drawing and extrusion operation.
3. Bending and other operations

IV. PROCESSING OF PLASTICS

1. Injection Moulding
2. Blow Moulding

NOTE: Minimum 10 of the above experiments are to be conducted

MATERIAL SCIENCE AND METALLURGY LAB**III Semester : MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME06	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW :

The main focus of the material science and metallurgy program is to provide undergraduates with a fundamental knowledge based associated with materials-processing, their properties, and their selection and application. Upon graduation, students would have acquired and developed the necessary background and skills for successful careers in the materials-related industries.

COURSE OUTCOMES:

At the end of the course students are able to

1. Prepare sample for the study of micro structure.
2. Find micro structure of various metals and alloys.
3. Determine the micro structure of heat treated steel.
4. Analyze significance of cooling rate in the formation of fine grains using jominy and quench test.
5. Find hardness of various treated and untreated steels.

LIST OF EXPERIMENTS:

1. Study of metallurgical microscope.
2. Standard method of specimen preparation and analysis of a metallurgical microscope.
3. Preparation and study of the Micro Structure of pure metals like Iron, Cu and Al.
4. Preparation and study of the Microstructure of mild steels, low carbon steels, high carbon steels.
5. Study of the Micro Structures of Cast Irons.
6. Study of the Micro Structures of Non-Ferrous alloys.
7. Study of the Micro structures of Heat treated steels.
8. Jominy End Quench Test.
9. Find out the hardness of various treated and untreated steels.
10. Hardeneability of steels.

MACHINE DRAWING**III Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME07	PC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW :

Machine Drawing gives representation of a machine component or machine by lines according to certain set rules. This course will focus on all the external and internal details of the machine components from which it can be manufactured.

COURSE OUTCOMES:

The course should enable the students to:

1. Explain different steps in producing drawings according to bureau of Indian standards (B.I.S.) as per SP:46
2. Draw various machine elements and simple parts.
3. Draw riveted joints couplings and bearings.
4. Draw assembled views for part drawings and engine parts.
5. Draw other machine parts such as screw jack, plumber block etc.

UNIT-I MACHINE DRAWING CONVENTIONS

Need for drawing conventions – introduction to ISI conventions:

1. Conventional representation of materials, common machine elements and parts such as screws, Nuts, bolts, keys, gears, webs, ribs.
2. Types of sections – selection of section planes and drawing of sections and auxiliary sectional views. Parts not usually sectioned.
3. Methods of dimensioning, general rules for sizes and placement of dimensions for holes, centers, curved and tapered features.
4. Title boxes, their size, location and details - common abbreviations and their liberal usage, Types of Drawings – working drawings for machine parts.

UNIT-II DRAWING OF MACHINE ELEMENTS AND SIMPLE PARTS

Selection of Views, additional views for the following machine elements and parts with every drawing proportions. Popular forms of Screw threads, bolts, nuts, stud bolts, tap bolts, set screws. Keys, cottered joints and knuckle joint.

UNIT-III RIVTED JOINTS, COUPLINGS AND BEARINGS

Riveted joints for plates, Shaft coupling, spigot and socket joint, Journal, pivot and collar and foot step bearings.

UNIT-IV ASSEMBLED VIEWS FOR PART DRWAINGS & ENGINE PARTS

Drawings of assembled views for the part drawings of the following using conventions and easy drawing proportions. Engine parts – stuffing boxes, cross heads, Eccentrics, Petrol Engine connecting rod, piston assembly.

UNIT-V	OTHER MACHINE PARTS & VALVES
Other machine parts - Screws jacks, Machine Vices Plummer block, Tailstock. Valves : Steam stop valve, spring loaded safety valve, feed check valve and air cock.	
Text Books:	
1. Machine Drawing –K.L.Narayana, P.Kannaiah & K.Venkata Reddy / New Age/Publishers. 2. Machine Drawing – Dhawan, S.Chand Publications. 3. Machine Drawing- Siddheswar.	
Reference Books:	
1. Machine Drawing – N.D.Bhatt, Charotar Publishing House. 2. Machine Drawing – P.S.Gill. 3. Machine Drawing – Luzzader.	
Web References:	
1 gt3.bme.hu/wp-content/uploads/2016/02/Narayana-Machine_Drawing	
E-Text Books:	
1 gt3.bme.hu/wp-content/uploads/2016/02/Narayana-Machine_Drawing.pdf 2 https://www.pdfdrive.com/engineering-drawing-books.html	
MOOC Course:	
1 https://www.mooc-list.com/tags/drawing 2 https://www.class-central.com/tag/engineering%20drawings	

GENDER SENSITIZATION**III Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC02	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	0	30	70	100

COURSE OBJECTIVES:

1. To provide a critical perspective on the socialization of men and women.
2. To introduce students to information about some key biological aspects of genders.
3. To expose the students to debates on the politics and economics of work.
4. To help students reflect critically on gender violence.
5. To expose students to more egalitarian interactions between men and women.

COURSE OUTCOMES

1. Students will have developed a better understanding of important issues related to gender in contemporary India. 2. Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender. This will be achieved through discussion of materials derived from research, facts, everyday life, literature and film. 3. Students will attain a finer grasp of how gender discrimination works in our society and how to counter it. 4. Students will acquire insight into the gendered division of labour and its relation to politics and economics. 5. Men and women students and professionals will be better equipped to work and live together as equals. 6. Students will develop a sense of appreciation of women in all walks of life. 7. Through providing accounts of studies and movements as well as the new laws that provide protection and relief to women, the textbook will empower students to understand and respond to gender violence.

UNIT-I UNDERSTANDING GENDER

Gender: Why Should We Study It? (Towards a World of Equals: Unit -1) Socialization: Making Women, Making Men (Towards a World of Equals: Unit -2) Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities.

UNIT-II GENDER BIOLOGY

Missing Women: Sex Selection and Its Consequences (Towards a World of Equals: Unit -4) Declining Sex Ratio. Demographic Consequences. Gender Spectrum: Beyond the Binary (Towards a World of Equals: Unit -10) Two or Many? Struggles with Discrimination.

UNIT-III GENDER LABOUR

Housework: the Invisible Labour (Towards a World of Equals: Unit -3) —My Mother doesn't Work.|| —Share the Load.|| Women's Work: Its Politics and Economics (Towards a World of Equals: Unit -7) Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.

UNIT-IV ISSUE OF VIOLENCE

Sexual Harassment: Say No! (Towards a World of Equals: Unit -6)
Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading: —Chupulu". Domestic Violence: Speaking Out (Towards a World of Equals: Unit -8) Is Home a Safe Place? -When Women Unite [Film].

Rebuilding Lives. Additional Reading: New Forums for Justice. Thinking about Sexual Violence (Towards a World of Equals: Unit -11) Blaming the Victim—I Fought for my Life.... - Additional Reading: The Caste Face of Violence	
UNIT-V	GENDER: CO-EXISTENCE
Just Relationships: Being Together as Equals (Towards a World of Equals: Unit -12) Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and Fathers. Additional Reading: Rosa Parks-The Brave Heart.	
Text Books:	
“Towards a World of Equals: A Bilingual Textbook on Gender”, A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote, Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad, Telangana State, 2015.	
Reference Books:	
1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012 2. Abdulali Sohaila. —I Fought For My Life...and Won. Available online at: http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdul/	

II B.TECH II SEMESTER SYLLABUS

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE64	ESC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course is intended to introduce the concept of electrical circuits and its components, concepts of diodes and transistors, and to impart the knowledge of various configurations, characteristics and applications.

COURSE OUTCOMES:

At the end of the course students are able to

- 1 Differentiate circuits with ideal, independent, controlled voltage and current sources and able to apply Kirchhoff's voltage and current laws to the analysis of electric circuits.
- 2 Apply concepts of electric network topology, nodes, branches, and loops to solve circuit problems, including the use of computer simulation.
- 3 Emphasize on basic laws and techniques to develop a working knowledge of the methods of analysis used in electric circuits.
- 4 Interpret to solve series and parallel magnetic circuits
- 5 Design various two port network parameters and relations between mutual and self inductances.

UNIT-I**ELECTRICAL CIRCUITS**

Basic definitions-Ohm's Law – Kirchhoff's Laws – simple problems. types of elements, types of sources, resistive networks, inductive networks, capacitive networks, series & parallel circuits, star to delta and delta to star transformations.

UNIT-II**DC MACHINES**

Principle of Operation of DC Motor, types of DC motor, Losses and Torque equation, DC Generator construction and working Principle, EMF Equation, working principle and Construction of transformer.

UNIT-III**AC MACHINES & INSTRUMENTS**

Principle of operation of 3-phase induction motor, slip and torque Equation, principle of operation of 3-phase Alternator. Basic principle of indicating instruments, permanent magnet moving coil and moving iron instruments.

UNIT-IV**SEMICONDUCTOR DEVICES**

Characteristics of PN Junction Diode – Zener Effect – Zener Diode and its Characteristics. Rectifiers, Half wave, Full wave and bridge Rectifiers – filters capacitor, inductor-Voltage Regulation.

UNIT-V	TRANSISTOR
Bipolar Junction Transistor – CB, CE, CC Configurations and Characteristics – Transistor Amplifier.	
Text Books:	
1. Basic Electrical Engineering by M.S.Naidu and S.Kamakshaiah TMH 2. Electronic Devices and circuits by J.Millman, C.C.Halkias and Satyabrata Jit 2ed.,	
Reference Books:	
1. Muthusubramanian R, Salivahanan S and Muraleedharan K A, "Basic Electrical, Electronics and Computer Engineering", Tata McGraw Hill, Second Edition, (2006). 2. Nagsarkar T K and Sukhija M S, "Basics of Electrical Engineering", Oxford press (2005). 3. Mehta V K, "Principles of Electronics", S.Chand & Company Ltd, (1994). 4. Mahmood Nahvi and Joseph A. Edminister, "Electric Circuits", Schaum' Outline Series, McGraw Hill, (2002).	
Web References:	
1. https://lecturenotes.in/subject/905/basic-electrical-and-electronics-engineering-beee 2. https://nptel.ac.in/downloads/108105053/	
E-Text Books:	
1. https://books.google.co.in/books?id=0ilzar-QB8IC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=NgZXYVz_0cEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.edx.org/learn/electrical-engineering	

KINEMATICS OF MACHINERY**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME08	PCC	L	T	P	C	CIE	SEE	Total
		4	0	0	4	30	70	100

COURSE OVERVIEW :

This course will deal with kinematic analysis of mechanisms and machines. It covers basic concepts and principles of various mechanisms, analysis of velocity and acceleration of different mechanisms, study of steering gear mechanisms, gears, gear trains, cams and followers.

COURSE OUTCOMES:

The course should enable the students to:

1. Explain basic concepts and principles of mechanisms.
2. Illustrate and Analyze velocity and acceleration in different mechanisms.
3. Illustrate and Analyze straight line and steering gear mechanisms.
4. Explain concepts of gears and gear trains.
5. Discuss and analyze various cams and followers.

UNIT-I**INTRODUCTION**

Introduction: Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematic pairs – Types of constrained motion – kinematic chain – Mechanism-Machine-Structure – Kleins criteria, Degrees of freedom, Grubler's Criterion, Grashof's law – inversion of mechanism – inversions of four bar chain, single and double slider crank chains.

UNIT-II**VELOCITY & ACCELERATION ANALYSIS**

Velocity in Mechanisms: Relative Velocity Method- Relative velocities of two bodies moving in straight line, velocity of a point on a link by relative velocity method, Velocities in four bar and single slider crank mechanisms, Rubbing velocity at a pin point.

Instantaneous Centre Method: Introduction, space and body centrodes, Velocity of a point on a link by instantaneous centre method, Properties of instantaneous centre, Number of instantaneous centres, Types of instantaneous centres, Kennedy's theorem, Method of locating instantaneous centres, Velocities in four bar and single slider crank mechanisms.

Acceleration in Mechanisms: Acceleration of a point on link, Acceleration in four bar and slider crank mechanisms, Coriolis component of acceleration.

UNIT-III**STRAIGHT LINE MOTION & STEERING GEAR MECHANISMS**

Straight line motion mechanisms: Types, Peaucellier, Hart's, Scott Russel, Tchebichef's, Watt's Grasshopper and Robert Mechanisms - Pantograph.

Steering gear mechanisms: Introduction, Condition for correct steering, Davis Steering gear mechanism, Ackerman's steering gear mechanism. Hooke's joint: Introduction, Velocity ratio, Maximum speed, condition for equal speeds of driven and driving shafts.

UNIT-IV	GEARS & GEAR TRAINS
Friction wheels and toothed gears – types – law of gearing, condition for constant velocity ratio for transmission of motion, form of teeth-cycloid and involute profiles, phenomena of interference, condition for minimum number of teeth to avoid interference, expressions for arc of contact and path of contact of pinion and gear. Gear trains: Introduction – Types – Simple, compound and reverted gear trains – Epicyclic gear train. Methods of finding train value or velocity ratio of Epicyclic gear trains, differential gear for an automobile.	
UNIT-V	CAMS
Cams: Introduction, classification of followers and cams; Analysis and construction of motion of the follower- Uniform velocity, Simple harmonic motion, Uniform acceleration and retardation, cycloidal motion. Cams with specified contours, Tangent cam with reciprocating roller follower, Circular arc cam with flat faced follower.	
Text Books:	
1. Theory of Machines / Thomas Bevan / Pearson / 3rd Edition 2. Theory of Machines / S.S. Rattan / Tata McGraw-Hill Education	
Reference Books:	
1 Kinematics and Dynamics of machinery / Robert L. Norton / Tata McGraw Hill 2 Mechanisms of machines / Cleghorn W.L. / Oxford University Press 3 Theory of Mechanism and Machines / Ghosh and Mallik / Affiliated East-West Pvt. Ltd. 4 Theory of Machines by R.S Khurmi & J.K Gupta, S.Chand Publications.	
Web References:	
1. http://nptel.ac.in/courses/112104121/ 2. https://freevidelectures.com/course/2359/kinematics-of-machines	
E-Text Books:	
1. https://www.pdfdrive.net/kinematics-of-machinery-e459890.html 2. https://www.pdfdrive.net/kinematics-of-machinery-e19480301.html 3. https://books.google.co.in/books	
MOOC Course	
1. https://cosmolearning.org/courses/kinematics-of-machines 2. https://nptel.ac.in/courses/112101096/	

THERMAL ENGINEERING-1**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME09	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course will make students to understand principles of Internal & External combustion engines, study testing and performance of compressors.

COURSE OUTCOMES:

The course should enable the students to:

1. Demonstrate the basic principles of operation of IC engines.
2. Analyze the combustion process and also how it does affect the performance of the IC engines.
3. Describe procedures of testing and evaluate the performance of IC engines.
4. Apply the thermodynamic principles in the design of an IC engines.
5. Formulate and perform the procedures required for the maintenance and operation of compressors.

UNIT-I**I.C. ENGINES**

Classification, Working principles, Valve and Port Timing Diagrams. Actual Cycles and Their Analysis: Comparison of Air Standard and Actual Cycles, Time Loss Factor, Heat Loss Factor, Exhaust Blow down-Loss due to Gas exchange process. Loss due to Rubbing Friction.

UNIT-II**COMBUSTION IN S.I AND CI ENGINES**

COMBUSTION IN S.I. ENGINES: Normal Combustion and abnormal combustion, Importance of flame speed and effect of engine variables, Type of Abnormal combustion, pre-ignition and knocking (explanation of) Fuel requirements and fuel rating, anti-knock additives, combustion chamber – requirements, types. COMBUSTION IN C.I. ENGINES: Four stages of combustion, Delay period and its importance, Effect of engine variables, Diesel Knock, open and divided combustion chambers and nozzles used – fuel requirements and fuel rating.

UNIT-III**SUPER CHARGING AND PERFORMANCE OF IC ENGINES**

Super Charging: Types of engine supercharging - Devices, Turbo charging. Performance of IC Engines: Measurement of engine power, analysis of engine performance. Factors effecting efficiency and pumping loss. Variable speed, Air/Fuel ratio. Heat balance test. BIS standards for testing and rating.

UNIT-IV	MODERN DEVELOPMENTS, EMISSIONS AND ALTERNATE FUELS
Modern Developments: Stratified charge engine, Dual-fuel engines and HCCI concept. Engine Emissions: SI and CI engine emissions, Harmful effects, Emissions measurement methods, methods for controlling engine emissions. Alternate Fuels For IC Engines: Need for use of alternate fuels, Use of alcohol fuels, Biodiesel, Biogas and Hydrogen in engine	
UNIT-V	COMPRESSORS
COMPRESSORS: Classification, positive displacement and roto dynamic machinery, Rotary blower, vanesealed compressor, mechanical details and principle of working, efficiency considerations. RECIPROCATING AIR COMPRESSORS: Principle of operation, work required, Isothermal efficiencyvolumetric efficiency, clearance, multi stage compression, under cooling, saving of work, minimum workcondition for stage compression..	
Text Books:	
1. V. Ganesan (2011), I.C. Engines, 3rd edition, Tata McGraw-Hill, New Delhi, India. 2. Mathur, Sharma (2008), IC Engines, 3 rd edition, DhanpatRai& Sons, New Delhi, India.	
Reference Books:	
1. Pulkrabek(2008), Engineering fundamentals of IC Engines, 2 nd edition, Pearson Education, New Jersey.Rudramoorthy(2003), Thermal Engineering, 5th edition, Tata McGraw-Hill, New Delhi, India. Engineering Mechanics /M. V Seshagiri Rao & D Rama Durgaiah / University Press 2. Tayal A.K, Engineering Mechanics, Umesh Publications 3. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3rd Edition, 2014.	
Web References:	
1 http://nptel.ac.in/courses/112104113/ 2 http://nptel.ac.in/courses/112105123/ 3 https://www.udemy.com/fundamentals-of-engineering-thermodynamics/	
E-Text Books:	
1. https://www.pdfdrive.net/engineering-thermal_engineering-e18770201.html 2. https://www.pdfdrive.net/engineering-thermal_engineering-e18762062.html 3. https://www.pdfdrive.net/engineering-thermal_engineering-e15057935.html	
MOOC Course	
1 https://onlinecourses.nptel.ac.in/ 2 https://www.coursera.org/learn/thermal_engineering-intro 3 https://swayam.gov.in/courses/4642-july-2018-engineering-thermal_engineering.	

FLUID MECHANICS AND HYDRAULIC MACHINES

IV Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME10	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course is intended to introduce the fundamentals of the fluid mechanics such as fluid and flow properties, fluid behaviour at rest and in motion and fundamental equations like mass, energy and momentum conservation of the fluid flow thereby developing an understanding of fluid dynamics in variety of fields. Applications of these basic equations have been highlighted for flow measurements. Hydraulic machinery plays an important role in the conversion of hydraulic energy to mechanical energy and vice-versa. Hydraulic turbines are used for meeting our day-today power demands. This course also covers different types of pumps are essential equipment in all the industries.

COURSE OUTCOMES:

The course should enable the students to

1. Describe the basic principles of fluid mechanics.
2. Identify various types of flows and formulate their governing equations.
3. Analyze the losses in pipe flows with the concepts of flow through pipes.
4. Design the working proportions of hydraulic turbines.
5. Explain the functioning and characteristic curves of pumps.

UNIT-I

FLUID STATICS

Fluid Statics: Dimensions and units, Physical properties of fluids-specific gravity, viscosity, surface tension, vapour pressure and their influence on fluid motion, atmospheric, gauge and vacuum pressures, measurement of pressure, piezometer, U-tube and differential manometers.

UNIT-II

FLUID KINEMATICS AND DYNAMICS

Fluid Kinematics: Stream line, path line, streak line and stream tube, classification of flows - steady and unsteady, uniform and non uniform, laminar and turbulent, rotational and irrotational flows, equation of continuity for one dimensional flow and three dimensional flows

Fluid dynamics: Surface and body forces, Euler's and Bernoulli's equations for flow along a stream line, momentum equation and its application on force on pipe bend

UNIT-III

CLOSED CONDUIT FLOW

Closed Conduit flow: Reynolds's experiment, Darcy Weisbach equation, minor losses in pipes, Pipes in series and pipes in parallel, Total energy line, hydraulic gradient line, Measurement of flow, Pitot tube, venturi meter, and orifice meter, flow nozzle

Basics of Turbo machinery: Hydrodynamic force of jets on stationary and moving plates, flat, inclined and curved vanes, jet striking centrally and at tip, velocity diagrams, work done and efficiency, flow over radial vanes.

UNIT-IV	HYDRAULIC TURBINES
Hydraulic turbines: classification of turbines, heads and efficiencies, impulse and reaction turbines, Pelton Wheel, Francis turbine and Kaplan turbine-working proportions, work done, efficiencies, hydraulic design, draft tube theory, functions and efficiency. Performance of hydraulic turbines: Geometric similarity, unit and specific quantities, characteristic curves, governing of turbines, selection of type of turbine, cavitation, surge tank, water hammer.	
UNIT-V	CENTRIFUGAL PUMPS AND RECIPROCATING PUMPS
Centrifugal pumps: Classification, working, work done, barometric head losses and efficiencies, specific speed, performance characteristic curves, NPSH. Reciprocating pumps: working, discharge, slip, indicator diagrams.	
Text Books:	
1. Modi and Seth, "Hydraulics, Fluid Mechanics and Hydraulic Machinery", Rajsons Publications, 20th Edition, 2013. 2. Yunus Cengel, John Cimbala, Fluid Mechanics, McGraw-Hill Education 3. Rajput, "Fluid Mechanics and Hydraulic Machines", S.Chand & Co, 6 th Edition, 1998.	
Reference Books:	
1. D.S. Kumar, "Fluid Mechanics and Fluid Power Engineering", Kotaria & Sons, 2013. 2. D. Rama Durgaiah, "Fluid Mechanics and Machinery", New Age International, 1st Edition, 2002. 3. Banga, Sharma, "Hydraulic Machines", Khanna Publishers, 6th Edition, 2001	
Web References:	
1. https://lecturenotes.in/subject/95/fluid-mechanics-and-hydraulic-machines-fmhm 2. https://www.smartworld.com/notes/fluid-mechanics-hydraulic-machinery-notes-pdf-fmhm-notes-pdf/	
E-Text Books:	
1. https://books.google.co.in/books?id=nCnifcUdNp4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=yCW_8APjwMkC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.coursera.org/learn/fluid_mechanics-statics 2. https://www.edx.org/course/fluid-mechanics	

DESIGN OF MACHINE MEMBERS-I**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME11	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course deals the systematic approach to design, standardization, design and manufacturing considerations, engineering materials properties and their selection, simple and compound stresses in machine elements, theories of failure under static and dynamic loading situations, Design of riveted joints, welded joints, bolted joints, keys and cotter joints, shafts, shaft couplings and springs.

COURSE OUTCOMES:

The course should enable the students to

1. Explain various types of design, various considerations for design and theories of failure.
2. Design riveted, welded and bolted joints.
3. Design keys, cotter and knuckle joints.
4. Design shafts and shaft couplings.
5. Design Mechanical Springs.

UNIT-I INTRODUCTION & FATIGUE

Introduction: General considerations & Manufacturing considerations for design of machine elements, Mechanical properties of materials, Static theories of failure and preferred numbers.

Fatigue: Types of Fatigue loading, S-N Curve – Endurance Limit, Factors effecting Endurance Limit, Stress Concentration - Theoretical stress concentration factor - Fatigue stress concentration factor - Notch sensitivity- Design for fluctuating stresses-endurance limit-Goodman line, Soderberg line and Gerber's Parabola.

UNIT-II DESIGN OF RIVTED , WELDED & BOLTED JOINTS

Riveted Joints: Cold riveting and hot riveting, types of riveted joints, terminology, methods of failure of riveted joints-strength equations-efficiency of riveted joints-eccentrically loaded riveted joints

Welded Joints: Types, Advantages, Dis advantages, Design of fillet welds-axial loads-circular fillet welds subjected to torsion and bending.

Design of bolts: Design of bolts with pre-stresses, bolts of uniform strength, bolted joints under eccentric loading.

UNIT-III	DESIGN OF KEYS, COTTER & KNUCKLE JOINTS
Keys, Cotter & Knuckle Joints: Design of key-stresses in keys-cotter joints-spigot & socket, sleeve & cotter, Gib & cotter joints & Knuckle Joints.	
UNIT-IV	DESIGN OF SHAFTS & SHAFT COUPLINGS
Design of shafts: Design of solid & hollow shafts for strength & rigidity- Design of shafts for complex loads-shaft sizes-Design of shaft for Belt drives.	
Shaft couplings: Rigid couplings-Muff, split muff and flange couplings-Flexible couplings-PIN Bush couplings	
UNIT-V	DESIGN OF MECHANICAL SPRINGS
Mechanical Springs: Stresses & Deflection of helical springs-extension of compression springs- springs for static & fatigue loading-natural frequency of helical springs- energy storage capacity-helical torsion springs	
Text Books:	
1. V. Bandari (2011), <i>A Text Book of Design of Machine Elements</i>, 3rd edition, Tata McGraw hill education (P) Ltd, New Delhi, India. 2. R. L. Norton (2006), <i>Machine Design (An Integrated approach)</i>, 2nd edition, Pearson Publishers, Chennai, India.	
Reference Books:	
1. Shigley, J.E, (2011), <i>Mechanical Engineering Design</i>, 9th Edition, Tata McGraw-Hill, New Delhi, India. 2. Dr.P.C Sharma, D.K Agarwal, Machine Design, S.K Kataria & Sons	
Web References:	
1. https://nptel.ac.in/downloads/112105125/ 2. http://mechteacher.com/machine-design-introduction/	
E-Text Books:	
1. https://books.google.co.in/books?id=d-eNe-VRc1oC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://www.pdfdrive.com/machine-design-r-s-kurmi-e32494497.html	
MOOC Course :	
1. https://www.mooc-list.com/course/machine-design-part-i-coursera 2. https://www.class-central.com/subject/mechanical-engineering	

MECHANICS OF SOLIDS LAB**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME12	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW :

The student will perform tests on materials in tension, compression, torsion, bending, and impact. These conditions and/or constraints are designed to reinforce classroom theory by having the student to perform required tests, analyze subsequent data, and present the results in a professionally prepared report. The machines and equipment used to determine experimental data include universal testing machines, torsion equipment, spring testing machine, compression testing machine, impact tester, hardness tester, etc.

COURSE OUTCOMES :

At the end of the course students are be able to

1. Analyze mechanical behaviour of test sample under tensile, compressive and shear loads.
2. Find deflection of simply supported and cantilever beams with different cross sections/materials.
3. Analyze the behaviour of a test sample under torsion.
4. Calculate stiffness and modulus of rigidity of helical coil spring.
5. Determine the capacity of material to absorb energy under impact loading.

SYLLABUS:

1. Direct tension test.
2. Deflection test on simply supported beam.
3. Deflection test on cantilever beam.
4. Torsion test.
5. Hardness test
 - I. Brinell's hardness test
 - II. Rockwell hardness test
6. Test on springs.
7. Compression test on cube.
8. Charpy-Impact test.
9. Izod- Impact test.
10. Punch shear test.

FLUID MECHANICS AND HYDRAULIC MACHINERY LAB**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME13	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OVERVIEW :

This course will provide a basic understanding of calibration of flow measuring devices such as venturi meter, orifice meter, notches, water meter, rotameter etc used in channels and pipes and Losses in pipes by major and minor loss apparatus. Measurements of critical Performance parameters like efficiency, flow rate etc of various hydraulic machines such as turbines, pumps are also measured in this lab. Students can perform tests and find the flow rate of equipment like venturimeter, orifice meter, and notches and can calibrate them. Darcy's as well as Chezy's coefficient of friction for different pipes can be found out in pipe friction apparatus. Also students can find out the performance of centrifugal and reciprocating pumps.

COURSE OUTCOMES:

At the end of course students are able to

1. Develop procedure for standardization of experiments.
2. Calibrate flow discharge measuring devices used in pipes.
3. Determine the major and minor losses in a given pipe.
4. Prove that the total head at any point along the fluid flow is same.
5. Test the performance of pumps and turbines.

LIST OF EXPERIMENTS:

1. Impact of jets on Vanes.
2. Performance Test on Pelton Wheel.
3. Performance Test on Francis Turbine.
4. Performance Test on Kaplan Turbine.
5. Performance Test on Single Stage Centrifugal Pump.
6. Performance Test on Multi Stage Centrifugal Pump.
7. Performance Test on Reciprocating Pump.
8. Calibration of Venturimeter.
9. Calibration of Orifice meter.
10. Determination of friction factor for a given pipe line.
11. Determination of loss of head due to sudden contraction in a pipeline.

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE65	ESC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW :

This course is intended to analyze a given network by applying various electrical laws and network theorems, response of electrical circuits for different excitations, the relation between basic electrical parameters, the performance characteristics of DC and AC electrical machines.

COURSE OUTCOMES:

At the end of the course student will be able to

- 1 Find the difference between Generator and Motor performance characteristics
- 2 Find the applications of the electrical machines with the experimental determination of the performance of the machines
- 3 Find the applications of the Transformers with the experimental determination of the performance of the Transformers
- 4 Apply the Basic components to real time applications.
- 5 Derive the characteristics of P-N junction diodes, Zener diode, transistors and rectifiers;
- 6 Calculate peak-peak voltage, frequency, current by using CRO.

LIST OF EXPERIMENTS**SECTION A: ELECTRICAL ENGINEERING**

1. Verification of KCL and KVL.
2. Magnetization characteristics of D.C. Shunt generator.
3. Speed control of DC motor.
4. Swinburne's Test on DC shunt machine.
5. Brake test on DC shunt motor.
6. OC and SC tests on Single-phase transformer.
7. Brake test on 3-phase Induction motor.
8. Regulation by an alternator by synchronous impedance method.

SECTION B: ELECTRONICS ENGINEERING

1. PN Junction Diode Characteristics (Forward bias, Reverse bias)
2. Transistor CE Characteristics (Input and Output)
3. Study of CRO.

4. Class A Power Amplifier
5. Zener Diode Characteristics
6. Transistor CE Characteristics
7. Rectifier without Filters (Full wave & Half wave)
8. Rectifier with Filters (Full wave & half wave).

Note: Total 12 experiments are to be conducted.

(Six experiments from PART-A, Six experiments from PART-B)

Note: Ten experiments should be performed by taking at least four experiments from each lab.

ENVIRONMENTAL STUDIES**IV Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC01	MC	L	T	P	C	CIE	SEE	Total

COURSE OVERVIEW :

This course is introduced to create the awareness about environmental problems among students, to impart basic knowledge about the environment and its allied problems. Also focus on developing an attitude of concern for the environment, motivating students to participate in environment protection and environment improvement.

COURSE OUTCOMES:

Upon successful completion of the course, the student should be able to

1. Explain the natural resources and their management.
2. Understanding the Classification and functioning of Ecosystems.
3. Remembering the Importance of biodiversity and its conservation.
4. Understanding the problems related to environmental pollution and management.
5. Apply the role of information technology, Analyze social issues and Acts associated with Environment.

UNIT-I**ECOSYSTEM**

Introduction, definition, scope and Importance of environmental studies. Classification, structure and function of an ecosystem, Food chains, food webs and ecological pyramids. Biogeochemical cycles, Bioaccumulation, Biomagnifications.

UNIT-II**NATURAL RESOURCES**

Classification of Resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources, Forest resources, Energy resources: renewable and non renewable energy sources.

UNIT-III**BIODIVERSITY**

Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation, Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife: co-existence and conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation.

UNIT-IV**ENVIRONMENTAL POLLUTION**

Environmental Pollution: Air Pollution: types of pollutants. Water pollution: Sources and types of pollution, Soil Pollution: Sources and types, Impacts of modern agriculture, degradation of soil. Noise Pollution: Sources and Health hazards, standards, composition and characteristics of e-Waste and its management. Global Environmental Problems: Ozone depletion and Ozone depleting substances (ODS).

UNIT-V	ENVIRONMENTAL POLICY & LEGISLATION
International conventions / Protocols: Earth summit, Kyoto protocol and Montréal Protocol, Legal aspects Air Act- 1981, Water Act, Forest Act, Wild life Act, Towards Sustainable Future: Concept of Sustainable Development, Environmental Education, urbanization ,Urban Sprawl. Concept of Green Building.	
Text Books:	
1. Erach Bharucha (2005)., Textbook of Environmental Studies for Undergraduate Courses, Hyderabad, Universities Press.	
Reference Books:	
2. Anubha Kaushik(2006).,Perspectives in Environmental Science, 3 rd Edition, New Delhi, New age international.	
3. Textbook of Environmental Studies by OVK Reddy	
Web References:	
https://info.stylee32.net/PDF%20Library/.../Environmental.pdf	
E-Text Books:	
https://www.pdfdrive.com/environmental-science-books.html	
MOOC Course :	
https://www.mooc-list.com/tags/environmental-studies	

III B.TECH I SEMESTER SYLLABUS

DYNAMICS OF MACHINERY**V Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME14	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course is intended to deal with the forces and their effects, while acting upon the machine parts in motion. The knowledge of this subject is very essential for an engineer in designing the various parts of a machine. Study of gyroscopes is very much helpful to learn the precession and stability of moving vehicle such as aircrafts, Naval ships and automobiles. This course helps to analyze and evaluate the force and torques that causes of motions in mechanisms. It provides knowledge on clutches, brakes and dynamometers, balancing of machines and application of vibrations

COURSE OUTCOMES:

After completing this course the student must:

1. Analyze the effect of gyroscopic couples on the stability of aeroplane, naval ship and automobiles moving in a curved path.
2. Analyze the frictional forces and torques transmitted by various components such as clutches, brakes and dynamometers.
3. Demonstrate governing systems that control the speed of engines using the inertia phenomena and centrifugal forces
4. Find the position and magnitude of unbalanced masses for rotating and reciprocating parts of the engine and analyse different types of governors.
5. Determine frequency of various machine members subjected to free and forced vibrations.

UNIT-I PRECESSION, STATIC & DYNAMIC FORCE ANALYSIS

Precession: Gyroscopes – effect of precession – motion on the stability of moving vehicles such as motorcycle – motorcar – aeroplanes and ships.

Static and Dynamic Force Analysis: Static force analysis of planar mechanisms – Analytical Method – D'Alembert's principle, Dynamic Analysis of 4-link mechanism, Single Slider Crank Mechanism.

UNIT-II FRICTION, BRAKES & DYNAMOMETERS

Friction: Pivots and collars – uniform pressure, uniform wear – friction circle and friction axis: Lubricated surfaces – boundary friction – film lubrication. Clutches – Types – Single plate, multi-plate, cone and centrifugal clutches.

Brakes and Dynamometers: Types of brakes: Simple block brake, band and block brake-internal expanding shoe brake-effect of braking of a vehicle. Dynamometers – absorption and transmission types.

UNIT-III	TURNING MOMENT DIAGRAM AND FLYWHEELS, GOVERNORS
Turning Moment Diagram And Flywheels: Engine Force Analysis – Piston Effort, Crank Effort, etc., Inertia Force in Reciprocating Engine – Graphical Method - Turning moment diagram –fluctuation of energy – flywheels and their design - Inertia of connecting rod- inertia force in reciprocating engines – crank effort and torque diagrams. Governors: Types of governors - Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung with auxiliary springs. Sensitiveness, isochronisms and hunting – stability of governors.	
UNIT-IV	BALANCING
Balancing of rotating masses- Analytical and graphical methods. Balancing of reciprocating masses- Primary, Secondary, and higher balancing. Examination of multi cylinder-in-line engines for primary and secondary balancing- locomotive balancing – Hammer blow – Swaying couple – variation of tractive effort.	
UNIT-V	VIBRATIONS
Vibrations: Types of vibratory motion, types of free vibrations, natural frequency of free longitudinal and transverse vibrations– natural frequency of free transverse vibrations over a simply supported shaft and fixed shaft due to concentrated and distributed loads. Critical or whirling speed of a shaft–Natural frequency of torsional vibrations of one, two and three rotor systems.	
Text Books:	
1. Theory of Machines / S.S. Rattan/ Tata McGraw-Hill education. 2. Theory of Machines/ Sadhu Singh/ Pearson	
Reference Books:	
1. Mechanism and Machine Theoryby JS Rao and RV Duggipati, New Age International Publishers 2. Theory of Machines by Shigley, Mc Graw Hill Publishers 3. Theory of Machines by Thomas Bevan, CBS Publishers	
Web References:	
1. https://nptel.ac.in/courses/112104114/ 2. https://freevideolectures.com/course/2364/dynamics-of-machines	
E-Text Books:	
1. https://books.google.co.in/books?id=T_WlquM4E14C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=YI7vAwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://nptel.ac.in/courses/112101096/ 2. https://www.mooc-list.com/course/mechanics-kinematics-and-dynamics-edx	

DESIGN OF MACHINE MEMBERS-II**V Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME15	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

The Design of machine members-II focus mainly on design of power transmitting elements like gears, connecting rod, crankpin, crankshafts, pistons, cylinders, bearings, belts, ropes, chain's, pulleys, Power screws and nuts.

COURSE OUTCOMES:

At the end of this course, the student shall

1. Identify design variables and performance factors in the study of journal bearings
2. Apply the basic principles for design of piston, connecting rod and crankshaft based on maximum bending and twisting moments
3. Design screws that are applied in hosting of heavy weights like screw jack etc.
4. Explain power transmission elements like belts, chains and ropes and able to design on stress criteria.
5. Apply the design concepts to evaluate the strength of the gear.

UNIT-I BEARINGS

Types of Journal bearings – lubrication – Bearing modulus –Full and partial bearings, Partial bearings – Clearance ratio – Heat dissipation of bearings, bearing materials – journal bearing design – Ball and roller bearings – Static loading of ball and roller bearings, bearing life.

UNIT-II ENGINE PARTS

Connecting rod: Thrust in connecting rod – stress due to whipping action on connecting rod ends – Cranks and Crank shafts, strength and proportions of over hung and center cranks. Pistons, Forces acting on piston-Construction Design and proportions of piston, Cylinder, Cylinder liners.

UNIT-III POWER SCREWS

Types of screw threads for power screws, Torque required to raise and lower the loads by square threaded screws, Efficiency of square threaded screws, Overhauling and self locking, Acme threads, Stresses in power screws, Design of screw jack.

UNIT-IV	BELTS & PULLEYS
Transmission of power by Belt and Rope drives, Transmission efficiencies, Belts-Flat and V types-Ropes-Pulleys for belt and rope drives, Materials, Chain drives.	
UNIT-V	GEARS
Spur gears-Helical gears-Load concentration factor-Dynamic load factor, Surface compressive strength, Bending strength- Design analysis of spur gears-Estimation of centre distance, module and face width, check for plastic deformation, check for dynamic and wear considerations.	
Text Books:	
1. V.Bandari (2011), A Text Book of Design of Machine Elements,3rd edition,Tata McGraw hill Education (P) Ltd., New Delhi,India. 2. R.L.Norton (2006), Machine Design (An Integrated approach), Pearson Publishers,Chennai,India. 3. K.Mahadevan, K.Balaveera Reddy, Design Data Handbook for Mechanical Engineers, CBS Publishers & Distributors Pvt Ltd.	
Reference Books:	
1. Dr.P.C Sharma, D.K Agarwal, Machine Design, S.K Kataria & Sons. 2. Richard Budynas, J.Keith Nisbett, Shigley's Mechanical Engineering Design, Mc Graw Hill	
Web References:	
1. http://nptel.ac.in/courses/112106137/	
E-Text Books:	
1. https://www.pdfdrive.com/machine-design-r-s-kurmi-e32494497.html 2. https://books.google.co.in/books?id=d-eNeVRc1oC&printsec=frontcover&source=gbs_ge_summary_r&cad=0	
MOOC Course:	
1. https://swayam.gov.in/nd1_noc19_me48/	

METROLOGY & MACHINE TOOLS**V Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME16	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course is introduced to impart knowledge of metal cutting principles and their application to various machining processes, concepts of limits, fits and tolerances, measurement and control.

COURSE OUTCOMES:

At the end of the course, the student would be able to

1. Identify techniques to minimize the errors in measurement.
2. Demonstrate methods and devices for measurement of length, angle, gear & thread parameters, surface roughness and geometric features of parts.
3. Explain working of lathe, shaper, planer, drilling, milling and grinding machines.
4. Comprehend speed and feed mechanisms of machine tools.
5. Estimate machining times for machining operations on machine tools.

UNIT-I LINEAR AND ANGULAR MEASUREMENTS

Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system. Interchangeability and selective assembly. Limit Gauges: Taylor's principle, Design of GO and NO GO gauges. Measurement of angles, Bevel protractor, and Sine bar. Measurements by using auto collimator.

UNIT-II SURFACE ROUGHNESS MEASUREMENT

Roughness, Waviness. CLA, RMS, Rz Values. Methods of measurement of surface finish, Talysurf. Screw thread measurement, Gear measurement; Machine Tool Alignment Tests on lathe, milling and drilling machines. Coordinate Measuring Machines: Types and Applications of CMM.

UNIT-III METAL CUTTING & LATHE MACHINES

Metal cutting: Introduction, elements of cutting process – Geometry of single point tools. Chip formation and types of chips. Engine lathe – Principle of working, types of lathe, specifications. Taper turning, – Lathe attachments. Capstan and Turret lathe – Single spindle and multi-spindle automatic lathes – tool layouts

UNIT-IV DRILLING AND BORING MACHINES

Drilling and Boring Machines – Principles of working, specifications, types, operations performed; twist drill. Types of Boring machines and applications. Shaping, slotting and planing machines –Principles of working – machining time calculations; Work Holding Devices-Jigs and Fixtures, Types and applications.

UNIT-V	MILLING & GRINDING MACHINES
Milling machines – Principles of working – Types of milling machines – Geometry of milling cutters methods of indexing. Grinding – theory of grinding – classification of grinding machines. Types of abrasives, bonds. Selection of a grinding wheel. Lapping, honing and broaching machines, comparison and Constructional features, machining time calculations	
Text Books:	
1. Machine Tool Practices/ Kibbe, John. Neely, T. White, Rolando O. Meyer/ Pearson 2. Fundamentals of Metal Machining and Machine Tools / Geoffrey Boothroyd/ McGraw Hill. 3. A Text Book of Metrology/M.Mahajan/Danpat Rai	
Reference Books:	
1. Principles of Machine Tools, Bhattacharyya A and Sen.G.C / New Central Book Agency. 2. Fundamentals of Dimensional Metrology / Connie Dotson / Thomson 3. Metrology & Measurements – Bewoor &Anand Kapoor	
Web References:	
1. https://nptel.ac.in/courses/112104028/ 2. https://nptel.ac.in/courses/112106179/ 3. https://nptel.ac.in/courses/112104250/	
E-Text Books:	
1. https://nptel.ac.in/courses/112103245/ 2. https://nptel.ac.in/courses/112103248/ 3. https://nptel.ac.in/courses/112106140/	

THERMAL ENGINEERING- II								
V Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME17	CORE	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OVERVIEW : The primary focus of this course is on application of laws of Thermodynamics to analyse steam and gas turbine cycles. It also emphasize on analysis of the major components of steam and gas turbine plants and their applications.								
COURSE OUTCOMES: At the end of the course students will be able to 1. Demonstrate rankine cycle, and Apply thermodynamic concepts on boilers. 2. Figure out different types and applications of steam nozzles, and condensers. 3. Classify and explain steam turbines. 4. Explain working of gas turbines. 5. Demonstrate the basic principles of operation of refrigeration and Air- conditioning systems.								
UNIT-I	BASIC CONCEPTS AND BOILERS							
Basic Concepts: Rankine cycle - Schematic layout, Thermodynamic Analysis, Concept of Mean Temperature of Heat addition, Methods to improve cycle performance – Regeneration & reheating-Cycles. Boilers:Classification – Working principles with sketches including H.P.Boilers – Mountings and Accessories – Working principles- Boiler horse power, Equivalent Evaporation, Efficiency and Heat balance.								
UNIT-II	STEAM NOZZLES & STEAM CONDENSORS							
Steam Nozzles: Stagnation Properties- Function of nozzle – Applications and Types- Flow through nozzles, Thermodynamic analysis – Assumptions -Velocity of nozzle at exit-Ideal and actual expansion in nozzle- Velocity coefficient- Condition for maximum discharge- Critical pressure ratio- Criteria to decide nozzle shape. STEAM CONDENSERS: Requirements of steam condensing plant, Classification of condensers, working principle of different types, vacuum efficiency and condenser efficiency, air leakage, sources and its affects, air pump, cooling water requirement.								
UNIT-III	STEAM TURBINES							
Impulse turbine: Mechanical details, Velocity diagram, effect of friction, power developed, axial thrust, blade or diagram efficiency, condition for maximum efficiency. De-Laval Turbine - its features. Methods to reduce rotor speed-Velocity compounding and pressure compounding, Velocity and Pressure variation along the flow. Reaction Turbine: Mechanical details, principle of operation, thermodynamic analysis of a stage, degree of reaction, velocity diagram, Parson's reaction turbine, condition for maximum efficiency.								

UNIT-IV	GAS TURBINES
Simple gas turbine plant , Ideal cycle, essential components, parameters of performance, actual cycle, regeneration, inter cooling and reheating, Closed and Semi-closed cycles, merits and demerits, Combustion chambers and turbines of Gas Turbine Plant- Brief Concepts.	
UNIT-V	REFRIGERATION AND AIR CONDITONING
REFRIGERATION: Reversed Carnot cycle – Performance Evaluation – Vapour Compression Cycle, Vapour absorption cycle – mechanical details, working principles, use of P-H charts for calculations. AIR- CONDITIONING: Introduction, classification and working principles of air-conditioning systems, requirements, schematic layout of a typical plant.	
Text Books:	
1. R. K. Rajput, Thermal Engineering, 18th edition, Lakshmi Publications, New Delhi, India. 2. V. Ganesan, Gas Turbines, 3rd edition, Tata McGraw-Hill education (P) Ltd, New Delhi, India. 3. CP Arora ,Refrigeration and Air Conditioning , TMH.	
Reference Books:	
1. P.L. Ballaney, 1966, Thermal Engineering, 25 edition, Khanna Publishers, New Delhi, India 2. Cohen, Rogers, SaravanaMuttoo (2011), Gas Turbines, 9th edition, Addison Wesley Longman, New Delhi, India. 3. R. Yadav (2011), Thermodynamics and Heat Engines, 7th Edition, Central Book Depot, Allahabad, India. 4. P. Khajuria, S. P. Dubey (2009), Gas Turbines and Propulsive Systems, 5th edition, Dhanpatra Publications, New Delhi, India. 5. Basic Refrigeration and Air-Conditioning by Ananthanarayanan, TMH	
Web References:	
1. http://nptel.ac.in/downloads/112101004/ 2. https://www.accessengineeringlibrary.com/browse 3. https://www.researchgate.net/publication/233903699	
E-Text Books:	
1. https://engineeringstudymaterial.net/ebook/introduction-to-thermal-systems-engineering 2. http://webserver.dmt.upm.es/~isidoro/tc3/Thermal_engineering.pdf	
MOOC Course:	
1. https://www.edx.org/learn/mechanical-engineering 2. https://www.coursera.org/browse/physical-science-and-engineering/mechanical-engineering 3. https://www.mooc-list.com/categories/engineering	

METROLOGY AND MACHINE TOOLS LAB**V Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME18	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OVERVIEW :

This course intends to understand the concepts of metrology and machine tools and to get practical exposure to the metrology equipment & Machine tools.

COURSE OUTCOMES:

At the end of the course students are able to:

1. Understand the cutting tool geometry, mechanism of chip formation and mechanics of orthogonal cutting.
2. Identify basic parts and operations of lathe, shaper, planer, drilling, boring, milling and grinding machines.
3. Select a machining operation and corresponding machine tool for a specific application in real time.
4. Perform various machining operations such as turning, threading, drilling, boring, shaping etc.
5. Inspect the dimensional and geometric features of a given component by selecting proper measuring instrument.

LIST OF EXPERIMENTS:**A. METROLOGY**

1. Measurement of length, height, diameter by vernier callipers and vernier height guage.
2. Use of gear tooth vernier callipers and checking the chordal thickness of spur gear.
3. Angle and taper measurements by bevel protractor, Sine bars, etc.
4. Thread measurement by two wire/ three wire method & tool makers microscope.
5. Measurement of length, diameter, depth and bore diameter by using micrometer (inside and outside) and dial bore indicator.

B. MACHINE TOOLS

1. Step turning, taper turning and Knurling operations on lathe machine
2. Thread cutting and Boring operations on -lathe machine.
3. Drilling and tapping by drilling Machine
4. Shaping and planning
5. Slotting and Milling operations
6. Cylindrical and Surface grinding
7. Grinding of tool angles.

THERMAL ENGINEERING LAB**V Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME19	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OUTCOMES:

At the conclusion of the course, students will have:

1. Examine the performance parameters of internal combustion engines.
2. Estimate the fuel properties like Viscosity, flash point & fire point
3. Determine the performance characteristics of reciprocating air compressor.
4. Illustrate the Valve timing and port timing diagrams of CI & SI Engines.
5. Interpret the emission characteristics of IC Engines.

LIST OF EXPERIMENTS:

1. Flash and fire points (open cup and closed cup method)
2. Viscosity determination by Red Wood and Saybolts method
3. I.C. Engines Valve / Port Timing Diagrams
4. I.C. Engines Performance Test 4 -Stroke Diesel Engines
5. I.C. Engines Performance Test on 2-Stroke Petrol
6. Morse test on Multi cylinder Petrol
7. I.C. Engines Heat Balance
8. I.C. Engines Air/Fuel Ratio and Volumetric Efficiency
9. Performance Test on Variable Compression Ratio Engine
10. Performance Test on Reciprocating Air – Compressor Unit
11. Dis-assembly / Assembly of Engines
12. Study of Boilers

THEORY OF MACHINES LAB**V Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME20	PCC	L	T	P	C	CIE	SEE	Total
		-	-	3	1.5	30	70	100

COURSE OUTCOMES:

At the end of course students are able to:

1. Evaluate the natural frequency of vibration systems.
2. Analyze the balancing of rotating masses and field balancing.
3. Determine the gyroscopic couple and explain its effects.
4. Analyze range and sensitivity of a governor and analyze cam and follower.
5. Perform forward and inverse kinematic analysis of robotic manipulator.

LIST OF EXPERIMENTS:

1. Determination of frequency of Torsional vibration
2. Determination of frequency of Transverse vibration of Cantilever beam
3. Determination of frequency of Transverse vibration of simply supported beam
4. Static and Dynamic balancing of rotating masses
5. Field balancing of the thin rotors system
6. Study the effect of reactive gyroscope
7. Demonstration of FFT analyzer
8. Forward and inverse kinematic analysis of a robot
9. Whirling speed of a shaft
10. Sensitivity of a governor
11. Cam and follower analysis

ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE**V Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC06	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	0	30	70	100

COURSE OVERVIEW :

The course aims at imparting basic principles of thought process, reasoning and inferencing. Sustainability is at the core of Indian Traditional Knowledge Systems connecting society and nature. The course focuses on introduction to Indian Knowledge System, Indian perspective of modern scientific world-view and basic principles of holistic health care system.

COURSE OUTCOMES:

Upon completion of the course, the students are expected to:

1. Demonstrate the concept of Traditional knowledge and its importance
2. Explain need and importance of protecting traditional knowledge.
3. Discuss various enactments related to the protection of traditional knowledge.
4. Explain the concepts of Intellectual property to protect the traditional knowledge.
5. Explain need of traditional knowledge in various sectors.

UNIT-I INTRODUCTION

Define traditional knowledge, nature and characteristics, scope and importance, kinds of traditional knowledge, the physical and social contexts in which traditional knowledge develop, the historical impact of social change on traditional knowledge systems. Indigenous Knowledge (IK), characteristics, traditional knowledge vis-à-vis indigenous knowledge, traditional knowledge Vs western knowledge traditional knowledge vis-à-vis formal knowledge

UNIT-II PROTECTION OF TRADITIONAL KNOWLEDGE

the need for protecting traditional knowledge Significance of TK Protection, value of TK in global economy, Role of Government to harness TK.

UNIT-III LEGAL FRAME WORK AND TK

The Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006, Plant Varieties Protection and Farmer's Rights Act, 2001 (PPVFR Act); B: The Biological Diversity Act 2002 and Rules 2004, the protection of traditional knowledge bill, 2016. Geographical indicators act 2003.

UNIT-IV	TRADITIONAL KNOWLEDGE AND INTELLECTUAL PROPERTY
Traditional knowledge and intellectual property: Systems of traditional knowledge protection, Legal concepts for the protection of traditional knowledge, Certain non IPR mechanisms of traditional knowledge protection, Patents and traditional knowledge, Strategies to increase protection of traditional knowledge, global legal FORA for increasing protection of Indian Traditional Knowledge.	
UNIT-V	TRADITIONAL KNOWLEDGE IN DIFFERENT SECTORS
Traditional knowledge in different sectors: Traditional knowledge and engineering, Traditional medicine system, TK and biotechnology, TK in agriculture, Traditional societies depend on it for their food and healthcare needs, Importance of conservation and sustainable development of environment, Management of biodiversity, Food security of the country and protection of TK. 139	
Text Books:	
1. Traditional Knowledge System in India, by Amit Jha, 2009. 2. Traditional Knowledge System and Technology in India by Basanta Kumar Mohanta and Vipin Kumar Singh, Pratibha Prakashan 2012.	
Reference Books:	
1. Traditional Knowledge System in India by Amit Jha Atlantic publishers, 2002 2. "Knowledge Traditions and Practices of India" Kapil Kapoor1, Michel Danino2	
Web References:	
1. https://shodhganga.inflibnet.ac.in/bitstream/10603/96925/9/09_chapter%203.pdf 2. https://en.wikipedia.org/wiki/Traditional_knowledge	
E-Text Books:	
1. https://books.google.co.in/books?id=N658caH-C74C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=GrV0DgAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	

III B.TECH II SEMESTER SYLLABUS

HEAT TRANSFER**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME22	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course will provide knowledge about application of conduction, convection and radiation heat transfer concepts to different practical applications.

COURSE OUTCOMES:

- 1 Demonstrate basic modes of heat transfer and compute temperature distribution in steady state and unsteady state heat conduction.
- 2 Analyze the heat transfer through variable thermal conductivity, system with heat generation and extended surfaces.
- 3 Interpret and analyze the free and forced convection heat transfer.
- 4 Comprehend the concepts of Boiling, Condensation and Heat exchangers.
- 5 Explain the principles of radiation heat transfer.

UNIT-I INTRODUCTION

Modes and mechanisms of heat transfer, Basic laws of heat transfer, Applications of heat transfer. Fourier rate equation, general heat conduction equations in Cartesian, Cylindrical and Spherical coordinates. Different forms of the field equation, steady, unsteady and periodic heat transfer – Initial and boundary conditions.

UNIT-II 1-D HEAT CONDUCTION

Homogeneous slabs, hollow cylinders and spheres, Overall heat transfer coefficient, Electrical analogy, and Critical radius of insulation. Systems with internal heat generation. Extended surfaces (Fins), Long, Short and insulated tips. Systems with negligible internal resistance, Significance of Biot and Fourier Numbers, Chart solutions of transient conduction systems.

UNIT-III CONVECTION

CONVECTIVE HEAT TRANSFER: Concepts of Continuity, Momentum and Energy Equations. Dimensional analysis-Buckingham's Pi Theorem - Application for developing non-dimensional correlation for convective heat transfer.

FORCED CONVECTION: External Flows, Concepts of hydrodynamic and thermal boundary layer and use of empirical correlations for Flat plates and Cylinders. Internal Flows, Concepts about Hydrodynamic and Thermal Entry Lengths, use of empirical correlations for Horizontal Pipe Flow and annulus flow.

FREE CONVECTION: Development of Hydrodynamic and thermal boundary layer along a vertical plate, Use of empirical relations for Vertical plates and pipes

UNIT-IV	BOILING & CONDENSATION, HEAT EXCHANGERS
BOILING & CONDENSATION: Regimes of Pool boiling and Flow boiling, Critical heat flux, Calculations on Nucleate Boiling. Film wise and drop wise condensation, Nusselt's theory of condensation on a vertical plate - Film condensation on vertical and horizontal cylinders using empirical correlations. HEAT EXCHANGERS: Classification of heat exchangers, overall heat transfer Coefficient and fouling factor, Concepts of LMTD and NTU methods, Problems using LMTD and NTU methods.	
UNIT-V	RADIATION HEAT TRANSFER
Emission characteristics , Laws of black-body radiation, Irradiation ,Total and monochromatic quantities, Laws of Planck, Wien, Kirchhoff, Lambert, Stefan and Boltzmann, Heat exchange between two black bodies , concepts of shape factor, Emissivity,heat exchange between grey bodies, radiation shields ,electrical analogy for radiation networks..	
Text Books:	
1. Yunus A. Cengel (2012), Heat Transfer a Practical Approach, 4th edition, Tata McGraw hill education (P) Ltd, New Delhi, India. 2. R. C. Sachdeva (2012), Fundamentals of Engineering, Heat and Mass Transfer, 3rd edition, New Age, New Delhi, India.	
Reference Books:	
1. Holman (2012), Heat Transfer (SI Units), 10th edition, Tata McGraw hill education (P) Ltd, New Delhi, India. 2. P. S. Ghoshdastidar (2012), Heat Transfer, 2nd edition, Oxford University Press, New Delhi, India. 3. Incropera, Dewitt (2012), Fundamentals of Heat Transfer, 6th edition, John Wiley, UK.	
Web References:	
1. http://nptel.ac.in/courses/112104113/ 2. http://nptel.ac.in/courses/112105123/ 3. https://www.udemy.com/fundamentals-of-engineering-heat-transfer	
E-Text Books:	
1. https://www.pdfdrive.net/heat-transfer-e18770201.html 2. https://www.pdfdrive.net/fundamentals-of-heat-transfer-e18762062.html 3. https://www.pdfdrive.net/heat-transfer-e15057935.html	
MOOC Course	
1. https://onlinecourses.nptel.ac.in/ 2. https://www.coursera.org/learn/heat-transfer-intro 3. https://swayam.gov.in/courses/4642-july-2018-conduction-and-convection	

CAD/CAM								
VI Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME23	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100
COURSE OVER VIEW : This course will provide an overview of how computers are being used in design, development of manufacturing plans and manufacture. It covers various concepts of CAD/CAM such as geometric modelling, NC control, NC part programming, group technology etc.								
COURSE OUTCOMES: At the end of the course students will be able to 1. Demonstrate basic concepts of CAD. 2. Explain wire frame modeling, surface modeling and solid modeling. 3. Discuss about NC, CNC, DNC, adaptive control systems, apt language, group technology and develop CNC programs. 4. Explain computer aided process planning and computer aided manufacturing resource planning. 5. Explain concepts of computer aided quality control and computer integrated manufacturing.								
UNIT-I	INTRODUCTION							
Fundamentals of CAD,CAM, Automation , design process, Application of computers for design, Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD, Design workstation, Graphic terminal, CAD software- definition of system software and application software ,CAD database and structure. Product Design and CAD- Comparison, benefits of using CAD in product design and product development cycle.								
UNIT-II	GEOMETRIC MODELING							
Introduction, Wire frame modeling: Definition, advantages, dis-advantages, wire frame entities- analytic entities and synthetic entities. Surface modeling: Definition, advantages, dis-advantages, surface entities- analytic entities and synthetic entities. Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities. Curve representation: Implicit and explicit forms of straight line, circle, ellipse, cubic spline and biezzer curve, differences between biezzer curve and , cubic spline curve.								

UNIT-III	NC CONTROL & GROUP TECHNOLOGY
NC Control: Introduction, Elements of NC system, NC part programming: Methods of NC part programming, manual part programming, Computer assisted part programming, Post Processor, Computerized part program. CNC, DNC and Adaptive Control Systems, APT language Group Technology: Part families, Parts classification and coding. Production flow analysis, Machine cell design using rank order cluster technique.	
UNIT-IV	COMPUTER AIDED PROCESS PLANNING & COMPUTER AIDED MANUFACTURING
Computer aided process planning: Difficulties in traditional process planning, Computer aided process planning: retrieval type and generative type. Computer aided manufacturing: Material resource planning, inputs to MRP, MRP output records, Benefits of MRP, Enterprise resource planning, Capacity requirements planning	
UNIT-V	COMPUTER AIDED QUALITY CONTROL AND CIM
Computer aided quality control: Automated inspection- Off-line, On-line, contact, Non-contact; Coordinate measuring machines, Machine vision. Computer Integrated Manufacturing: CIM system, Benefits of CIM, Dis-advantages of CIM, Structured query language, Networking configurations-star, ring and bus networks.	
Text Books:	
1. CAD/CAM Principles and Applications by P.N.Rao, TMH 2. CAD/CAM Concepts and Applications by Alavala, PHI	
Reference Books:	
1. CAD/CAM by Groover M.P., Pearson education 2. CAD/CAM Theory and Practice, / Ibrahim Zeid, TMH 3. CAD/CAM/CIM by Radhakrishnan and Subramanian, New Age 4. Principles of Computer Aided Design and Manufacturing by Farid Amirouche, Pearson 5. Computer Numerical Control Concepts and programming by Warren S Seames, Thomson.	
Web References:	
1. https://www.slideshare.net/AbhayGore/cadcam-cae-notes 2. https://www.allintuworld.in/download/cad-cam-materials-notes/	
E-Text Books:	
1. https://books.google.co.in/books?id=mzm9Wuul4mQC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=hEkjURbiN7MC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	

MOOC Course:

1. <https://www.mooc-list.com/course/computer-aided-design-cad-saylororg>
2. <https://www.lynda.com/CAD-training-tutorials/1665-0.html>
3. <https://nptel.ac.in/courses/112102101/>

HEAT TRANSFER LAB**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME24	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW :

Heat Transfer laboratory provides fundamental and industrial knowledge about modes of heat transfer, like conduction, convection and radiation, and their application.

COURSE OUTCOMES:

At the conclusion of the course, students will be able to:

1. Apply the principles of various modes of heat transfer to evaluate thermal conductivity of different metal and non metal objects under steady state conditions
2. Apply the principles of various modes of heat transfer to evaluate thermal conductivity of different objects under unsteady state conditions
3. Estimate heat transfer coefficients in forced convection, free convection, condensation and correlate with theoretical values
4. Obtain variation of temperature along the length of the pin fin under forced and free convection
5. Apply the principles of radiation to evaluate the surface emissivity of a test plate and Stefan-Boltzmann's constant and compare with theoretical values

LIST OF EXPERIMENTS:

1. Composite Slab Apparatus – Overall heat transfer co-efficient.
2. Heat transfer through lagged pipe.
3. Heat Transfer through a Concentric Sphere
4. Thermal Conductivity of given metal rod.
5. Heat transfer through pin-fin
6. Experiment on Transient Heat Conduction
7. Heat transfer in forced convection apparatus.
8. Heat transfer in natural convection
9. Parallel and counter flow heat exchanger.
10. Emissivity apparatus.
11. Stefan Boltzmann Apparatus.
12. Heat transfer in drop and film wise condensation.
13. Critical Heat flux apparatus.
14. Study of heat pipe and its demonstration.

PRODUCTION DRAWING AND CAD/CAM LAB**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME25	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW :

This course is framed to have hands on experience on various software's such as AUTOCAD, CATIA AND ANSYS so that students are able to model and analyze various mechanical engineering systems.

COURSE OUTCOMES:

At the end of the course the students are able to

1. Draft various part drawings using AutoCAD or CATIA.
2. Model various machine parts and assemble using CATIA OR Pro-E or UNIGRAPHICS.
3. Use CATIA OR Pro-E or UNIGRAPHICS software for assembly of machine parts.
4. Use ANSYS software to solve engineering field problems (static, dynamic and heat transfer).
5. Perform different machining operations on CNC Lathe and CNC Mill.

SYLLABUS**A. CAD**

1. **DRAFTING:** Development of part drawings for various components in the form of orthographic and isometric. Representation of Dimensioning and tolerances scanning and plotting.
2. **PART MODELING:** Generation of various 3D Models through Protrusion, revolve, shell sweep Creation of various features. Study of parent child relation. Feature based and Boolean based modeling surface and assembly modeling.
3. **ANALYSIS:**
 - a. Static analysis
 - b. Dynamic analysis
 - c. Heat transfer Analysis

Any Two Software Packages from the following should be used:

AUTOCAD, Pro-E, I-DEAS, CATIA, UNIGRAPHICS, ANSYS AND NASTRAN

B. CAM

1. Perform Turning operation for a given model by using CNC lathe.
2. Perform Step turning operation for a given model by using CNC lathe.
3. Perform profile turning operation for a given model by using CNC lathe.
4. Perform different operations on CNC Mill.

C. PRODUCTION DRAWING

Any two softwares from the following can be used:

AutoCAD, CATIA, Pro-E and Fusion 360.

SYLLABUS**UNIT - I**

Conventional representation of Materials – conventional representation of parts – screw joints, welded joints, springs, gears, electrical, hydraulic and pneumatic circuits – methods of indicating notes on drawings.

UNIT - II

Limits and Fits: Types of fits, exercises involving selection / interpretation of fits and estimation of limits from tables.

UNIT - III

Geometric Dimensioning and Tolerancing (GD&T): Need for GD&T, Features and rules of GD&T, datums control, Adding GD&T to a drawing/design, Form tolerances, Orientation tolerances, Profile tolerances, Location Tolerances, Runout tolerances.

UNIT - IV

Surface roughness and its indication: Definitions – finishes obtainable from various manufacturing processes, recommended surface roughness on mechanical components. Heat treatment and surface treatment symbols used on drawings.

UNIT - V

Detailed and Part drawings: Drawing of parts from assembly drawings with indications of size, tolerances, roughness, form and position errors etc. Part drawing using computer aided drafting by CAD software

TEXT BOOKS

1. Production and Drawing – K.L. Narayana & P. Kannaiah/ New Age.
2. machine Drawing with Auto CAD- Pohit and Ghosh, PE .

REFERENCE BOOKS

1. Geometric dimensioning and tolerancing- James D. Meadows/ B.S Publications.
2. Engineering Metrology, R.K. Jain, Khanna Publications

ADVANCED ENGLISH COMMUNICATIONS LAB**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS07	HSMC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW:

The English Language Lab is considered essential at Third year level. At this stage the students need to prepare themselves for their careers which may require them to listen to, read, speak and write in English both for their professional and interpersonal communication in the globalised context. The proposed course should be an integrated theory and lab course to enable students to use 'good' English and perform the following: Gather ideas and information, to organise ideas relevantly and coherently, engage in debates, participate in group discussions, face interviews, write project/research reports/technical reports, make oral presentations, write formal letters, transfer information from non-verbal to verbal texts and vice versa and to take part in social and professional communication.

COURSE OUTCOMES:

1. Organize the ideas coherently from the text.
2. Participate in debates, group discussions.
3. Write project/research reports/technical reports/formal letters.
4. Make oral presentations.

SYLLABUS

The following course content to conduct the activities is prescribed for the Advanced Communication Skills (ACS) Lab:

Week -1**Activities on Fundamentals of Inter-personal Communication**

Starting a conversation - responding appropriately and relevantly - using the right body language - Role Play in different situations & Discourse Skills- using visuals.

Week -2**Activities on Building Vocabulary**

Synonyms and antonyms, word roots, one-word substitutes, prefixes and suffixes, study of word origin, business vocabulary, analogy, idioms and phrases, collocations & usage of vocabulary

Week -3**Activities on Reading Comprehension**

General Vs Local Comprehension, Reading for facts, guessing meanings from context, Scanning and Skimming.

Week -4

Activities on Reading for Specific Purposes

Inferring meaning, Critical reading & Effective goggling.

Week-5**Activities on Writing Skills- Technical Reports**

Structure and presentation of different types of writing - letter writing/ Resume writing/ e-correspondence

Week-6**Activities on Writing Skills**

Technical report writing/ Portfolio writing - planning for writing - improving one's writing.

Week- 7**Activities on Presentation Skills**

Oral presentations (individual and group) through JAM sessions and Seminars.

Week- 8**Activities on Presentation Skills Using ICT**

PPTs and written presentations through posters/ projects/ reports/ e-mails/ assignments etc.

Week- 9**Activities on Group Discussion**

Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation- Concept and process

Week-10**Interview Skills**

Pre-interview planning, opening strategies, answering strategies, interview through tele-conference & video-conferencing and Mock Interviews.

Books Recommended:

1. Raman, M & Sharma, S. (2009). Technical Communication. Oxford University Press.
2. Rani. S. (2011). Advanced Communication Skills Laboratory Manual. Pearson Education.
3. Anderson, V. (2007). Technical Communication. Cengage Learning pvt. Ltd.
4. Kelly M. Quintanilla & Shawn T. Wahl. (2011). Business and Professional Communication: Keys for Workplace Excellence. Sage South Asia Edition. Sage Publications.
5. Stev. D & David T. Mc Mahan. (2012). The Basics of Communication: A Relational Perspective. Sage South Asia Edition. Sage Publications.
6. Mc Murrey. D & Buckley. J. (2012). Handbook for Technical Communication Cengage Learning.
7. Sen. L. (2009). Communication Skills. PHI Learning Pvt Ltd.
8. Vishvamohan, A. (2009). English for Technical Communication for Engineering Students.

Tata Mc Graw Hill.

9. Books on TOFEL/ GRE/ GMAT/ CAT/ IELTS by Barron's/ DELTA/ Cambridge University Press.
10. Tomalin, B & Thomas, B. (2009). International English for Call Centers. Macmillan Publishers.

CONSTITUTION OF INDIA

V Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4MC03	MC	L	T	P	C	CIE	SEE	Total
		2	-	-	0	30	70	100

COURSE OVERVIEW:

The Constitution of India is the supreme law of India. It reflects the idea of “Constitutionalism” – a modern and progressive concept historically developed by the thinkers of “liberalism” – an ideology which has been recognized as one of the most popular political ideology and result of historical struggles against arbitrary use of sovereign power by state. The historic revolutions in France, England, America and particularly European Renaissance and Reformation movement have resulted into progressive legal reforms in the form of “constitutionalism” in many countries. The Constitution of India was made by borrowing models and principles from many countries including United Kingdom and America. The Constitution of India is not only a legal document but it also reflects social, political and economic perspectives of the Indian Society. This course will give complete picture of constitution of india.

COURSE OUTCOMES:

Students will be able to:

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy, its institutions and processes at the local, state and union levels.
5. Explain the functions and responsibilities of election commission of India and union public service commission.

U NIT-I HISTORY

History of Making of the Indian Constitution: Introduction to the constitution of India, the making of the constitution and salient features of the constitution.

UNIT-II PHILOSOPHY

Philosophy of the Indian Constitution: Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures

UNIT-III UNION GOVERNMENT

Union Government: Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.

UNIT-IV	STATE GOVERNMENT
State Government: State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)	
UNIT-V	ELECTION COMMISSION
Election Commission: Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.	
Text Books:	
1. The constitution of India, P.M.Bakshi, Universal Law Publishing Co., 2. The Constitution of India, 1950 (Bare Act), Government Publication. 3. Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015. 4. M. P. Jain, Indian Constitution Law, 7th Edn., Lexis Nexis, 2014.	
Reference Books:	
1. D.D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2015. 2. Indian constitution at work, NCERT	
Web References:	
1. https://www.constitution.org/cons/india/const.html 2. https://www.india.gov.in/sites/upload_files/npi/files/coi_part_full.pdf	
E-Books:	
1. https://books.google.co.in/books?id=rA39_9XtbBkC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=3mrKDQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
Mooc Course:	
1. http://nludelhi.ac.in/images/moocs/moocs-courses.pdf 2. https://mhrd.gov.in/online-courses 3. https://www.classcentral.com/tag/constitutional-law	

IV B.TECH I SEMESTER SYLLABUS

COMPOSITE MATERIALS**VII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME26	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

The course is framed to cover basic and important knowledge of composites such as classification of composite materials, design, fabrication and testing of composite materials.

COURSE OUTCOMES:

At the end of the course, the students will be able to

1. Explain and differentiate composite materials based on specific applications.
2. Describe the various fabrication methods of polymer composites
3. Choose suitable manufacturing route for generating metal and ceramic matrix composites
4. Demonstrate the various methods available to test the Composites through NDT methods.
5. Elucidate the purpose and significance of Multi-Functional Composites

UNIT- I INTRODUCTION

INTRODUCTION: Introduction to Composites, function of the matrix and reinforcement in composites. Classification: polymer matrix composites, metal matrix composites, ceramic matrix composites, carbon-carbon composites, fiber reinforced composites, particulate reinforced composites and nature-made composites. Reinforcement types: Fiber Glass, Silica, Kevlar, carbon, boron, silicon carbide, and boron carbide.

UNIT- II POLYMER COMPOSITES: MANUFACTURING METHODS

Thermoset Composites Manufacturing- Lay-up processes, Spray up process, Fiber placement process, Resin transfer moulding, Vacuum assisted resin transfer moulding, Compression moulding process, Filament winding. Thermoplastic Composite manufacturing- Sheet moulding, Injection moulding, sheet moulding, Calendaring, Extrusion, Blow moulding, rotational moulding, Thermoforming.

UNIT- III METAL AND CERAMIC COMPOSITES: MANUFACTURING METHODS

Metal Matrix Composites- Solid state methods- hot isostatic pressing (HIP), Foil diffusion bonding. Liquid state methods- Stir casting, Squeeze casting, Pressure infiltration; Ceramic Matrix Composites- sintering, CVD. Manufacturing Process Selection: Cost, performance, size shape, rate of production. Steps for process selection

UNIT- IV	COMPOSITES DESIGN AND TESTING
COMPOSITES DESIGN AND TESTING: Laminate theory, Rule of mixtures, symmetry and balance. Non-destructive testing of Composites- Visual inspection, Tap testing, Ultrasonic inspection, X-ray inspection, Thermography.	
UNIT- V	MULTI-FUNCTIONAL POLYMER NANOCOMPOSITES
MULTI-FUNCTIONAL POLYMER COMPOSITES: Introduction, necessity Carbon Nanotubes based Polymer Nanocomposites - Fabrication routes: Ultrasonication, In-Situ method. Grafting CNTs onto fibre surfaces - spraying method, transfer printing, chemical grafting. Properties and Applications Graphene based polymer composites - Introduction, properties and applications	
Text Books:	
1. Mazumdar, S., 2001. Composites manufacturing: materials, product, and process engineering. CRC press. 2. Balasubramanian, M., 2013. Composite materials and processing. CRC press. 3. Campbell Jr, F.C. ed., 2003. Manufacturing processes for advanced composites. Elsevier. 4. Bafekrpour, E. (2019). Advanced Composite Materials: Properties and Applications. Berlin, Boston: De Gruyter.	
Reference Books:	
1. Mechanics of Composite Materials, Madhujit Mukhopadhyay, Universities Press.	
Web References:	
1. https://www.slideshare.net/COLLINSBEN1/composites-lecture-notes 2. https://link.springer.com/content/pdf/bfm%3A978-1-4020-8772-1%2F1.pdf 3. http://ocw.uc3m.es/ciencia-e-oin/materials-science-and-engineering/lecture-notes-1/Chapter_8_1.pdf	
E-Text Books:	
1. https://books.google.co.in/books?id=D3Wta8e07t0C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=5Q6oUTF00RqC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://online.stanford.edu/courses/aa256-mechanics-composites 2. https://www.edx.org/course/composite-materials-overview-for-engineers 3. https://www.coursera.org/courses?query=materials	

INSTRUMENTATION AND CONTROL SYSTEMS

VII Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME27	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW :

This course will focus on the basic characteristics of a typical instrument, Identification of errors and their types that would occur in a instrument, Identification of properties used for evaluating the thermal systems and the concept of transducer, various types and their character

COURSE OUTCOMES:

At the end of the course, the student will be able to

1. Explain constructional details and working principles of various instruments and their purpose
2. Demonstrate the instruments used for temperature, pressure and level.
3. Explain measurement of flow, speed, acceleration and vibration measuring instruments.
4. Explain about equipment used for the measurement of stress-strain, force, torque and power.
5. Discuss about humidity measurement instruments and elements of control system.

UNIT- I INTRODUCTION & MEASUREMENT OF DISPLACEMENT

Introduction: Definition – Basic principles of measurement – Measurement systems, generalized configuration and functional description of measuring instruments – examples. Static and Dynamic performance characteristics – sources of errors, Classification and elimination of errors.

Measurement of Displacement: Theory and construction of various transducers to measure displacement – Piezo electric, Inductive, capacitance, resistance, ionization and Photo electric transducers, Calibration procedures.

UNIT- II MEASUREMENT OF TEMPERATURE & PRESSURE

Measurement of Temperature: Various Principles of measurement-Classification: Expansion Type: Bimetallic Strip- Liquid in glass Thermometer; Electrical Resistance Type: Thermistor, Thermocouple, RTD. Measurement of Pressure: Different principles used- Classification: Manometers, Dead weight pressure gauge, Tester (Piston gauge), Bourdon pressure gauges; Bellows – Diaphragm gauges. Low pressure measurement – Thermal conductivity gauges, ionization pressure gauges, Mcleod pressure gauge.

UNIT- III MEASUREMENT OF LEVEL, FLOW AND SPEED

Measurement of Level: Direct methods – Indirect methods – Capacitive, Radioactive, Ultrasonic, Magnetic, Cryogenic Fuel level indicators – Bubbler level indicators.

Flow measurement: Rotameter, magnetic, Ultrasonic, Turbine flow meter.

Measurement of Speed: Mechanical Tachometers, Electrical tachometers, Non- contact type-Stroboscope

UNIT- IV	MEASUREMENT OF STRESS-STRAIN, ACCELERATION & VIBRATION
Stress-Strain measurements: Various types of stress and strain measurements , electrical strain gauge , gauge factor , method of usage of resistance strain gauge for bending compressive and tensile strains, usage for measuring torque, Strain gauge Rosettes. Measurement of Acceleration and Vibration: Different simple instruments – Principles of Seismic instruments – Vibrometer and accelerometer using this principle- Piezo electric accelerometer.	
UNIT- V	MEASUREMENT OF HUMIDITY & ELEMENTS OF CONTROL SYSTEMS
Measurement of Humidity: Moisture content of gases, Sling Psychrometer, Absorption Psychrometer, Dew point meter. Elements of Control Systems: Introduction, Importance – Classification – Open and closed systems Servomechanisms – Examples with block diagrams – Temperature, speed and position control systems- Transfer functions- First and Second order mechanical systems	
Text Books:	
1. D.S. Kumar(2011), Mechanical Measurements and Controls, 4th edition, Metropolitan Book Co. Pvt Ltd., New Delhi, India. 2. K. Tayal (2004), Instrumentation and mechanical Measurements, 2nd edition, Galgotia Publications, New Delhi, India.	
Reference Books:	
1. Chennakesava R. Alavala(2010), Principles of Industrial Instrumentation and Control Systems, 1st edition, Cengage Learning, New Delhi, India. 2. C. Nakra, K. K. Choudhary (2010), Instrumentation, measurement and analysis, 4th, Tata McGraw-Hill, New Delhi, India.	
Web References:	
1. http://nptel.ac.in/courses/112107145/ 2. http://nptel.ac.in/courses/112107078/	
E-Text Books:	
1. https://books.google.co.in/books?id=8ve7z2XRBy8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=X-gADDWI-NiC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://onlinecourses.nptel.ac.in/noc18_me05/preview 2. https://www.edx.org/course/introduction-to-control-system-design-a-first-look	

COMPOSITE MATERIALS LAB**VII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME28	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OBJECTIVES:

Composite Material Lab is intended to teach the basics of composites manufacturing including polymer matrix composites and metal matrix composites. Experiments are framed to guide the learning process and impart technical knowledge in selecting the matrix and reinforcement material for fabricating the composites.

COURSE OUTCOMES :

At the end of the course, students will be able:

1. Fabricate metal matrix and polymer matrix composites.
2. Determine mechanical properties of laminated composites.
3. Determine voids and flaws of laminated composites.
4. Prepare metal matrix composites using stir casting machine.
5. Find mechanical properties of MMCs.

LIST OF EXPERIMENTS:

1. Fabrication of a laminated composite using hand lay-up process.
2. Study on the flexural properties of laminated composites using three point bend fixture.
3. Determining the Barcol Hardness of Laminated Composites.
4. Detecting the Voids and Flaws in Laminated Composites using Ultrasonic Flaw Detector.
5. Preparation of Aluminium Matrix composites using Stir Casting Machine.
6. Determining the tensile properties of MMCs using UTM.
7. Determining the hardness of MMCs using Rockwell Hardness Tester.
8. Study on the flexural properties of MMCs using three point bend fixture.
9. Studying the Impact properties of MMCs using drop weight test machine.
10. Detecting the Voids and Flaws in MMCs using Ultrasonic Flaw Detector

INSTRUMENTATION LAB**VII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME29	PC C	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OVERVIEW:

This course intended to learn calibration of pressure gauges, calibration of resistance temperature detector for temperature measurement, thermister and RTD, thermocouple for temperature measurement, calibration of photo and magnetic speed pickups for the measurement of speed, measurement of angular displacement using capacitive transducer, measurement of strain gauge, calibration of LVDT transducer for displacement measurement and calibration of Mcleod gauge for low pressure.

COURSE OUTCOMES :

At the end of the course, the student will be able to

1. Calibrate Pressure gauge and McLeod gauge for measurement of pressure.
2. Calibrate transducer, strain gauge, resistance temperature detector for temperature measurement.
3. Calibrate LVDT for measurement of displacement and Transducer for measurement of angular displacement.
4. Calibrate Photo and Magnetic Speed Pickup equipment for measurement of speed, Seismic pickup equipment for measurement of vibration amplitude of an engine.
5. Calibrate rotameter for flow measurement.

LIST OF EXPERIMENTS:

1. Calibration of Pressure Gauges.
2. Calibration of transducer for temperature measurement.
3. Study and calibration of LVDT transducer for displacement measurement.
4. Calibration of strain gauge for temperature measurement.
5. Calibration of thermocouple for temperature measurement.
6. Calibration of capacitive transducer for angular displacement.
7. Study and calibration of photo and magnetic speed pickups for the measurement of speed.
8. Calibration of resistance temperature detector for temperature measurement.
9. Study and calibration of a rotameter for flow measurement.
10. Study and use of a Seismic pickup for the measurement of vibration amplitude of an engine bed at various loads.
11. Study and calibration of McLeod gauge for low pressure.

PROFESSIONAL OPEN ELECTIVES

FINITE ELEMENT METHODS**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME35	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

The aim of this course to enable students to understand fundamental theory of the FEA method, generate the governing FE equations for systems governed by partial differential equations, understand the use of the basic finite elements for structural applications using truss, beam, frame, and plane elements; understand the application and use of the FE method for heat transfer problems.

COURSE OUTCOMES:

At the end of the course students will

1. Demonstrate basic concepts of FEM, formulate and solve finite element equations for 1-D elements.
2. Derive stiffness matrices, analyze trusses and beams subjected to different load conditions
3. Solve 2-d and axi-symmetric problems using constant strain triangular elements.
4. Formulate and solve finite element equations of heat transfer problems, Gaussian one-point and two point formulae.
5. Analyze axial bar, beam and truss subjected to dynamic loads and demonstrate basic features of various CAE software's.

UNIT-I INTRODUCTION

Basic concept, historical background, applications of FEM, Basic equations of elasticity-strain-displacement relations and stress-strain relations, Convergence requirements, Principle of Minimum Potential Energy-Spring Problems.

UNIT-II ONE DIMENSIONAL PROBLEMS & ANALYSIS OF TRUSSES**Classes: 12**

ONE DIMENSIONAL PROBLEMS: Formulation of Stiffness Matrix for a Bar Element by the Principle of Minimum Potential Energy, Properties of Stiffness Matrix, Characteristics of Shape Functions, Quadratic shape functions. Problems on uniform stepped and tapered bars subjected to point loads, body force and temperature loads.

ANALYSIS OF TRUSSES: Local and Global co-ordinates, Transformation matrix, Stiffness Matrix for Plane Truss, Stress-Strain Calculations and Calculation of reaction forces.

UNIT-III ANALYSIS OF BEAMS & 2-D PROBLEMS**Classes: 12**

ANALYSIS OF BEAMS: Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, slope and deflection. Simple problems-Point loads & UDL.

2-D PROBLEMS: CST-Stiffness matrix and load vector, Isoperimetric element representation, Shape functions, convergence requirements, Problems. Shape functions of Two dimensional four noded isoperimetric element

and convergence requirements.

UNIT-IV	AXI-SYMMETRIC SOLIDS & HEAT TRANSFER	Classes: 11
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AXI-SYMMETRIC SOLIDS: Axi-symmetric solids subjected to Axi-symmetric loading with triangular elements- simple problems.

STEADY STATE HEAT TRANSFER ANALYSIS: One Dimensional Finite Element analysis of fin and composite slabs.

UNIT-V	DYNAMIC ANALYSIS	Classes: 11
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DYNAMIC ANALYSIS: Lumped and consistent mass methods, Evaluation of Eigen values and Eigen vectors for a stepped bar, truss and beam.

Text Books:

1. Introduction to Finite Elements in Engineering by Chandrupatla, Ashok and Belegundu, Prentice, Hall, Pearson.
2. The Finite Element Methods in Engineering by SS Rao, Pergamon.

Reference Books:

1. Finite Element Methods: Basic Concepts and applications by Alavala, PHI
2. Finite Element Method by Zincowitz, Mc Graw Hill
4. Finite Element Analysis by P. Seshu, PHI
5. Finite Element Analysis by Hutton, TMH
6. Finite Element Analysis by Bathe, PHI
7. Finite Element Method by Krishna Murthy, TMH

Web References:

1. http://web.mit.edu/16.810/www/16.810_L4_CAE.pdf
2. <https://www.simscale.com/blog/2016/10/what-is-finite-element-method/>

E-Text Books:

1. https://books.google.co.in/books?id=0FU8DwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=xbugmEwQz8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
3. https://books.google.co.in/books?id=wzZ7BkOgMnAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.classcentral.com/course/npTEL-basics-of-finite-element-analysis-i-5234>
2. <https://www.open.edu/openlearn/science-maths-technology/introduction-finite-element-analysis/content->

[section-0?active-tab=description-tab](#)

NANOTECHNOLOGY

VI Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME36	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Nano Technology is one of the core subjects of multidisciplinary nature. This has extensive applications in the field of energy; electronics Etc., built to specifications by manufacturing matter on the atomic scale, the Nano products would exhibit an order of magnitude improvement in strength, toughness, and efficiency. The course will give insight into many aspects of nanotechnology and their applications in the prospective of materials science

COURSE OUTCOMES:

At the end of the course student will be able to

1. Demonstrate various concepts of nanotechnology and explain effect of size on different properties of nano materials.
2. Explain properties, microstructure and defects in nano materials.
3. Explain how to synthesize nano materials
4. Enumerate and explain various characterization methods of nano materials
5. Explain applications of nanotechnology.

UNIT-I INTRODUCTION

Defining nanotechnology, Importance of Nano-technology, nanoscale comparison, Challenges in nanotechnology. Nanoscience vs Nanotechnology, Classification-According to their origin, dimensions & structural configuration.

Size Effect: Structural differences on the nanoscale, Factors Influencing Properties of nanomaterials, Surface Area and Energy, effect of size on optical, electrical, physical & chemical properties

UNIT-II UNIQUE PROPERTIES OF NANO MATERIALS

Microstructure and Defects in Nanocrystalline Materials: Dislocations, Twins, stacking faults and voids, Grain Boundaries, triple, and disclinations, Effect of Nano-dimensions on Materials Behavior: Elastic properties, Melting Point, Diffusivity, Grain growth characteristics, enhanced solid solubility, Magnetic Properties: Soft magnetic Nanocrystalline alloy, Permanent magnetic Nanocrystalline materials, Giant Magnetic Resonance, Electrical Properties, Optical Properties, Thermal Properties, and Mechanical Properties.

UNIT-III SYNTHESIS OF NANO MATERIALS

Physical Methods- Mechanical Methods, Methods based on Evaporation, Sputter Deposition, Chemical Vapour Deposition, Electric Arc Deposition, Ion Beam Technique, Molecular Beam Epitaxy.

Chemical Methods-synthesis of Metal nano particles by colloidal route, Sol-Gel Method, Hydrothermal synthesis, Sonochemical Synthesis, Microwave Synthesis

UNIT-IV NANO MATERIALS CHARACTERIZATION

Nanomaterials characterization: Instrumentation Fractionation principles of Particle size measurements, Particle size and its distribution, XRD, Zeta potential Microscopies SEM, TEM, Atomic Force Microscopy, Scanning and Tunneling Microscopy

UNIT-V APPLICATIONS OF NANO MATERIALS

Nano-medical and healthcare applications, Textiles, Paints, Energy, Defence and Space Applications, Future transportation benefits

Text Books:

1. Text Book of Nano Science and Nano Technology – B.S. Murthy, P. Shankar, Baldev Raj, B.B. Rath and James Munday, University Press-IIIM.
2. Introduction to Nanotechnology – Charles P. Poole, Jr., and Frank J. Owens, Wiley India Edition, 2012.

Reference Books:

1. Nano: The Essentials by T. Pradeep, Mc Graw- Hill Education.
2. Nanomaterials, Nanotechnologies and Design by Michael F. Ashby, Paulo J. Ferreira and Daniel L. Schodek.
3. Transport in Nano structures- David Ferry, Cambridge University press 2000
4. Nanofabrication towards biomedical application: Techniques, tools, Application and impact – Ed. Challa S., S. R. Kumar, J. H. Carola.
5. Carbon Nanotubes: Properties and Applications- Michael J. O'Connell
6. Electron Transport in Mesoscopic systems – S. Dutta, Cambridge University press

Web References:

1. <https://www.understandingnano.com/introduction.html>
2. <https://www.understandingnano.com/>

E-Text Books:

1. https://books.google.co.in/books?id=erjymgEACAAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=kVOtnMY3jzAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.coursera.org/courses?query=nanotechnology>
2. <https://www.edx.org/learn/nanotechnology>

COMPUTATIONAL FLUID DYNAMICS**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME37	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

The primary focus of this course is to differentiate between different types of partial differential equations, to apply appropriate numerical techniques and solve heat transfer and fluid flow problems using different numerical techniques

COURSE OUTCOMES:

At the end of the course, the student should be able to

1. Demonstrate modes of heat transfer, methods to solve a physical problem, differentiate analytical and numerical methods.
2. Differentiate between different types of Partial Differential Equations and to apply appropriate numerical techniques
3. Solve 1-D and 2-D steady state heat conduction problems
4. Explain explicit, implicit, semi implicit methods and application of FDM to solve 1D transient heat conduction equations
5. Formulate and solve governing equations for incompressible flow

UNIT-I INTRODUCTION

Review of Modes of Heat Transfer – Governing Equations – Initial and boundary conditions Methods to solve a physical problem –Relative advantages and disadvantages of experimental, analytical and numerical methods – Scope of CFD – Its applications and limitations - Brief comparison between different numerical methods, viz., FDM, FEM & FVM Methods to solve a system of simultaneous Linear Algebraic Equations, - Direct Method – Banded Matrices – Thomas algorithm / TDMA - iterative schemes of Matrix Inversion.

UNIT-II CLASSIFICATION OF PDE & FDM

Classification of PDE – Elliptic, parabolic and hyperbolic PDE as governing equations – Examples and their physical significance

FDM – Discretization of Partial Derivative Terms using Taylor's series of approximation – Finite Difference Formulae – Application and implementation aspects of finite-difference equations –Consistency Application of FDM to elliptic equations, viz., Laplace Equations – solution of 1D steady state heat conduction using FDM – Systems with heat generation –Systems with variable thermal conductivity - Fins

UNIT-III	1-D & 2-D STEADY STATE HEAT CONDUCTION	
Application of FDM to solve 1D steady state heat conduction in Curvilinear geometry – Singularities – Treatment of singularities. Application of FDM to solve 2D steady state heat conduction– with and without heat generation and subjected to different boundary conditions		
UNIT-IV	PARABOLIC EQUATIONS & 1-D TRANSIENT HEAT CONDUCTION	Classes: 12
Parabolic Equations – Use of Explicit, implicit and semi implicit methods – Errors and Stability analysis - application of FDM to solve 1D transient heat conduction equations – ADI Scheme – Treatment and Implementation		
UNIT-V	NUMERICAL METHODS FOR INCOMPRESSIBLE FLOW	Classes: 12
Numerical methods for incompressible flow – Governing equations –Difficulties in solving N-S equations – Stream function and Vorticity method – Advantages and disadvantages – treatment of boundary conditions – Determination of Pressure for viscous flows – Disadvantages – Staggered Grid – SIMPLE algorithm for pressure liked equations.		
Text Books:		
1. Computational Fluid Flow and Heat Transfer/ Muralidharan&Sundarajan/ Narosa Publications 2. Finite Difference Method in Heat Transfer – NecatiOzisik, CRC Press 3. Computer Simulation of Flow and Heat Transfer – Ghoshdastidar, Tata McGraw Hill 4. Numerical Methods – Chapra and Canale, TMH 5. Numerical Methods –Balaguruswamy/TMH		
Reference Books:		
1. Computational Fluid Dynamics basics with applications- John.D, Anderson / Mc grow hill. 2. Computational Methods for Fluid Dynamics –Firziger&peric/springer.		
Web References:		
1. https://www.sciencedirect.com/topics/materials-science/computational-fluid-dynamics 2. https://nptel.ac.in/courses/112105045/		
E-Text Books:		
1. https://books.google.co.in/books?id=IIUkqI-HNbQC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=ZqJVFIsUsTsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.mooc-list.com/tags/fluid-dynamics 2. https://www.coursera.org/lecture/digital-thread-implementation/computational-fluid-dynamics-HXWVG		

PRODUCT LIFE CYCLE MANAGEMENT**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME38	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course aims to provide the students with knowledge about how a Product Lifecycle Management (PLM) system is used to structure and manage the information which guides the product during its lifecycle. The course identifies different stakeholders which both generate and consume information related to the product and its manufacturing system over the lifecycle. The course also presents an overview of integration of PLM with other applications.

COURSE OUTCOMES:

At the end of the course, the student should be able to

1. Explain basic concepts of product life cycle management.
2. Demonstrate product development approaches.
3. Explain elements of product modelling.
4. Discuss in detail the concept of product data management.
5. Discuss about integration of PLM with other applications.

UNIT-I INTRODUCTION TO PLM

Background, Overview, Need, Benefits, Concept of Product Life Cycle. Components / Elements of PLM, Emergence of PLM, Significance of PLM, Customer Involvement. Product Data and Product Workflow, Company's PLM vision, The PLM Strategy, Principles for PLM strategy, Preparing for the PLM strategy, Developing a PLM strategy, Strategy identification and selection, Change Management for PLM

UNIT-II PRODUCT DEVELOPMENT

Product Development Approaches: Bottom-up design, Top-down design, Front-loading design workflow, Design in context, Modular design. Concurrent engineering, partnership with supplier, collaborative and Internet based design, work structuring and team deployment, Product and process systemization, problem identification and solving methodologies, improving product development solutions

UNIT-III PRODUCT MODELING

Product Modelling - Definition of concepts - Fundamental issues - Role of Process chains and product models - Types of product models - model standardization efforts - types of process chains - Industrial demands. Foundation technologies and standards (e.g. visualization, collaboration and enterprise application integration),

UNIT-IV PRODUCT DATA MANAGEMENT

Product Data Management (PDM) - Benefits and Terminology, PDM functions, definition and architectures of PDM systems, product data interchange, portal integration, PDM acquisition and implementation. Information authoring tools (e.g., MCAD, ECAD, and technical publishing), Core

functions (e.g., data vaults, document and content management, workflow and program management), Functional applications (e.g., configuration Management).

UNIT-V	INTEGRATION OF PLM WITH OTHER APPLICATIONS, PLM SOFTWARES
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Different ways to integrate PLM systems, Transfer file, Database integration, System roles, ERP, Optimization of ERP for PLM and CAD.

Different ways to integrate PLM systems, Transfer file, Database integration, System roles, ERP, Optimization of ERP for PLM and CAD.

PLM Softwares-Basic features and modules of ENOVIA and Windchill.

Text Books:

1. Grieves, Michael. Product Lifecycle Management, McGraw-Hill, 2006. ISBN 0071452303
2. Saaksuori Antti / ImmonenAnselmie, product Life Cycle Management Springer, Dreamtech, 3-540-25731-4.

Reference Books:

1. Stark, John. Product Lifecycle Management: Paradigm for 21st Century Product Realisation, Springer-Verlag, 2004. ISBN 1852338105.
2. Fabio Giudice, Guido La Rosa, Product Design for the environment-A life cycle approach, Taylor & Francis 2006.

Web References:

1. <https://www.slideshare.net/anandsubramaniam/product-life-cycle-management>
2. <http://productlifecyclestages.com/>
3. <https://rxrev.com/2018/02/windchill-vs-enovia/>

E-Text Books:

1. https://books.google.co.in/books?id=q9AdtdDeuPsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=CiHbLm6twJMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.cimdata.com/en/education/plm-basics-e-learning-course>
2. <https://www.cimdata.com/en/education/plm-certificate-program>

INDUSTRIAL ENGINEERING								
VI Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME39	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OVERVIEW: This course will give insight of various concepts of industrial engineering like qualitative tools, principles, plant layout, plant location, material handling, operations management, SQC, TQM and project management.								
COURSE OUTCOMES: At the end of the course students will be able to 1. Explain basic concepts and principles of industrial engineering. 2. Enumerate and explain plant layouts, material handling objectives, considerations and material handling devices. 3. Explain about plant location, factors affecting plant location, inventory functions, ABC & VED analysis and evaluate EOQ. 4. Explain control charts and TQM. 5. Demonstrate PERT, CPM and determine critical path.								
UNIT-I	INTRODUCTION							
Introduction: Definition of industrial engineering (I.E), development, applications, role of an industrial engineer, differences between production management and industrial engineering, quantitative tools of I.E and productivity measurement. Taylor's principles, theory X and theory Y, Fayol's principles of management.								
UNIT-II	PLANT LAY OUT & MATERIAL HANDLING							
Factors governing plant location, types of production layouts, advantages and disadvantages of process layout and product layout, applications, quantitative techniques for optimal design of layouts, plant maintenance, preventive and breakdown maintenance. Material Handling: Introduction, Functions, Objectives, Engineering & Economic considerations, Guiding Principles, Material handling devices.								
UNIT-III	OPERATIONS MANAGEMENT							
Plant location, definition, factors affecting the plant location, comparison of rural and urban sites-methods for selection of plant- Matrix approach. Inventory – functions, types, Determination of Economic Order Quantity (EOQ), ABC and VED analysis. Inventory Control Systems-Continuous review system periodical review system. Stores Management and Stores Records. Purchase management, duties of purchase of manager, JIT System.								

UNIT-IV	STATISTICAL QUALITY CONTROL & TQM
STATISTICAL QUALITY CONTROL: Quality control, its importance, SQC, attribute sampling inspection with single and double sampling, Control charts - X and R - charts X AND S charts and their applications, numerical examples. TOTAL QUALITY MANAGEMENT: zero defect concept, quality circles, implementation, applications, ISO quality systems. six sigma - definition, basic concepts	
UNIT-V	PROJECT MANAGEMENT
PROJECT MANAGEMENT: PERT, CPM - differences & applications, critical path, determination of floats, importance, project crashing, smoothing and numerical examples.	
Text Books:	
1. T.R Banga, N.K Agarwal, S.C Sharma, Industrial Engineering and Management, Khanna Publishers. 2. Industrial Engineering and Production Management, Martand Telsang, S.Chand & Company Ltd. New Delhi.	
Reference Books:	
1. Operations Management by J.G Monks, McGrawHill Publishers. 2. Principles of Management by Koontz O' Donnel, McGraw Hill Publishers. 3. Industrial Engineering and Management by Raju, Cengage Publishers.	
Web References:	
1. https://nptel.ac.in/courses/112107143/ 2. https://www.scribd.com/document/337992425/Industrial-Engineering-Lecture-notes	
E-Text Books:	
1. https://books.google.co.in/books?id=SyG1A0YLxPMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=Q4SUIQXdpqQC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=eqfOEDSmcc8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.classcentral.com/tag/industrial-engineering	

FRACTURE MECHANICS

VI Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME40	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Fracture mechanics is a failure theory that determines material failure by energy criteria, possibly in conjunction with strength (or yield) criteria and considers failure to be propagating throughout the structure rather than simultaneous throughout the entire failure zone or surface. This course will impart the knowledge on basic concepts and principles of fracture mechanics, elastic fracture mechanics, Elastic-plastic fracture mechanics and applications of fracture mechanics.

COURSE OBJECTIVES:

The objectives of this course are to :

1. Study different types of fractures
2. Study the stress field of elastic crack and its solution.
3. Study about the crack growth and crack arrest
4. Study about the elastic-plastic fracture mechanics
5. Study about the application of fracture mechanics

COURSE OUTCOMES:

On completion of the course the student will be able to

1. Describe the crack and its effect on the service
2. Solve the elastic crack problems
3. Demonstrate energy principle and concepts of fatigue crack growth.
4. Solve problems of elastic plastic fracture mechanics.
5. Explain applications of fracture mechanics.

UNIT-I INTRODUCTION

Introduction: Crack in a Structure – Griffith Criterion – Cleavage fracture – Ductile fracture – Fatigue Cracking. Service failure analysis.

UNIT-II ELASTIC CRACK

Elastic Crack: Elastic Crack tip stress field – Solution to crack problems. Effect of finite size stress intensity factor – Special cases – Irwin plastic zone correction. Actual shape of plastic zone – Plane stress – Plane strain

UNIT-III	ENERGY PRINCIPLE & FATIGUE CRACK GROWTH
Energy Principle: Energy release rate – Criterion for crack growth – Crack resistance curve – Principles of crack arrest – Crack arrest in practice. Fatigue Crack Growth: Fatigue crack growth test, stress intensity factor, factors affecting stress intensity factor – Variable amplitude service loading, retardation model.	
UNIT-IV	ELASTIC PLASTIC FRACTURE MECHANICS
Elastic Plastic Fracture Mechanics: Elastic plastic fracture concept – Crack tip opening displacement – J-integral technique; Determination of J-using FEM.	
UNIT-V	APPLICATION OF FRACTURE MECHANICS
Application of Fracture Mechanics: Fracture design – Selection of materials – fatigue crack growth rate curve – Stress intensity factor range – Use of crack growth law.	
Text Books:	
1. David Broek – Elementary Engineering Fracture Mechanics: Sifted off an Noordhoff Internal Publishers – 1978. 2. John M. Barson and Stanely T. Rolfe: Fracture and Fatigue Control in Structures – Prentice Hall, Inc. USA 1987.	
Reference Books:	
1. Jean Cemative and Jean Louis Chboche Mechanics of Solid Materials, Cambridge University Press, Cambridge, 1987. 2. Prashant Kumar, “Elements of Fracture Mechanics”, Wheeler Publications, 1999	
Web References:	
1. http://faculty.olin.edu/~jstolk/failure2008/Lecture%20notes/Griffith%20%20fracture%20Roylance%20MIT%20notes.pdf 2. https://www.slideshare.net/NguyenVinhPhu/untitled-42330424	
E-Text Books:	
1. https://books.google.co.in/books?id=zTRBTjCv110C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=M062CgAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://ocw.mit.edu/courses/materials-science-and-engineering/3-35-fracture-and-fatigue-fall-2003/ 2. https://www.my-mooc.com/en/mooc/mechanical-behavior-materials-part-3-mitx-3-032-3x-0/	

REFRIGERATION AND AIR-CONDITIONING**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME41	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Refrigeration may be defined as the process of achieving and maintaining a temperature below that of the surroundings, the aim being to cool some product or space to the required temperature. Air conditioning is the process of altering the properties of air to more comfortable conditions, typically with the aim of distributing the conditioned air to an occupied space, such as a building or a vehicle, to improve the thermal comfort and indoor air quality. This course covers analysis of various refrigeration cycles, vapour compression and absorption refrigeration systems, psychrometry, analysis of air conditioning loads and various air conditioning systems.

COURSE OUTCOMES:

At the end of the course students are able

1. Demonstrate basic concepts of refrigeration.
2. Explain elements and working of vapour compression refrigeration system
3. Analyze vapour absorption refrigeration system.
4. Discuss basic concepts of air conditioning, psychrometric properties and processes.
5. Analyze air conditioning loads.

UNIT-I INTRODUCTION TO REFRIGERATION

Introduction: Necessity and applications, unit of refrigeration, C.O.P and Mechanical refrigeration. Air Refrigeration: Air Refrigeration Cycles-reversed Carnot cycle, Bell-Coleman cycle analysis, Air Refrigeration systems-merits and demerits, analysis.

UNIT-II VAPOUR COMPRESSION REFRIGERATION

Working principle and essential components of the plant – simple vapour compression refrigeration cycle – COP – representation of cycle on T-S and p-h charts – effect of sub cooling and super heating – cycle analysis – actual cycle influence of various parameters on system performance – use of p-h charts – numerical problems.

UNIT-III VAPOUR ABSORPTION SYSTEMS

Vapour Absorption systems: Simple Vapour Absorption Refrigeration System, Absorbent – Refrigerant combinations, Water-Ammonia Systems, Water Lithium Bromide System, Contrast between the two systems, Modified Version of Aqua-Ammonia System with Rectifier and Analyser Assembly. Brief description of Refrigerants

UNIT-IV	INTRODUCTION TO AIR CONDITIONING AND PSYCHROMETRY
Introduction to Air-Conditioning, Basic Definition, Classification, ASHRAE Nomenclature pertaining to Air-Conditioning, Applications of Air-Conditioning. Psychrometry – Air-water vapour mixtures, Psychrometric Properties, Psychrometric or Air-Conditioning processes, Psychrometric Chart.	
UNIT-V	ANALYSIS OF AIR CONDITIONING LOADS AND AIR CONDITIONING SYSTEMS
Mathematical Analysis of Air-Conditioning Loads, Related Aspects, Numerical Problems, Different Air-Conditioning Systems-Central – Station Air-Conditioning System, Unitary Air-Conditioning System, Window Air-Conditioner and Packaged Air-Conditioner, Components related to Air-Conditioning Systems.	
Text Books:	
1. Roy J. Dossat, Principles of Refrigeration, Wiley Limited 2. Arora, C.P., "Refrigeration and Air Conditioning", 3rd edition, McGraw Hill, New Delhi, 2010. 3. R.S. Khurmi & J.K. Gupta., "A textbook of Refrigeration and Air Conditioning", S Chand Publications, 2010. 4. Domkundwar Arora., "Course In Refrigeration & Air Conditioning", Dhanpat Rai & Co.(p) Ltd-Delhi.	
Reference Books:	
1. Dossat R.J., Principles of refrigeration, John Wiley, S.I. Version, 2001. 2. Klaus D. Timmerhaus and Thomas M. Flynn, Cryogenic Process Engineering, Plenum Press, New York, 1989. 3. Robert W. Vance, Cryogenic Technology, John Wiley & Sons, Inc., New York, London.	
Web References:	
1. https://lecturenotes.in/subject/149/refrigeration-and-air-conditioning-rac 2. http://www.unep.fr/ozonaction/information/mmcfiles/7799_e_University_Course_Lecture_%20Notes.pdf	
E-Text Books:	
1. https://books.google.co.in/books?id=JyGeRoZly80C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=7CHZAgAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=vYxeBAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. http://www.i-know.com/refrigeration-training-courses.aspx 2. https://www.tpctraining.com/collections/air-conditioning-and-refrigeration-training	

DESIGN FOR MANUFACTURING**VI Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME42	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

DFM techniques are focused on individual parts and components with a goal of reducing or eliminating expensive, complex or unnecessary features which would make them difficult to manufacture. This course will give insight on design for machining, casting, welding, brazing, forging, extrusion, sheet metal and assembly.

COURSE OUTCOMES:

At the end of the course students will be able to

1. Demonstrate steps in design process, design rules for manufacturability, material selection considerations and developments in material technology.
2. Demonstrate various machining processes and explain design considerations for machining and casting.
3. Describe various welding processes and explain design considerations for welding, brazing and forging.
4. Explain design considerations for extrusion and sheet metal work.
5. Explain objectives and design considerations for assembly.

UNIT-I INTRODUCTION

INTRODUCTION: Design philosophy steps in Design process — General Design rules for manufacturability — basic principles of designing for economical production, Materials-parameters to be considered for the selection of materials for design—criteria for material selection, material selection process for new product design, developments in material technology.

UNIT-II DESIGN FOR MACHINING & CASTING

Overview of various machining processes — general design rules for drilling, milling and turning —General design recommendations for machined parts.

Metal Casting: Appraisal of various casting processes, selection of casting process, – general design considerations for casting — casting tolerances — use of solidification simulation in casting design — product design rules for sand casting.

UNIT-III	DESIGN FOR WELDING, BRAZING & FORGING
Appraisal of various welding processes, Factors in design of weldments — general design guidelines — pre and post treatment of welds — effects of thermal stresses in weld joints — design of brazed joints. Design considerations for Forging,	
UNIT-IV	DESIGN FOR EXTRUSION & SHEET METAL WORK
Extrusion-Design recommendations, design guidelines, Sheet Metal Work-cost effective design principles for Punching, Blanking and bending, Design principles for successful deep drawing, Keeler Goodman Forming Line Diagram	
UNIT-V	DESIGN FOR ASSEMBLY
Design for assembly: objectives, link to design, General design rules for manual assembly, Development of systematic design for assembly methodology, Assembly efficiency, Effect of part symmetry, part thickness, size, weight on handling time, Effect of chamfer design on insertion operations, Estimation of insertion time.	
Text Books:	
1. Geoffrey Boothroyd, "Assembly Automation and Product Design", Marcel Dekker Inc., NY, 1992. 2. Geoffrey Boothroyd, "Hand Book of Product Design" Marcel and Dekken, N.Y. 1990.	
Reference Books:	
1. Engineering Design – Material & Processing Approach – George E. Deiter, McGraw Hill Intl. 2nd Ed. 2000.	
Web References:	
1. https://news.ewmfg.com/blog/manufacturing/dfm-design-for-manufacturing 2. https://quality-one.com/dfm-dfa/	
E-Text Books:	
1. https://books.google.co.in/books?id=ApHLBQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=GHFsBV_49cC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=u1mRDwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.mooc-list.com/tags/design 2. https://www.mooc-list.com/tags/manufacturing-processes	

INDUSTRIAL ROBOTICS**VII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME43	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Today robot finds applications in industries, medical and other fields. For example, in eye surgery (replacement of retina), where a cylindrical portion needs to be replaced, the operation is best done by robots. Mobile robots like walking machines, hopping machines are examples of robots, Nuclear and power plants uses fish like robots which move inside pipes for purpose of inspection. This course will give an insight on various types of industrial robots, their kinematic and kinetic aspects, different types of grippers, mechanics of grippers, trajectory planning etc.

COURSE OBJECTIVES:

The objectives of this course are to

1. Study about basic components, architecture, anatomy and specifications of robot.
2. Study about grippers, mechanics of grippers and homogeneous transformation
3. Study D-H notation, direct and indirect kinematics
4. Study Trajectory planning techniques, lagrangian-euler and newton-euler formulations
5. Study feedback components and applications of robots.

COURSE OUTCOMES:

At the end of the course, the student will be able to

1. Demonstrate different types of robots, specifications of robots and different end effectors used in robots.
2. Analyze gripper forces and evaluate composite rotation and transformation matrices.
3. Model forward, inverse and differential kinematics of robot manipulators.
4. Explain trajectory planning techniques; formulate lagrangian-euler and newton-euler equations.
5. Explain about actuators, feedback components used in robots and industrial applications.

UNIT-I INTRODUCTION

Introduction: Automation and Robotics, Asimov's laws, Robot Architecture, Components, , Anatomy of robot, Factors to be considered in the selection of robot, present and future applications, Specifications - Degree of freedom, Pay load, Parts per hour, Accuracy, Repeatability, Speed, Work space, Work volume, Work envelope, classification of robots based on configuration and control systems
End effectors: Mechanical and Non-mechanical grippers, Types of actuation mechanisms, requirements for the design of grippers, considerations for the selection of grippers.

UNIT-II	MOTION ANALYSIS
Motion Analysis: Basic Rotation Matrices, Euler Angles, Composite Rotation Matrices. Homogeneous transformations as applicable to rotation and translation – problems.	
UNIT-III	MANIPULATOR & DIFFERENTIAL KINEMATICS
Manipulator Kinematics: D-H notation, D-H method of Assignment of frames, D-H Transformation Matrix, joint coordinates and world coordinates, Forward kinematics of 2R, RP, 3R and SCARA manipulators and Inverse kinematics of 2R, RP and 3R manipulators. Differential Kinematics: Introduction to differential kinematics, Jacobian, Jacobian singularities, Problems.	
UNIT-IV	TRAJECTORY PLANNING & DYNAMICS
Trajectory Planning: Definition of Trajectory planning, Path, Trajectory, Knot points, Steps involved in trajectory planning, Trajectory planning techniques-Joint space and Cartesian space techniques, Cubic polynomial trajectory. Dynamics: Lagrange – Euler and Newton – Euler formulations for 2 DOF R-R manipulator.	
UNIT-V	ACTUATORS, FEEDBACK COMPONENTS & APPLICATIONS
Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors, comparison of Actuators, Feedback Components: Position sensors – potentiometers, resolvers, optical encoders, Velocity sensor, Contact Sensors-Touch sensors, Tactile and Range sensors, Force and Torque sensors, Proximity sensor, Inductive sensor. Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading- Processing - spot and continuous arc welding & spray painting - Assembly and Inspection.	
Text Books:	
1. Industrial Robotics by Groover M P, Pearson Edu. 2. Robotics and Control by Mittal R K & Nagrath I J, TMH. 3. Theory of Applied Robotics (kinematics,Dynamics and Control-Jazar, Springer.	
Reference Books:	
1. Fu K S, Robotics, McGraw Hill. 2. Spong and Vidyasagar, Robot Dynamics and Control, John Wiley 3. Asada and Slotine, Robot Analysis and control, Wiley Inter-Science 4. John J Craig, Introduction to Robotics, Pearson Education	

Web References:

1. https://www.sciencedaily.com/terms/industrial_robot.htm
2. <https://www.robotics.org/robotics/industrial-robot-industry-and-all-it-entails>

E-Text Books:

1. https://books.google.co.in/books?id=dr9IAI7wucUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=rESVUHwMcvYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.classcentral.com/tag/robotics>
2. <https://www.mooc-list.com/tags/robotics>

TOTAL QUALITY MANAGEMENT

VII Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME44	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Total Quality Management (TQM) is an approach that organizations use to improve their internal processes and increase customer satisfaction. When it is properly implemented, this style of management can lead to decreased costs related to corrective or preventative maintenance, better overall performance, and an increased number of happy and loyal customers. This course is developed to impart knowledge on principles, tools and techniques of TQM, control charts, quality cost and systems.

COURSE OUTCOMES:

At the end of the course students will be able to

1. Demonstrate basic concepts of total quality management.
2. Explain principles of total quality management.
3. Explain seven tools of quality, management tools and bench marking.
4. Enumerate and describe various control charts, quality costs and quality management.
5. Explain quality systems for TQM.

UNIT-I INTRODUCTION

Introduction – Need for quality – Evolution of quality – Definitions of quality – Dimensions of product and service quality – Basic concepts of TQM – TQM Framework – Contributions of Deming, Juran and Crosby – Barriers to TQM – Quality statements – Customer focus – Customer orientation, Customer satisfaction, Customer complaints, Customer retention

UNIT-II TQM PRINCIPLES

Leadership – Strategic quality planning, Quality Councils – Employee involvement – Motivation, Empowerment, Team and Teamwork, Quality circles Recognition and Reward, Performance appraisal – Continuous process improvement – PDCA cycle, 5S, Kaizen – Supplier partnership – Partnering, Supplier selection, Supplier Rating.

UNIT-III TQM TOOLS & TECHNIQUES

The seven traditional tools of quality – New management tools – Six sigma: Concepts, Methodology, applications to manufacturing, service sector including IT – Bench marking – Reason to bench mark, Bench marking process – FMEA – Stages, Types.

UNIT-IV	CONTROL CHARTS & COST OF QUALITY
Control Charts – Process Capability – Concepts of Six Sigma – Quality Function Development (QFD) – Taguchi quality loss function – TPM – Concepts, improvement needs – Performance measures. The Cost of Quality: Definition of the Cost of Quality, Quality Costs, Measuring Quality Costs, use of Quality Cost information, Accounting Systems and Quality Management.	
UNIT-V	QUALITY SYSTEMS
Need for ISO 9000 – ISO 9001-2008 Quality System – Elements, Documentation, Quality Auditing – QS 9000 – ISO 14000 – Concepts, Requirements and Benefits – TQM Implementation in manufacturing and service sectors.	
Text Books:	
1. Total Quality Management / Joel E. Ross/Taylor and Franscis Limited 2. Total Quality Management/P. N. Mukherjee/PHI	
Reference Books:	
1. Beyond TQM / Robert L.Flood 2. Statistical Quality Control / E.L. Grant. 3. Total Quality Management:A Practical Approach/H. Lal 4. Quality Management/Kanishka Bedi/Oxford University Press/2011 5. Total Engineering Quality Management/Sunil Sharma/Macmillan	
Web References:	
1. https://lecturenotes.in/subject/495/total-quality-management-tqm 2. https://lecturenotes.in/notes/20800-note-for-total-quality-management-tqm-by-engineering-kings	
E-Text Books:	
1. https://books.google.co.in/books?id=h7US1hlqV_IC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=0vfDUNyUZIUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=q8glBgAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.coursera.org/lecture/supply-chain-management/total-quality-management-wLrvy 2. http://amitymooc.com/home/quality_management.php	

AUTOMOBILE ENGINEERING**VII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME45	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW:

This course is designed to introduce basic lay-out of an automobile, operation of engine cooling, lubrication, ignition, electrical and air conditioning systems, principles of transmission, suspension, steering and braking system and latest developments in automobiles.

COURSE OUTCOMES:

At the end of the course, the student will be able to:

1. Explain the basic lay-out of an automobile, types of automobile engines, basic concepts like engine construction, turbo charging, super charging, oil filters, pumps etc.
2. Discuss fuels systems of S.I and C.I engines
3. Explain the operation of engine cooling, lubrication, ignition, electrical and air conditioning systems.
4. Demonstrate the principles, elements and working of transmission and braking systems
5. Explain the elements and working of steering and suspension systems.

UNIT-I INTRODUCTION

Components of four wheeler automobile , chassis and body , power unit , power transmission , rear wheel drive, front wheel drive, 4 wheel drive , types of automobile engines, engine construction, turbo charging and super charging , engine lubrication, splash and pressure lubrication systems, oil filters, oil pumps , crank case ventilation , engine service, reborning, decarburization, Nitriding of crank shaft.

UNIT-II FUEL SYSTEM AND C.I. ENGINES

FUEL SYSTEM: S.I. Engine: Fuel supply systems, Mechanical and electrical fuel pump, filters, carburettor, types, air filters, petrol injection.

C.I. ENGINES: Requirements of diesel injection systems, types of injection systems, fuel pump, nozzle, spray formation, injection timing, testing of fuel pumps. Emission from Automobiles, Pollution standards National and international, Pollution Control, Techniques, Multipoint fuel injection for SI Engines.

UNIT-III COOLING SYSTEM AND IGNITION SYSTEM

COOLING SYSTEM : Cooling Requirements, Air Cooling, Liquid Cooling, Thermo, water and Forced Circulation System, Radiators, Types, Cooling Fan - water pump, thermostat, evaporating cooling , pressure sealed cooling , antifreeze solutions.

IGNITION SYSTEM: Function of an ignition system, battery ignition system, constructional features of storage, battery, auto transformer, contact breaker points, condenser and spark plug, Magneto coil ignition system, electronic ignition system using contact breaker, electronic ignition using contact triggers, spark advance and retard mechanism.

UNIT-IV	TRANSMISSION SYSTEM AND BRAKING SYSTEM
TRANSMISSION SYSTEM: Clutches, principle, types, cone clutch, single plate clutch, multi plate clutch, magnetic and centrifugal clutches, fluid fly wheel, gear boxes, types, sliding mesh, construct mesh, synchro mesh gear boxes, epicyclic gear box, over drive torque converter. Propeller shaft, Hotch , Kiss drive, Torque tube drive, universal joint, differential rear axles , types , wheels and tiers. BRAKING SYSTEM: Mechanical brake system, Hydraulic brake system, Master cylinder, wheel cylinder tandem master cylinder Requirement of brake fluid, Pneumatic and vacuum brakes	
UNIT-V	STEERING SYSTEM AND SUSPENSION SYSTEM
STEERING SYSTEM: Steering geometry, camber, castor, king pin rake, combined angle toe in, center point steering. Types of steering mechanism , Ackerman steering mechanism, Davis steering mechanism, steering gears , types, steering linkages. SUSPENSION SYSTEM: Objects of suspension systems, rigid axle suspension system, torsion bar, shock absorber, Independent suspension system.	
Text Books:	
1. Kirpal Singh (2012), Automobile Engineering - Vol. 1 &2, 12th edition, standard publishers, New Delhi, India. 2. William Crouse (2012), Automobile Engineering (SIE), 10th edition, Tata McGraw hill education (P) Ltd, New Delhi, India..	
Reference Books:	
1. B.S. Narang (2011), Automobile Engineering, 5th edition, Karman publishers, New Delhi, India 2. J. B. Gupta (2012), Automobile Engineering, satya prakashan, New Delhi, India. 3. A Text book Automobile Engineering, Rajput.	
Web References:	
1. https://nptel.ac.in/courses/112104225/22 2. https://nptel.ac.in/courses/108108078/ 3. https://nptel.ac.in/courses/112105051/	
E-Text Books:	
1. https://www.ebooks.com/en-us/subjects/mechanical-engineering -ebooks/10357 2. https://www.springer.com/in/book/9783319723709 3. https://www.pdfdrive.com/ -e14705840.html	
MOOC Course	
1. https://www.mooc-list.com/tags/ 2. https://www.edx.org/learn/ 3. https://www.coursera.org/courses?query	

ADDITIVE MANUFACTURING**VII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME46	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW:

This course will introduce fundamentals for additive manufacturing (AM), various types of liquid based, solid based and powder based AM technologies, Pre-processing, processing, post-processing errors in AM, different types of data formats and software's used in AM. and applications of AM in design analysis, aerospace, automotive, biomedical and other fields

COURSE OUTCOMES:

The course should enable the students to

1. Demonstrate fundamental concepts of additive manufacturing.
2. Classify and explain liquid based additive manufacturing systems.
3. Classify and explain solid based additive manufacturing systems.
4. Analyze the capabilities and limitations of powder based Additive Manufacturing system.
5. Enumerate applications of additive manufacturing.

UNIT-I INTRODUCTION

Need for Additive Manufacturing(AM), Historical development, Fundamentals of Additive Manufacturing, AM Process Chain, Advantages and Limitations of AM, Commonly used Terms, Classification of AM process, Fundamental Automated Processes: Distinction between AM and CNC, other related technologies.

UNIT-II LIQUID BASED ADDITIVE MANUFACTURING SYSTEMS

Stereo lithography Apparatus (SLA): Models and specifications, Process, working principle, photopolymers, photo polymerization, Layering technology, laser and laser scanning, Applications, Advantages and Disadvantages, Case studies.

Solid ground curing (SGC): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies.

Polyjet: Process, Principle, working principle, Applications, Advantages and Disadvantages, Case studies

UNIT-III SOLID BASED ADDITIVE MANUFACTURING SYSTEMS

Laminated Object Manufacturing (LOM): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies.

Fused Deposition Modeling (FDM): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies.

MultiJet Modelling (MJM): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies.

UNIT-IV	POWDER BASED ADDITIVE MANUFACTURING SYSTEMS
Selective laser sintering (SLS): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies. Three dimensional Printing (3DP): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies. Laser Engineered Net Shaping (LENS): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies. Electron Beam Melting (EBM): Models and specifications, Process, working principle, Applications, Advantages and Disadvantages, Case studies	
UNIT-V	ADDITIVE MANUFACTURING APPLICATIONS
Application – Material Relationship, Application in Design, Application in Engineering, Analysis and Planning, Aerospace Industry, Automotive Industry, Jewelry Industry, Coin Industry, GIS application, Arts and Architecture. Medical and Bioengineering Applications: Planning and simulation of complex surgery, Customised Implants & Prosthesis, Design and Production of Medical Devices, Forensic Science and Anthropology, Visualization of Biomolecules.	
Text Books:	
1. Chua C.K., Leong K.F., and Lim C.S. (2010), Rapid prototyping: Principles and applications, Third Edition, World Scientific Publishers, New Jersey 2. Gebhardt A. (2003), Rapid prototyping, Hanser Gardener Publications. 3. Ian Gibson, David Rosen, and Brent Stucker. (2015), Additive Manufacturing Technologies, Springer Publications.	
Reference Books:	
1. Liou L.W. and Liou F.W. (2007), Rapid Prototyping and Engineering applications : A tool box for prototype development, CRC Press. 2. Kamrani A.K. and Nasr E.A. (2006), Rapid Prototyping: Theory and practice, Springer. 3. Hilton P.D. and Jacobs P.F. (2000), RapidTooling: Technologies and Industrial Applications, CRC press. 4. DouglasBryden. (2014), CAD and Prototyping for Product Design (Portfolio Skills), Laurence King Publishing. 5. Barnatt, C. (2013), 3D Printing: The Next Industrial Revolution, CreateSpace Independent Publishing Platform.	
Web References:	
1. https://nptel.ac.in/courses/112104204/47 2. https://nptel.ac.in/courses/112106249/23 3. https://nptel.ac.in/courses/113108071/31 4. https://additivemanufacturing.mit.edu/	
E-Text Books:	
1. http://www.tntconf.org/2015/Abstracts/2015_Ortega_Amaya%20_aortega@aimplas.es_TNT2015OrtegaAmaya.pdf 2. https://www.explainingthefuture.com/3dp_chapter1.pdf 3. home.iitk.ac.in/~nsinha/Additive_Manufacturing%20I.pdf	

MOOC Course
1. https://nptel.ac.in/courses/112104204/47 2. https://nptel.ac.in/courses/112106249/23 3. https://engineering.purdue.edu/ProEd/certifications/additive-manufacturing

1. <https://nptel.ac.in/courses/112104204/47>
2. <https://nptel.ac.in/courses/112106249/23>
3. <https://engineering.purdue.edu/ProEd/certifications/additive-manufacturing>

OPERATIONS RESEARCH								
VII Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME47	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OVERVIEW: Operations research (OR) is an analytical method of problem-solving and decision-making that is useful in the management of organizations. In operations research, problems are broken down into basic components and then solved in defined steps by mathematical analysis. This course gives insight of Linear Programming, Transportation, assignment problems, sequencing etc.								
COURSE OUTCOMES: At the end of course students will be able to 1. Describe types of models and solve linear programming problem. 2. Solve transportation and assignment problems. 3. Analyze sequencing and replacement models and apply them for optimization. 4. Apply gaming theory for optimal decision making and analyze inventory models to optimize the cost. 5. Model and solve problems using dynamic programming								
UNIT-I	INTRODUCTION & ALLOCATION							
Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications. ALLOCATION: Linear Programming Problem - Formulation – Graphical solution – Simplex method – Artificial variables techniques: Two–phase method, Big-M method; Duality Principle.								
UNIT-II	TRANSPORTATION & ASSIGNMENT PROBLEMS							
TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy. ASSIGNMENT PROBLEM: Formulation – Optimal solution - Variants of Assignment Problem; Traveling Salesman problem.								
UNIT-III	SEQUENCING & REPLACEMENT							
SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines – Job shop sequencing – two jobs through ‘m’ machines REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely- Group Replacement.								

UNIT-IV	THEORY OF GAME & INVENTORY
THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points- 2 x 2 games –m x 2 & 2 x n games - graphical method – m x n games - dominance principle. INVENTORY: Introduction – Single item, Deterministic models – Types - Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.	
UNIT-V	WAITING LINES & DYNAMIC PROGRAMMING
WAITING LINES: Introduction – Terminology-Single Channel – Poisson arrivals and Exponential Service times – with infinite population and finite population models– Multichannel – Poisson arrivals and exponential service times with infinite population. DYNAMIC PROGRAMMING: Introduction – Terminology- Bellman’s Principle of Optimality – Applications of dynamic programming- shortest path problem – linear programming problem.	
Text Books:	
1. Operation Research by J.K.Sharma, MacMilan. 2. Operations Researchby ACS Kumar, Yesdee	
Reference Books:	
1. Operations Research: Methods and Problems by Maurice Saseini, Arhur Yaspan and Lawrence Friedman 2. Operations Research by A.M.Natarajan, P.Balasubramaniam, A. Tamilarasi, Pearson Education. 3. Operations Research by Wagner, PHI Publications. 4. Introduction to O.Rby Hillier & Libermann,TMH.	
Web References:	
1. http://www.cs.toronto.edu/~stacho/public/IEOR4004-notes1.pdf 2. http://www.mathcity.org/msc/notes/operation_research 3. http://biobharati.com/admin/img/gall/1467763196_BCA-504.pdf	
E-Text Books:	
1. https://books.google.co.in/books?id=6khDDAAAOBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=cwoWJrgCrP8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=rj6bBMVzfPsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. http://www.orcomplete.com/internet/enesbilgin/open-courses-on-operations-research 2. https://www.coursera.org/learn/wharton-operations	

FLEXIBLE MANUFACTURING SYSTEMS

VII Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME48	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

A flexible manufacturing system (FMS) gives manufacturing firms an advantage to quickly change a manufacturing environment to improve process efficiency and thus lower production cost. However, upfront costs may be greater for installing specialized equipment that allows for flexibility and customization. This course imparts knowledge of FMS evolution, objectives, applications and focuses on FMS layout, processing stations material handling systems etc.

COURSE OUTCOMES:

At the end of the course, the student shall be able to

1. Demonstrate basic concepts of flexible manufacturing systems.
2. Discuss concepts of manufacturing cells and unattended machining.
3. Explain in detail about manufacturer's driving force..
4. Discuss about software for FMS.
5. Explain FMS planning, hardware and implementation.

UNIT-I INTRODUCTION

Introduction: Evolution of FMS, Definition, FMA Layout configurations, -Inline layout, loop layout, loader layout, open field layout, robot configured layout, general FMS considerations, functions of FMS, FMS Justification, Cell/FMS Justification Flow chart.

UNIT-II MANUFACTURING CELLS & UNATTENDED MACHINING

Manufacturing cells: Introduction, Classification of manufacturing cells-Stand alone NC machine cell, Spindle NC machine cell, Integrated multi machine cell, FMS
Unattended Machining: Introduction, unattended turning center features and requirements, unattended machining center features and requirements, cellular Vs flexible manufacturing.

UNIT-III MANUFACTURERS DRIVING FORCE

QUALITY: Definition, characteristics of quality, Quality improvements, Importance of Quality to cells and systems, Quality of team work, Methods for getting and keeping quality under control-statistical process control and taguchi methods. Just In-Time (JIT) Manufacturing, Benefits of JIT, JIT Relation to FMS, Primary JIT principles.

UNIT-IV SOFTWARE FOR FMS

Software for FMS: Introduction, flexibility targets, system concepts-sequential flow system, single station system, random flow system, software for design functions-capacity planning, simulation and knowledge based system, software for extrinsic functions-production scheduling, process planning, tool management, maintenance planning/reporting, software for intrinsic functions-production control, production monitoring,

machine/process control, Machine diagnostics.

Software Specifications-absolute specifications and functional specifications, FMS database- Database layout and sub systems.

UNIT-V	FMS PLANNING, HARDWARE & IMPLEMENTATION
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FMS Planning: CAD Considerations FMS planning, CAM Considerations for FMS planning.

Hardware: FMS hardware configurations and considerations, Programmable logic controllers(PLC'S)- components of PLC, advantages of PLC, Cell controllers, Communication networks-star network, ring network and bus network.

FMS Implementation: Acceptance testing, Performance goals and expectations, Maintenance concerns and continued support.

Text Books:

1. William W Luggen, "Flexible Manufacturing Cells and System" Prentice Hall of Inc New Jersey, 1991
2. Reza A Maleki "Flexible Manufacturing system" Prentice Hall of Inc New Jersey, 1991
3. John E Lenz "Flexible Manufacturing" marcel Dekker Inc New York , 1989.

Reference Books:

1. Groover, M.P "Automation, Production Systems and Computer Integrated Manufacturing", Prentice Hall

Web References:

1. <https://www.slideshare.net/anandkhare/flexible-manufacturing-system-24793372>
2. <https://slideplayer.com/slide/4597142/>

E-Text Books:

1. https://books.google.co.in/books?id=ceRtRNRUYXoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=OOGq1kf8xRUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.classcentral.com/course/nptel-industrial-automation-and-control-5222>

RENEWABLE ENERGY SOURCES

VII Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME49	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Renewable energy is the energy produced from sources that do not deplete or can be replenished within a human's life time. This course focuses on various renewable energy sources such as solar energy, wind energy, geothermal energy etc.

COURSE OUTCOMES:

At the end of the course, the student shall be able to

1. Classify renewable energy sources and explain environmental impact of solar energy, solar radiation, their measurement and instruments.
2. Classify concentrating collectors and explain in detail about various solar gadgets.
3. Explain about wind mill, types of wind mills, its construction, wind energy conversion systems and principles of bio mass.
4. Explain in detail geothermal and ocean energies.
5. Demonstrate the concepts of direct energy conversion.

UNIT-I INTRODUCTION

Introduction – Renewable energy sources – Limitations – Classifications – Comparison – Global and Indian energy scenario. Solar Energy: Introduction – Environmental impact of solar power – extraterrestrial and terrestrial solar radiation – solar radiation on tilted surface – Solar Radiation Measurement and Instruments – Data and estimation.

UNIT-II SOLAR ENERGY

Flat plate and concentrating collectors – classification of concentrating collectors – orientation and thermal analysis – advanced collectors. Solar Gadgets: Solar furnace, Solar air heater, Solar water heater, Solar cooker, solar still, solar pond, solar pump, solar dryer, solar cell. Solar space heating and cooling system – Storage of solar energy.

UNIT-III	WIND ENERGY & BIO MASS
Wind energy utilization: Introduction – Origin and nature of winds – Types of wind mills and its construction – Wind energy conversion systems – performance characteristics . Bio-Mass: Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Bio-gas digesters, combustion characteristics of bio-gas, applications and economic aspects.	
UNIT-IV	GEO THERMAL ENERGY & OCEAN ENERGY
Geothermal Energy: Resources, types of wells, methods of harnessing the energy, potential in India. Ocean Energy: OTEC, Principles utilization, setting of OTEC plants, thermodynamic cycles. Tidal and wave energy: Potential and conversion techniques, mini-hydel power plants, and their economics.	
UNIT-V	DIRECT ENERGY CONVERSION
Direct Energy Conversion: Need, limitations, principles of DEC. Thermo-electric generators, seebeck, peltier and Joule-Thomson effects – applications, MHD generator principle, Fuel cells: Overview; Classification of fuel cells; Operating principles. Bio-fuels, Urban waste to energy conversion	
Text Books:	
1. Rai G.D, <i>Non-Conventional Energy Sources</i>, Khanna Publishers, New Delhi, 2011. 2. Twidell, J.W. & Weir, A., <i>Renewable Energy Sources</i>, EFN Spon Ltd., UK, 2006	
Reference Books:	
1. Ashok V Desai, <i>Non-Conventional Energy</i>, Wiley Eastern Ltd, New Delhi, 2003. 2. Mittal K M, <i>Non-Conventional Energy Systems</i>, Wheeler Publishing Co. Ltd, New Delhi, 2003. 3. Khan B H, <i>Non-Conventional Energy Resources</i>, Tata McGraw Hill, 2nd Edn. New Delhi, 2009. 4. Ramesh R & Kumar K U, <i>Renewable Energy Technologies</i>, Narosa Publishing House, New Delhi, 2004 5. Tiwari. G.N., <i>Solar Energy – Fundamentals Design, Modelling & Applications</i>, Narosa Publishing House, New Delhi, 2002.	
Web References:	
1. http://www.afdc.energy.gov/fuels/natural_gas.html 2. https://cleancities.energy.gov/ 3. http://www.firmgreen.com/ 4. https://www.renewableresourcescoalition.org/alternative-energy-sources/ 5. https://www.edx.org/learn/renewable-energy	
E-Text Books:	
1. https://www.sciencedirect.com/book/9780126561531/renewable-energy 2. https://www.pdfdrive.com/renewable-energy-resources-e14705840.html 3. https://www.pdfdrive.com/search?q=renewable&pagecount=&pubyear=&searchin=	
MOOC Course:	
1. https://www.coursera.org/learn/renewable-energy-entrepreneurship 2. https://www.coursera.org/learn/wind-for-renewable-energies 3. https://online-learning.tudelft.nl/courses/sustainable-energy-design-a-renewable-future/	

MECHANICAL VIBRATIONS**VII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME50	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Vibrations are oscillations in mechanical dynamic systems. Although any system can oscillate when it is forced to do so externally, the term “vibration” in mechanical engineering is often reserved for systems that can oscillate freely without applied forces. Sometimes these vibrations cause minor or serious performance or safety problems in engineered systems. This course provides essential concepts of single degree of freedom, multi degree of freedom systems, vibration measuring equipment and numerical methods applications in vibration problems.

COURSE OUTCOMES:

At the end of the course, the student will be able to,

1. Model and analyze the response of undamped and damped vibrations of single degree of freedom systems.
2. Explain principal modes, undamped and damped vibrations of two degree of freedom systems.
3. Model multi degree of freedom systems, solve eigen value problems, torsional vibrations of multi-rotor systems and geared systems and discrete- time systems
4. Explain concepts of frequency domain vibration analysis.
5. Explain various instruments used for measuring vibration and application of numerical method to solve vibration problems.

UNIT-I SINGLE DEGREE OF FREEDOM SYSTEMS

Single Degree of Freedom Systems: Undamped and damped free vibrations; forced vibrations coulomb damping; Response to excitation; rotating unbalance and support excitation; vibration isolation and transmissibility- Response to Non Periodic Excitations: unit impulse, unit step and unit Ramp functions; response to arbitrary excitation, The Convolution Integral; shock spectrum; System response by the Laplace Transformation method.

UNIT-II TWO DEGREE OF FREEDOM SYSTEMS & VIBRATION MEASURING INSTRUMENTS

Two degree freedom systems: Principal modes- undamped and damped free and forced vibrations; undamped vibration absorbers.

UNIT-III MULTI DEGREE OF FREEDOM SYSTEMS

Multi degree of freedom systems: Matrix formulation, stiffness and flexibility influence coefficients; Eigen value problem; normal modes and their properties; Free and forced vibration by Modal analysis; Method of matrix inversion; Torsional vibrations of multi- rotor systems and geared systems; Discrete- Time systems

UNIT-IV	FREQUENCY DOMAIN VIBRATION ANALYSIS
Frequency Domain Vibration Analysis: Over view, machine-train monitoring parameters-Data base development-vibration data acquisition-trending analysis-failure- node analysis-signature analysis-root cause analysis.	
UNIT-V	VIBRATION MEASURING INSTRUMENTS & NUMERICAL METHODS
Vibration measuring instruments: Vibrometers, velocity meters & accelerometers	
Numerical Methods: Raleigh's stodola's, Matrix iteration, Rayleigh- Rftz Method and Hoizer's methods.	
Text Books:	
1. Mechanical Vibrations/Groover/Nem Chand and Bros. 2. Elements of Vibration Analysis I Meirovitch! TMH, 2001. 3. Mechanical Vibrations NP Singh/Danapathi Ral & Sons. 4. Mechanical Vibrations! SS Rao/ Pearson, 2009Mth Edition.	
Reference Books:	
1. Mechanical Vibrations/Debabrata Nag/Wiley. 2. Vibration problems in Engineering / S.P. Timoshenko. 3. Mechanical Vibrations and sound engineering/ A.G.Ambekar/ PHI. 4. Theory and Practice of Mechanical Vibrations/JS Rao & K. Gupta! New Age Intl. Publishers/Revised 2 Edition.	
Web References:	
1. https://slideplayer.com/slide/5023002/ 2. http://oaktrust.library.tamu.edu/handle/1969.1/93266	
E-Text Books:	
1. https://books.google.co.in/books?id=G2UyBTji18oC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. http://160592857366.free.fr/joe/ebooks/Mechanical%20Engineering%20Books%20Collection/VIBRATION/mechVib%20theory%20and%20applications.pdf	
MOOC Course:	
1. https://nptel.ac.in/courses/112103112/ 2. https://ocw.mit.edu/courses/mechanical-engineering/2-003sc-engineering-dynamics-fall-2011/mechanical-vibration/	

POWER PLANT ENGINEERING**VIII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME51	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW:

This course is framed to cover aspects of coal based thermal power plants, Diesel, Gas turbine and combined cycle power plants and nuclear power plants. It also covers economic and environmental issues of power plant.

COURSE OUTCOMES:

Upon completion of the course, the student will be able to:

1. Demonstrate concepts of coal based thermal power plants.
2. Explain principle of working and components of diesel, gas turbine and combined cycle power plants.
3. Explain principle of working and components of nuclear power plant.
4. Demonstrate generation of power from renewable energy.
5. Illustrate economic and environmental aspects of power plant.

UNIT-I COAL BASED THERMAL POWER PLANTS

Coal Based Thermal Power Plant: Layout of modern coal power plant, Super Critical Boilers, FBC Boilers, Turbines, Condensers, Steam & Heat rate, Subsystems of thermal power plants – Fuel and ash handling, Draught system, Feed water treatment. Binary Cycles and Cogeneration systems.

UNIT-II DIESEL, GAS TURBINE AND COMBINED CYCLE POWER PLANTS

Diesel Power Plant: Introduction, IC Engines, types, construction, Plant layout with auxiliaries, fuel supply system, air starting equipment, air starting equipment.

GAS TURBINE PLANT: Introduction, classification, construction Layout with auxiliaries, Principles of working of closed and open cycle gas turbines. Combined Cycle Power Plants and comparison.

UNIT-III NUCLEAR POWER PLANTS

Basics of Nuclear Engineering, Layout and subsystems of Nuclear Power Plants, Working of Nuclear Reactors : Boiling Water Reactor (BWR), Pressurized Water Reactor (PWR), CANada Deuterium-Uranium reactor (CANDU), Breeder, Gas Cooled and Liquid Metal Cooled Reactors. Safety measures for Nuclear Power plants.

UNIT-IV	POWER FROM RENEWABLE ENERGY
Hydro Electric Power Plants – Classification, Typical Layout and associated components including Turbines. Principle, Construction and working of Wind, Tidal, Solar Photo Voltaic (SPV), Solar Thermal, Geo Thermal, Biogas and Fuel Cell power systems.	
UNIT-V	ECONOMIC AND ENVIRONMENTAL ISSUES OF POWER PLANT
ECONOMIC AND ENVIRONMENTAL ISSUES: Capital cost, investment of fixed charges, operating costs, general arrangement of power distribution, Load curves, load duration curve. Definitions of connected load, Maximum demand, demand factor, average load, load factor, diversity factor - related exercises. Effluents from power plants and Impact on environment, pollutants and pollution standards, Methods of Pollution control.	
Text Books:	
1. Rajput (2011), A Text Book of Power Plant Engineering, 4 th edition, Laxmi Publications, New Delhi, India.	
2. Elanchezhian, L. Saravana Kumar, B. Vijaya Ramkanth (2007), Power plant Engineering, 1 st edition, I.K International Publishing House, New Delhi, India	
Reference Books:	
1. Arora and S. Domkundwar (2008), A Course in Power Plant Engineering, 5th edition, Dhanpat Rai & Co. Delhi.	
2. P. K. Nag (2008), Power Plant Engineering, 3 rd edition, Tata McGraw- Hill Publishing Company Ltd., New Delhi.	
Web References:	
1. https://nptel.ac.in/courses/112104225/22	
2. https://nptel.ac.in/courses/108108078/	
3. https://nptel.ac.in/courses/112105051/	
E-Text Books:	
1. https://www.ebooks.com/en-us/subjects/mechanical-engineering -ebooks/10357	
2. https://www.springer.com/in/book/9783319723709	
3. https://www.pdfdrive.com/ -e14705840.html	
MOOC Course	
1. https://www.mooc-list.com/tags/	
2. https://www.edx.org/learn/	
3. https://www.coursera.org/courses?query	

MACHINE TOOL DESIGN								
VIII Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME52	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OVERVIEW: This course is intended to introduce Design of machine tool structures, Design and analysis of systems for specified speeds and feeds and selection of subsystems for achieving high accuracy in machining.								
COURSE OUTCOMES: At the end of the course, the student will be able to, 1. Explain basic concepts of machine tool drives and mechanisms. 2. Design and analyze systems for specified speeds and feeds. 3. Design machine tool structures. 4. Design Guide ways, Power Screws and Spindles: 5. Design Spindles and Spindle Supports.								
UNIT-I	INTRODUCTION							
Introduction to Machine Tool Drives and Mechanisms: Introduction to the course, Working and Auxiliary Motions in Machine Tools, Kinematics of Machine Tools, Motion Transmission.								
UNIT-II	REGULATION OF SPEEDS AND FEEDS							
Regulation of Speeds and Feeds: Aim of Speed and Feed Regulation, Stepped Regulation of Speeds, Multiple Speed Motors, Ray Diagrams and Design Considerations, Design of Speed Gear Boxes, Feed Drives, Feed Box Design.								
UNIT-III	DESIGN OF MACHINE TOOL STRUCTURES							
Design of Machine Tool Structures: Functions of Machine Tool Structures and their Requirements, Design for Strength, Design for Rigidity, Materials for Machine Tool Structures, Machine Tool Constructional Features, Beds and Housings, Columns and Tables, Saddles and Carriages.								
UNIT-IV	DESIGN OF GUIDEWAYS AND POWER SCREWS							
Design of Guideways, Power Screws and Spindles: Functions and Types of Guideways, Design of Guideways, Design of Aerostatic Slideways, Design of Anti-Friction Guideways, Combination Guideways, Design of Power Screws.								

UNIT-V	DESIGN OF SPINDLES AND SPINDLE SUPPORTS
Design of Spindles and Spindle Supports: Functions of Spindles and Requirements, Effect of Machine Tool Compliance on Machining Accuracy, Design of Spindles, Antifriction Bearings.	
Text Books:	
1. Machine Tool Design and Numerical Control by N.K. Mehta, TMH, New Delhi, 2010. 2. Principles of Machine Tools by G.C. Sen and A. Bhattacharya, New Central Book Agency, 2009. 3. Design of Machine Tools by D. K Pal, S. K. Basu, 5th Edition, Oxford IBH, 2008.	
Reference Books:	
1. Machine Tool Design by N. S. Acherkhan, Vol. I, II, III and IV, MIR publications, 1968. 2. Tool Design by Cyril Donaldson, 5th Edition, Mc Graw Hill.	
Web References:	
1. https://shodhganga.inflibnet.ac.in/bitstream/10603/64298/11/11_chapter%203.pdf 2. https://pdfs.semanticscholar.org/611f/c947cda5ed8fef1fa444a6208fb3ad8e0cbf.pdf	
E-Text Books:	
1. https://books.google.co.in/books?id=gTxanqEACAAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=Sk6ynQEACAAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.coursera.org/specializations/digital-manufacturing-design-technology	

AUTOMATION IN MANUFACTURING								
VIII Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME53	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OVERVIEW: This course is designed to impart knowledge of automation in production systems, automated flow lines concept of line balancing, material handling and storage systems, cellular manufacturing and group technology concepts, automate assembly systems and inspection.								
COURSE OUTCOMES: At the end of the course, the student will be able to, 1. Illustrate the basic concepts of automation. 2. Demonstrate the importance of automated material handling and storage systems. 3. Analyze various transfer lines no internal storage and with storage buffers. 4. Explain the concepts of automated assembly systems. 5. Discuss group technology and cellular manufacturing.								
UNIT-I	INTRODUCTION							
Introduction to Automation: Automation in Production Systems-Automated Manufacturing Systems, Computerized Manufacturing Support Systems, Reasons for Automation, Automation Principles and Strategies. Manufacturing operations, Production Concepts and Mathematical Models. Costs of Manufacturing Operations, Basic Elements of an Automated Systems, Advanced Automation Functions, Levels of automation.								
UNIT-II	AUTOMATED FLOW LINES & LINE BALANCING							
Automated Flow Lines: General terminology, Analysis of Transfer Lines with no Internal Storage, Analysis of Transfer lines with Storage Buffers. Line Balancing: line balancing methods, ways of improving line balance								
UNIT-III	AUTOMATED MATERIAL HANDLING & STORAGE SYSTEMS							
Automated Material Handling & Storage Systems: Overview of Material Handling Equipment, Considerations in Material Handling System Design, the 10 Principles of Material Handling. Material Transport Systems, Automated Guided Vehicle Systems, Conveyor Systems, Analysis of Material Transport Systems. Storage Systems, Storage System Performance, Storage Location Strategies, Automated Storage Systems, Engineering Analysis of Storage Systems.								

UNIT-IV	GROUP TECHNOLOGY & CELLULAR MANUFACTURING
Group Technology: Part Families, Part classification and coding-Optiz and Multiclass Systems, Production flow analysis, Manufacturing and Product design applications. Cellular manufacturing: Objectives, Composite part concept, Machine cell design, Quantitative analysis - grouping parts and machines by rank order clustering.	
UNIT-V	AUTOMATED ASSEMBLY SYSTEMS & INSPECTION
Automated Assembly Systems: Fundamentals of Automated Assembly Systems, Design for Automated Assembly, and Quantitative Analysis of Assembly Systems. Inspection: Types of Inspection, Inspection Procedure, Automated Inspection, Off-line and Online Inspection, Contact Vs Non contact Inspection techniques, Coordinate measuring machines(CMM)- Construction, Operation and Programming, CMM Applications and Benefits.	
Text Books:	
1. Automation, Production systems and computer integrated manufacturing, Mikel P. Groover/ Pearson Education. 2. CAD CAM: Principles, Practice and Manufacturing Management / Chris Mc Mohan, Jimmie Browne / Pearson edu. (LPE)	
Reference Books:	
1. Automation, Buckinghsm W, Haper& Row Publishers, New York, 1961 2. Automation for Productivity, Luke H.D, John Wiley & Sons, New York, 1972.	
Web References:	
1. https://nptel.ac.in/courses/108105063/pdf/L-01(SM)(IA&C)%20((EE)NPTEL).pdf 2. https://nptel.ac.in/courses/112103174/pdf/mod1.pdf	
E-Text Books:	
1. https://aerocastle.files.wordpress.com/2012/08/automationproduction-systems-and-cim-groover2001.pdf 2. https://books.google.co.in/books?id=H977GaxHwaAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://www.classcentral.com/course/nptel-industrial-automation-and-control-5222 2. https://www.mooc-list.com/tags/automation	

VALUE ENGINEERING

VIII Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME54	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course helps students to learn various value analysis techniques in order to reduce the cost of production in the industry.

COURSE OUTCOMES:

At the end of the course , the student will be able to

1. Discuss the concepts of value engineering, identify the advantages, applications.
2. Discuss various phases, selection and evaluation of value engineering projects.
3. Describe about analysis function, evaluation of costs, worth and value.
4. Explain various value engineering techniques.
5. Appraise the value engineering operation in maintenance and repair activities.

UNIT-I INTRODUCTION & ORGANIZATION

INTRODUCTION: Value engineering concepts, advantages, applications, problem recognition, and role in productivity, criteria for comparison, element of choice.

ORGANIZATION: Level of value engineering in the organization, size and skill of VE staff, small plant, VE activity, unique and quantitative evaluation of ideas.

UNIT-II VALUE ENGINEERING JOB PLAN

VALUE ENGINEERING JOB PLAN: Introduction, Value Engineering Job Plan Steps-General phase, Information phase, Function phase, Creation phase, Evaluation phase, Investigation phase, Implementation/Recording phase. Techniques employed in value engineering job plan, Logic of the Job plan.

UNIT-III ANALYSIS FUNCTION

ANALYSIS FUNCTION: Anatomy of the function, use esteem and exchange values, basic vs. secondary vs. unnecessary functions. Approach of function, Evaluation of function, determining function, classifying function, evaluation of costs, evaluation of worth, determining worth, evaluation of value

UNIT-IV VALUE ENGINEERING TECHNIQUES

VALUE ENGINEERING TECHNIQUES : Selecting products and operation for value engineering action, value engineering programmes, determining and evaluating function(s) assigning rupee equivalents, developing alternate means to required functions, decision making for optimum alternative, use of decision matrix, queuing theory and Monte Carlo method make or buy, measuring profits, reporting results, Follow up, Use of advanced technique like Function Analysis System.

UNIT-V	VERSATILITY OF VALUE ENGINEERING
VERSATILITY OF VALUE ENGINEERING: Value engineering operation in maintenance and repair activities, value engineering in non hardware projects. Initiating a value engineering programme Introduction, training plan, career development for value engineering specialties	
Text Books:	
1. Alphonse Dell'Isola, "Value Engineering: Practical Applications for Design, Construction, Maintenance & Operations", R S Means Co., 1997. 2. Richard Park, "Value Engineering: A Plan for Invention", St. Lucie Press, 1999. 3. Del L. Younker, "Value Engineering analysis and methodology", Marcel Dekker Inc, New York, 2004. 4. Miles, L.D., "Techniques of Value Analysis and Engineering", McGraw Hill second Edition, 1989.	
Reference Books:	
1. Anil Kumar Mukhopadhyaya, "Value Engineering: Concepts Techniques and applications", SAGE Publications 2010. 2. Khanna, O.P., "Industrial Engineering and Management", Dhanpat Rai & Sons, 1993.	
Web References:	
1. https://www.academia.edu/6158369/Value_Engineering_Basic_Concepts_Methodology_Implementation 2. https://www.slideshare.net/farzadfarahani/value-engineering-42414994 3. https://nptel.ac.in/courses/112107143/38	
E-Text Books:	
https://minds.wisconsin.edu/bitstream/handle/1793/5978/1556.pdf?sequence=1&isAllowed=y	
MOOC Course	
https://www.mooc-list.com/tags/value-creation	

PRODUCT DESIGN AND PROCESS PLANNING**VIII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME55	PEC	L	T	P	C	CIE	SEE	Total
		3		-	3	30	70	100

COURSE OVERVIEW:

This course is intended to impart knowledge of product design principles, ergonomic considerations in product design, process parameters, process planning and economics of process planning and GD & T.

COURSE OUTCOMES:

At the end of the course students will be able to

1. Explain the basic concepts of product and process design.
2. Demonstrate ergonomic considerations for product design.
3. Analyze various process parameters
4. Discuss process planning and economic aspects of process planning.
5. Explain geometric dimensioning and tolerancing..

UNIT-I INTRODUCTION

Product design and process design functions, selection of a right product, essential factors of product design, Morphology of design, sources of new ideas for products, evaluation of new product ideas. Product innovation procedure-Flow chart, Qualifications of product design Engineer, Criteria for success/failure of a product.

UNIT-II PRODUCT DESIGN & ERGONOMICS

Product Design: Introduction, Factors affecting design of product, Product Basic product design rules for casting, forging, machining, sheet metal and welding, Product Design and Production Cost- Estimating in machine shop, Estimation in welding shop, Estimation in forging shop Estimation in Forging Shop.

Ergonomics: Man- machine considerations, ease of maintenance. Ergonomic considerations in product design-Anthropometry, Design of controls, man-machine information exchange.

UNIT-III PROCESS PARAMETERS

Process Parameters: Factors affecting speed, feeds and depth of cut, Surface cutting speed-Cutting speeds for turning, boring, milling and drilling, Spindle speeds for turning, boring, milling and drilling, Stroke speeds for shaping and planing, Feed rates for turning and boring, Feed rates for milling, Feed rates for drilling, Depth of cut for turning and boring, Depth of cut for milling, Depth of cut for drilling, Depth of cut for planing and shaping, Depth of cut for grinding, Example problems.

UNIT-IV	PROCESS PLANNING & ECONOMICS OF PROCESS PLANNING	
Process Planning: Design and Manufacture cycle, Process Planning, Process Planning-Design/Manufacture Interface, Process Planning activities, Manual Process Planning-Traditional and Workbook approaches, Computer Aided Process Planning (CAPP)- Variant and Generative approaches, Advantages of CAPP. Economics of Process Planning: Manufacturing costs-Variable and fixed costs, Job /Batch costing, Marginal costing, Criteria for Make or Buy Decision, Analysis for Make or Buy Decisions-Simple cost analysis, Economic analysis, and Break-even analysis.		
UNIT-V	GEOMETRIC DIMENSIONING AND TOLERANCING (GD&T)	Classes: 12
Introduction, GD&T Benefits, Tolerance Zone & its conversions, Important terms-Feature, Feature of Size, Datum, Datum feature, Datum triangle, Datum letter, Feature control frame, Modifiers-Maximum material condition, Least material condition, Projected tolerance zone, Radius, Controlled Radius; Geometric characteristic symbols- Form controls, Orientation controls, Profile controls, Location controls and Runout controls.		
Text Books:		
1. Niebel, B.W., and Draper, A.B., Product design and process Engineering, Mc Graw Hill Kogalkusha Ltd., Tokyo, 1974. 2. Chitale, A.K, and Gupta, R.C., Product Design and Manufacturing, Prentice Hall of India Pvt. Ltd., New Delhi, 2004. 3. Mahajan, M. Industrial Engineering and Production Management, Dhanpath Rai & Co., 2000		
Reference Books:		
1.R.Kesavan, C.Elanchezian, B.Vijaya Ram nath, Process planning & Cost estimation, New Age International Publishers. 2.Karl T.Urlich,Steven D Eppinger,Product design and development, Tata McGraw-Hill Publishing Company Limited		
Web References:		
1. https://www.slideshare.net/zimbar/product-and-process-planning 2. https://www.citeman.com/74-process-planning-and-process-design.html		
E-Text Books:		
1. https://books.google.co.in/books?id=bxGjoNst2PYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=A9-ZXblNrPoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=R7GkqkbZbPIC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 4. https://books.google.co.in/books?id=CfKlFk_uhzoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://nptel.ac.in/noc/individual_course.php?id=noc19-me51 2. https://www.classcentral.com/course/nptel-product-design-and-manufacturing-10068		

TRIBOLOGY								
VIII Semester: MECHANICAL ENGINEERING								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME56	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OVERVIEW: This course is framed to cover basic concepts of tribology, hydrodynamic theory of lubrication, friction and power losses in journal bearings, air lubricated bearing, bearing oil pads and materials.								
COURSE OUTCOMES: At the end of the course students will be able to 1. Demonstrate various properties of fluid and explain the basic concepts of hydrostatic lubrication. 2. Explain hydrodynamic theory of lubrication. 3. Explain friction and power losses in journal bearings. 4. Discuss in detail about air lubricated bearings. 5. Explain types of bearing oil pads and bearing materials.								
UNIT-I	INTRODUCTION Study of various parameters: Viscosity, flow of fluids, viscosity and its variation, absolute and kinematic viscosity, temperature variation, viscosity index, determination of viscosity, different viscometers used. Hydrostatic lubrication: Hydrostatic step bearing, application to pivoted pad thrust bearing and other applications, hydrostatic lifts, hydrostatic squeeze films and its application to journal bearing.							
UNIT-II	HYDRODYNAMIC THEORY OF LUBRICATION Hydrodynamic theory of lubrication: Various theories of lubrication, petroffs equation, Reynold's equation in two dimensions –Effects of side leakage – Reynolds equation in three dimensions, Friction in sliding bearing, hydrodynamic theory applied to journal bearing, minimum oil film thickness, oil whip and whirl anti-friction bearing.							
UNIT-III	FRICTION AND POWER LOSSES IN JOURNAL BEARINGS Friction and power losses in journal bearings: Calibration of friction loss, friction in concentric bearings, bearing modulus, Sommer-field number, heat balance, practical consideration of journal bearing design considerations.							
UNIT-IV	AIR LUBRICATED BEARING Air lubricated bearing: Advantages and disadvantages, application to Hydrodynamic journal bearings, hydrodynamic thrust bearings. Hydrostatic bearing Analysis including compressibility effect. Study of current concepts of boundary friction and dry friction.							

UNIT-V	TYPES OF BEARING OIL PADS & BEARING MATERIALS
Types of bearing oil pads: Hydrostatic bearing wick oiled bearings, oil rings, pressure feed bearing, partial bearings –externally pressurized bearings. Bearing materials: General requirements of bearing materials, types of bearing materials.	
Text Books:	
1. Fundamentals of Tribology by Basu, SenGupta and Ahuja, PHI 2. Tribology in Industry by Sushil Kumar Srivatsava, S. Chand &Co.	
Reference Books:	
1. Tribology by B.C. Majumdar	
Web References:	
1. http://web.iitd.ac.in/~hirani/mcl748.pdf 2. https://ocw.mit.edu/courses/mechanical-engineering/2-800-tribology-fall-2004/lecture-notes/	
E-Text Books:	
1. https://books.google.co.in/books?id=DJDv5jVwC2UC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=otw4UKI631UC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://ocw.mit.edu/courses/mechanical-engineering/2-800-tribology-fall-2004/ 2. https://www.classcentral.com/course/nptel-friction-and-wear-of-materials-principles-and-case-studies-13027 3. https://nptel.ac.in/courses/112102014/	

MECHATRONICS**VIII Semester: MECHANICAL ENGINEERING**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME57	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This program, developed in direct response to industrial demand for engineers with multi-disciplinary skills, is a combination of mechanical, electronics, control, computer and systems design engineering streams. The program allows engineers to design construct and run factory production lines and automated processes, where they use their skills in computers, micro-controllers, programmable logic controllers, programming, industrial sensors, hydraulic, pneumatic and electric drives, design of mechanical structures and mechanisms and knowledge of manufacturing processes.

COURSE OUTCOMES:

At the end of the course, the student will be able to

1. Demonstrate basic concepts of mechatronics.
2. Explain actuators and elements of motion control.
3. Classify and discuss about sensors.
4. Explain elements and working of various transducers.
5. Explain concepts of PLC's and artificial intelligence.

UNIT-I INTRODUCTION

Introduction: Evolution of Mechatronics, Modules in Mechatronic Systems, Design process, Control Systems-Open and Closed loop systems, Analog and digital control systems, Mechatronics in Manufacturing –Adaptive and distributed control systems – Modelling and simulation of mechatronics

UNIT-II ACTUATORS & MOTION CONTROL

Actuators and Motion Control: Characteristics of Mechanical, electrical, Hydraulic and pneumatic actuators and their limitations. Control parameters and system objectives. Mechanical configurations, Popular control system configurations, S-curve, Motor/load inertia machining, design with linear studies.

UNIT-III SENSORS

Sensors: Definition of sensor, Classification of sensors by function , Classification of performance, Performance parameters, Classification of sensors by output, Pressure sensors, flow sensors, Hall-effect sensors-Linear and threshold type, Light sensors-photo diodes, Proximity Sensors-Inductive and Ultrasonic type; Optical sensor.

UNIT-IV TRANSDUCERS

Transducers: Difference between transducer and sensor, Transducer principle, Photo electric transducers-photo emissive transducer, photo conductive transducer, photovoltaic transducer; Inductive transducers, Capacitive Transducers, pyro electric transducers, piezo electric transducer, Ionization transducers, strain gauge, load cells and flow transducers.

UNIT-V	PLCs & ARTIFICIAL INTELLIGENCE
PLCs: Programmable Logic Controllers Architecture of Programmable Logic Controllers – Input/output modules – programming methods – Timers and counters – Master control – Branching – Data handling – Analog input/output – Selection of PLC and troubleshooting. AI: Intelligent Mechatronics and Case Studies Fuzzy logic control and Artificial Neural Networks in mechatronics – Algorithms – Computer based instrumentation – Real-time Data Acquisition and Control – Software integration – Man Machine interface – Vision system.	
Text Books:	
1. Introduction to Mechatronics and Measurement Systems, Tata McGraw Hill 2. Ganesh S.Hegde, Mechatronics, University Science Press.	
Reference Books:	
1. Designing Intelligent Machines, Michel B. Histan and David G. Alciatore, Open University London 2. Control Sensors and Actuators, ICW. Desiha, Prentice Hall	
Web References:	
1. https://lecturenotes.in/subject/137/mechatronics-mech 2. https://sites.google.com/site/profnarendralakal/Home/presentations/mtrx	
E-Text Books:	
1. https://books.google.co.in/books?id=NCEeONKWzX4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=z8yEUou06cEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course:	
1. https://ocw.mit.edu/courses/mechanical-engineering/2-737-mechatronics-fall-2014/ 2. https://nptel.ac.in/courses/112103174/ 3. https://www.igi-global.com/chapter/massive-online-open-course-assisted-mechatronics-learning/137327	

UNCONVENTIONAL MACHINING PROCESSES

VIII Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME58	PEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course covers operating principles of unconventional machining processes which helps to perform research and development activities.

COURSE OUTCOMES:

At the end of the course student will be able to

1. Explain need for unconventional machining processes, elements and working ultrasonic machining process.
2. Demonstrate mechanical material removal processes.
3. Explain thermal metal removal processes.
4. Discuss electron beam machining and laser beam machining processes.
5. Explain chemical material removal processes.

UNIT-I INTRODUCTION

INTRODUCTION: Need for non-conventional machining processes, Comparison between conventional and unconventional machining processes, Classification of non-conventional machining processes, considerations in process selection-shape, materials, process parameters, effects on equipment and tooling, economic considerations

Ultrasonic machining: Elements of ultrasonic machining, Mechanics of material removal-Grain throwing and Grain hammering models, Process parameters, Applications, Recent developments, simple problems.

UNIT-II MECHANICAL MATERIAL REMOVAL PROCESSES

ABRASIVE JET MACHINING: Elements of abrasive jet machining, working principle, advantages, limitations, applications.

Water Jet Machining: Working principle, elements & their functions, dis-advantages, applications and mechanics of water jet machining, advantages,

Abrasive Water Jet Machining – basic principles, components, process parameters advantages and disadvantages, applications.

UNIT-III THERMAL MATERIAL REMOVAL PROCESSES

THERMAL MATERIAL REMOVAL PROCESSES: Electro Discharge Machining(EDM)-principle of operation, elements of EDM, Power delivered, MRR, surface finish, process parameters problems on R-C generator, Layers formed on machined components during EDM, Flushing methods in EDM, Different circuits used in EDM, Factors to be considered for tool selection in EDM, Dielectric fluids used in EDM, Advantages of EDM, Limitations of EDM, Applications of EDM. Wire EDM: working, advantages, limitations and applications, Electric Discharge Grinding: Working, process parameters.

UNIT-IV	ELECTRON BEAM MACHINING & LASER BEAM MACHINING
ELECTRON BEAM MACHINING: Elements, Generation of electron beam and working, process parameters, advantages, dis-advantages, applications, MRR, specific power consumption. LASER BEAM MACHINING: Elements, working, laser materials, cutting speed, accuracy, thermal features, advantages, dis-advantages, applications.	
UNIT-V	CHEMICAL MATERIAL REMOVAL PROCESSES
CHEMICAL MATERIAL REMOVAL PROCESSES: Chemical Machining, Electro chemical machining-elements, working, accuracy, surface finish, economics, advantages, dis-advantages, applications, MRR, problems, Electro chemical grinding, Electro chemical honing and Electro chemical deburring.	
Text Books:	
1. Advanced machining processes by VK Jain, Allied publishers. 2. Non Traditional Manufacturing Processes by Gary F Benedict, CRC Press.	
Reference Books:	
1. MEMS & Microsystems – Design and Manufacture by Tai-Ran Hsu, Tata McGraw Hill 2. Modern Machining Process by Pandey P.C. and Shah H.S., TMH 3. New Technology by Bhattacharya A, the Institution of Engineers, India 1984. 4. Non-Traditional Machining by P.K.Mishra, New Age. 5. Micro Machining of Engineering Materials Edited by J.Mc Geough, CRC Press.	
Web References:	
1. http://www.engineershandbook.com/MfgMethods/nontraditionalmachining.htm 2. https://www.mech4study.com/2017/04/non-traditional-machining-process-requirements-types-advantages-and-disadvantages.html	
E-Text Books:	
1. https://books.google.co.in/books?id=9nl1ngEACAAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=ZdQEZ5pyCv4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false	
MOOC Course	
1. https://www.classcentral.com/university/iitm 2. https://www.classcentral.com/course/coursera-advanced-manufacturing-process-analysis-8154 3. https://www.mooc-list.com/tags/manufacturing-processes	

**OPEN ELECTIVES
OFFERED BY
AERONAUTICAL DEPARTMENT**

FABRICATION PROCESSES**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE17	OEC	L	T	P	C	CIE	SEE	Total
		3	1	-	3	30	70	100

COURSE OBJECTIVES:

Students will learn various fabrication processes and able to select suitable fabrication process required for manufacturing aircraft components.

COURSE OUTCOMES:

The students should be able to

1. Demonstrate various metal casting processes used in aircraft industry.
2. Explain metal joining processes used in aircraft industry.
3. Discuss bulk deformation and surface finishing processes used in aircraft industry.
4. Explain various sheet metal forming processes used in aircraft industry.
5. Suggest suitable molding process for plastics for the given application

UNIT-I	METAL CASTING PROCESSES	Classes: 12
Sand casting – Sand moulds – Types of patterns – Pattern materials – Pattern allowances – Types of Moulding sand- Properties – Core making – Moulding machines – Types of moulding machines – Melting furnaces – Working principle of Special casting processes – Shell, investment casting – Lost Wax process – Centrifugal casting –Sand Casting defects.		
UNIT-II	METAL JOINING TECHNIQUES	Classes: 12
Fusion welding processes – Types of Gas welding – Equipment's used – Flame characteristics – Filler and Flux materials – Arc welding equipment – Electrodes – Coating and specifications – Principles of Resistance welding – Spot/butt, seam welding – Percussion welding – Gas metal arc welding – Flux cored – Submerged arc welding –Electro slag welding – TIG welding – Principle and application of special welding processes – Plasma arc welding – Thermit welding – Electron beam welding – Friction welding –Weld defects – Brazing and soldering process.		
UNIT-III	BULK DEFORMATION AND SURFACE FINISHING PROCESSES	Classes: 12
Hot working and cold working of metals – Forging processes – Open and close die forging – Characteristics of the process – Types of Forging Machines – Typical forging operations – Rolling of metals –Types of Rolling mills –Tube piercing – Defects in rolled parts – Principles of Extrusion – Types of Extrusion – Hot and Cold extrusion – Principle of rod and wire drawing – Equipments used - surface roughness and measurements - grinding - honing - lapping - polishing - burnishing - thermal spray coating.		
UNIT-IV	SHEET METAL FORMING PROCESSES	Classes: 12
Sheet metal characteristics – Typical shearing operations, bending and drawing operations –Formability of sheet metal –Working principle and application of special forming processes – Hydro forming –Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics		
UNIT-V	FORMING AND SHAPING OF PLASTICS	Classes: 12

Types of plastics – Characteristics of forming and shaping processes – Moulding of Thermoplastics – Working principles– Injection moulding – Plunger and screw machines – Blow moulding –Extrusion – Typical industrial applications – Thermoforming – Processing of Thermosets – Working principles and typical applications – Compression moulding – Transfer moulding – Bonding of Thermoplastics –Induction and Ultrasonic methods.

Text Books:

1. Hajra Choudhury , Elements of Workshop Technology, Vol. I and II, Media Promotors Pvt. Ltd., Mumbai, 2007
2. Serope Kalpak Jain, Steven R, Schmid, Manufacturing Engineering and Technology, Pearson Education, Inc. 4th Edition, 2009

Reference Books:

1. Elements of Manufacturing Processes, B.S. Magendran Parashar & R.K. Mittal, Prentice Hall of India, 2008
2. Manufacturing Technology, P.N. Rao, Tata Mc Graw-Hill Publishing Limited, 2010.
3. A text book of production technology, P.c. Sharma, S. Chand and Company, 2010
4. Manufacturing Process – Begman, John Wiley & Sons, VIII Edition, 1999

Web References:

1. <https://www.velaction.com/fabrication/>
2. <https://www.boconline.co.uk/en/.../fabrication-processes/fabrication-processes.html>

E-Text Books:

1. https://books.google.co.in/books?id=8i-xJCYnln8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=jeb9wdpFx2oC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://nptel.ac.in/courses/112107144/>
2. <https://nptel.ac.in/courses/112105127/>
3. <https://nptel.ac.in/courses/112104195/>
4. <https://nptel.ac.in/courses/112107078/>

FUNDAMENTALS OF AVIONICS

V Semester: OPEN ELECTIVE -I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE18	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

The purpose of this subject is to provide the students with the theoretical background and engineering applications. It gives an overview of fundamental concepts of aviation such as basic principles of avionics, flight deck and display systems, audio and communication systems, ranging and landing systems, positioning systems etc.

COURSE OUTCOMES:

At the end of the course the students are able to

- 1 Explain the instrumentation used in avionics.
- 2 Classify various ranges of the communication techniques used in aircraft.
- 3 Distinguish between network systems, controlling parts & surfaces
- 4 Compare various principles of navigation systems
- 5 Build phenomena of auto pilot control system

UNIT-I	BASICS & FLIGHT DECK AND DISPLAY SYSTEMS	Classes: 8
BASICS: Basic principles of Avionics, Typical avionics sub system in civil/ military aircraft and space vehicles. FLIGHT DECK AND DISPLAY SYSTEMS: Flight deck display technologies, CRT, LED, LCD, Touch screen, Head up display, Electronic instrumentation systems.		
UNIT-II	AUDIO AND COMMUNICATION SYSTEMS	Classes: 10
AUDIO AND COMMUNICATION SYSTEMS: Aircraft audio systems, basic audio transmitter and receiver principles, VHF communication system, UHF communication systems.		
UNIT-III	RANGING, LANDING AND POSITIONING SYSTEMS	Classes: 12
RANGING AND LANDING SYSTEMS: VHF Omnidirectional Range, VOR receiver principles, distance measuring equipment, principles of operation, Instrument landing system, and localizer and glide slope. POSITIONING SYSTEM: Global positioning system principles, triangulation, position accuracy, applications in aviation		
UNIT-IV	NAVIGATION AND SURVEILLANCE SYSTEMS	Classes: 12
INERTIAL NAVIGATION SYSTEM: Principle of Operation of INS, navigation over earth, components of inertial Navigation systems, accelerometers, gyros and stabilized platform. SURVEILLANCE SYSTEM: ATC surveillance systems principles and operation interrogation and replay standards, Collision avoidance system, ground proximity warning system		

UNIT-V	AUTO FLIGHT SYSTEM	Classes: 8
AUTO FLIGHT SYSTEM: Automatic flight control systems fly by wire and fly by light technologies, flight director systems, flight management systems..		
Text Books:		
1. N. S. Nagaraja(1996), Elements of electronic navigation, 2nd edition, Tata McGraw Hill, New Delhi. 2. Janes W. Wasson, Jeppesen Sandersen(1994), Avionic systems Operation and maintenance,		
Reference Books:		
1. Albert Hel Frick (2010), Principle of Avionics, 6th edition, Avionics Communications Inc, India. 2. H. J. Pallet (2010), Aircraft Instrumentation and Integrated systems, Pearson Education, New Delhi. 3. J. Powell (1998), Aircraft Radio Systems, Pitman publishers, London		
Web References:		
1. https://www.britannica.com/technology/avionics 2. https://www.pdc.com/aviation/		
E-Text Books:		
1. https://books.google.co.in/books?id=0BI6ioB8kM8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=Dm5mQfFZtSkC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.canvas.net/browse/erau/courses/aviation-101		

INTRODUCTION TO JETS AND ROCKETS**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE19	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

Students will

1. Illustrate an overview of aerospace propulsion system.
2. Identify the foundation in fundamentals of thermodynamics.
3. Compare the ideal components and characteristics of jet engine
4. Interpret the performance of nozzles
5. Simplify the ideal performance analysis of rocket engines.
6. Select appropriate fuel for aerospace application.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Explain basic concepts of aerospace propulsion.
- 2 Describe principles of jet propulsion.
- 3 Discuss about the aspects of ramjet engines, scream jet engines and nozzles.
- 4 Formulate Equations of rocket motion in free space, Tsiokovsky's equation.
- 5 Describe about types, applications of rockets and the types of fuel in aviation and aerospace engineering.

UNIT-I	INTRODUCTION TO AEROSPACE PROPULSION	Classes: 12
What is Propulsion system? Propulsive Systems – Evolution, Development, Growth and Challenges. Fundamentals of Thermodynamics – Variables, Thermodynamic Process, Introduction to IC Engines and Reciprocating Engines, Propellers and Working of Propellers.		
UNIT-II	PRINCIPLES OF JET PROPULSION	Classes: 12
Fundamentals of jet propulsion, Working Principle, Analysis of Ideal Jet Engine cycle, Engine components- merit-significance- ideal component characteristics, Classification – turbo jet, turbo fan, turbo prop and Ramjet engines. Basic Problems based on Engine Cycle.		
UNIT-III	RAMJET, SCRAMJET ENGINES AND NOZZLES	Classes: 12
Speed limitations of gas turbines, Basics of Ramjets, Combustors for liquid fuel ramjet engines, Combustion Instability and its Suppression, Solid fuel Ramjet Engines, SCRAM jet engines, Applications of RAM Jet and SCRAM Jet Engines to Missiles with Examples, Nozzles- Types of Nozzles, Converging-Diverging Nozzle, Variable Nozzle and Effects of Pressure Ratios on Engine Performance.		
UNIT-IV	ROCKET THEORY	Classes: 12
Applications of Rockets, Types of Rockets, Basics of Thermal Rocket Engine-Thermodynamics and Ideal Performance Analysis, Equations of motion-Rocket Motion in free space, Tsiokovsky's equation, Rocket Parameters, Burnout range, Burnout Velocity. Practical Problems		

UNIT-V	PROPELLANT ROCKETS	Classes: 12
Solid Propulsion-Solid Propellant Rockets, Basic Configuration and Performance, Propellant Grain and Configuration, Propellant Characteristics Combustion Chamber, Ignition Process Liquid Propulsion - Design consideration of liquid rocket combustion chamber, injector, and propellant feed lines, valves, propellant tank outlet and helium pressurized and turbine feed systems-.BIO Fuels and Impact on the Atmosphere, Aviation turbine fuels - Requirements of aviation fuels of kerosene type.		
Text Books:		
1. Mechanics and Thermodynamics of Propulsion – Philip G Hill & Carl R Peterson , Pearson Publication – 2 nd Edt 2. Rocket Propulsion Elements, Sutton, G.P., John Wiley, 1993. 3. Fuels and Combustion, Sharma, S.P. and Mohan .C., Tata McGraw Hill Publishing Co, Ltd., 1984		
Reference Books:		
1. The Jet Engine – Rolls Royce 2. Gas Turbines and Jet and Rocket Propulsion, M. L. Mathur, R. P. Sharma, Standard Publishers Distributors.		
Web References:		
1. http://engineeringvideolectures.com/course/24 2. http://engineeringvideolectures.com/course/28		
E-Text Books:		
1. https://books.google.co.in/books?id=xI6-fZyHWUOC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://online.stanford.edu/courses/aa283-aircraft-and-rocket-propulsion		

INTRODUCTION TO AIRCRAFT INDUSTRY

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE25	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES

Students will be enabled to acquire knowledge on various types of aerospace industry, the role of regulatory bodies and its business context, the organisational structure of airlines and airports, importance of air navigation services and its types, concepts of air safety and security for ensuring passenger safety.

COURSE OUTCOMES:

At the end of the course the students are able to:

- 1 Discuss the types of aerospace industry and evolution of flying machine to airline alliances.
- 2 Explain standardisation and the importance of regulatory bodies.
- 3 Outline the organisational structure of airlines and airports.
- 4 Demonstrate types of air navigation services and air traffic control services.
- 5 Explain the concepts of air safety and security.

UNIT-I	AIRCRAFT INDUSTRY OVERVIEW	Classes: 12
Flying machine to airline alliances, Types of Aerospace Industry, Introduction to ages of engineering, Aerospace Manufacturing, Introduction to the space environment & human space exploration.		
UNIT-II	REGULATORY AND BUSINESS CONTEXT	Classes: 12
Regulatory bodies, Introduction to standardisation, Standards and recommended practices (SARPS), Freedoms of the air, Role of governments, Major agreements and treaties, Business side of the industry – Key performance indicators (revenue passenger kilometre, yield etc.), Industry characteristics of passenger airlines.		
UNIT-III	AIRLINES AND AIRPORTS	Classes: 12
Introduction to airlines, Organisational structure, Types of airline personnel, Flight crew and cabin crew, Role of training with airlines, Organisational culture of an airline Introduction to airports – Airport personnel, Processing passengers and freight, Security		
UNIT-IV	AIR NAVIGATION SERVICES	Classes: 12
Introduction to air navigation services, Types of navigation services, Air traffic control (ATC) services, Structure of controlled airspace, Air traffic and control services, Classifications of airspace, Flight information services and alerting services, Air navigation facilities and air navigation service (ANS) providers		
UNIT-V	AIR SAFETY AND SECURITY	Classes: 12
Introduction to air safety and security, Manufacturer responsibilities, Maintenance procedures, Airside safety, Safety in the air, Issues in air safety, Accident and incident investigation, Security – Response of International Civil Aviation Organization (ICAO).		

Text Books:

1. Peter Belobaba, Amedeo Odoni and Cynthia Barnhart (2015), The Global Airline Industry (Aerospace Series), John Willey & Sons, Ltd, United Kingdom.
2. Connor R Walsh, Airline Industry – Strategies, Operations and Safety, Nova Science Publishers, New York.
3. Shevell (2004), Fundamentals of Flight, 2nd edition, Pearson Education Limited, New Delhi
4. Allan Seabridge, Ian Moir (2008), Aircraft Systems: Mechanical, Electrical and Avionics Subsystems Integration, 3rd edition, John Willey & Sons, New York.

Reference Books:

1. Andrew R Thomas (2011), Soft Landing: Airline Industry Strategy, Service and Safety, Apress, New York
2. Ahmed Abdelghany, Khaled Abdelghany (2009), Modeling Applications in the Airline Industry, Ashgate Publishing Ltd, England.
3. Kermode, A.C. (1989), Flight without Formulae, 5th edition, Pearson Education Limited, New Delhi.
4. Raymer Daniel (2002), Aircraft Design: A Conceptual Approach, AIAA Publisher, USA.

Web References:

1. <https://www.investopedia.com/features/industryhandbook/airline.asp>
2. http://web.mit.edu/airlines/analysis/analysis_airline_industry.html
3. <https://www.ukessays.com/essays/commerce/an-introduction-to-the-airline-industry-business-essay.php>

E-Text Books:

1. https://books.google.co.in/books?id=frfBDQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=7llzqsl3bXIC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://online-learning.tudelft.nl/courses/introduction-aeronautical-engineering/>
2. <https://ocw.mit.edu/courses/aeronautics-and-astronautics/>

NON DESTRUCTIVE TESTING METHODS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE26	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

Students will learn the Importance of NDT, basic principles of various NDT methods, discontinuities in different product forms, applications and limitations of NDT methods.

COURSE OUTCOMES:

At the end of the course students are able to

1. Describe the various types of NDT tests carried out on components.
2. Explain liquid penetrate testing and magnetic particle testing.
3. Explain ultrasonic method of testing the materials.
4. Discuss about radiography uses in engineering.
5. Demonstrate principles of special NDT methods such as acoustic emission testing, infrared thermography, laser shearography etc.

UNIT-I	INTRODUCTION	Classes: 12
Introduction to non-destructive testing (NDT) - importance of NDT - types of NDT techniques - ASME, ASTM, AWS, BIS, SAE standard sample specifications - visual testing (direct and remote visual inspection) - principle and types of liquid penetrant tests (LPT) - properties of liquid penetrants and developers - advantages and limitations of LPT - applications of LPT.		
UNIT-II	MAGNETIC PARTICLE TESTING	Classes: 12
Introduction to magnetic particle testing (MPT) - magnetization methods - dry particle and wet fluorescent particle techniques - demagnetization - advantages and limitations of MPT - magnetic flux leakage testing - principle, instrumentation and applications of electromagnetic induction techniques and eddy current testing (ECT) method.		
UNIT-III	ULTRASONIC TESTING	Classes: 12
Introduction to ultrasonic testing (UT) - characteristics of ultrasonic waves - principle of UT – UTprobes - UT inspection methods (pulse echo, transmission and phased array techniques, PAUT) -types of scanning and displays - application of UT for welded parts.		
UNIT-IV	RADIOGRAPHY TESTING	Classes: 12
Introduction to radiography testing (RT) - sources of X-rays and Gamma rays - characteristics of Xraysand Gamma rays (absorption, scattering) - filters and screens - film radiography and digitalradiography (shadow formation, exposure factors, film handling and storage) - inverse square law -exposure charts - penetrameters - safety issues.		

UNIT-V	SPECIAL TECHNIQUES	Classes: 12
Acoustic emission testing (AET) principle, advantages, limitations - instrumentation and application of AET - infra red thermography (IRT) - contact and non-contact inspection methods - pressure and leak detection - LASER shearography - acoustic holography.		
Text Books:		
1. Baldev Raj, T. Jayakumar, M. Thavasimuthu, "Practical Non-Destructive Testing", Narosa Publishing, London, 2012. 2. Paul E. Mix, "Introduction to Non Destructive Testing", A Training Guide, Wiley- Interscience, New Jersey, USA, June 2005.		
Reference Books:		
1. ASM Metals Handbook, V-17, "Non-Destructive Evaluation and Quality Control", American Society of Metals, Metals Park, Ohio, USA, 2001 2. W.T. Mc Gonnagle, "Non-Destructive Testing", McGraw Hill Book Co., USA, 2013. 3. Louis Cartz, "Non-Destructive Testing", ASM International, Metals Park Ohio, US, 2007. 4. Barry Hull and Vernon John, "Non Destructive Testing", ELBS/Macmillan, Hampshire, UK, 2015.		
Web References:		
1. https://nptel.ac.in/reviewed_pdfs/114106035/lec35.pdf 2. http://studentsfocus.com/me6019-ndtm-notes-non-destructive-testing-and-materials-lecture-handwritten-notes-mech-8th-sem/ 3. https://lecturenotes.in/notes/14391-note-for-non-destructive-testing-ndt-by-malko-trar		
E-Text Books:		
1. https://books.google.co.in/books?id=Ja6Wga74pYQC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=qXcCKsL2IMUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://awo.aws.org/online-courses/non-destructive-testing-courses/ 2. http://helliendtd.com/Online-Courses.html		

FUNDAMENTALS OF FINITE ELEMENT METHODS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE27	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the concept of numerical analysis of structural components.
2. To understand the basic knowledge of Isoparametric Element Formulation.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate an understanding of the principles and theories of finite element method.
2. Understand the concepts of discretization, approximations, mathematical tools, FEM tools and error control
3. Analyze 2D problems using bar, beam and truss elements.
4. Demonstrate the basic concepts of isoparametric elements and its application in aircraft.
5. Apply Finite element methods for heat transfer and dynamic loading problems.

UNIT-I	INTRODUCTION TO FEM & 1-D PROBLEMS	Classes: 12
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Basic concept, historical background, application of FEM, general description, comparison of FEM with other methods. Basic equations of elasticity, Stress-strain relations, Strain- Displacement relations.
ONE DIMENSIONAL PROBLEM: Finite element modeling and shape functions. Minimum potential energy principle, assembly of Global stiffness matrix and load vector. Treatment of boundary conditions.

UNIT-II	ANALYSIS OF BEAMS	Classes: 12
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Hermite shape functions-Element stiffness matrix for two nodes, two degrees of freedom per node beam element, load vector, calculation of deflection, slope, shear force and bending moments.

UNIT-III	2-D PROBLEMS	Classes: 12
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CST-Stiffness matrix and load vector, Isoparametric element representation, Shape functions, convergence requirements, Problems.

UNIT-IV	STEADY STATE HEAT TRANSFER ANALYSIS	Classes: 12
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One Dimensional Finite Element analysis of fin and composite slabs.

UNIT-V	DYNAMIC ANALYSIS	Classes: 12
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Formulation of finite element model, element matrices, Lumped and consistent mass matrices-evaluation of Eigen values and Eigen vectors for a stepped bar.

Text Books:

1. R. Tirupathi Chandrapatla (2011), Introduction to Finite Elements in Engineering, 4rd edition, Pearson Education, India.
2. S. S. Rao (2012), The Finite Element Methods in Engineering, 5th edition, Elsevier, USA.
3. V. David. Hutton (2010), Fundamentals of finite elements analysis, 1st edition, Tata McGraw-Hill education (P) Ltd, New Delhi, India

Reference Books:

1. Chennakesava R. Alavala (2009), Finite elements methods, 1st edition, second reprint, Prentice Hall of India publishers, New Delhi, India.
2. J. N. Reddy (2010), An introduction to Finite Element Method, 3rd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
3. Kenneth H. Huebner, Donald L. Riddewhirst, Douglas E. Smith, Ted G. Byrom (2009), The Finite Element Method for Engineers, 3 edition, John Wiley & sons (ASIA) Pvt. Ltd., New York.
- 4.

Web References:

3. http://web.mit.edu/16.810/www/16.810_L4_CAE.pdf
4. <https://www.simscale.com/blog/2016/10/what-is-finite-element-method/>

E-Text Books:

4. https://books.google.co.in/books?id=0FU8DwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
5. https://books.google.co.in/books?id=xbugmEwQz8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
6. https://books.google.co.in/books?id=wzZ7BkQgMnAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

3. <https://www.classcentral.com/course/nptel-basics-of-finite-element-analysis-i-5234>
4. <https://www.open.edu/openlearn/science-maths-technology/introduction-finite-element-analysis/content-section-0?active-tab=description-tab>

GUIDANCE AND CONTROL OF AEROSPACE VEHICLES

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE40	OEC	L	T	P	C	CIE	SEE	Total
		3	1	-	3	30	70	100

COURSE OBJECTIVES:

The purpose of this subject is to provide the students with the theoretical background and engineering applications.

1. To introduce the concepts of Navigation, guidance and control
2. To familiarize with various ways in which aerospace vehicles are guided and controlled
3. The dynamic objectives which students also learn to achieve by designing flight control systems.
4. Familiarize with the control principles of rockets and missiles
5. To give insight into the manoeuvres of the space craft

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Formulate the navigational equations of the space vehicle
2. Describe the guidance of the vehicle with state feed back
3. Explain the automatic control and guidance of the aircraft
4. Evaluate the control techniques of the rockets and missiles
5. Describe major manoeuvres of the space aircraft.

UNIT-I	NAVIGATION	Classes: 13
Introduction, Basic Principles and Definitions; Dead reckoning and Position Fixing, Celestial, Radio, Inertial Navigation; Principle and Construction of Accelerometers, Mechanical Gyros and Ring Laser Gyros, Inertial Measurement Units, Navigation Equations, Sensor Error Models, Kalman Filter, Attitude Heading Reference System, GPS, Terrain Reference Navigation.		
UNIT-II	GUIDANCE	Classes: 11
Optimal Terminal Guidance of Interceptors, Optimal Terminal Guidance - planar and non-planar, Robust and Adaptive Guidance, Guidance with State Feedback , Guidance with Normal Acceleration Input , Minimum Energy Orbital Transfer.		
UNIT-III	CONTROL OF AIRCRAFT	Classes: 13
Powered Flying Controls, Helicopter Flight Controls, Fly-by-Wire Flight Control, Control laws, Redundancy and Failure Survival, Digital Implementation, Fly-by-Light Flight Control, Auto Pilot, Flight Management Systems, Unmanned Aerial Vehicle.		
UNIT-IV	CONTROL OF ROCKETS AND MISSILES	Classes: 12
Open-loop and Closed Loop Control Systems, Multi-variable Optimization, Optimal Control of Dynamic Systems, Hamiltonian and Minimum Principle and Jacobi-Bellman Equation, Linear Time-Varying System with Quadratic Performance Index..		

UNIT-V	CONTROL OF SPACECRAFT	Classes: 12
Launch of Satellite/ Spacecraft, Terminal Control of Spacecraft Attitude, Optimal Single-Axis Rotation of Spacecraft, Multi-axis Rotational Manoeuvres of Spacecraft, Spacecraft Control Torques, Rocket Thrusters, Reaction Wheels, Momentum Wheels and Control Moment Gyros, Torque.		
Text Books:		
1. Tewari, A.—Advanced Control of Aircraft, Spacecraft and RocketsII, John Wiley & Sons, Ltd, Chichester, UK, 2011 2. Nelson R. C - Flight Stability and Automatic Control, SIE edition, McGraw Hill, New York, 2007. 3. R.P.G Collinson —Introduction to Avionics SystemsII, Springer; 3rd ed. edition, 2011.		
Reference Books:		
1. Noton,M. —Spacecraft navigation and Guidancell, Springer-Verlag, Germany, 1998 2. Mc. Cormic 2. B. W - Aerodynamics, Aeronautics and Flight Mechanics, Wiley India Pvt. Ltd, USA, 2010. 3. Richard H. Battin —An Introduction to the Mathematics and Methods of AstrodynamicsII, AIAA, 1999.		
Web References:		
1. https://nptel.ac.in/courses/101108056/ 2. https://nptel.ac.in/courses/101108054/5 3. http://www.srmuniv.ac.in/sites/default/files/downloads/Aircraft_ctrl_Systems.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=CqhmXXdTlIC&printsec=frontcover&source=gbs_g_e_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://ocw.mit.edu/courses/aeronautics-and-astronautics/16-885j-aircraft-systems-engineering-fall-2005/video-lectures/lecture-16/ 2. https://www.courses.com/massachusetts-institute-of-technology/aircraft-systems-engineering/16		

WIND TUNNEL TECHNIQUES

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE41	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To interpret the basic concepts of measurement of forces and moments on models during the wind tunnel testing.
2. To understand the application of various types of wind tunnels.
3. To learn the basic measurement procedure involving wind tunnel testing.

COURSE OUTCOMES:

Students will have

1. Demonstrate basic concepts of aerodynamics and to identify the type of wind tunnel
2. Explain flow visualization techniques.
3. Describe concepts of low speed and high speed wind tunnels
4. Discuss about measurement and balancing of loads on model
5. Explain different types of equipment's for measuring pressure and velocity

UNIT-I	WIND TUNNELS	Classes: 12
Wind Tunnel, layouts and nomenclature, Types of Wind Tunnels – continuous and intermittent -closed circuit and open circuit - closed jet and open jet test section – application. Special purpose tunnels - Smoke Tunnels – Water Tunnels – Spin tunnel, automobile wind tunnel and environmental wind tunnel Important parameters of flow similarity. types of flow similarities for compressible and incompressible flows Model power consideration.		
UNIT-II	FLOW VISUALIZATION TECHNIQUES	Classes: 12
Path – Streak – Stream and Timelines; Techniques: Smoke, Tuft, Streaks, Surface oil flow. Pressure measurements: Manometers – U-Tube, Inclined and Precession. Bourdon Gauge and Pressure Transducer – Strain Gauge, Semi conductor – Absolute and Differential. Velocity Measurements: Pivot Tube – Static and Total. Calibration of test section: Test section flow calibration and Boundary Layers.		
UNIT-III	MEASUREMENTS OF FORCES AND MOMENTS	Classes: 12
Forces, moments and Reference Frames – Balances – Internal and External - Requirements and Specifications – Fundamentals of Model Installations. Boundary correction, types of blockages: 2-d.		
UNIT-IV	HIGH SPEED WIND TUNNELS	Classes: 12
Supersonic Wind Tunnels and - Classification - Runtime - Compressors - Charging Times - nozzle Mass Flows - Starting Loads - Model Size – Calibration. Hypersonic Wind Tunnels: Classification – Runtime – Vacuum Tanks – Vacuum pumps – Evacuation Times. Shock Tube: Driver – driven – Vacuum Pumps – Diaphragm.		

UNIT-V	HIGH SPEED FLOW VISUALIZATIONS AND MEASUREMENTS	Classes: 12
Schlieren and Shadow Graph – Pressure sensitive Paints – Temperature sensitive Paints – Force Measurements – Strain Gauge Balances – Pressure Measurements.		
Text Books:		
1. Rae, W.H. and Pope, A. —Low Speed Wind Tunnel Testingll, John Wiley Publication, 1999 2. Pope, A., and Goin, L., —High Speed wind Tunnel Testingll, John Wiley Publication , 1999 3. Pope, J B Barlow —low speed wind tunnel testing — 3 edition j.w publication		
Reference Books:		
1. John D. Anderson, Jr., "Fundamentals of Aerodynamics", Third edition, McGraw-Hillpublications, 2001 2. E L Houghton and PW Carpenter, "Aerodynamics for Engineering students", Fourth edition,Edward Arnold publications, 1993. 3. L.M Miline Thomson, —Theoretical Aerodynamicsll, 1996 McGraw-Hill,New Delhi. 4. R. Halmshaw (1991), Non-Destructive Testing, 2nd edition, Edward Arnold, New York		
Web References:		
1. https://lecturenotes.in/materials/22861-note-of-wind-tunnel-techniques-ppt-by-bibhuti-charan-ojha 2. https://www.scribd.com/document/223552935/Wind-Tunnel-Technique-Notes-for-Aeronautical-Engineers-Tp-1		
E-Text Books:		
1. https://books.google.co.in/books?id=I9NIqi66UxC&printsec=frontcover&source=qbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://nptel.ac.in/courses/101106040		

INTRODUCTION TO AEROSPACE TECHNOLOGY**VII Semester: OPEN ELECTIVE - III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE42	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To introduce the basic concepts of Aerospace technology and the current developments in the field.
2. To provide knowledge on the basic principles on which the development of aerodynamics, Structures, propulsion and satellite systems.

COURSE OUTCOMES:

Students should able to

1. Compare the atmosphere conditions of different altitudes for spacecraft system
2. Analyze how lift, drag and thrust are generated and understand which components constitute them
3. Analyze the flight performance parameters with respective stability condition
4. Distinguish the different systems used in a satellite mission
5. Design lighter than air vehicle using Catia software

UNIT-I	HISTORY OF FLIGHT- THE AEROSPACE ENVIRONMENT	Classes: 12
Balloons and dirigibles, heavier than air aircraft, commercial air transport, introduction of jet aircraft, helicopters, missiles, conquest of space, commercial use of space, exploring solar system and beyond. Earth's atmosphere, the temperature extremes of space, laws of gravitation, low earth orbit, microgravity, benefits of microgravity. The near earth radiative environment. The magnetosphere. Environmental impact on spacecraft. Meteoroids and micrometeoroids, space debris. Planetary environments		
UNIT-II	AERODYNAMICS AND FLIGHT VEHICLE PROPULSION	Classes: 12
Anatomy of the airplane, helicopter, launch vehicles and missiles, space vehicles. Static forces and moments on the vehicle. Understanding engineering models. Aerodynamics of wings and bodies. Generation of lift. Sources of drag. Force and moment coefficients, centre of pressure. Thrust for flight, the propeller, the jet engine, rocket engines- description, principles of operation. Governing equations.		
UNIT-III	FLIGHT VEHICLE PERFORMANCE AND STABILITY	Classes: 12
Performance parameters. Performance in steady flight, cruise, climb, range, endurance; accelerated flight- symmetric manoeuvres, turns, sideslips, take off and landing. Flight vehicle stability- longitudinal, lateral and directional- static, dynamic; trim, control. Handling qualities of airplanes		
UNIT-IV	SATELLITE SYSTEMS ENGINEERING- HUMAN SPACE EXPLORATION	Classes: 12
Satellite missions, an operational satellite system, elements of satellite, satellite subsystems. Satellite structures, mechanisms and materials. Power systems. Communication and telemetry. Thermal control. Attitude determination and control. Propulsion and station keeping. Space missions. Mission objectives. Case studies. Human space flight missions- goals, historical background. The Soviet and US missions. The Mercury, Gemini, Apollo (manned flight to the moon), Skylab, Apollo-Soyuz, Space Shuttle. International Space Station, extravehicular activity		

UNIT-V	INTRODUCTION TO ENGINEERING DESIGN, AIR TRANSPORTATION	Classes: 12
Design as a critical component of engineering education- as a skill- the process, design thinking, design drawing. Design for mission, performance and safety requirements. Concurrent engineering. Computer aided engineering, design project. Example: the lighter-than – air vehicle student design project at MIT. Air Transportation Systems- civil, military- objectives- principal constituents- the vehicle, the ground facilities, the organization- role. Regulation- national and international. Indian effort- civil and military- in the field of Aerospace Engineering.		
Text Books:		
1. Newman, D., Interactive Aerospace Engineering and Design, (with software and reference material on CD), McGraw-Hill, 2002, ISBN 0-07-112254-0. 2. Anderson, J.D., Introduction to Flight, fifth edition, Tata McGraw-Hill, 2007, ISBN: 0-07-006082		
Reference Books:		
1. Russell Mikel, Aerospace and Aeronautical Engineering, Willford press, 2017. 2. Ajoy kumar Kundu, Mark A Price and David Riordan, Conceptual Design: An Industrial Approach, Wiley-Blackwell, 2019.		
Web References:		
1. https://www.docsity.com/en/study-notes/engineering/aerospace-engineering/ 2. https://cosmolearning.org/courses/introduction-aerospace-engineering-i/		
E-Text Books:		
1. https://books.google.co.in/books?id=frfBDQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=FO_iJrc8V5qC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.shortcoursesportal.com/studies/56427/introduction-to-aeronautical-engineering.html 2. https://www.coursebuffet.com/sub/aerospace-engineering		

**OPEN ELECTIVES
OFFERED BY
COMPUTER SCIENCE AND ENGINEERING
DEPARTMENT**

FUNDAMENTALS OF DATABASES								
V SEMESTER - OPEN ELECTIVE – I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS43	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: The course should enable the students to 1. Understand the basic database concepts, data models including the architecture and operation of the relational data model. 2. Make a study of Structured Query Language (SQL) and construct simple and moderately advanced database queries using SQL. 3. Conceptualize relational database design and depict a database system using ERdiagram. 4. Know the fundamental concepts of transaction processing- concurrency control techniques and recovery procedure. 5. Develop a database application using commercial databases								
COURSE OUTCOMES: At the end of the course students are able to 1. Design effective database schema for realistic applications. 2. Design a database using ER modelling. 3. Make use the basic concepts of database systems in database design. 4. Apply SQL queries to interact with database. 5. Apply normalization on database design to eliminate anomalies.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction :Relational model, File System vs. Database System — Data Models – Database System Architecture – Relational Databases - Relational Model - Keys -- Integrity Constraints - Relational Algebra								
UNIT-II	SQL FUNDAMENTALS						Classes: 10	
SQL FUNDAMENTALS :SQL fundamentals - SQL Data Definition - Basic Structure of SQL Queries - Additional Basic Operations - Set Operations - Aggregate Functions - Nested Subqueries - Join Operations - Views								
UNIT-III	DATABASE DESIGN						Classes: 15	
DATABASE DESIGN : Entity-Relationship Model – ER Diagrams - Mapping ER Model to Relational Model - Functional Dependencies Normalization - First Normal Form - Second Normal Form - Third Normal Form								
UNIT-IV	TRANSACTION MANAGEMENT						Classes: 13	
TRANSACTION MANAGEMENT: Transaction Concepts - Transaction Recovery – ACID Properties - Concurrency Control – Need for Concurrency Control – Serializability - Locking Protocols – Two Phase Locking – Deadlock – Recovery								
UNIT-V	APPLICATION DEVELOPMENT						Classes: 10	
APPLICATION DEVELOPMENT :Commercial Databases - Oracle/MySQL - PL/SQL – Stored Procedures - Functions - Triggers - Database Security using SQL commands								
Text Books:								

1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Sixth Edition, Tata McGraw Hill, 2010.

Reference Books:

1. C.J.Date, A.Kannan, S.Swamynathan, "An Introduction to Database Systems", Eighth Edition, Pearson Education, 2006.
2. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Sixth Edition, Pearson Education, 2010.
3. Raghu Ramakrishnan, Johannes Gehrke "Database Management Systems", Fourth Edition, Tata McGraw Hill, 2010.
4. G.K.Gupta, "Database Management Systems", Tata McGraw Hill, 2011.
5. Carlos Coronel, Steven Morris, Peter Rob, "Database Systems: Design, Implementation and Management", Ninth Edition, Cengage Learning, 2011
6. B. A. Forouzan, R. F. Gillberg, "C Programming and Data Structures", Cengage Learning, India, 3rd Edition, 2014.

Web References:

1. <http://iips.icci.edu.iq/images/exam/databases-ramaz.pdf>
2. https://www.w3schools.com/sql/sql_intro.asp

E-Text Books:

1. <http://www.freebookcentre.net/Database/dbCategory.html>

MOOC Course

1. <https://swayam.gov.in/course/1322-fundamentals-of-database-system>

INTRODUCTION TO C++								
V SEMESTER - OPEN ELECTIVE – I								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS44	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To familiarize the students with the different techniques pertaining to problem solving skills. 2. To familiarize the students with basic concepts of object oriented programming. 3. To facilitate students in understanding the syntax and semantics of the “C++” language as well as basic constructs offered by the language 4. To facilitate students in solving real world problems using Object Oriented approach. 5. To impart the concepts of File handling techniques 6. To familiarize students with Generic Programming								
COURSE OUTCOMES: Students will be able to 1. Demonstrate different basic concepts of object oriented programming. 2. Analyze the performance of OOP system with POP in terms of the complexity. 3. Apply OOP concepts and design a new system. Analyze various Inheritance & polymorphism techniques to design a system. 4. Use file handling mechanism in C++ and design data base applications. 5. Design C++ programs using Templates and analyze the performance of OOP & POP program in terms of reusability.								
UNIT-I	OBJECT ORIENTED PROGRAMMING						Classes: 10	
Introduction to Object Oriented Programming, Structure of C++ program, Tokens, Keywords, Identifiers and Constants, Basic Data Types, Operators and Type Conversions. Functions: Returning values from functions. Reference arguments, Overloaded function, Inline function, Default arguments, returning by reference.								
UNIT-II	OBJECT, CLASSES, CONSTRUCTORS AND DESTRUCTORS						Classes: 12	
Object and Classes: Specifying a Class, Member function, Nesting of Member functions, Arrays in a Class, Memory Allocation for Objects, Static Members and Member Functions, Array of Objects, Friend Functions, Pointers to Members. Constructors and Destructors: Parameterized Constructor, Copy Constructor, Dynamic Constructor, Destructors.								
UNIT-III	OPERATOR OVERLOADING AND INHERITANCE						Classes: 12	
Operator Overloading: Overloading unary, binary operators, data conversion, pitfalls of operators overloading and conversion keywords, Explicit and Mutable. Inheritance: Derived class and Base class, Single Inheritance, Multiple Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Virtual Base Classes, Abstract Classes, Constructor in Derived Classes.								
UNIT-IV	POINTER, STREAMS AND FILES						Classes: 13	
Pointer: Introduction, Pointers to Objects, this Pointer, Pointers to Derived Class, Virtual Functions, Pure Virtual Functions. Streams and Files: C++ Streams, Stream Classes, Unformatted IO Operations, Formatted Console IO Operations, Managing Output with Manipulators.								

UNIT-V	FILES, TEMPLATES AND EXCEPTION HANDLING	Classes: 13
Files: Classes for File Handling, File Pointer, Updating a File, Error Handling in Files. Templates and Exception Handling: Class Templates, Function Templates, Exception Handling Mechanism, And Components of STL: Containers, Algorithms, Iterators.		
Text Books:		
1. Object Oriented Programming with C++, <i>E Balaguruswamy</i> , Third Edition, and McGraw Hill Publications.		
Reference Books:		
1. Object Oriented Programming in C++ by <i>Robert Lafore</i> Techmedia Publication. 2. Object Oriented Programming in C++ <i>Saurav Sahay</i> Oxford University Press.		
Web References:		
1. https://www.mycplusplus.com/tutorials/cplusplus-programming-tutorials/introduction-cplusplus-lecture-notes/ 2. https://lecturenotes.in/notes/6206-notes-for-object-oriented-programming-using-cpp-oop-by-swarnalata-rath		
E-Text Books:		
1. https://books.google.co.in/books?id=KbiDKvExnsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=rMjcBAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.my-mooc.com/en/mooc/introduction-c-microsoft-dev210x-0/ 2. https://www.coursera.org/learn/c-plus-plus-a		

CORE JAVA PROGRAMMING**V SEMESTER - OPEN ELECTIVE – I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS45	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To introduce the object oriented programming concepts.
2. To understand object oriented programming concepts, and apply them in solving problems.
3. To introduce the principles of inheritance and polymorphism; and demonstrate how they relate to the design of abstract classes
4. To introduce the implementation of packages and interfaces
5. To introduce the concepts of exception handling and multithreading.
6. To introduce the design of Graphical User Interface using applets and swing controls.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate object oriented programming concepts, and apply them in solving problems.
2. Demonstrate principles of inheritance and polymorphism and how they relate to the design of abstract classes
3. Explain concepts of packages, exception handling and multithreading.
4. Develop multithreaded applications with synchronization
5. Design GUI based applications

UNIT-I	OPEN CONCEPTS AND JAVA PROGRAMMING	Classes: 11
Oops concepts- Procedural and objects oriented programming paradigms, classes and objects, Data abstraction, encapsulation, polymorphism, inheritance, benefits of inheritance. Java Programming- History of java, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expression, type conversion and casting, control flow- block scope, condition statements, loops, break, and continue statements, simple java stand alone programs, arrays, constructors, methods, parameter passing, static keyword, access control, this pointer, overloading methods and constructor, recursion, garbage collection, exploring string class		
UNIT-II	INHERITANCE AND POLYMORPHISM	Classes: 12
Inheritance- Inheritance hierarchies super and sub classes, member access rules, super keyword, final keyword, the Object class. Polymorphism- dynamic binding, method overriding, abstract classes and methods. Interface- Interfaces vs. Abstract classes, defining an interface, implementing interfaces, implementing multiple inheritance using interfaces, extending interfaces. Inner classes- Uses of inner classes, local inner classes, anonymous inner classes, static inner classes.		
UNIT-III	PACKAGES, EXCEPTION HANDLING AND FILES	Classes: 12
Packages- Defining, Creation and Accessing Packages, Understanding CLASSPATH, importing packages. Exception handling- Types of errors, benefits of exception handling, classification of exceptions- exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, creating own exception sub classes. Files- streams- byte streams, character streams, text input/output, binary input/output, File management		

UNIT-IV	MULTITHREADING AND AWT CONTROLS	Classes: 13
Multithreading- Difference between multiprocessing and multithreading, thread states, creating threads, interrupting threads, thread priorities, synchronizing threads, inter-thread communication, producer consumer problem. AWT CONTROLS: The AWT class hierarchy, user interface components- Labels, Button, Text Components, Check Box, Check Box Group, Choice, List Box, Panels – Scroll Pane, Menu, ScrollBar. Working with Frame class, Colour, Fonts and layout managers.		
UNIT-V	GUI PROGRAMMING WITH JAVA AND EVENT HANDLING	Classes: 12
GUI Programming with Java-Introduction to Swing, Hierarchy for Swing components, Swing vs.AWT, Containers- JFrame, JApplet, JDialog, JPanel, JButton, JLabel, JTextField, JtextArea Event Handling- Events, Event sources, Event classes, Event Listeners, Relationship between Event sources and Listeners, Delegation event model, Examples: handling a button click, handling mouse events, Adapter classes.		
Text Books:		
1. Java Fundamentals – A comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH 2. Java A Beginner's Guide Sixth Edition , Herbert Schildt		
Reference Books:		
1. Java for Programmers, P.J. Deitel and H.M. Deitel, Pearson education (OR) Java: How to Program P.J.Deitel and H.M. Deitel, PHI. 2. Object Oriented Programming through Java, P. Radha Krishna, Universities Press. 3. Thinking in Java, Bruce Eckel, Peason Education 4. Programming in Java, S. Malhotra and S. Choudary, Oxford Univ. Press.		
Web References:		
1. http://www.math.hcmuns.edu.vn/~hvt hao/courses/java_programming/lecture_notes/ 2. http://people.alari.ch/derino/Teaching/Java/JavaLectureNotes-Derin.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=prwTLvCJkh0C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=8qFDDAAAOBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. http://moocfi.github.io/courses/2013/programming-part-1/ 2. https://www.edx.org/learn/java		

INTRODUCTION TO COMPUTER NETWORKS								
VI SEMESTER - OPEN ELECTIVE – II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS46	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To introduce the fundamental various types of computer networks. 2. To demonstrate the TCP/IP and OSI models with merits and demerits. 3. To explore the various layers of OSI Model.								
COURSE OUTCOMES: By the end of this course, student will be able to 1. Explore the basic concepts of Computer Networks. 2. Analyze the internal implementation of Physical Layer, Data link layer and Network Layer. 3. Analyze the internal implementation of Transport Layer and Application layer. 4. Solve problems with respect to various layers of reference model. 5. Administrate a network and flow of information								
UNIT-I	INTRODUCTION TO COMPUTER NETWORKS AND PHYSICAL LAYER						Classes: 12	
INTRODUCTION TO COMPUTER NETWORKS: Uses of Computer Networks, Network Hardware, Network Software, Reference Models (OSI, TCP/IP), Example Networks. PHYSICAL LAYER- Basis for Data Communication, Guided Transmission Media, Wireless Transmission, Communication Satellites, Digital Modulation And Multiplexing, The Public Switched Telephone Network.								
UNIT-II	DATA LINK LAYER						Classes: 12	
DATA LINK LAYER- Data Link Layer Design Issues, Error Detection and Correction, Elementary Data Link Protocols, Sliding Window Protocols. The Channel Allocation Problem, Multiple Access Protocols.								
UNIT-III	NETWORK LAYER						Classes:12	
NETWORK LAYER- Virtual Circuit and Datagram Networks, Inside A Router, The Internet Protocol (IP): Forwarding and Addressing in The Internet, Routing Algorithms, Routing in The Internet, Broadcast And Multicast Routing.								
UNIT-IV	TRANSPORT LAYER						Classes: 12	
TRANSPORT LAYER: Transport-Layer Services, Multiplexing and Demultiplexing, Principles of Reliable Data Transfer, Connection-Oriented Transport: TCP, Principles of Congestion Control, TCP Congestion Control.								
UNIT-V	APPLICATION LAYER						Classes: 12	
APPLICATION LAYER: Principles of Network Applications, The Web and HTTP, File Transfer: FTP, Electronic Mail In The Internet, DNS—The Internet’s Directory Service, Peer-To-Peer Application.								

Text Books:

1. Andrew S Tanenbaum David J Wetherall, Computer Networks, 5th Edition, Pearson Education.
2. Kurose, Ross (2010), Computer Networking: A top down approach, Sixth Edition, Pearson Education, India.

Reference Books:

1. Behrouz A. Forouzan, Data communication and Networking, 4th Edition, Mc Graw-Hill, India.
2. Larry L. Peterson, Bruce S. Davie, Computer Networks: A Systems Approach, MK Publishers.

Web References:

1. <https://www.cse.iitk.ac.in/users/dheeraj/cs425/>
2. <http://www2.kios.ucy.ac.cy/christos/courses/ece360/lecture-notes.html>

E-Textbooks:

1. https://books.google.co.in/books?id=KpOb37EHETcC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=b2HyGSu46lQC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.classcentral.com/subject/computer-networking>
2. <https://www.udacity.com/course/computer-networking--ud436>

INTRODUCTION TO LINUX PROGRAMMING

VI SEMESTER - OPEN ELECTIVE – II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS47	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand principles of operating system including File handling utilities, Security by file permissions, Process utilities, Disk utilities, Networking Commands, Basic Linux commands, Scripts and filters.
2. To familiarize fundamentals of the Bourne again shell (bash), shell programming, pipes, input and output redirection Control structures, arithmetic in shell interrupt processing, functions, debugging shell scripts.
3. To impart fundamentals of file concepts kernel support for file, File structure related system calls (file API's).
4. To facilitate students in understanding Inter process communication.
5. To facilitate students in understanding semaphore and shared memory.

COURSE OUTCOMES:

At the end of the course students are able to

1. Use various Linux commands that are used to manipulate system operations at admin level and a prerequisite to pursue job as a Network administrator.
2. Write Shell Programming using Linux commands.
3. Design and write application to manipulate internal kernel level Linux File System.
4. Develop IPC-API's that can be used to control various processes for synchronization.
5. Develop Network Programming that allows applications to make efficient use of resources available on different machines in a network.

UNIT-I	INTRODUCTION TO LINUX AND LINUX UTILITIES	Classes: 12
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INTRODUCTION TO LINUX AND LINUX UTILITIES: A brief history of LINUX, architecture of LINUX, features of LINUX, introduction to vi editor.

Linux commands- PATH, man, echo, printf, script, passwd, uname, who, date, stty, pwd, cd, mkdir, rmdir, ls, cp, mv, rm, cat, more, wc, lp, od, tar, gzip, file handling utilities, security by file permissions, process utilities, disk utilities, networking commands, unlink, du, df, mount, umount, find, unmask, ulimit, ps, w, finger, arp, ftp, telnet, rlogin. Text Processing utilities and backup utilities , tail, head , sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk, cpio

UNIT-II	INTRODUCTION TO SHELLS AND FILTERS	Classes: 13
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Introduction to Shells: Linux Session, Standard Streams, Redirection, Pipes, Tee Command, Command Execution, Command-Line Editing, Quotes, Command Substitution, Job Control, Aliases, Variables, Predefined Variables, Options, Shell/Environment Customization.

Filters: Filters and Pipes, Concatenating files, Display Beginning and End of files, Cut and Paste, Sorting, Translating Characters, Files with Duplicate Lines, Count Characters, Words or Lines, Comparing Files

UNIT-III	GREP, SED, UNIX FILE STRUCTURE AND FILE MANAGEMENT	Classes: 12
GREP: Operation, grep Family, Searching for File Content. Sed :Scripts, Operation, Addresses, commands, Applications, grep and sed. UNIX FILE STRUCTURE: Introduction to UNIX file system, inode (Index Node), file descriptors, system calls and device drivers. FILE MANAGEMENT :File Structures, System Calls for File Management – create, open, close, read, write, lseek, link, symlink, unlink, stat, fstat, lstat, chmod, chown, Directory API – opendir, readdir, closedir, mkdir, rmdir, umask.		
UNIT-IV	PROCESS AND SIGNALS	Classes: 12
PROCESS AND SIGNALS: Process, process identifiers, process structure: process table, viewing processes, system processes, process scheduling, starting new processes: waiting for a process, zombie processes, orphan process, fork, vfork, exit, wait, waitpid, exec, signals functions, unreliable signals, interrupted system calls, kill, raise, alarm, pause, abort, system, sleep functions, signal sets. File locking: creating lock files, locking regions, use of read and write with locking, competing locks, other lock commands, deadlocks.		
UNIT-V	INTER PROCESS COMMUNICATION AND SOCKETS	Classes: 11
INTER PROCESS COMMUNICATION: Pipe, process pipes, the pipe call, parent and child processes, and named pipes: fifos, semaphores: semget, semop, semctl, message queues: msgget, msgsnd, msgrcv, msgctl, shared memory: shmget, shmat, shmdt, shmctl, ipc status commands. INTRODUCTION TO SOCKETS: Socket, socket connections - socket attributes, socket addresses, socket, connect, bind, listen, accept, socket communications.		
Text Books:		
1. W. Richard. Stevens (2005), Advanced Programming in the UNIX Environment, 3rd edition, Pearson Education, New Delhi, India. 2. Unix and shell Programming Behrouz A. Forouzan, Richard F. Gilberg. Thomson		
Reference Books:		
1. Linux System Programming, Robert Love, O'Reilly, SPD. 2. Advanced Programming in the UNIX environment, 2nd Edition, W.R.Stevens, Pearson Education. 3. UNIX Network Programming, W.R. Stevens, PHI. UNIX for Programmers and Users, 3rd Edition, Graham Glass, King Ables, Pearson Education		
Web References:		
1. https://www.smartworld.com/notes/linux-programming-pdf-notes-lp-pdf-notes/ 2. https://lecturenotes.in/subject/455/linux-programming-lp		
E-Textbooks:		
1. https://books.google.co.in/books?id=wuzDziOMeMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=TKjATVAXNlwC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		

MOOC Course
1. https://www.classcentral.com/course/edx-c-programming-using-linux-tools-and-libraries-11538 2. https://www.udacity.com/course/linux-command-line-basics--ud595

1. <https://www.classcentral.com/course/edx-c-programming-using-linux-tools-and-libraries-11538>
2. <https://www.udacity.com/course/linux-command-line-basics--ud595>

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**VI SEMESTER - OPEN ELECTIVE – II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS48	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To learn the difference between optimal reasoning vs human like reasoning
2. To understand the notions of state space representation, exhaustive search, heuristic search along with the time and space complexities
3. To learn different knowledge representation techniques
4. To understand the applications of AI viz Game Playing, Theorem Proving, Expert Systems, Machine Learning and Natural Language Processing

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate fundamental concepts of artificial intelligence.
2. Select a search algorithm for a problem and characterize its time and space complexities.
3. Discuss various approaches to knowledge representation.
4. Apply AI techniques to solve problems of Game Playing, Expert Systems, Machine Learning and Natural Language Processing
5. Outline advanced knowledge representation techniques.

UNIT-I	INTRODUCTION	Classes:10
Introduction, History, Intelligent Systems, Foundations of AI, Sub areas of AI, Applications. Problem Solving – State-Space Search and Control Strategies: Introduction, General Problem Solving, Characteristics of Problem, Exhaustive Searches, Heuristic Search Techniques, Iterative-Deepening A*, Constraint Satisfaction. Game Playing, Bounded Look-ahead Strategy and use of Evaluation Functions, Alpha-Beta Pruning		
UNIT-II	LOGIC CONCEPTS AND LOGIC PROGRAMMING & KNOWLEDGE REPRESENTATION	Classes:12
Logic Concepts and Logic Programming: Introduction, Propositional Calculus, Propositional Logic, Natural Deduction System, Axiomatic System, Semantic Tableau System in Propositional Logic, Resolution Refutation in Propositional Logic, Predicate Logic, Logic Programming. Knowledge Representation: Introduction, Approaches to Knowledge Representation, Knowledge Representation using Semantic Network, Extended Semantic Networks for KR, Knowledge Representation using Frames.		
UNIT-III	EXPERT SYSTEM AND APPLICATIONS & UNCERTAINTY MEASURE – PROBABILITY THEORY	Classes:13
Expert System and Applications: Introduction, Phases in Building Expert Systems, Expert System Architecture, Expert Systems Vs Traditional Systems, Truth Maintenance Systems, Application of Expert Systems, List of Shells and Tools. Uncertainty Measure – Probability Theory: Introduction, Probability Theory, Bayesian Belief Networks, Certainty Factor Theory, Dempster-Shafer Theory.		
UNIT-IV	MACHINE-LEARNING PARADIGMS & ARTIFICIAL NEURAL NETWORKS	Classes:13
Machine-Learning Paradigms: Introduction. Machine Learning Systems. Supervised and Unsupervised Learning. Inductive Learning. Learning Decision Trees, Deductive Learning. Clustering, Support Vector Machines. Artificial Neural Networks: Introduction, Artificial Neural Networks, Single-Layer Feed Forward Networks, Multi-		

Layer Feed-Forward Networks, Radial-Basis Function Networks, Design Issues of Artificial Neural Networks, Recurrent Networks.		
UNIT-V	ADVANCED KNOWLEDGE REPRESENTATION TECHNIQUES	Classes:12
Advanced Knowledge Representation Techniques: Case Grammars, Semantic Web Natural Language Processing: Introduction, Sentence Analysis Phases, Grammars and Parsers, Types of Parsers, Semantic Analysis, Universal Networking Knowledge.		
Text Books:		
1. Saroj Kaushik. Artificial Intelligence. Cengage Learning. 2011 2. Russell, Norvig: Artificial intelligence, A Modern Approach, Pearson Education, Second Edition. 2004		
Reference Books:		
1. Rich, Knight, Nair: Artificial intelligence, Tata McGraw Hill, Third Edition 2009. 2. Introduction to Artificial Intelligence by Eugene Charniak, Pearson. 3. Introduction to Artificial Intelligence and expert systems Dan W. Patterson. PHI. 4. Artificial Intelligence by George Fluger Pearson fifth edition.		
Web References:		
1. http://zsi.tech.us.edu.pl/~nowak/bien/BIEN_introduction.pdf 2. https://epub.uni-regensburg.de/13629/1/ubr06078_ocr.pdf 3. https://lecturenotes.in/subject/128/artificial-intelligence-ai		
E-Textbooks:		
1. https://books.google.co.in/books?id=DDNHzcN6jasC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=YmH1tXFA14MC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.edx.org/course/artificial-intelligence-1 2. https://www.appliedaicourse.com/		

SOFT COMPUTING								
VII SEMESTER - OPEN ELECTIVE – III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS49	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: Students will 1. Learn the various soft computing frame works 2. Be familiar with design of various neural networks 3. Be exposed to fuzzy logic 4. Learn genetic programming.								
COURSE OUTCOMES: At the end of the course students are able to 1. Demonstrate fundamental concepts of Neural Networks & Feed Forward Networks 2. Use fuzzy logic in computing and data analytics. 3. Use Genetic Algorithms to solve optimization problems. 4. Design hybrid systems to suit a given real life problem 5. Familiarize with neural network toolbox, Fuzzy logic toolbox etc.								
UNIT-I	NEURAL NETWORKS						Classes: 14	
Fundamentals of Neural Networks–History-Architectures-Learning methods-XOR problem-Delta rule-derivation-Back propagation-applications-parameters in BPN-Associative memory–Hetero associative-BAM-energy function-problems-applications of associative memories-ART1-ART2-applications of adaptive Networks.								
UNIT-II	FUZZY LOGIC						Classes: 10	
Fuzzy set theory–crisp sets–fuzzysets–crisp relations–Fuzzy relations–Fuzzy systems-Crisp logic–predicate logic–fuzzy logic-fuzzy based systems-Defuzzification methods–applications.								
UNIT-III	GENETIC ALGORITHMS						Classes: 14	
Fundamentals of GA–creation of off springs–encoding–fitness function-reproduction–crossover-Insertion & deletion-mutation-bitwise operators–applications.								
UNIT-IV	HYBRID SYSTEMS						Classes: 12	
Hybrid systems–Neuro Fuzzy–Neuro Genetic–fuzzy Genetic hybrids-GA based weight determination and applications-fuzzy BPN–simplified fuzzy ARTMAP.								
UNIT-V	FPROGRAMMING USING MATLAB						Classes: 10	
Using Neural Network toolbox–Using Fuzzy Logic toolbox-Using Genetic Algorithm & directed search toolbox.								

Text Books:
1. Rajasekaran.S and VijayalakshmiPai.G.A, "Neural Networks, Fuzzy Logic and Genetic Algorithms", PHI, 2011
Reference Books:
1. Timothy J.Ross,"Fuzzy Logic with Engineering applications ", John Wiley and Sons, 2010. 2. Jang.J.S.R, Sun.C.T, Mizutani.E, "Neuro fuzzy and Soft Computing ", PHI Learning Pvt. Ltd., 2012. 3. Davis E.Goldberg, "Genetic Algorithms: Search, Optimization and Machine Learning ", Addison Wesley, N.Y., 1989.
Web References:
1. http://www.myreaders.info/html/soft_computing.html 2. https://en.wikipedia.org/wiki/Soft_computing
E-Text Books:
1. http://freecomputerbooks.com/Introduction-to-Soft-Computing.html 2. https://www.shroffpublishers.com/books/9789351106159/ 3. https://bookboon.com/en/introduction-to-soft-computing-ebook
MOOC Course
1. https://onlinecourses.nptel.ac.in/noc18_cs13/course 2. https://swayam.gov.in/course/4574-introduction-to-soft-computing

INTRODUCTION TO CLOUD COMPUTING**VII SEMESTER - OPEN ELECTIVE – III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS50	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the Virtualization paradigms
2. To learn the Cloud Computing fundamentals and its importance to various organizations.
3. To analyze the concepts of IaaS, PaaS, SaaS, Public and Private Clouds.
4. To develop applications in cloud security.

COURSE OUTCOMES:

At the end of the course, student will be able to:

1. Demonstrate concepts of system modeling, clustering and virtualization.
2. Describe the infrastructure of cloud computing including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, etc.
3. Discuss about monitoring, management and applications of cloud computing.
4. Analyze the core issues of cloud computing such as security, privacy, and interoperability.
5. Identify problems, analyze, and evaluate various cloud computing solutions.

UNIT-I	INTRODUCTION	Classes: 12
System Modeling, Clustering And Virtualization: Distributed system models and enabling technologies, computer clusters for scalable parallel computing, virtual machines and virtualization of clusters and data centers.		
UNIT-II	FOUNDATIONS	Classes: 15
Foundations: Introduction to cloud computing, migrating into cloud, enriching the integration of service paradigm for cloud era, the enterprise cloud computing paradigm.		
UNIT-III	INTER PROCESS COMMUNICATION	Classes: 17
Inter Process Communication: Infra Structure As Service (IaaS) & Platform And Software Service(PAAS/SAAS): Virtual machine provisioning and migration services, on the management of virtual machines for cloud infrastructure, enhancing cloud computing environments using a cluster as service. Secure distributed data storage in cloud computing. Aneka, comet cloud, T-systems, work flow engine for clouds, understanding scientific applications for cloud environments		
UNIT-IV	MONITORING, MANAGEMENT AND APPLICATIONS	Classes: 10
Monitoring, Management And Applications : An Architecture for federated cloud computing ,SLA management in cloud computing, performance prediction for HPC on clouds, best practices in Architecting cloud applications in the AWS cloud, building content delivery networks using clouds, resource cloud mashups.		

UNIT-V	GOVERNANCE AND CASE STUDIES	Classes: 10
Governance and case studies: organizational readiness and change management in cloud age, data security in cloud, legal issues in cloud computing, achieving production readiness for cloud services.		
Text Books:		
1. Cloud computing: principles and paradigms by rajkumar buyya, James Broberg and Andrzej M.Goscinski,Wiley,2011. 2. Distributed and cloud computing , kai Hwang, Geofferyu C.fox, jack J.dongarra, Elsevier, 2012 .		
Reference Books:		
1. Cloud Computing :A practical approach, Anthony T.velte, Toby J.velte, Robert Isenpeter, Tata McGraw Hill,2011. 2. Enterprise Cloud Computing, Gautam Shroff, Cambridge University press,2010. 3. Cloud computing: implementation, management and security, john W .Ritting house ,James F. Ransome ,CRC press,rp2012. 4. Cloud applications architectures: building Applications and infrastructure in the cloud,George Reese , O reilly,SPD,rp2011. 5. Cloud security and privacy: An Enterprise perspective on Risks and compliance,im Mather, Subra Kumaraswamy, Shahed Latif,Oreilly,SPD,rp2011		
Web References:		
1. https://cs.uwaterloo.ca/~a78khan/courses-offered/cs446/2010_05/lecture-slides/16_CloudComputing.pdf 2. https://www.smartworld.com/notes/cloud-computing-pdf-notes-cc/ 3. http://michaelkona.blogspot.com/p/introduction-to-cloud-computing_4.html		
E-Textbooks:		
1. https://books.google.co.in/books?id=F8VoBAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=8es62w1MOVcC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.coursera.org/learn/cloud-computing 2. https://www.classcentral.com/subject/cloud-computing		

WEB PROGRAMMING								
VII SEMESTER - OPEN ELECTIVE – III								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS51	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To learn the basics of server side scripting using HTML 2. To explain web application development procedures 3. To impart servlet technology for writing business logic 4. To facilitate students to connect to databases using JDBC 5. To familiarize various concepts of application development using XML								
COURSE OUTCOMES: At the end of the course students are able to 1. Use HTML, CSS to develop web pages. 2. Apply concepts of Java. 3. Build JDBC connectivity. 4. Identify the basic building blocks of XML and describe the applications of web services. 5. Write server side programming using applets.								
UNIT-I	SCRIPTING						Classes: 11	
Web page Designing using HTML, Scripting basics- Client side and server side scripting. Java Script-Object, names, literals, operators and expressions- statements and features-events - windows - documents - frames - data types - built-in functions- Browser object model - Verifying forms.-HTML5- CSS3- HTML 5 canvas - Web site creation using tools.								
UNIT-II	JAVA						Classes: 11	
Introduction to object oriented programming-Features of Java – Data types, variables and arrays – Operators – Control statements – Classes and Methods – Inheritance. Packages and Interfaces –Exception Handling – Multithreaded Programming –Input/Output – Files – Utility Classes – String Handling.								
UNIT-III	JDBC						Classes: 12	
JDBC Overview – JDBC implementation – Connection class – Statements – Catching Database Results, handling database Queries. Networking– Inet Address class – URLclass- TCP sockets - UDP sockets, Java Beans –RMI.								
UNIT-IV	APPLETS						Classes: 14	
Java applets- Life cycle of an applet – Adding images to an applet – Adding sound to an applet, Passing parameters to an applet, Event Handling, Introducing AWT: Workingwith Windows Graphics and Text. Using AWT Controls, Layout Managers and Menus.Servlet – life cycle of a servlet, The Servlet API, Handling HTTP Request and Response, using Cookies, Session Tracking. Introduction to JSP.								
UNIT-V	XML AND WEB SERVICES						Classes: 12	
Xml – Introduction-Form Navigation-XML Documents- XSL – XSLT- Web services-UDDI-WSDL-Java web services – Web resources.								

Text Books:

1. Java™ : The Complete Reference, Seventh Edition, Herbert Schildt
2. J2EE: The complete Reference by James Keogh
3. Java EE and HTML5 Enterprise Application Development (Oracle Press) by John Brock (Author), Arun Gupta (Author), Geertjan Wielenga (Author)
4. Struts: The Complete Reference, 2nd Edition by James Holmes

Reference Books:

1. ASP.NET 4 Unleashed by Stephen Walther (Author), Kevin Scott Hoffman (Author), Nate Dudek (Author)
2. Microsoft Visual C# 2013 Step by Step by John Sharp

Web References:

1. <http://cgi.csc.liv.ac.uk/~martin/teaching/comp519/notes.php>
2. <https://lecturenotes.in/notes/24802-note-for-web-programming-wp-by-ashutosh-jaiswal>
3. <http://web.stanford.edu/class/cs193x/lectures/01/lecture01.pdf>

E-Textbooks:

1. https://books.google.co.in/books?id=PZhIDwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=fj1Pd8uxNLEc&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.classcentral.com/subject/web-development>
2. <https://www.coursera.org/courses?query=web%20programming>

INTRODUCTION TO NETWORK SECURITY								
VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS52	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To understand the basic components of security and mathematics of cryptography. 2. To demonstrate various traditional symmetric key ciphers 3. To Analyze DES and AES its variants. 4. To demonstrate private and public-key cryptosystem, Digital signature and attacks on it.								
COURSE OUTCOMES: At the end of the course students able to 1. Identify and explain various types of security threats/vulnerabilities attacks, consequences and security mechanisms 2. Discuss concepts of advanced and data encryption standards. 3. Apply various Cryptographic and Authentication techniques for secure transmission of data. 4. Compare different security algorithms with respect feasibility, cost and security service 5. Choose the most suitable security mechanisms based on the nature of data and complexity of algorithm.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction: Security Goals, Attacks, Services and Mechanism, Techniques, Mathematics of Cryptography, Integer arithmetic, Modular Arithmetic, Matrices, Linear Congruence, Primes, Primality Testing, Factorization, Chinese Remainder Theorem. Traditional Symmetric key Ciphers: Introduction, Substitution Ciphers, Transposition Ciphers, Streams and Block Ciphers.								
UNIT-II	SYMMETRIC CIPHERS						Classes: 12	
Advanced Encryption Standards: AES Structure, AES Transformation Function, AES Key Expansion, AES Implementation. Data Encryption Standards: Introduction, DES Structure, DES Analysis, Multiple DES, Security of DES, Differential Cryptanalysis, Linear Cryptanalysis of DES, Encipherment Using Modern Symmetric Key Ciphers Use of Modern Block Ciphers, Use of Stream Ciphers								
UNIT-III	ASYMMETRIC CIPHERS, DATA INTEGRITY ALGORITHMS						Classes: 12	
Asymmetric Key Cryptography and Message Integrity: Introduction, RSA Cryptosystem, Message Integrity, Message Authentication Cryptographic Hash Functions, Introduction, SHA-512, Digital Signature Comparison, Attacks on Digital Signature. User Authentication Protocols: Remote User Authentication Principles, Remote User Authentication using Symmetric Encryption, Kerberos								
UNIT-IV	MUTUAL TRUST						Classes: 10	
Key Management And Distribution : Symmetric key Distribution Using Symmetric and Asymmetric Encryption, Distribution of Public Key, X.509 Certificate , Public Key Infrastructure. Network Access Control : Network Access Control, Extensible Authentication Protocol, IEEE802.1,								

UNIT-V	NETWORK SECURITY	Classes: 10
Transport-Level Security: Transport-Level Security, HTTPS, Secure Shell(SSH) Wireless Network Security: Web Security ,Mobile Device Security, IEEE802.11 Wireless LAN Overview, IEEE802.11i Wireless LAN Security		
Text Books:		
1. Cryptography and Network Security”by Behrouz A. Forouzan, Special Indian Edition, 2008 2. Cryptography and Network Security”by William Stallings, 6th Edition, Pearson Education/PHI, 2014		
Reference Books:		
1. “Cryptography and Network Security”by Atul Kahate, Tata McGraw Hill, 2003. 2. "RSA Security's official guide to cryptography" by Steve Burnett, Stephen Paine, RSA Press, Tata McGraw Hill Edition, 2001		
Web References:		
1. www.open.edu/openlearn/science.../network-security/content-section---references www.cse.iitm.ac.in/~chester/courses/16e_cns/slides/01_Introduction.pdf williamstallings.com/Crypto/Crypto4e.html 2. pubs.sciepub.com/iteces/3/1/1/ 3. https://www.sanfoundry.com/best-reference-books-cryptography-network-security/ 4. https://dl.acm.org/citation.cfm?id=1824151 5. web.mit.edu/6.857/OldStuff/Fall03/references.html		
E-Text Books:		
1. http://williamstallings.com/Crypto3e.html 2. http://www.cacr.math.uwaterloo.ca/hac/		
MOOC Course		
1. https://www.mooc-list.com/tags/network-security 2. nptel.ac.in/courses/106105031/ 3. https://onlinecourses.nptel.ac.in/noc18_cs07/course		

DATA ANALYTICS								
VIII SEMESTER - OPEN ELECTIVE – IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS53	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To explore the fundamental concepts of data analytics 2. Formulate and use appropriate models of data analysis to solve hidden solutions to business-related challenges.								
COURSE OUTCOMES: At the end of the course students are able to 1. Demonstrate the ability to use technical skills in Descriptive and prescriptive modelling to support business decision-making 2. Apply the knowledge of classification and regression techniques for various data analysis. 3. Examine the results of Classification for further predictions. 4. Make use of regression techniques for predicting future outcomes on various data 5. Illustrate the concepts of data analytics for data visualization and bigdata analysis.								
UNIT-I	DESCRIPTIVE STATISTICS						Classes: 10	
Descriptive Statistics, Introduction to the course, Descriptive Statistics, Probability Distributions, Inferential Statistics, Inferential Statistics through hypothesis tests, Permutation & Randomization Test								
UNIT-II	REGRESSION AND ANOVA						Classes: 12	
Regression & ANOVA, Regression , ANOVA(Analysis of Variance), Machine Learning: Introduction and Concepts, Differentiating algorithmic and model based frameworks. Regression: Ordinary Least Squares, Ridge Regression, Lasso Regression, K Nearest Neighbours Regression & Classification								
UNIT-III	SUPERVISED LEARNING WITH REGRESSION AND CLASSIFICATION TECHNIQUES - I						Classes: 13	
Supervised Learning with Regression and Classification techniques -1, Bias-Variance Dichotomy, Model Validation Approaches, Logistic Regression, Linear Discriminant Analysis, Quadratic Discriminant Analysis Regression and Classification Trees, Support Vector Machines								
UNIT-IV	SUPERVISED LEARNING WITH REGRESSION AND CLASSIFICATION TECHNIQUES - 2						Classes: 12	
Supervised Learning with Regression and Classification techniques -2, Ensemble Methods: Random Forest Neural Networks, Deep learning, Unsupervised Learning and Challenges for Big Data Analytics Clustering Associative Rule Mining Challenges for big data analytics.								
UNIT-V	PRESCRIPTIVE ANALYTICS						Classes: 10	
Prescriptive analytics, Creating data for analytics through designed experiments, Creating data for analytics through Active learning, Creating data for analytics through Reinforcement learning								

Text Books:

1. Hastie, Trevor, et al. The elements of statistical learning. Vol. 2. No. 1. New York: springer, 2009.
2. Montgomery, Douglas C., and George C. Runger. Applied statistics and probability for engineers. John Wiley & Sons, 2010

Reference Books:

1. Data Analytics- Models and Algorithms for Intelligent Data Analysis, Author: Thomas Runkler ISBN-10: 3834825883
2. Data Analytics, Anil Maheshwari, McGraw Hill Education; First edition, 9352604180.

Web References:

1. <https://www.ee.columbia.edu/~cylin/course/bigdata/EECS6893-BigDataAnalytics-Lecture4.pdf>
2. <https://intellipaat.com/blog/what-is-data-analytics/>

E-Textbooks:

1. https://books.google.co.in/books?id=PaylsdmgkIC&printsec=frontcover&source=gbg_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=xhADQAAQBAJ&printsec=frontcover&source=gbg_e_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.edx.org/learn/data-analysis>
2. <https://www.mooc-list.com/categories/statistics-data-analysis>

.NET & C# PROGRAMMING**VIII SEMESTER - OPEN ELECTIVE – IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS54	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To impart basic concepts of .NET Framework Components.
2. To familiarize the C# Core Concepts.
3. To facilitate students in understanding OOP Concepts.
4. To facilitate students in understanding Exception Handling.
5. To impart the concepts of Delegates, Events.

COURSE OUTCOMES:

At the end of the course, students will

1. Explain the purpose of the .net framework, and use c# and visual studio 2010 to build .net framework applications.
2. Outline the syntax of basic c# programming constructs a, use new types (enumerations, classes, and structures), and enumerate the differences between reference types and value types.
3. Create new reference types using inheritance; control the visibility and lifetime of members in a type.
4. Manage the lifetime of objects and control the use of resources, catch, handle and throw exceptions.
5. Exemplify the .net delegate type & event handling.

UNIT-I	INTRODUCTION TO .NET	Classes: 10
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Introduction - The Philosophy of .NET, An Initial Look at the .NET Platform, Introducing the Building Blocks of the .NET Platform (the CLR, CTS, and CLS), An Overview of .NET Assemblies, Understanding the Common Type System, Understanding the Common Language Specification, Understanding the Common Language Runtime, The Assembly/Namespace/Type Distinction, The Platform-Independent Nature of .NET.

UNIT-II	CORE C# PROGRAMMING CONSTRUCTS	Classes: 13
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Core C# Programming Constructs, Part I - The Anatomy of a Simple C# Program, An Interesting Aside: Some Additional Members of the System.Environment Class, The System.Console Class, System Data Types and Corresponding C# Keywords, Working with String Data, Narrowing and Widening Data Type Conversions, Understanding Implicitly Typed Local Variables, C# Iteration Constructs, Decision Constructs and the Relational/Equality Operators.

Core C# Programming Constructs, Part II - Methods and Parameter Modifiers, Understanding C# Arrays, Understanding the enum Type, Understanding the Structure Type, Understanding Value Types and Reference Types, Understanding C# Nullable Types.

UNIT-III	OBJECT-ORIENTED PROGRAMMING WITH C#	Classes: 15
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Understanding Encapsulation - Introducing the C# Class Type, Understanding Constructors, The Role of the this Keyword, Understanding the static Keyword, Defining the Pillars of OOP, C# Access Modifiers, The First Pillar: C#'s Encapsulation Services, Understanding Automatic Properties, Understanding Object Initialization Syntax, Working with Constant Field Data, Understanding Partial Types.

Understanding Inheritance and Polymorphism - The Basic Mechanics of Inheritance, The Second Pillar of OOP: The Details of Inheritance, Programming for Containment/Delegation, The Third Pillar of OOP: C#'s Polymorphic Support, Understanding Base Class/Derived Class Casting Rules, The Master Parent Class: System.Object.

UNIT-IV	STRUCTURED EXCEPTION HANDLING	Classes: 12
Understanding Structured Exception Handling - Ode to Errors, Bugs, and Exceptions, The Role of .NET Exception Handling, The Simplest Possible Example, Configuring the State of an Exception, System-Level Exceptions (System.SystemException), Application-Level Exceptions (System.ApplicationException), Processing Multiple Exceptions, Who Is Throwing What?, The Result of Unhandled Exceptions.		
UNIT-V	WORKING WITH INTERFACES, DELEGATES AND EVENTS	Classes: 14
Working with Interfaces - Understanding Interface Types, Defining Custom Interfaces, Implementing an Interface, Invoking Interface Members at the Object Level, Interfaces As Parameters, Interfaces As Return Values, Arrays of Interface Types, Implementing Interfaces Using Visual Studio, Explicit Interface Implementation, Designing Interface Hierarchies, The IEnumerable and IEnumerator Interfaces, The ICloneable Interface, The IComparable Interface. Delegates, Events - Understanding the .NET Delegate Type, The Simplest Possible Delegate Example, Sending Object State Notifications Using Delegates, Understanding Generic Delegates, Understanding C# Events, Understanding C# Anonymous Methods.		
Text Books:		
1. Andrew Troelsen, Pro C-Hash and the .NET 4 Platform, 5th edition, Springer (India) Private Limited, New Delhi, India., 2010		
Reference Books:		
1. E. Balagurusamy (2004), Programming in C#, 5 th edition, Tata McGraw-Hill, New Delhi, India. 2. Herbert Schildt (2004), The Complete Reference: C#, Tata McGraw-Hill, New Delhi, India. 3. Simon Robinson, Christian Nagel, Karli Watson, Jay GI (2006), Professional C#, 3rd edition, Wiley & Sons, India.		
Web References:		
1. https://lecturenotes.in/notes/19333-note-for-c-and-net-c-net-by-anna-superkings 2. https://www.guru99.com/c-sharp-tutorial.html		
E-Textbooks:		
1. https://books.google.co.in/books?id=JUNqMPIZ4yMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=ZmBzoG--oNAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.gatrainingshub.com/c-net-online-training.php 2. https://www.abhishekluv.in/online-one-on-one-csharp-training-and-coaching		

**OPEN ELECTIVES
OFFERED BY
ELECTRONICS AND COMMUNICATION
ENGINEERING DEPARTMENT**

LOGIC DESIGN**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC59	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To learn how to design a digital circuit.
2. To create minimal realizations of single and multiple output Boolean functions.
3. To design and analyze combinational circuits using medium scale integrated (MSI) components, including arithmetic logic units.
4. To derive state diagrams and state transition tables for synchronous systems.
5. To study the characteristics and performance of digital circuits built using various MOS technologies

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain how digital circuit of large complexity can be built in a methodological way, starting from Boolean logic and applying a set of rigorous techniques.
2. Create minimal realizations of single and multiple output Boolean functions.
3. Design and analyze combinational circuits using medium scale integrated (MSI) components, including arithmetic logic units.
4. Derive state diagrams and state transition tables for synchronous systems.
5. Outline the characteristics and performance of digital circuits built using various MOS technologies

UNIT-I	BOOLEAN ALGEBRA	Classes: 09
Boolean Algebra: Axiomatic Definition of Boolean Algebra, Basic Theorems and Properties of Boolean Algebra, Boolean Functions, Canonical and Standard Forms, Digital Logic Gates, IC Digital Logic Families		
UNIT-II	SIMPLIFICATION OF BOOLEAN FUNCTIONS	Classes: 09
Simplification of Boolean Functions: The Map Method, Two & Three Variable Maps, Four Variable Map, Five & Six Variable Maps, Product of Sum Simplification, NAND & NOR Implementation, Two-Level Implementations, Don't-Care Conditions, Tabulation Method, Determination of Prime Implicants, Selection of Prime Implicants.		
UNIT-III	COMBINATIONAL LOGIC	Classes: 09
Combinational Logic: Design Procedure, Adders, Subtractors, Code Conversion, Multilevel NAND Circuits, Multilevel NOR Circuits, Exclusive-Or and Equivalence Functions. Combinational Logic with MSI And LSI: Binary Parallel Adder, Decimal Adder, Magnitude Comparator, Decoders, Multiplexers, ROM, PLA.		
UNIT-IV	SEQUENTIAL CIRCUIT DESIGN AND ANALYSIS	Classes: 09
Sheet metal characteristics – Typical shearing operations, bending and drawing operations –Formability of sheet metal –Working principle and application of special forming processes – Hydro forming –Explosive forming – Magnetic pulse forming – Peen forming – Super plastic forming – Process characteristics		

UNIT-V	DIGITAL INTEGRATED CIRCUITS	Classes: 09
Digital Integrated Circuits: Introduction, Bi-Polar Transistor Characteristics, Integrated Injection Logic, Transistor-Transistor Logic (TTL), Emitter-Coupled Logic(ECL), MOS, Complementary MOS.		
Text Books:		
1. M.Morris Mano, —Digital DesignII, 2nd Edition, 1997, PHI. (Unit I, II, III & IV).		
Reference Books:		
1. Zvi Kohavi, —Switching and Automata TheoryII, 2nd Edition, 1978, McGraw-Hill. 2. Thomas L. Floyd —Digital FundamentalsII, 10th Edition, Pearson Education India. 3. Stephen Brown, Zvonko Vrsanec, IIFundamentals of Digital Logic with Verilog DesignII, 2nd Edition, McGraw-Hill.		
Web References:		
1. http://www.freebookcentre.net/electronics-ebooks-download/logic design.html 2. http://coen.boisestate.edu/sml00/sml00-courses/ece-332-logic designs-fall07/lecture-notes/ 3. http://www.freebookcentre.net/electronics-ebooks-download/logic design-Lecture-Notes.html		
E-Text Books:		
1. http://gen.lib.rus.ec/book/logic design 2. http://www.faadooengineers.com/threads/logicdesign -by-RAMESH-GANOKAR-ebook-pdf-download 3. https://e.edim.co/123389964/logic design.pdf		
MOOC Course		
1. https://www.mooc-list.com/tags/logic design 2. https://www.coursera.org/courses?query=logicedesign		

PRINCIPLES OF COMMUNICATIONS**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC60	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OUTCOMES:

1. To learn basic concepts of Communication technology
2. To study the Models of Communication
3. To Understand Analog and Digital Signal transmission
4. To Understand Computer Networking
5. To learn various types of Communication Media
6. To understand application of computer in data transmission and Security.

COURSE OUTCOMES:

At the end of the course students will

1. Describe the basic concepts of Communication technology
2. Explain the Models of Communication
3. Discourse Analog and Digital Signal transmission and Computer Networking
4. Explain the types of Communication Media
5. Explain the Application of computer in data transmission and Security.

UNIT-I	FUNDAMENTALS OF ANALOG COMMUNICATION	Classes: 10
Principles of amplitude modulation, AM envelope, frequency spectrum and bandwidth, modulation index and percent modulation, AM Voltage distribution, AM power distribution, Angle modulation FM and PM waveforms, phase deviation and modulation index, frequency deviation and percent modulation, Frequency analysis of angle modulated waves. Bandwidth requirements for Angle modulated waves.		
UNIT-II	DIGITAL COMMUNICATION	Classes: 09
Introduction, Shannon limit for information capacity, digital amplitude modulation, frequency shift keying, FSK bit rate and baud, FSK transmitter, BW consideration of FSK, FSK receiver, phase shift keying binary phase shift keying QPSK, Quadrature Amplitude modulation, bandwidth efficiency, carrier recovery – squaring loop, Costas loop, DPSK.		
UNIT-III	DIGITAL TRANSMISSION	Classes: 10
Introduction, Pulse modulation, PCM, PCM sampling, sampling rate, signal to quantization noise rate, companding analog and digital percentage error, delta modulation, adaptive delta modulation, differential pulse code modulation, pulse transmission – Inter symbol interference, eye patterns.		
UNIT-IV	SPREAD SPECTRUM AND MULTIPLE ACCESS TECHNIQUES:	Classes: 08
Introduction, Pseudo noise sequence, DS spread spectrum with coherent binary PSK, processing gain, FH spread spectrum, multiple access techniques wireless communication, TDMA and CDMA in wireless communication systems, source coding of speech for wireless communications.		

UNIT-V	SATELLITE AND OPTICAL COMMUNICATION	Classes: 08
Satellite Communication Systems ,Keplers Law, LEO and GEO Orbits, footprint, Link model Optical Communication Systems-Elements of Optical Fiber Transmission link, Types, Losses, Sources and Detectors		
Text Books:		
- Wayne Tomasi, —Advanced Electronic Communication SystemsII, 6/e, Pearson Education, 2007. - Simon Haykin, —Communication SystemsII, 4th Edition, John Wiley & Sons., 2001.		
Reference Books:		
H. Taub, D L Schilling, G Saha , II Principles of CommunicationII 3/e, 2007. - B.P. Lathi, II Modern Analog And Digital Communication systemsII, 3/e, Oxford University Press, 2007 - Blake, —Electronic Communication SystemsII, Thomson Delmar Publications, 2002. - Martin S. Roden, —Analog and Digital Communication SystemII, 3rd Edition, PHI, 2002. B. Sklar, II Digital Communication Fundamentals and ApplicationsII 2/e Pearson Education 2007.		
Web References:		
https://nptel.ac.in/courses/117106090/ https://lecturenotes.in/notes/15347-note-for-principles-of-communication-systems-pcs-by-sunil-s-harakannanavar		
E-Text Books:		
https://books.google.co.in/books?id=6Zunu4Acfg8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false https://books.google.co.in/books?id=tZiEj2WxioC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
https://onlinecourses.nptel.ac.in/noc18_ee26/preview		

MEASUREMENTS AND INSTRUMENTATION**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC61	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the measuring methods and instruments of electrical quantities.
2. To understand, design aspects and performance criterion of measuring instruments.
3. To understand the working principle of various transducers.
4. To aware about the advances instrumentation.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply knowledge of electronic instruments for measurement of electrical quantities.
2. Demonstrate the principles and operations of instruments.
3. Select and use latest hardware for measurements.
4. Identify the various instruments for various measurements
5. Use the instruments in laboratory and real life

UNIT-I	MEASUREMENT CONCEPTS	Classes: 10
Measurement concepts : Quantities of Measurement- Performance Characteristics – Static and dynamic characteristics – units and standards of measurements – errors in measurements- accuracy and precision, statistical analysis – moving coil, moving iron types Instruments –DC Voltmeter multi-meter & Calibrations		
UNIT-II	DIGITAL INSTRUMENTS	Classes: 08
Digital instruments: Digital multimeters – Digital frequency meter-digital tacho meter-digital PH meter-digital phase meter-digital capacitance meter-Microprocessor based instruments- IEEE 488 bus		
UNIT-III	SIGNAL GENERATORS CATHODE RAY OSCILLOSCOPES	Classes: 08
Signal generators: Function generators – pulse and square wave generators, RF signal generators –Sweep generators – Frequency synthesizer Cathode ray oscilloscopes: Basic block diagram – operation-applications special Oscilloscopes: delayed time base oscilloscopes, sampling oscilloscope		
UNIT-IV	MEASUREING INSTRUMENTS AND BRIDGES	Classes: 09
Measuring instruments: Introduction-field strength meter –Q- meter- LCR Bridge-Transistor tester Bridge: Wheatstone's bridges for resistance measurements-Maxwell's bridge for inductance measurements- Schering's bridge for capacitance measurements measurement		

UNIT-V	TRANSDUCERS & DATA ACQUISITION SYSTEMS	Classes: 10
Transducer: Introduction-electrical transducers-selecting a transducers-active and passive transducers with examples Data acquisition systems: Elements of a digital data acquisition system – interfacing of transducers – multiplexing –data loggers –computer controlled instrumentation		
Text Books:		
1.Albert D.Helfrick and William D.Cooper – Modern Electronic Instrumentation and Measurement Techniques, Pearson / Prentice Hall of India, 2007. 2.Ernest O. Doebelin, Measurement Systems- Application and Design, TMH, 2007.		
Reference Books:		
1.Joseph J.Carr, Elements of Electronics Instrumentation and Measurement, Pearson Education, 2003. 2.Alan. S. Morris, Principles of Measurements and Instrumentation, 2nd Edition,Prentice Hall of India, 2003. 3.David A. Bell, Electronic Instrumentation and measurements, Prentice Hall of India Pvt Ltd, 2003. 4.B.C. Nakra and K.K. Choudhry, Instrumentation, Measurement and Analysis, 2nd Edition, TMH, 2004. 5.James W. Dally, William F. Riley, Kenneth G. McConnell, Instrumentation for Engineering Measurements, 2nd Edition, John Wiley, 2003.		
Web References:		
1. https://www.smartworld.com/notes/electronic-measurements-and-instrumentation-pdf-notes-emi-notes/ 2. http://www.crectirupati.com/sites/default/files/lecture_notes/EMI_0.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=bLJDDAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=mxVRJRYCi-OC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://ocw.tudelft.nl/courses/electronic-instrumentation/ 2. https://www.schneider-electric.com/en/.../86485-measurement-and-instrumentation/		

FUNDAMENTALS OF INTEGRATED CIRCUITS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC62	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students:

1. To understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. To learn the concepts of modeling of Delay techniques and MOSLayers.
3. To learn the concepts of Technology Scaling of MOStransistors.
4. To understand the concepts of testing of combinational and sequential circuits

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Enumerate semiconductor materials, devices, fabrication steps and concepts of crystal growth.
2. Explain silicon oxidation and photolithography.
3. Analyze various etching techniques.
4. Analyze the Ion implantation process and metallization.
5. Evaluate various IC technologies such as Bipolar, MOSFET and MESFET.

UNIT-I	INTRODUCTION	Classes: 09
Semiconductor materials, Semiconductor Devices, Semiconductor process technology, Basic fabrication steps. Crystal Growth: Silicon Crystal Growth from melt, Silicon Float-Zone Process, GaAs Crystal Growth Techniques, Material Characterization.		
UNIT-II	SILICON OXIDATION	Classes: 09
Thermal oxidation, Impurity Redistribution during oxidation, masking properties of silicon dioxide, oxide quality, oxide thickness characterization, Photolithography.		
UNIT-III	ETCHING	Classes: 09
Wet chemical etching, Dry etching. Diffusion: Basic Diffusion Process, Extrinsic Diffusion, Lateral Diffusion.		
UNIT-IV	ION IMPLANTATION	Classes: 09
Range of Implanted ions, Implant Damage and annealing, Implantation-related Process. Film Deposition: Epitaxial growth techniques, Structures and defects in epitaxial layers, Dielectric deposition, Poly-silicon deposition, Metallization.		
UNIT-V	PROCESS INTEGRATION	Classes: 09
Passive Components, Bipolar Technology, MOSFET technology, MESFET technology, MEMS technology. IC Manufacturing: Electrical testing, Packaging.		

Text Books:

1. Gary S. May, Simon M. Sze, "Fundamentals of Semiconductor Fabrication", John Wiley Inc., 2004.
2. C.Y. Chang and S.M.Sze (Ed), "ULSI Technology", McGraw Hill Companies Inc, 1996

Reference Books:

1. S.K. Ghandhi, "VLSI Fabrication Principles", John Wiley Inc., New York, 1983.
2. S.M. Sze (Ed), "VLSI Technology", 2nd Edition, McGraw Hill, 1988
3. The Science and Engineering of Microelectronic Fabrication, Stephen Cambell, Oxford University Press, 2001

Web References:

1. <https://nptel.ac.in/courses/108108111/>
2. https://www.electronics-tutorials.ws/opamp/opamp_1.html

E-Text Books:

1. https://books.google.co.in/books/.../The_Science_and_Engineering_of_Microelectronics
2. https://books.google.co.in/books?id=K2FpuYEchakC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
3. https://books.google.co.in/books?id=aByz--9D63wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://online.stanford.edu/courses/ee214b-advanced-integrated-circuit-design>
2. <https://www.edx.org/course/circuits-electronics-1-basic-circuit-mitx-6-002-1x-0>

SIGNALS TRANSMISSION THROUGH LINEAR SYSTEMS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC63	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand various fundamental characteristics of signals and systems.
2. To study the importance of transform domain.
3. To analyze and design various systems.
4. To study the effects of sampling.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Analyze various signals and systems by applying Laplace and Z-transforms.
2. Apply Fourier series, Fourier transform and inverse Fourier transform techniques to find magnitude and phase spectrum of various signals.
3. Apply convolution and correlation operations to various signals and systems.
4. Design IIR and FIR analog and digital filters.
5. Analyze multirate Digital Signal Processing.

UNIT-I	TIME DOMAIN SIGNALS	Classes: 09
Introduction: Definitions of signals and systems, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sync, Signum, and Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.		
UNIT-II	SPECTRAL DOMAIN REPRESENTATION OF SIGNALS	Classes: 09
Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transforms, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.		
UNIT-III	CONVOLUTION AND CORRELATION	Classes: 09
Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.		
UNIT-IV	DIGITAL FILTERS	Classes: 09
IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method. FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.		
UNIT-V	MULTI RATE DIGITAL SIGNAL PROCESSING	Classes: 09

Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.

Text Books:

1. John G. Proakis, Dimitris G. Manolakis , Digital Signal Processing, Pearson Prentice Hall, 2007.
2. B.P. Lathi , Signals, Systems & Communications ,BS Publications 2013 .
3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.

Reference Books:

1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed.,
2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008.
3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007

Web References:

1. https://web.sonoma.edu/eese/courses/ee442/archives/sp2019/lectures/lecture03_sig_transmission.pdf
2. <https://lecturenotes.in/subject/44/digital-signal-processing-dsp>

E-Text Books:

1. <http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59>
2. https://books.google.co.in/books?id=m04fl_7pa1QC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.mooc-list.com/course/digital-signal-processing-coursera>
2. <http://nptel.ac.in/courses/117102060/>

FUNDAMENTALS OF VLSI DESIGN**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC64	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. To designing of stick diagrams and layouts for MOS transistors.
3. To learn the concepts of modeling of Delay techniques and MOS layers.
4. To learn the concepts of Technology Scaling of MOS transistors.
5. Understand the concepts of testing of combinational and sequential circuits and also the scan of design techniques

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Illustrate the Basic NMOS, CMOS & BiCMOS circuits and their process technology.
2. Design stick diagrams and layouts for MOS transistors.
3. Describe the concepts of modeling of Delay techniques and MOS layers.
4. Explain the concepts of Technology Scaling of MOS transistors.
5. Discuss the concepts of testing of combinational and sequential circuits and also the scan of design techniques

UNIT-I	REVIEW OF MICROELECTRONICS AND INTRODUCTION TO MOS TECHNOLOGY	Classes: 09
Introduction MOS and related VLSI technology – NMOS-CMOS-BICMOS-GaAs Technologies – thermal aspects of processing – production of E beam masks.		
UNIT-II	MOS AND BICMOS CIRCUIT DESIGN PROCESS	Classes: 09
MOS layers – stick diagrams – design rules and layout – 2m meter – 1.2 m meter CMOS rules – Layout diagrams – Symbolic diagrams.		
UNIT-III	BASIC CIRCUIT CONCEPTS	Classes: 09
Sheet resistance – Area capacitance of layers – delay unit – wiring capacitances – choice of layers.		
UNIT-IV	SCALING OF MOS CIRCUITS	Classes: 09
Scaling of MOS circuits: Scaling modes I – Scaling function for device parameters – Limitation of Scaling Subsystem design process: Architectural issues – switch logic – examples of structural design (Combinational logic)– design of ALU subsystem – commonly used storage elements – aspects of design rules.		
UNIT-V	TEST AND TESTABILITY	Classes: 09
Test and Testability: Design for testability built in self test (BIST) – teaching combinational logic – testing sequential logic – practical design for test guide lines – scan design techniques – etc.		

Text Books:

1. Basic VLSI design by Douglas A, Pucknell, Kamran Eshraghian, Prantice Hall, 1996 3rd edition.
2. Essentials of VLSI circuits and systems – Kamran Eshraghian, Eshraghian Douglas and A. Pucknell, PHI, 2005 Edition

Reference Books:

1. Mead, C.A and Conway, L.A., Introduction to VLSI Systems, Wesley – Wesley.
2. Modern VLSI Design – Wayne Wolf, Pearson Education, 3rd Edition, 1997.

Web References:

1. <http://coen.boisestate.edu/smluo/smluo-courses/ece-332-vlsi-fall07/lecture-notes/>
2. <http://www.freebookcentre.net/electronics-ebooks-download/Vlsi design-Lecture-Notes.html>

E-Text Books:

1. <http://gen.lib.rus.ec/book/Vlsi design>
2. <http://www.faadooengineers.com/threads/Vlsi design -by-kamran eshragghan-ebook-pdf-download>
3. <https://e.edim.co/123389964/Vlsi design.pdf>
4. <http://www.freebookcentre.net/electronics-ebooks-download/Vlsi .html>

MOOC Course

1. <https://www.mooc-list.com/tags/Vlsi design>
2. <https://www.coursera.org/courses?query=Vlsi design>

INTRODUCTION OF MICROPROCESSORS AND MICROCONTROLLERS**VII Semester: OPEN ELECTIVE -III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC65	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The course should enable the students to

1. Understand the basic 8&16 bit microprocessor architectures and its functionalities.
2. Understand Assembly language programming skills of processor.
3. Interface different peripheral devices with Micro processors or Micro controllers.
4. Develop an assembly language program for specified application
5. Develop systems using different micro processors and micro controllers.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate the basic 8&16 bit microprocessor architectures and its functionalities.
2. Describe Assembly language programming skills of processor.
3. Interface different peripheral devices with Micro processors or Micro controllers.
4. Develop an assembly language program for specified application
5. Develop systems using different micro processors and micro controllers.

UNIT-I	MICROPROCESSOR ARCHITECTURE	Classes: 09
Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization, Signal descriptions of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupt service routine.		
UNIT-II	INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086	Classes: 09
Instruction set, Assembler directives, Addressing modes, programming model, Programs involving logical, branch and call instructions, Sorting.		
UNIT-III	INTERFACING WITH 8086	Classes: 09
8255 PPI, various modes of operation and interfacing to 8086, Stepper motor interfacing, D/A & A/D converter, Memory interfacing to 8086, DMA controller (8257).		
UNIT-IV	INTRODUCTION TO MICROCONTROLLERS	Classes: 09
Overview of 8051 microcontroller, Architecture, I/O ports, Memory organization, addressing modes and instruction set of 8051, Simple programs.		
UNIT-V	8051 REAL TIME CONTROL	Classes: 09
Programming Timer interrupts, programming external hardware interrupts, Programming the serial communication interrupts, programming 8051 timers and Counters.		

Text Books:

1. D.V.Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006.
2. Kenneth.J.Ayala. The 8051 microcontroller, 3rd edition, Cengage learning, 2010
3. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006.
4. R. S. Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.

Reference Books:

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,
3. Microcontrollers and application, Ajay. V. Deshmukh, TMGH, 2005.

Web References:

1. <https://www.smartworld.com/notes/microprocessor-and-microcontroller-pdf-notes-mpmc-notes-pdf/>
2. <https://www.slideshare.net/gokulvlsi/microprocessor-microcontroller-notes>

E-Text Books:

1. https://books.google.co.in/books?id=Rs6r3Ghjn1wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. <https://e.edim.co/123389964/A.K. Ray and K.M. Bhurchandi-Advanced Microprocessors and Peripherals 3e-Tata Mcgraw Hill.pdf>
3. <http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html>

MOOC Course

1. <https://www.mooc-list.com/tags/microprocessors>
2. <https://www.coursera.org/courses?query=microprocessor>

FUNDAMENTALS OF IMAGE PROCESSING

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC66	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the Fundamental concepts of Digital Image Processing.
2. To provide a detailed approach towards image processing applications like enhancement, segmentation, and compression.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate basic digital image processing concepts.
2. Explain Image enhancement techniques.
3. Implement the spatial and frequency domain image transforms on enhancement and restoration of images
4. Describe the need for compression and evaluate the basic compression algorithms
5. Explain fundamental concepts of video coding and video segmentation.

UNIT-I	INTRODUCTION	Classes: 10
Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighborhood, adjacency, connectivity, distance measures.		
UNIT-II	IMAGE ENHANCEMENT	Classes: 12
Image Enhancements and Filtering-Gray level transformations, histogram equalization and specifications, pixel-domain smoothing filters – linear and order-statistics, pixel-domain sharpening filters – first and second derivative, two-dimensional DFT and its inverse, frequency domain filters – low-pass and high-pass		
UNIT-III	IMAGE SEGMENTATION	Classes: 12
Image Segmentation- Detection of discontinuities, edge linking and boundary detection, thresholding – global and adaptive, region-based segmentation.		
Wavelets and Multi-resolution image processing- Uncertainty principles of Fourier Transform, Time-frequency localization, continuous wavelet transforms, wavelet bases and multi-resolution analysis, wavelets and Subband filter banks, wavelet packets.		
UNIT-IV	IMAGE COMPRESSION	Classes: 08
Image Compression-Redundancy–inter-pixel and psycho-visual; Lossless compression – predictive, entropy; Lossy compression- predictive and transform coding; Discrete Cosine Transform; Still image compression standards – JPEG and JPEG-2000.		

UNIT-V	VIDEO PROCESSING	Classes: 12
Fundamentals of Video Coding- Inter-frame redundancy, motion estimation techniques – full-search, fast search strategies, forward and backward motion prediction, frame classification – I, P and B; Video sequence hierarchy – Group of pictures, Elements of a video encoder and decoder; Video coding standards – MPEG and H.26X. Video Segmentation- Temporal segmentation–shot boundary detection, hard-cuts and soft-cuts; spatial segmentation – motion-based; Video object detection and tracking.		
Text Books:		
1. R.C. Gonzalez and R.E. Woods, Digital Image Processing, Second Edition, Pearson Education 3rd edition 2008 2. Anil Kumar Jain, Fundamentals of Digital Image Processing, Prentice Hall of India. 2nd edition 2004		
Reference Books:		
1. Murat Tekalp , Digital Video Processing" Prentice Hall, 2nd edition 2015		
Web References:		
1. https://nptel.ac.in/courses/106105032/ 2. https://lecturenotes.in/notes/17316-note-for-digital-image-processing-dip-by-annapurna-mishra		
E-Text Books:		
1. https://books.google.co.in/books/about/Digital_Image_Processing.html?id=a62xQ2r_f8wC 2. https://books.google.co.in/books?id=JeDGn6Wmf1kC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=a62xQ2r_f8wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://online.stanford.edu/courses/ee368-digital-image-processing 2. https://www.coursera.org/learn/computervision-imageprocessing		

TELEVISION ENGINEERING

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC67	OEC	L	T	P	C	CIE	SEE	Total
		4	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To familiarize the students with Television transmitters and receive, and TV signal transmission.
2. To make them understand different signal processing step monochrome television,
3. To introduce colour television transmitters and receivers

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate basic concepts of Television Engineering.
2. Explain different elements and principles of monochrome TV receiver.
3. Discuss Sync Separation and Detection.
4. Explain speech recognition architecture, acoustics, language models and applications.
5. Describe speech synthesis technologies.

UNIT-I	INTRODUCTION	Classes: 14
TV transmitter and receivers, synchronization. Geometric form and aspect ratio, image continuity, interlaced scanning, picture resolution Composite video signal, TV standards. Camera tubes: image Orthicon Plumbicon, vidicon, silicon Diode Array vidicon, Comparison of camera tubes Monochrome TV camera, TV Signal Transmission and Propagation: Picture Signal transmission positive and negative modulation. VSB transmission. sound signal transmission. standard channel BW, TV transmitter, TV signal propagation interference, TV broadcast channels, TV transmission Antennas.		
UNIT-II	MONOCHROME TV RECEIVER	Classes: 12
RF tuner. IF subsystem. video amplifier. sound section. sync separation and processing. deflection circuits. scanning circuits, AGC, noise cancellation, video and inter carrier sound signal detection, vision IF subsystem of Black and White receivers, Receiver sound system: FM detection, FM Sound detectors, and typical applications.		
UNIT-III	SYNC SEPARATION AND DETECTION	Classes: 12
TV Receiver Tuners. Tuner operation, VHF and UHF tuners, digital tuning techniques. remote control of receiver functions. Sync Separation, AFC and Deflection Oscillators: Synchronous separation, k noise in sync pulses. separation of frame and line sync pulses. AFC, single ended AFC circuit, Deflection Oscillators, deflection drive IOs, Receiver Antennas, Picture Tubes.		

UNIT-IV	SPEECH RECOGNITION	Classes: 10
Large Vocabulary Continuous Speech Recognition: Architecture of large vocabulary continuous speech recognition system – acoustics and language models – n-grams, context dependent sub-word units; Applications and present status.		
UNIT-V	SPEECH SYNTHESIS	Classes: 12
Color Receiver: Introduction to colour receiver, Electron tuners, IF subsystem, Y-signal channel, Chroma decoder, Separation of U & V Color, Phasors, synchronous demodulators, Sub carrier generation, raster circuits. Digital TV: Introduction to Digital TV, Digital Satellite TV, Direct to Home Satellite TV, Digital TV Transmitter, Digital TV Receiver, Digital Terrestrial TV, LCD TV, LED TV, CCD Image Sensors, HDTV.		
Text Books:		
1. Television and Video Engineering- A.M.Dhake, 2nd Edition. 2. Modern Television Practice Principles, Technology and Service R .R.Gallatin, New Age International Publication, 2002. 3. Monochrome and Colour TV- R.R. Gulati, New Age International Publication, 2002.		
Reference Books:		
1. Colour Television Theory and Practice-S.P.Bal, TMH, 1994. 2. Basic Television and Video Systems-B.Grob and C.E.Herndon, McGraw Hill, 1999.		
Web References:		
1. http://www.sasurieengg.com/e-course-material/ECE/IV-Year%20Sem%207/EC2034%20Television%20and%20Video%20Engineering.pdf 2. http://www.gcebagur.ac.in/television-and-video-engineering-notes		
E-Text Books:		
1. https://books.google.co.in/books?id=xEl7svssilYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=U9aH92iLEZAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://nptel.ac.in/courses/109103018/34		

INTRODUCTION TO SIGNAL PROCESSING**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC68	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand various signals and systems.
2. Understand various transformation techniques and apply them to find spectral domain representation of various signals.
3. To know how to apply convolution and correlation operations between various signals and systems.
4. To learn how to develop the digital filter designs.
5. To Understand Multirate digital signal processing concepts.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Identify various signals and systems.
2. Demonstrate various transformation techniques and apply them to find spectral domain representation of various signals.
3. Apply convolution and correlation operations between various signals and systems.
4. Develop the digital filter designs.
5. Discuss Multirate digital signal processing concepts.

UNIT-I	TIME DOMAIN SIGNALS	Classes: 09
Introduction, Definitions of signal and system, Elementary Continuous-time and Discrete time signals: Unit impulse, Unit step, Ramp, Exponential, Sinusoidal, Sinc, Signum, Parabolic. Classification of various signals and systems, Sampling and types of Sampling, Laplace and Z-transforms.		
UNIT-II	SPECTRAL DOMAIN REPRESENTATION OF SIGNALS	Classes: 09
Introduction, Fourier series, Fourier transform, Properties, Magnitude Spectrum and Phase spectrum. Discrete Fourier transform, properties, Interrelationships between Fourier, Laplace and Z-Transforms, problems.		
UNIT-III	CONVOLUTION AND CORRELATION	Classes: 09
Introduction, System transfer function, Convolution in continuous time domain, Convolution in discrete time domain: Linear convolution and Circular Convolution, System Response, Auto correlation function and properties, Cross Correlation function and properties, Problems.		

UNIT-IV	DIGITAL FILTERS	Classes: 09
IIR Filters: IIR Analog filter approximations - Butterworth and Chebyshev, Design of IIR Digital filters from analog filters: Impulse invariant techniques, Bilinear transformation method. FIR Filters: Characteristics of FIR Digital Filters, Frequency response, Design of FIR Filters: Fourier Method. Window Techniques, Comparison of IIR & FIR filters.		
UNIT-V	MULTIRATE DIGITAL SIGNAL PROCESSING	Classes: 09
Introduction. Down sampling, Decimation. Up sampling, Interpolation, Sampling Rate Conversion, conversion of band pass signals. Concept of re-sampling. Applications of multi rate signal processing.		
Text Books:		
1. John G. Proakis, Dimitris G. Manolakis, Digital Signal Processing, Pearson Prentice Hall, 2007. 2. B.P. Lathi , Signals, Systems & Communications ,BS Publications 2013 . 3. Sophocles J. Orfanidis, Introduction to Signal Processing, Pearson Prentice Hall, 1998.		
Reference Books:		
1. A.V. Oppenheim, Signals and Systems - A.S. Willsky and S.H. Nawab, 2 Ed., 2. Li Tan, Digital Signal Processing - Fundamentals and Applications - Elsevier. 2008. 3. Robert J. Schilling. Sandra L, Harris, Fundamentals of Digital Signal Processing using Matlab - Thomson. 2007		
Web References:		
1. http://www.freebookcentre.net/special-books-download/An-Introduction-to-Signal-Processing-(T.-O'Haver).html 2. https://www.sciencedirect.com/science/article/pii/B9780123965028000012 3. Applied Digital Signal Processing (1st Edition)		
E-Text Books:		
1. http://gen.lib.rus.ec/book/index.php?md5=69531D2C9E6A3A5111E2CC1977F70F59 2. http://www.freebookcentre.net/special-books-download/Introduction-to-DSP-(Digital-Signal-Processing).html		
MOOC Course		
1. https://www.mooc-list.com/course/digital-signal-processing-coursera 2. http://nptel.ac.in/courses/117102060/		

INTRODUCTION TO MOBILE COMMUNICATION**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC69	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand impairments due to multi path fading channel.
2. To understand the fundamental techniques to overcome the different fading effects.
3. To understand Co-channel and Non Co channel interference
4. To become familiar with cell coverage for signal and traffic, diversity techniques and mobile antennas.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate basic concepts of cellular and mobile radio systems.
2. Discuss about Co-channel and Non Co channel interference
3. Explain multi access and modulation schemes.
4. Explain the concepts of frequency management and channel assignment.
5. Discuss about handoff initiation, different types of handoffs, their advantages, dropped call rates and their evaluation.

UNIT-I	INTRODUCTION TO CELLULAR MOBILE RADIO SYSTEMS	Classes: 11
Limitations of Conventional Mobile Telephone Systems, Basic Cellular Mobile System, First, Second, Third and Fourth Generation Cellular Wireless Systems, Uniqueness of Mobile Radio Environment- Fading -Time Dispersion Parameters, Coherence Bandwidth, Doppler Spread and Coherence Time. Fundamentals of Cellular Radio System Design: Concept of Frequency Reuse, Co-Channel Interference, Co-Channel Interference Reduction Factor, Desired C/I From a Normal Case in a Omni Directional Antenna System, System Capacity, Trunking and Grade of Service, Improving Coverage and Capacity in Cellular Systems - Cell Splitting, Sectoring, Microcell Zone Concept.		
UNIT-II	CO-CHANNEL INTERFERENCE	Classes: 09
Measurement Of Real Time Co-Channel Interference, Design of Antenna System, Antenna Parameters and Their Effects, Diversity Techniques-Space Diversity, Polarization Diversity, Frequency Diversity, Time Diversity. Non-Co-Channel Interference: Adjacent Channel Interference, Near End Far End Interference, Cross Talk, Effects on Coverage and Interference by Power Decrease, Antenna Height Decrease, Effects of Cell Site Components.		
UNIT-III	MULTIPLE ACCESS & MODULATION SCHEMES	Classes: 10
Signal Reflections in Flat And Hilly Terrain, Effect of Human Made Structures, Phase Difference Between Direct and Reflected Paths, Constant Standard Deviation, Straight Line Path Loss Slope, General Formula for Mobile Propagation Over V'later and Flat Open Area, Near and Long Distance Propagation, Path Loss From a Point to Point Prediction Model in Different Conditions, Merits of Lee Model. Cell Site and Mobile Antennas: Space Diversity Antennas, Umbrella Pattern Antennas, Minimum Separation of Cell Site Antennas, Mobile Antennas.		
UNIT-IV	FREQUENCY MANAGEMENT AND CHANNEL ASSIGNMENT	Classes: 07
Numbering And Grouping, Setup Access And Paging Channels, Channel Assignments to Cell Sites and Mobile Units, Channel Sharing and Borrowing, Sectorization, Overlaid Cells, Non Fixed Channel Assignment.		

UNIT-V	HANDOFFS AND DROPPED CALLS	Classes: 08
Handoff Initiation, Types of Handoff, Delaying Handoff, Advantages of Handoff, Power Difference Handoff, Forced Handoff, Mobile Assisted and Soft Handoff, inter system Handoff, Introduction to Dropped Call Rates and their Evaluation.		
Text Books:		
1. Mobile Cellular Telecommunications — W.C.Y. Lee, Mc Graw Hill, 2nd Edn., 1989. 2. Wireless Communications – Theodore. S. Rapport, Pearson Education, 2nd Edn., 2002. 3. Mobile Cellular Communication – Gottapu sashibhushana Rao, Pearson, 2012.		
Reference Books:		
1. Principles of Mobile Communications — Gordon L. Stuber, Springer International, 2nd Edn., 2001 2. Modern Wireless Communications-Simon Haykin, Michael Moher, Pearson Education, 2005 3. Wireless Communications Theory and Techniques, Asrar U. H .Sheikh, Springer, 2004.		
Web References:		
1. http://bu.edu.eg/portal/uploads/Engineering,%20Shoubra/Electrical%20Engineering/2461/crs-12001/E-716-A lec01.pdf 2. http://nikithasnotes.weebly.com/lecture-notes-in-wireless-mobile-communications.html		
E-Text Books:		
1. https://books.google.co.in/books?id=ZZeYb-GASbMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=OnDGbzMRcbwC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. http://nptel.ac.in/courses/117102062/		

BASIC EMBEDDED SYSTEMS DESIGN**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EC70	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To learn Basic functions, Structure, Concepts and Applications of Embedded systems.
2. To learn simple programs to solve well-defined problems on an Embedded platform
3. To understand various communication interfaces available.
4. To study different processors for various applications.
5. To study about the tools used to develop in an embedded environment and implement real time applications on embedded platform

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explains basic concepts and applications of Embedded systems.
2. Build simple programs to solve well-defined problems on an Embedded platform
3. Enumerate various communication interfaces available.
4. Identify different processors for various applications.
5. Explains about the tools used to develop in an embedded environment and implement real time applications on embedded platform

UNIT-I	INTRODUCTION TO EMBEDDED SYSTEMS	Classes: 12
Definition and Classification – Overview of Processors and hardware units in an embedded system – Software embedded into the system – Exemplary Embedded Systems – Embedded Systems on a Chip (SoC) and the use of VLSI designed circuits.		
UNIT-II	INTRODUCING THE 8051 MICROCONTROLLER FAMILY	Classes: 10
Introduction, What's in a name, The external interface of the Standard 8051, Reset requirements, Clock frequency and performance, Memory issues, I/O pins, Timers, Interrupts, Serial interface, Power consumption, Conclusions		
UNIT-III	DEVICES AND BUSES FOR DEVICES NETWORK	Classes: 12
I/O Devices - Device I/O Types and Examples – Synchronous - Iso-synchronous and Asynchronous Communications from Serial Devices - Examples of Internal Serial-Communication Devices - UART and HDLC - Parallel Port Devices - Sophisticated interfacing features in Devices/Ports- Timer and Counting Devices - '12C', 'USB', 'CAN' and advanced I/O Serial high speed buses- ISA, PCI, PCI-X, cPCI and advanced buses.		

UNIT-IV	PROGRAMMING CONCEPTS AND EMBEDDED PROGRAMMING IN C, C++	Classes: 14
Programming in assembly language (ALP) vs. High Level Language - C Program Elements, Macros and functions -Use of Pointers - NULL Pointers - Use of Function Calls – Multiple function calls in a Cyclic Order in the Main Function Pointers – Function Queues and Interrupt Service Routines Queues Pointers – Concepts of EMBEDDED PROGRAMMING in C++ - Objected Oriented Programming – Embedded Programming in C++, 'C' Program compilers – Cross compiler – Optimization of memory codes.		
UNIT-V	REAL TIME OPERATING SYSTEMS	Classes: 12
Tasks and task states, tasks and data, semaphores, and shared data; message queues, mailboxes and pipes, timer functions, events, memory management, interrupt routines in an RTOS environment. Semaphores, queues, hard real-time scheduling considerations, saving memory and power, an example RTOS like uC-OS (open source).		
Text Books:		
1. Rajkamal, Embedded Systems Architecture, Programming and Design, TATA McGraw-Hill, First reprint Oct. 2003 2. Shibu K.V, Introduction to Embedded Systems, Mc Graw Hill.		
Reference Books:		
1. Steve Heath, Embedded Systems Design, Second Edition-2003, Newnes, 2. David E.Simon, An Embedded Software Primer, Pearson Education Asia, First Indian Reprint 2000.		
Web References:		
1. https://nptel.ac.in/downloads/108105057/ 2. https://www.slideshare.net/arunkumargowdru/embedded-system-design-notes-written-by-arun-kumar-g-associate-professor-dept-of-ec-stjit-ranebennur-karnataka-india		
E-Text Books:		
1. https://books.google.co.in/books?id=HV-dbKHt0mOC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=BjNZXwH7HlkC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://nptel.ac.in/courses/108102045/		

**OPEN ELECTIVES
OFFERED BY
ELECTRICAL AND ELECTRONICS
ENGINEERING DEPARTMENT**

ELECTRICAL WIRING AND SAFETY MEASURES**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE52	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand basic concepts of electrical wiring
2. To learn Indian electricity rules
3. To study causes electrical failures and their prevention.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate basic concepts of electrical installations.
2. Explain importance of earthing, factors affecting earth resistance and methods of earthing.
3. Discuss causes of electrical accidents and their prevention.
4. Demonstrate general guidelines and principles of power circuit design.
5. Outline Indian electricity rules as per 1956 act.

UNIT-I	BASICS OF ELECTRICAL INSTALLATIONS	Classes: 12
Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.		
UNIT-II	EARTHING	Classes: 08
Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.		
UNIT-III	SAFETY & PREVENTION OF ACCIDENTS	Classes: 08
Definition of terminology used in safety, I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation. Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.		
UNIT-IV	RESIDENTIAL BUILDING ELECTRIFICATION	Classes: 10
General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub-circuits. Method of drawing single line diagram & wiring diagram.		

UNIT-V	R ENDS& SUBSTATION AND METERS	Classes: 12
Rule 28 : Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises. Rule 31: Cut-out on consumer's premises. Rule 46: Periodical inspection & testing of consumer's installation. Rule 47: Testing of consumer's installation. Rule 54: Declared voltage of supply to consumer. Rule 55: Declared frequency of supply to consumer. Rule 56: Sealing of meters & cut-outs. Rule 77: Clearances above ground of the lowest conductor. Rule 79: Clearances between conductors & trolley wires. Rule 87: Lines crossing or approaching each other. Rule 88: Guarding.		
Text Books:		
1. K.B. Raina, S.K.Bhattacharya Electrical Design;Estimating and costing New Age International (p) Limited, New Delhi Surjit Singh. 2. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B.Gupta 3.A course in Electrical Installation, Estimating & costing S.K.Kataria&sons,S.L. Uappal . 4. Electrical wiringEstimating and costing Khanna Publication. ,A.K.Sawhney		
Reference Books: 1. Electrical Machine Design Danpat Rai & co. 2. The Electricity Rule 2005Universal Law Publishing Co. Pvt. Ltd. N. AlagapanS. Ekambaram 3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi ,Surjit Singh 4. TarlokSibgh Installation, Commissioning & Maintenance of Electrical Equipment S.K.Kataria& Sons 5. B.V.S.Rao Operation & Maintenance of Electrical Machines Vol I & II Media Promoters & Publisher Ltd. Mumbai		
Web References:		
1. https://electrical-engineering-portal.com > Technical Articles 2. https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/ 3. https://www.cpwd.gov.in/Publication/Internal2013.pdf		
E-Text Books:		
1. https://books.google.co.in/books?isbn=0323170064 2. https://www.jove.com/science../electrical-safety-precautions-and-basic-equipment		
MOOC Course		
1. https://nptel.ac.in/courses/103106071/5 2. https://nptel.ac.in/courses/108108099/28 3. https://nptel.ac.in/courses/124107001/		

ELECTRICAL MATERIALS**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE53	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To understand the importance of various materials used in electrical engineering and obtain a qualitative analysis of their behaviour and applications.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate fundamental requirements, properties, characteristics and applications of conductive materials.
2. Explain general concepts and applications of semi conductors.
3. Describe properties and characteristics of dielectrics and insulators.
4. Explain behaviour of magnetic materials.
5. Outline optical properties of semiconductors and applications of photo sensitive materials.

UNIT-I	CONDUCTORS	Classes: 10
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Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminium, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.

UNIT-II	SEMICONDUCTORS	Classes: 08
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General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semi-conductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, Semi-conductors and their applications, superconductors.

UNIT-III	DIELECTRICS AND INSULATORS	Classes: 12
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Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.

UNIT-IV	MAGNETIC MATERIALS	Classes: 10
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magneto-striction.		
UNIT-V	OPTICAL PROPERTIES OF SOLIDS	Classes: 10
Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).		
Text Books:		
1. "Electrical Engineering Materials", Dekker, PHI Pbs. 2. "Electrical Engineering Materials", Indulkar, S.Chand		
Reference Books:		
1. "Electrical Engineering Materials", Tareev 2. "Electrical Engineering Materials", Yu. Koritsky. 3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs		
Web References:		
1. https://physics.info/dielectrics/ 2. https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265.web.mit.edu/course/6/6.732/www/6.732-pt2.pdf		
E-Text Books:		
1. https://easyengineering.net/electrical-engineering-materials-by-dekker/ 2. https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/		
MOOC Course		
1. https://nptel.ac.in/courses/108108076/ 2. https://nptel.ac.in/courses/112104203/3 3. https://onlinecourses.nptel.ac.in/noc18_ee14/		

NEW TRENDS IN ELECTRICAL ENERGY**V Semester: OPEN ELECTIVE -I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE54	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the fundamental principles of wireless power transfer (WPT) for cable-free transfer of power.
2. To study theories for inductive power transfer (IPT) based on the coupled inductor model and low-order circuit compensation.
3. To understand specific converter topologies for lighting and battery charging applications.
4. To learn technology trends in the adoption of WPT for key consumer applications.

COURSE OUTCOMES:

The students should be able to

1. Demonstrate concepts of basic circuit theory.
2. Describe the characteristics of power transfer through coupled inductors and the significance of leakage inductance.
3. Analyze and design appropriate compensation circuits and efficient power converters for WPT applications.
4. Outline technical requirements for applications involving solid-state loads and battery loads using WPT technologies.
5. Explain the factors affecting adoption of WPT in consumer applications including lightings, charging of smart phones and electric vehicles.

UNIT-I	BASIC CIRCUIT THEORY	Classes: 10
Review of transformers. Leakage inductance. Circuit compensation principles. Low-order compensations; series and parallel compensations. Resonance and operating frequency. Efficiency equation.		
UNIT-II	POWER CONVERTERS FUNDAMENTALS	Classes: 10
DC-DC converters. AC-DC converters and inverters. PWM and soft switching principles. Basic topologies with transformers. Input, output and transfer characteristics of power converters. Incorporation of leaky transformer. Control methods.		
UNIT-III	COMPENSATION CONFIGURATIONS	Classes: 10
Types of compensation for inductor power transfer. Characteristics for various termination requirements. Design for load-independence output voltage and output current. Efficiency optimization.		
UNIT-IV	APPLICATIONS	Classes: 10
Circuit requirements for various loading conditions. Characteristics of LED loads, resistors and battery loads. Appropriate compensation design. Lighting systems. Battery charging profiles. Electric vehicle charging. Energy efficiency metric for charging.		

UNIT-V	TECHNOLOGY TRENDS	Classes: 10
Demand for safe power transfer and durable operation. Portable and smart devices. Mobile communication devices. IoT devices and systems. Sensors. Solid-state lighting development. Battery technologies. Electric vehicle development. Renewable source integration trends. Future trends and demand for wireless power transfer.		
Text Books:		
1. C. T. Rim and C. Mi, Wireless Power Transfer for Electric Vehicles and Mobile Devices, New York: IEEE Press-Wiley, 2017. 2. J. I. Agbinya, Wireless Power Transfer, River Publishers, 2015		
Reference Books:		
1. Z. Huang, S. C. Wong, and C. K. Tse, "Design of a single-stage inductive power-transfer converter for efficient EV battery charging," IEEE Transactions on Vehicular Technology, vol. 66, no. 7, pp. 5808-5821, July 2017. 2. L. Xu, Q. Chen, X. Ren, S. C. Wong, and C. K. Tse, "Self-oscillating resonant converter with contactless power transfer and integrated current sensing transformer," IEEE Transactions on Power Electronics, vol. 32, no. 6, pp. 4839-4851, June 2017. 3. W. Zhang, S. C. Wong, C. K. Tse, and Q. Chen, "Load-independent duality of current and voltage outputs of a series or parallel compensated inductive power transfer converter with optimized efficiency," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, pp. 137-146, March 2015. 4. J. Hou, Q. Chen, X. Ren, X. Ruan, S. C. Wong, and C. K. Tse, "Precise characteristics analysis of series/series-parallel compensated contactless resonant converter," IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 3, no. 1, 101-110, March 2015.		
Web References:		
1. https://www.researchgate.net/.../267512972_Wireless_Power_Transmission_Trends 2. https://ieeexplore.ieee.org/document/6403386 3. https://www.cse.wustl.edu/~jain/cse574-14/ftp/power/		
E-Text Books:		
1. https://books.google.co.in/books/about/Wireless_Power_Transfer.html?id... 2. https://www.intechopen.com/books/wireless-power-transfer-principles-and-engineering-explorations		
MOOC Course:		
1. https://ocw.mit.edu/.../6-452-principles-of-wireless-communications-spring-2006/ 2. www.iitq.ac.in/e_mobility/WPT.html 3. https://nptel.ac.in/courses/117102062/ 4. https://nptel.ac.in/courses/117106108/150 5. https://nptel.ac.in/courses/117104099/2		

POWER PLANT ENGINEERING**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE55	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To introduce students to different aspects of power plant engineering.
2. To familiarize the students to the working of power plants based on different fuels.
3. To expose the students to the principles of safety and environmental issues

COURSE OUTCOMES:

1. Demonstrate the resources for development of power in India.
2. Explain working principle and working of diesel and gas turbine power plants.
3. Discuss the working principle and basic components of the hydro electric plants.
4. Describe the working principle and basic components of the nuclear power plant
5. Outline economic and environmental considerations of power plants.

UNIT-I	INTRODUCTION TO THE SOURCES OF ENERGY	Classes: 10
Resources and Development of Power in India. Steam Power Plant: Plant Layout, Working of different Circuits, Fuel and handling equipment, types of coals, coal handling, choice of handling equipment, coal storage, Ash handling systems. Combustion Process: Properties of coal — overfeed and underfeed fuel beds, traveling grate stokers, spreader stokers, retort stokers, pulverized fuel burning system and its components, combustion needs and draught system, cyclone furnace, design and construction, Dust collectors, cooling towers and heat rejection. Corrosion and feed water treatment.		
UNIT-II	DIESEL POWER PLANT & GAS TURBINE POWER PLANT	Classes: 10
DIESEL POWER PLANT: Introduction — IC Engines, types, construction— Plant layout with auxiliaries fuel supply system, air starting equipment, lubrication and cooling system-super charging. GAS TURBINE PLANT: Introduction — classification – construction — Layout with auxiliaries — Principles of working of closed and open cycle gas turbines. Combined Cycle Power Plants and comparison. Direct Energy Conversion: Solar energy, Fuel cells, Thermo electric and Thermo ionic, MHD generation.		
UNIT-III	HYDRO ELECTRIC POWER PLANT	Classes: 08
Water power — Hydro logical cycle / flow measurement — drainage area characteristics — Hydro graphs — storage and Pounding — classification of dams and spill ways. Hydro Projects And Plant: Classification — Typical layouts — plant auxiliaries — plant operation pumped storage plants. Power From Non-Conventional Sources: Utilization of Solar- Collectors- Principle of Working, Wind Energy — types — HAWT, VAWT Tidal Energy.		
UNIT-IV	NUCLEAR POWER STATION	Classes: 10
Nuclear fuel — breeding and fertile materials — Nuclear reactor — reactor operation. Types of Reactors: Pressurized water reactor. Boiling water reactor. sodium-graphite reactor. fast Breeder Reactor, homogeneous Reactor, Gas cooled Reactor, Radiation hazards and shielding radioactive waste disposal.		
UNIT-V	ECONOMIC AND ENVIRONMENTAL CONSIDERATIONS OF POWER PLANT	Classes: 12

Capital cost, investment of fixed charges, operating costs, general arrangement of power distribution, Load curves, load duration curve. Definitions of connected load, Maximum demand, demand factor, average load, load factor, diversity factor — related exercises. Effluents from power plants and Impact on environment — pollutants and pollution standards — Methods of Pollution control.

Text Books:

1. Power Plant Engineering! P.C.Sharma / S.K.Kataria Pub.
2. A Course in Power Plant Engineering: I Arora and S. Domkundwar.

Reference Books:

1. A Text Book of Power Plant Engineering I Rajput I Laxmi Publications.
2. Power Plant Engineering: P.K.Nag/ II Edition /TMH.
3. An Introduction to Power Plant Technology I G.D. Rai/Khanna Publishers.
4. Power plant Engg I Elanchezhian/ I.K. International Pub.
5. Power plant Engineering! Ramalingam / Scietech Publishers.

Web References:

1. www.cbspd.co.in/power-plant-engineering-9788123919461-black-veatch.html
2. <https://www.semcomaritime.com/en-en/references/power-generation>

E-Text Books:

1. <https://www.intechopen.com/books/subject/power-plant-engineering>
2. <https://books.google.co.in/books?isbn=1482221519>

MOOC Course

1. <https://nptel.ac.in/courses/108105058/8>
2. www.nptelvideos.in/2012/11/energy-resources-and-technology.html
3. www.ignou.ac.in/upload/Unit-2-58.pdf

ANALYSIS OF LINEAR SYSTEMS**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE56	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To develop ability to analyze linear systems and signals
2. To develop critical understanding of mathematical methods to analyze linear systems and signals

COURSE OUTCOMES:

The students should be able to

1. Formulate and solve state equations for electrical networks.
2. Apply Fourier series and Fourier transforms.
3. Apply Laplace Transforms.
4. Prove band limited signal impulse sampling, natural and flat top sampling theorems and describe other sampling theory concepts.
5. Obtain the response for discrete systems using Z-Transforms.

UNIT-I	STATE VARIABLE ANALYSIS	Classes: 08
Choice of state variables in Electrical networks-Formulation of state equations for Electrical networks Equivalent source method. Network topological method – Solution of state equations-Analysis of simple networks with state variable approach.		
UNIT-II	FOURIER SERIES AND FOURIER TRANSFORM REPRESENTATION	Classes: 12
Introduction. Trigonometric form of Fourier series. Exponential form of Fourier series. Wave symmetry. Fourier integrals and transforms. Fourier transform of a periodic function . Properties of Fourier Transform . Parseval's theorem , Fourier transform of some common signals, Fourier transform relationship with Laplace Transform. Applications of Fourier series and Fourier Transform Representation: Introduction. Effective value. and average values of non sinusoidal periodic waves. currents, Power Factor, Effects of harmonics, Application in Circuit Analysis, Circuit Analysis using Fourier Series.		
UNIT-III	LAPLACE TRANSFORM APPLICATIONS	Classes: 10
Application of Laplace transform Methods of Analysis – Response of RL, RC, RLC Networks to Step, Ramp, and impulse functions, Shifting Theorem – Convolution Integral – Applications Testing of Polynomials: Elements of realisability – Hurwitz polynomials-positive real functions-Properties-Testing-Sturm's Test, examples. Network Synthesis: Network synthesis: Synthesis of one port LC networks-Foster and Cauer methods-Synthesis of RL and RC one port networks-Foster and Cauer methods		
UNIT-IV	SAMPLING	Classes: 10
Sampling theorem – Graphical and Analytical proof for Band Limited Signal impulse sampling, natural and Flat top Sampling, Reconstruction of signal from its samples, effect of under sampling – Aliasing, Introduction to Band Pass sampling, Cross correlation and auto correlation of functions, properties of correlation function, Energy density spectrum, Power density spectrum, Relation between auto correlation function and Energy / Power spectral density function.		
UNIT-V	Z-TRANSFORMS	Classes: 10

Fundamental difference between continuous and discrete time signals, discrete time complex, exponential and sinusoidal signals. periodicity of discrete time complex exponential. concept of Z Transform of a discrete sequence. Distinction between Laplace, Fourier, and Z-Transforms. Region of convergence in Z-Transforms, constraints on ROC for various classes of signals, Inverse Z-Transform properties of Z-Transforms.

Text Books:

1. B. P. Lathi", "Signals, Systems and Communications", BS Publications 2003.
2. "Umesh Sinha" "Network Analysis and Synthesis", Satya Prakashan Publications, 2013.

Reference Books:

1. "A. N. Tripathi", "Linear System Analysis", New Age International, 2nd Edition 1987.
2. "D. Roy Chowdhary", "Network and Systems", New Age International, 2005.
3. "Gopal G Bhise, Prem R. Chadha", Engineering Network Analysis and Filter Design, Umesh Publications 2009.
4. "A. Cheng", linear system analysis, Oxford publishers, 1999

Web References:

1. <https://archive.org/details/introductiontoli00brow>
2. <https://ieeexplore.ieee.org/iel5/9/24171/01101971.pdf>

E-Text Books:

1. www.cds.caltech.edu/~murray/books/AM08/pdf/am08-complete_04Mar10.pdf
2. https://books.google.co.in/books?id=gMLIAm-TmpkC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. www.nptelvideos.in/2012/11/estimation-of-signals-and-systems.html
2. <https://nptel.ac.in/courses/108104100/6>

NEURAL NETWORKS AND FUZZY LOGIC**VI Semester: OPEN ELECTIVE -II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE57	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

COURSE OUTCOMES:

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback network.
3. Demonstrate the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

UNIT-I	INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS	Classes: 12
Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCulloch-PiUs Model, Historical Developments, Potential Applications of ANN, Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application		
UNIT-II	SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS	Classes: 10
Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training. Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.		
UNIT-III	ASSOCIATIVE MEMORIES-I	Classes: 08
Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).		

UNIT-IV	ASSOCIATIVE MEMORIES-II	Classes: 10
Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications		
UNIT-V	FUZZY LOGIC	Classes: 10
Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions. Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, Defuzzification methods.		
Text Books:		
1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI. 2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publications.		
Reference Books:		
1. Artificial Neural Networks, B. Yegnanarayana, PHI. 2. Artificial Neural Networks, Zaruda, PHI. 3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI. 4. Fuzzy Logic and Neural Networks, M. Amirhavalli, Scitech Publications India Pvt. Ltd. 5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education. 6. Neural networks by satish Kumar, TIVIH, 2004 7. Neural Networks, Simon Hakens, Pearson Education. 8. Neural Engineering, C. Eliasmith and CH. Anderson, PHI.		
Web References:		
1. users.monash.edu/~app/CSE5301/Lnts/LaD.pdf 2. https://engineering.purdue.edu/~tsoukala/rational.html 3. https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf		
E-Text Books:		
1. https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book 2. www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf 3. www.vssut.ac.in/lecture_notes/lecture1423723637.pdf		
MOOC Course		
1. https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07 2. https://nptel.ac.in/courses/108104049/16 3. https://nptel.ac.in/courses/117105084/ 4. https://nptel.ac.in/syllabus/127105006		

ELECTRICAL ENGINEERING MATERIALS**VII Semester: OPEN ELECTIVE -III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE58	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To understand the importance of various materials used in electrical engineering, their behaviour and applications.

COURSE OUTCOMES:

1. Describe properties and characteristics of conductive and resistive materials.
2. Outline semiconductor materials and technologies.
3. Explain about various types of dielectric materials. their properties in various conditions.
4. Discuss about magnetic materials and their behaviour
5. Describe optical properties of semi conductors and applications of photo sensitive materials.

UNIT-I	CONDUCTORS	Classes: 10
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.		
UNIT-II	SEMICONDUCTORS	Classes: 08
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semi-conductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, Semi-conductors and their applications, superconductors.		
UNIT-III	DIELECTRICS AND INSULATORS	Classes: 12
Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials, mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.		
UNIT-IV	MAGNETIC MATERIALS	Classes: 10
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magneto-striction.		

UNIT-V	OPTICAL PROPERTIES OF SOLIDS	Classes: 10
Photo emission, photo emission materials, electro luminescence junction diode, photo emitters, photo transistor, photo resistors, inijunction lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).		
Text Books:		
1. "Electrical Engineering Materials", Dekker, PHI Pbs. 2. "Electrical Engineering Materials", Indulkar, S.Chand		
Reference Books:		
1. "Electrical Engineering Materials", Tareev 2. "Electrical Engineering Materials", Yu. Koritsky. 3. "Electrical Engineering Materials", R.K.Rajput, Laxmi Pbs		
Web References:		
1. https://physics.info/dielectrics/ 2. https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265 3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf		
E-Text Books:		
1. https://easyengineering.net/electrical-engineering-materials-by-dekker/ 2. https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/		
MOOC Course		
1. https://nptel.ac.in/courses/108108076/ 2. https://nptel.ac.in/courses/112104203/3 3. https://onlinecourses.nptel.ac.in/noc18_ee14/		

NON-CONVENTIONAL POWER GENERATION**VII Semester: OPEN ELECTIVE -III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE59	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. Understand basic concepts of power generation.
2. Understand basic working principles of nuclear power generation systems.
3. Apply knowledge of solar power generation systems in design and implementation to obtain clean energy.

COURSE OUTCOMES:

The students should be able to

1. Describe working principles of solar radiation.
2. Classify and explain solar energy collectors; mention their applications.
3. Analyze photovoltaic systems and related technologies for energy conversion.
4. Discuss in detail the solar cell applications.
5. Analyze cost of solar panels and collectors; Describe about various alternative energy sources.

UNIT-I	PRINCIPLES OF SOLAR RADIATION	Classes: 08
Role and potential of new and renewable source, the solar energy option, Environmental impact of solar power, physics of the sun, the solar constant, extraterrestrial and terrestrial solar radiation, solar radiation on tilted surface, instruments for measuring solar radiation and Sun shine, solar radiation data.		
UNIT-II	SOLAR ENERGY COLLECTORS	Classes: 10
Flat plate and concentrating collectors, classification of concentrating collectors, orientation and thermal analysis, advanced collectors.		
STORAGE AND APPLICATIONS: Different methods of solar energy storage, Sensible, latent heat and stratified storage, solar ponds. Solar Applications- solar heating /cooling technique, solar distillation and drying.		
UNIT-III	PHOTO VOLTAICS (PV)	Classes: 12
Fundamentals of solar cells, types of solar cells, semiconducting materials, band gap theory, absorption of photons, excitations and photoemission of electrons, band engineering.		
PV CELL PROPERTIES: Solar cell properties and design, p-n junction photodiodes, depletion region, electrostatic field across the depletion layer, electron and holes transports, device physics, charge carrier generation, recombination and other losses, I-V characteristics, output power.		
UNIT-IV	SOLAR CELL APPLICATIONS	Classes: 10
PV cell interconnection, module structure and module fabrication, Equivalent circuits, load matching, efficiency, fill factor and optimization for maximum power, Design of stand-alone PV systems, system sizing, device structures, device construction, DC to AC conversion, inverters, on-site storage and grid connections.		

UNIT-V	COST ANALYSIS AND ALTERNATIVE ENERGY SOURCES	Classes: 10
Cost analysis and pay back calculations for different types of solar panels and collectors, installation and operating costs, Environmental and safety issues, protection systems, performance monitoring. ALTERNATIVE ENERGY SOURCES: Solar Energy: Types of devices for Solar Energy Collection, Thermal Storage System. Control Systems, Wind Energy, Availability, Wind Devices, Wind Characteristics, Performance of Turbines and systems.		
Text Books:		
1. G. D. Rai (2009), Non-Conventional Energy Sources, 4th edition, Khanna Publishers, New Delhi. 2. Martin A. Green (2008), Solar Cells: Operating Principles, Technology and system Applications, 1st edition, Prentice Hall, New Delhi.		
Reference Books:		
1. Sukatme (2008), Solar Energy, 3rd Edition, McGraw Hill Companies, New Delhi. 2. D. Yogi gosuami, Frank Kreith, Jan F. Kreider (2000), Principles of Solar Engineering, 2nd edition, Taylor & Francis, USA.		
Web References:		
1. https://nptel.ac.in/courses/Webcourse-contents/IISc-BANG/notused/Non-Conventional%20Energy%20Systems-/Learning%20Material%20-%20NCES.pdf 2. https://www.ukessays.com/essays/environmental-sciences/promoting-non-conventional-energy-resource		
E-Text Books:		
1. https://books.google.co.in/books?id=y-xYsi5469UC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=Vps3P-S85o8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://nptel.ac.in/courses/108102047/ 2. https://onlinecourses.nptel.ac.in/noc19_ge11/course 3. https://nptel.ac.in/downloads/108108078/		

SOLAR ENERGY AND APPLICATIONS

VII Semester: OPEN ELECTIVE -III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE60	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To Study about solar modules and PV system design and their applications
2. To Deal with grid connected PV systems
3. To Discuss about different energy storage systems

COURSE OUTCOMES:

The students should be able to

1. Demonstrate characteristics of sunlight, behaviour of solar cells, cell properties and PV cell interconnection.
2. Explain about standalone PV systems, their design and sizing.
3. Describe the issues in grid connected PV systems
4. Discuss about the modelling of different energy storage systems and their performances
5. Outline different applications of solar energy

UNIT-I	INTRODUCTION	Classes: 10
Characteristics of sunlight –behaviour of solar cells – cell properties – Semi conductor devices and PN Junction solar cells-PV cell interconnection		
UNIT-II	STAND ALONE PV SYSTEM	Classes: 10
Solar modules – storage systems – power conditioning and regulation - MPPT- protection – stand alone PV systems design – sizing		
UNIT-III	GRID CONNECTED PV SYSTEMS	Classes: 10
PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs		
UNIT-IV	ENERGY STORAGE SYSTEMS	Classes: 10
Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage		
UNIT-V	APPLICATIONS	Classes: 10
Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.		

Text Books:

1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd., 2015.
2. Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007, Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa, 1994

Reference Books:

1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011.
2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 S.P. Sukhatme, "Solar Energy", Tata McGraw Hill, 1987.

Web References:

1. <https://www.renewableenergyworld.com/ugc/articles/2011/11/main-elements-of-pv-system.html>
2. <https://thegrid.rexel.com/en-us/knowledge/energy-efficiency/w/wiki/82/solar-pv-systems-overview-of-the-main-components>

E-Textbooks:

1. https://books.google.co.in/books?id=37zYCgAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=swz6UhzaFbkC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.edx.org/course/solar-energy-photovoltaic-pv-systems>
2. <https://www.coursera.org/learn/photovoltaic-solar-energy>
3. <https://online-learning.tudelft.nl/courses/solar-energy/>

INSTRUMENTATION AND CONTROL**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE61	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

COURSE OUTCOMES:

The students should be able to

1. Identify the instruments suitable for typical measurements.
2. Explain principle of operation of potentiometers and describe different ratios and errors in current and potential transformers.
3. Apply the knowledge about transducers and instrument transformers to use them effectively.
4. Improve the system performance by selecting a suitable controller for a specific application
5. Apply various time domain techniques to assess the system performance.

UNIT-I	INTRODUCTION TO MEASURING INSTRUMENTS	Classes: 10
Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.		
UNIT-II	POTENTIOMETERS & INSTRUMENT TRANSFORMERS	Classes: 08
Principle and operation of D.C. Crompton's potentiometer—standardisation — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardisation — applications; Instrument Transformers: Current Transformer and Potential Transformer — Ratio and phase angle errors.		
UNIT-III	TRANSDUCERS & OSCILLOSCOPES	Classes: 12
Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation. gauge factor. Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes.		
CRO: Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO- Measurement of phase and frequency-lissajous patterns.		

UNIT-IV	CLASSIFICATION OF CONTROL SYSTEMS	Classes: 10
Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Different examples of control systems- Classification of control systems Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse Response and transfer functions . Block diagram algebra – Representation by Signal flow graph – Reduction using mason's gain formula.		
UNIT-V	TIME RESPONSE ANALYSIS	Classes: 10
Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.		
Text Books:		
1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd. 2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications. 3. I. J. Nagrath and M. Gopal", "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009 4. "B. C. Kuo", "Automatic Control Systems", John wiley and sons, 8th edition, 2003.		
Reference Books:		
1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications. 2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd. 3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications. 4. N. K. Sinha", "Control Systems", New Age International (P) Limited Publishers, 3rdEdition, 1998. 5. "NISE", "Control Systems Engineering", John wiley, 6th Edition, 2011. 6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.		
Web References:		
1. http://www.crmagnetics.com/assets/technical-references/analysis_of_ct_error.pdf 2. http://www.yourelectrichome.com/2011/08/instrument-transformer.html		
E-Text Books:		
1. https://books.google.co.in/books?id=qhAlgmUX2YEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=1kzWPb4ok2EC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://nptel.ac.in/courses/108105064/ 2. https://nptel.ac.in/courses/108101037/ 3. https://nptel.ac.in/courses/107106081		

ENERGY AUDIT AND MANAGEMENT SYSTEMS**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE62	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand energy efficiency, scope, conservation and technologies.
2. To design energy efficient lighting systems.
3. To estimate/calculate power factor of systems and propose suitable compensation techniques.
4. To understand energy conservation in HVAC systems.
5. To understand life cycle costing analysis and return on investment on energy efficient technologies.

COURSE OUTCOMES:

The students should be able to

1. Explain energy efficiency, conservation and various technologies.
2. Design energy efficient lighting systems.
3. Calculate power factor of systems and propose suitable compensation techniques.
4. Explain energy conservation in HVAC systems.
5. Calculate life cycle costing and return on investment on energy efficient technologies.

UNIT-I	ENERGY AUDIT AND MANAGEMENT ENERGY AUDIT	Classes: 12
Basic Principles of Energy Audit and management Energy audit – Definitions – Concept – Types of audit – Energy index – Cost index – Pie charts – Sankey diagrams – Load profiles – Energy conservation schemes and energy saving potential – Numerical problems – Principles of energy management –Initiating, planning, controlling, promoting, monitoring, reporting – Energy manager – Qualities and functions – Language – Questionnaire – Check list for top management.		
UNIT-II	LIGHTING	Classes: 10
Modification of existing systems – Replacement of existing systems – Priorities: Definition of terms and units – Luminous efficiency – Polar curve – Calculation of illumination level – Illumination of inclined surface to beam – Luminance or brightness – Types of lamps – Types of lighting – Electric lighting fittings (luminaries) – Flood lighting – White light LED and conducting Polymers – Energy conservation measures.		
UNIT-III	POWER FACTOR AND ENERGY INSTRUMENTS	Classes: 08
Power factor – Methods of improvement – Location of capacitors – Power factor with non linear loads – Effect of harmonics on Power factor – Numerical problems. Energy Instruments – Watt-hour meter – Data loggers – Thermocouples – Pyrometers – Lux meters – Tong testers – Power analyzer.		

UNIT-IV	SPACE HEATING AND VENTILATION	Classes: 08
Ventilation – Air-Conditioning (HVAC) and Water Heating: Introduction – Heating of buildings – Transfer of Heat–Space heating methods – Ventilation and air-conditioning – Insulation–Cooling load – Electric water heating systems – Energy conservation methods.		
UNIT-V	ECONOMIC ASPECTS AND ANALYSIS	Classes: 12
Economics Analysis – Depreciation Methods – Time value of money – Rate of return – Present worth method – Replacement analysis – Life cycle costing analysis – Energy efficient motors (basic concepts) Computation of Economic Aspects : Calculation of simple payback method – Net present worth method – Power factor correction – Lighting – Applications of life cycle costing analysis – Return on investment		
Text Books:		
1. Energy management by W.R. Murphy & G. McKay Butter worth, Elsevier publications. 2012 2. Energy efficient electric motors by John .C. Andreas, Marcel Dekker Inc Ltd–2nd edition, 1995		
Reference Books:		
1. Electric Energy Utilization and Conservation by S C Tripathy, Tata McGraw hill publishing company Ltd. New Delhi. 2. Energy management by Paul o' Callaghan, Mc-Graw Hill Book company–1st edition, 1998. 3. Energy management hand book by W.C.Turner, John wiley and sons. 4. Energy management and conservation –k v Sharma and pvenkateshaiah-I K International Publishing House pvt.ltd,2011.		
Web References:		
1. https://www.docsity.com/en/energy-management-and-audit-bureau-of-energy-efficiency-lecture-notes/247696/ 2. http://shodhganga.inflibnet.ac.in/bitstream/10603/46067/11/11_chapter%201.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=E3YawQ1f3JkC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=I8PwBm2E7U0C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=w6y-0n3DxWgC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://nptel.ac.in/courses/108106022/ 2. https://www.dnvgi.ae/training/energy-environmental-management-systems-training-6909		

ENERGY STORAGE SYSTEMS**VIII Semester: OPEN ELECTIVE -IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4EE63	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

To enable the student to understand the need for energy storage, devices and technologies available and their applications

COURSE OUTCOMES:

The students should be able to

1. Demonstrate basic concepts of electrical energy storage technologies.
2. Describe needs for electrical energy storage.
3. Analyze the characteristics of energy from various sources
4. Classify various types of energy storage and various devices used for the purpose
5. Identify various real time applications.

UNIT-I	ELECTRICAL ENERGY STORAGE TECHNOLOGIES	Classes: 12
Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.		
UNIT-II	NEEDS FOR ELECTRICAL ENERGY STORAGE	Classes: 08
Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies. The roles from the viewpoint of a utility. The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.		
UNIT-III	FEATURES OF ENERGY STORAGE SYSTEMS	Classes: 08
Classification of EES systems, Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES), Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H ₂), Synthetic natural gas (SNG).		
UNIT-IV	TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS	Classes: 10
Electrical storage systems, Double-layer capacitors (DLC), Superconducting magnetic energy storage (SMES), Thermal storage systems, Standards for EES, Technical comparison of EES technologies.		
UNIT-V	APPLICATIONS	Classes: 12
Present status of applications, Utility use (conventional power generation, grid operation & service), Consumer use (uninterruptable power supply for large consumers), New trends in applications, Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems, Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA – aggregation of many dispersed batteries.		

Text Books:

1. "James M. Eyer, Joseph J. Iannucci and Garth P. Corey ", "Energy Storage Benefits and Market Analysis", Sandia National Laboratories, 2004.
2. The Electrical Energy Storage by IEC Market Strategy Board.

Reference Books:

1. "Jim Eyer, Garth Corey", Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.

Web References:

1. https://www.smartgrid.gov/files/sandia_energy_storage_report_sand2010-0815.pdf
2. <https://nptel.ac.in/courses/108103009/download/M9.pdf>

E-Text Books:

1. <https://www.geni.org/globalenergy/research/energy-storage-technologies/Energy-Storage-Technologies.pdf>
2. https://books.google.co.in/books?id=uDFPAAAAMAAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. https://pdhonline.com/courses/e320/e320_new.htm
2. <https://ocw.tudelft.nl/course-lectures/introduction-energy-storage/>

**OPEN ELECTIVES
OFFERED BY
INFORMATION TECHNOLOGY DEPARTMENT**

FUNDAMENTALS OF DATA STRUCTURES

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT11	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To impart the basic concepts of data structures and algorithms.
2. To understand concepts linked lists and their applications.
3. To understand basic concepts about stacks, queues and their applications.
4. To understand basic concepts of trees, graphs and their applications.
5. To write algorithms for sorting and searching and hashing.
6. To use advanced data structures like B-Trees, AVL-trees etc., for efficient problem solving.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Formulate new solutions for problems or improve existing code using data structures and algorithms.
2. Implement basic data structures such as arrays, linked lists, stacks and queues.
3. Solve problem involving graphs, trees and heaps
4. Apply Algorithms for solving problems like sorting, searching, and hashing.
5. Implement advanced data structures such as B-Trees, Red-Black, and AVL-Trees

UNIT-I	INTRODUCTION TO DATA STRUCTURES	Classes: 12
Basic concepts- Algorithm Specification-Introduction, Recursive algorithms, Data Abstraction, Performance analysis- time complexity and space complexity, Asymptotic Notation-Big O, Omega and Theta notations Introduction to Linear and Non Linear data structures-Singly Linked Lists- Operations-Insertion, Deletion, Concatenating singly linked lists, Circularly linked lists-Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion, Deletion.		
UNIT-II	STACKS AND QUEUES	Classes: 10
Stacks-Stack ADT, definition, operations, array and linked implementations in C, applications-infix to postfix conversion, Postfix expression evaluation. Queues-Queue ADT, definition and operations, array and linked Implementations in C, Circular queues- Insertion and deletion operations, Dequeue (Double ended queue)ADT, array and linked implementations in C.		
UNIT-III	TREESANDGRAPHS	Classes: 14
Trees – Terminology, Representation of Trees, binary tree ADT, Properties of Binary Trees, Binary Tree Representations-array and linked representations, Binary Tree traversals. Max Priority Queue-ADT- implementation-Max Heap-Definition, Insertion into a Max Heap, Deletion from a		

Max Heap. Graphs , Introduction, Definition, Terminology, Graph ADT, Graph Representations- Adjacency matrix, Adjacency lists, Graph traversals- DFS and BFS.		
UNIT-IV	SEARCHING AND SORTING	Classes: 12
Searching- Linear Search, Binary Search, Comparison of search techniques. Sorting- Insertion Sort, Selection Sort, Radix Sort, Quick sort, Merge Sort, Heap Sort, Comparison of Sorting methods.		
UNIT-V	BINARY SEARCH TREES	Classes: 12
Search Trees- Binary Search Trees, Definition, Operations- Searching, Insertion and Deletion, AVL Trees- Definition and Examples, Insertion into an AVL Tree, B-Trees, Definition, B-Tree of order m, operations- Insertion and Searching, Comparison of Search Trees. Pattern matching algorithm- The Knuth-Morris-Pratt algorithm, Tries (examples only)		
Text Books:		
1. Fundamentals of Data Structures II, Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press. 2. Fundamentals of Data structures in C, 2nd Edition, E.Horowitz, S.Sahni and Susan Anderson- Freed, Universities Press		
Reference Books:		
1. Algorithms, Data Structures, and Problem Solving with C++II, Illustrated Edition by Mark Allen Weiss, Addison- Wesley Publishing Company 2. How to Solve it by Computer II, 2nd Impression by R.G.Dromey, Pearson Education		
Web References:		
1. https://nptel.ac.in/courses/106103069/1 2. https://ece.uwaterloo.ca/~dwharder/aads/Lecture_materials/ 3. http://www.cs.yale.edu/homes/aspnes/classes/223/notes.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=nB_ZAqAAQBAJ&printsec=frontcover&source=qbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=ZgGzBt074e8C&printsec=frontcover&source=qbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.coursera.org/learn/data-structures 2. https://www.edx.org/course/algorithms-and-data-structures-4 3. https://www.edx.org/course/foundations-of-data-structures-2		

SOFTWARE ENGINEERING PRINCIPLES

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT12	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To be familiar with basic software engineering methods and practices, and its applications.
2. To learn the implementation of software engineering layered technology and process framework.
3. To understand the concepts of software measurement and software risks.
4. To be familiar with software requirements and the SRS documents.
5. To learn the role of project management including planning, scheduling, risk management.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate basic concepts of software engineering.
2. Describe various process models, software and engineering process requirements.
3. Describe different system models and design concepts.
4. Discuss basic object oriented design concepts and perform user interface design.
5. Explain strategies of risk management and quality concepts.

UNIT-I	INTRODUCTION TO SOFTWARE ENGINEERING	Classes: 11
The evolving role of software, Changing Nature of Software, Software myths. A Generic view of process: Software engineering- A layered technology, a process framework, The Capability Maturity Model Integration (CMMI), Process patterns, process assessment, personal and team process models.		
UNIT-II	PROCESS MODELS	Classes: 17
The waterfall model, Incremental process models, Evolutionary process models, The Unified process. Software Requirements: Functional and non-functional requirements, User requirements, System requirements, Interface specification, the software requirements document. Requirements engineering process: Feasibility studies, Requirements elicitation and analysis, Requirements validation, Requirements management.		
UNIT-III	SYSTEM MODELS	Classes: 15
Context Models, Behavioral models, Data models, Object models, structured methods. Design Engineering: Design process and Design quality, Design concepts, the design model. Creating an architectural design: Software architecture, Data design, Architectural styles and patterns, Architectural Design.		
UNIT-IV	OBJECT-ORIENTED DESIGN	Classes: 17

Objects and object classes, An Object-Oriented design process, Design evolution. Performing User interface design: Golden rules, User interface analysis and design, interface analysis, interface design steps, Design evaluation. Testing Strategies: A strategic approach to software testing, test strategies for conventional software, Black-Box and White-Box testing, Validation testing, System testing, the art of Debugging.		
UNIT-V	RISK MANAGEMENT	Classes: 16
Reactive vs. Proactive Risk strategies, software risks, Risk identification, Risk projection, Risk refinement, RMMM, RMMM Plan. Quality Management: Quality concepts, Software quality assurance, Software Reviews, Formal technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.		
Text Books:		
1. Software Engineering a practitioner's Approach, Roger S Pressman, 6th edition. Mc. Graw Hill International Edition. 2. Software Engineering, Ian Somerville, 7th edition, Pearson education.		
Reference Books:		
1. Software Engineering, A Precise Approach, Pankaj Jalote, Wiley India, 2010. 2. Software Engineering: A Primer, Waman S Jawadekar, Tata McGraw-Hill, 2008 3. Fundamentals of Software Engineering, Rajib Mall, PHI, 2005		
Web References:		
1. https://courses.cs.washington.edu/courses/cse403/01au/lectures/ 2. https://lecturenotes.in/notes/1594-note-for-software-engineering-se-by-sushri-rout		
E-Text Books:		
1. https://books.google.co.in/books?id=r-SH73Cuc-IC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=sbicEWSPe40C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.edx.org/course/software-engineering-introduction-ubcx-softeng1x 2. https://www.edx.org/course/software-engineering-essentials 3. https://www.classcentral.com/tag/software-engineering		

COMPUTER ORGANIZATION AND OPERATING SYSTEMS

V Semester: OPEN ELECTIVE - I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT13	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

The course objectives are:

1. To have a thorough understanding of the basic structure and operation of a digital computer.
2. To discuss in detail the operation of the arithmetic unit including the algorithms & implementation of fixed-point and floating-point addition, subtraction, multiplication & division.
3. To study the different ways of communicating with I/O devices and standard I/O interfaces.
4. To study the hierarchal memory system including cache memories and virtual memory.
5. To demonstrate the knowledge of functions of operating system memory management scheduling, file system and interface, distributed systems, security and dead locks.
6. To implement a significant portion of an Operating System

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain the basic structure and operation of a digital computer
2. Describe the different ways of communicating with I/O devices and standard I/O interfaces
3. Demonstrate the functions of operating system memory management scheduling, file system and interface, distributed systems, security and dead locks.
4. Explain the hierarchal memory system including cache memories and virtual memory.
5. Describe principles of deadlock.

UNIT-I	BASIC STRUCTURE OF COMPUTERS	Classes: 24
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Computer Types, Functional UNIT, Basic Operational Concepts, Bus Structures, Software, Performance, Multiprocessors and Multi Computers, Data Representation, Fixed Point Representation, Floating - Point Representation.

Register Transfer Language and Micro Operations: Register Transfer Language, Register Transfer Bus and Memory Transfers, Arithmetic Micro Operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit, Instruction Codes, Computer Registers Computer Instructions - Instruction Cycle. Memory - Reference Instructions, Input - Output and Interrupt, STACK Organization, Instruction Formats, Addressing Modes, Data Transfer and Manipulation.

UNIT-II	MICRO PROGRAMMED CONTROL	Classes: 13
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Control Memory, Address Sequencing, Micro Program Examples, Design of Control Unit, Hard Wired Control, Micro Programmed Control.

Input-Output Organization: Peripheral Devices, Input-Output Interface, Asynchronous Data Transfer Modes, Priority Interrupt, Direct Memory Access, Input-Output Processor (IOP).

UNIT-III	OPERATING SYSTEMS OVERVIEW	Classes: 09
Overview of Computer Operating Systems Functions, Protection and Security, Distributed Systems, Special Purpose Systems, Operating Systems Structures-Operating System Services and Systems Calls, System Programs, Operating System Generation. Principles of Deadlock: System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock.		
UNIT-IV	MEMORY MANAGEMENT	Classes: 13
Swapping, Contiguous Memory Allocation, Paging, Structure of the Page Table, Segmentation, Virtual Memory, Demand Paging, Page-Replacement Algorithms, Allocation of Frames, Thrashing Case Studies - UNIX, Linux, Windows.		
UNIT-V	PRINCIPLES OF DEADLOCK	Classes: 15
System Model, Deadlock Characterization, Deadlock Prevention, Detection and Avoidance, Recovery from Deadlock. File System Interface: The Concept of a File, Access Methods, Directory Structure, File System Mounting, File Sharing, Protection. Allocation Methods, Free-Space Management.		
Text Books:		
1. Computer Organization - Carl Hamacher, Zvonks Vranesic, SafeaZaky, 5th Edition, McGraw Hill. 2. Computer System Architecture - M. moris mano, 3rd edition, Pearson 3. Operating System Concepts - Abreham Silberchatz, Peter B. Galvin, Greg Gagne, 8th Edition, John Wiley.		
Reference Books:		
1. Computer Organization and Architecture - William Stallings 6th Edition, Pearson 2. Structured Computer Organization - Andrew S. Tanenbaum, 4th Edition, PHI 3. Fundamentals of Computer Organization and Design - Sivaraama Dandamudi, Springer Int. Edition 4. Operating Systems - Internals and Design Principles, Stallings, 6th Edition - 2009, Pearson Education. 5. Modern Operating Systems, Andrew S Tanenbaum 2nd Edition, PHI 6. Principles of Operating System, B. L. Stuart, Cengage Learning, India Edition		
Web References:		
1. https://nptel.ac.in/courses/106103068/pdf/coa.pdf 2. https://www.smartworld.com/notes/computer-organization-pdf-notes-co/ 3. https://nptel.ac.in/downloads/106108101/		

E-Text Books:

1. https://books.google.co.in/books?id=AlHgbyyElHoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=8eh2ROK2BK4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.classcentral.com/course/nptel-computer-organization-and-architecture-a-pedagogical-aspect-9824>
2. <https://www.classcentral.com/course/nptel-introduction-to-operating-systems-6559>

BASICS OF PYTHON PROGRAMMING**VI Semester: OPEN ELECTIVE - II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT14	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To describe the basic elements of the Python language and the Python interpreter and discuss the differences between Python and other modern languages.
2. To describe Python dictionaries and demonstrate the use of dictionary methods.
3. Define, analyze and code the basic Python conditional and iterative control structures and explain how they can be nested and how exceptions can be used.
4. To Explain and demonstrate methods of error handling and Python exceptions.
5. To demonstrate the understanding of —magic methods through use of these in the context of a Python application.

COURSE OUTCOMES:

Up on successful completion of the course, the student is able to

1. Write and debug Python programs which make use of the fundamental control structures and method-building techniques common to all programming languages.
2. Use data types, input, and output, iterative, conditional, and functional components of the language in his or her programs.
3. Use object-oriented programming techniques to design and implement a clear, well-structured Python program.
4. Use and design classes and objects in programs.
5. Outline the specific features of Python which made it more powerful programming language.

UNIT-I	INTRODUCTION TO PYTHON	Classes: 08
Overview, Basic. Python Installation, Comments in Python, Concept of Indentation in python.		
UNIT-II	DATA TYPES	Classes: 08
Tuples, Lists More advanced data types (dictionary, string), Python operators, control flows, Loops, Functions.		
UNIT-III	MODULES AND PACKAGES	Classes: 08
File handling in python, Module and Packages, Object oriented programming.		

UNIT-IV	ADVANCED FUNCTIONS	Classes: 08
Exceptions, sorting, advanced function: map, filter, and reduce.		
UNIT-V	INTRODUCTION TO STANDARD LIBRARIES	Classes: 08
Multi-Processing And Multi-Threading, Introduction To Standard Libraries (pandas, Turtle, numpy, os).		
Text Books:		
1. Learning Python, by Shroff Pub& Dist., O'relly publications, Publication Year: 2013.		
Reference Books:		
1. Python Programming for Beginners: Python Programming Languageby Joseph Joyner		
Web References:		
1. https://mothicse.blogspot.com/2017/07/r16-python-programming-lecture-notes.html 2. http://scipy-lectures.org/intro/language/python_language.html		
E-Text Books:		
1. https://books.google.co.in/books?id=aJQILILxRmAC&printsec=frontcover&redir_esc=y#v=onepage&q&f=false 2. https://books.google.co.in/books?id=tN7JAceyJdUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.coursera.org/learn/python-programming-introduction 2. https://www.edx.org/course/introduction-to-computer-science-and-programming-using-python-2		

HUMAN COMPUTER INTERACTION**VI Semester: OPEN ELECTIVE - II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT15	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

To learn

1. The human components functions.
2. The Computer components functions.
3. The Interaction between the human and computer components.
4. Paradigms
5. Interaction design basics
6. HCI in the software process
7. Design rules
8. Implementation supports
9. Evaluation techniques

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Explain the human components functions regarding interaction with computer
2. Explain computer components functions regarding interaction with human
3. Demonstrate understanding of Interaction between the human and computer components.
4. Describe about specification methods and interface-building tools.
5. Implement Interaction design basics

UNIT-I	INTRODUCTION	Classes: 12
Importance of user Interface – definition, importance of good design, Benefits of good design. A brief history of Screen design, The graphical user interface – popularity of graphics, the concept of direct manipulation, graphical system, Characteristics, Web user – Interface popularity, characteristics- Principles of user interface.		
UNIT-II	DESIGN PROCESS	Classes: 15
Human interaction with computers, importance of human characteristics human consideration, Human interaction speeds, understanding business junctions. Screen Designing : Design goals – Screen planning and purpose, organizing screen elements, ordering of screen data and content – screen navigation and flow – Visually pleasing composition – amount of information – focus and emphasis – presentation information simply and meaningfully – information retrieval on web – statistical graphics – Technological consideration in interface design.		
UNIT-III	WINDOWS	Classes: 12

New and Navigation schemes selection of window, selection of devices based and screen based controls. Components – text and messages, Icons and increases – Multimedia, colors, uses problems, choosing colors.		
UNIT-IV	SOFTWARE TOOLS	Classes: 08
Specification methods, interface – Building Tools.		
UNIT-V	INTERACTION DEVICES	Classes: 08
Keyboard and function keys – pointing devices – speech recognition digitization and generation – image and video displays – drivers.		
Text Books:		
1. The essential guide to user interface design, Wilbert O Galitz, Wiley Dreamtech. 2. Designing the user interface. 3rd Edition Ben Shneidermann , Pearson Education Asia.		
Reference Books:		
1. Human – Computer Interaction. ALAN DIX, JANET FINCAY, GRE GORYD, ABOWD, RUSSELL BEALG, PEARSON. 2. Interaction Design PRECE, ROGERS, SHARPS. Wiley Dreamtech, 3. User Interface Design, Soren Lauesen , Pearson Education.		
Web References:		
1. https://www.studocu.com/en/document/anna-university/human-computer-interaction/lecture-notes/hci-notes-all-units-1/3707695/view 2. http://www.cs.bham.ac.uk/~rxb/Teaching/SSC%20HCI%202008-9/HCIlecturenotes.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=8GbKBQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=WuQbERqXR10C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.classcentral.com/course/coursera-human-computer-interaction-306 2. https://www.udacity.com/course/human-computer-interaction--ud400		

SOFTWARE TESTING FUNDAMENTALS**VI Semester: OPEN ELECTIVE - II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT16	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

To understand the objectives and principles of software testing, software testing methodologies, test management and software test automation.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate basic concepts of software testing.
2. Describe software testing methods and apply various testing techniques
3. Design and conduct a software test process for a software testing project
4. Design solutions for various software testing problems by selecting appropriate software test model
5. Implement various practice oriented software testing projects

UNIT-I	INTRODUCTION	Classes: 08
Basics of software testing, Testing objectives, Principles of testing, Test Life Cycle, Types of testing, Software defect tracking.		
UNIT-II	TESTING METHODOLOGIES	Classes: 12
White Box And Black Box Testing, Static Testing, Static Analysis Tools, Structural Testing, Unit/Code functional, testing, Code coverage testing, Code complexity testing, Black Box testing, Requirements based testing.		
UNIT-III	INTEGRATION TESTING	Classes: 10
Integration, System, and Acceptance Testing Top down and Bottom up integration, Functional versus Non-functional testing, Deployment testing, Beta testing, Scalability testing, Reliability testing, Stress testing, Acceptance testing		
UNIT-IV	TEST SELECTION & MINIMIZATION FOR REGRESSION TESTING	Classes: 10
Test Selection & Minimization for Regression Testing Regression testing, Regression test process, Initial Smoke or Sanity test, Selection of regression tests, Execution Trace, Dynamic Slicing, Test Minimization, Tools for regression testing, Ad hoc Testing: Pair testing, Exploratory testing, Iterative testing, Defect seeding.		
UNIT-V	TEST MANAGEMENT AND AUTOMATION TEST PLANNING	Classes: 10
Test Management and Automation Test Planning, Management, Execution and Reporting, Software Test Automation: Scope of automation, Design & Architecture for automation, Generic requirements for test tool		

framework, Test tool selection.

Text Books:

1. S. Desikan and G. Ramesh, "Software Testing: Principles and Practices", Pearson Education.
2. Aditya P. Mathur, "Fundamentals of Software Testing", Pearson Education.

Reference Books:

1. Naik and Tripathy, "Software Testing and Quality Assurance", Wiley
2. K. K. Aggarwal and Yogesh Singh, "Software Engineering", New Age International Publication.

Web References:

1. <https://www.cs.colorado.edu/~kena/classes/5828/s12/lectures/05-introtesting.pdf>
2. <https://lecturenotes.in/notes/14690-note-for-software-testing-st-by-jntu-heroes>

E-Text Books:

1. https://books.google.co.in/books?id=zUm8My7SiakC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=PjoesdBMp0UC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.edx.org/course/software-testing-fundamentals>
2. <https://www.guru99.com/software-testing.html>

E - COMMERCE**VII Semester: OPEN ELECTIVE - III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT17	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To identify the major categories and trends of e-commerce applications.
2. To identify the essential processes of an e-commerce system.
3. To identify several factors and web store requirements needed to succeed in e-commerce.
4. To understand the main technologies behind e-commerce systems and how these technologies interact.
5. To discuss the various marketing strategies for an online business.
6. To define various electronic payment types and associated security risks and the ways to protect against them

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate basic concepts of electronic commerce.
2. Define various electronic payment types and associated security risks and the ways to protect against them
3. Describe concepts of intra organizational commerce.
4. Discuss various digital document types and marketing strategies for online business.
5. Explain concepts of consumer search, resource discovery and multimedia.

UNIT-I	ELECTRONIC COMMERCE	Classes: 08
Electronic Commerce - Frame work, anatomy of E-Commerce applications, E-Commerce Consumer applications, and E-Commerce organization applications, Consumer Oriented Electronic commerce - Mercantile Process models.		
UNIT-II	ELECTRONIC PAYMENT SYSTEMS	Classes: 08
Electronic payment systems – Digital Token-Based, Smart Cards, Credit Cards, Risks in Electronic Payment systems. Inter-Organizational Commerce – EDI, EDI Implementation, Value added networks.		
UNIT-III	INTRA ORGANIZATIONAL COMMERCE	Classes: 08
Intra Organizational Commerce – work Flow, Automation Customization and internal Commerce, Supply chain Management.		
UNIT-IV	CORPORATE DIGITAL LIBRARY	Classes: 11
Corporate Digital Library - Document Library, digital Document types, corporate Data Warehouses. Advertising and Marketing - Information based marketing, Advertising on Internet, on-line marketing process, market research.		
UNIT-V	CONSUMER SEARCH AND RESOURCE DISCOVERY & MULTIMEDIA	Classes: 12

Consumer Search And Resource Discovery: Information search and Retrieval, Commerce Catalogues, Information Filtering

Multimedia - key multimedia concepts, Digital Video and electronic Commerce, Desktop video processing, Desktop video conferencing.

Text Books:

1. Frontiers of electronic commerce – Kalakata, Whinston, Pearson.
2. E-Commerce fundamentals and applications Hendry Chan, Raymond Lee, Tharam Dillon, Ellizabeth Chang, John Wiley.

Reference Books:

1. E-Commerce, S.Jaiswal – Galgotia.
2. E-Commerce, Efrain Turbon, Jae Lee, David King, H.Michael Chang.
3. Electronic Commerce – Gary P.Schneider – Thomson.
4. E-Commerce – Business, Technology, Society, Kenneth C.Taudon, Carol

Web References:

1. <https://www.smartzworld.com/notes/e-commerce-pdf-notes-lecture-notes/>
2. <https://www.studocu.com/en/document/university-of-rajasthan/e-commerce/lecture-notes/e-commerce-notes-pdf-lecture-notes-university-level/1911743/view>

E-Text Books:

1. https://books.google.co.in/books?id=cheOQeh8rk4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=MwEB8LuK0P0C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://www.edx.org/learn/ecommerce>
2. <https://alison.com/course/e-commerce-and-selling-online>

CYBER FORENSICS

VII Semester: OPEN ELECTIVE - III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT18	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

To enable students to understand the usage of computers in forensic, and how to use various forensic tools for a wide variety of investigations.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate the basic concepts of computer forensics.
2. Explain importance of evidence collection and data seizure.
3. Analyze and validate computer forensics.
4. Describe use of current computer forensic tools.
5. Discuss about file systems, Microsoft file structures, disk encryption, start-up tasks and virtual machines.

UNIT-I	COMPUTER FORENSICS FUNDAMENTALS	Classes: 15
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What is Computer Forensics?, Use of Computer Forensics in Law Enforcement, Computer Forensics Assistance to Human Resources/Employment Proceedings, Computer Forensics Services, Benefits of Professional Forensics Methodology, Steps taken by Computer Forensics Specialists Types of Computer Forensics Technology: Types of Military Computer Forensic Technology, Types of Law Enforcement — Computer Forensic Technology — Types of Business Computer Forensic Technology Computer Forensics Evidence and Capture: Data Recovery Defined — Data Back-up and Recovery — The Role of Back-up in Data Recovery — The Data-Recovery Solution.

UNIT-II	EVIDENCE COLLECTION AND DATA SEIZURE	Classes: 15
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Why Collect Evidence? Collection Options — Obstacles — Types of Evidence — The Rules of Evidence — Volatile Evidence — General Procedure — Collection and Archiving — Methods of Collection — Artifacts — Collection Steps — Controlling Contamination: The Chain of Custody Duplication and Preservation of Digital Evidence: Preserving the Digital Crime Scene — Computer Evidence Processing Steps — Legal Aspects of Collecting and Preserving Computer Forensic Evidence Computer Image Verification and Authentication: Special Needs of Evidential Authentication — Practical Consideration — Practical Implementation.

UNIT-III	COMPUTER FORENSICS ANALYSIS AND VALIDATION	Classes: 12
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Determining what data to collect and analyze, validating forensic data, addressing data-hiding techniques, and performing remote acquisitions, Network Forensics: Network forensics overview, performing live acquisitions, developing standard procedures for network forensics, using network tools, examining the honeynet project. Processing Crime and Incident Scenes: Identifying digital evidence, collecting evidence in private-sector incident scenes, processing law enforcement crime scenes, preparing for a search, securing a computer incident or crime scene, seizing digital evidence at the scene, storing digital evidence, obtaining a digital hash, reviewing a case.

UNIT-IV	CURRENT COMPUTER FORENSIC TOOLS	Classes: 15
Evaluating computer forensic tool needs, computer forensics software tools, computer forensics hardware tools, validating and testing forensics software E-Mail Investigations: Exploring the role of e-mail in investigation, exploring the roles of the client and server in e-mail, investigating e-mail crimes and violations, understanding e-mail servers, using specialized e-mail forensic tools. Cell phone and mobile device forensics: Understanding mobile device forensics, understanding acquisition procedures for cell phones and mobile devices.		
UNIT-V	WORKING WITH WINDOWS AND DOS SYSTEMS	Classes: 08
Understanding file systems, exploring Microsoft File Structures, Examining NTFS disks, Understanding whole disk encryption, windows registry, Microsoft start-up tasks, MS-DOS startup tasks, virtual machines.		
Text Books:		
1. Computer Forensics, Computer Crime Investigation by John R. Vacca, Firewall Media, New Delhi. 2. Computer Forensics and Investigations by Nelson, Phillips Enfinger, Steuart, CENGAGE Learning		
Reference Books:		
1. Real Digital Forensics by Keith J. Jones, Richard Bejtich, Curtis W. Rose, Addison Wesley Pearson Education 2. Forensic Compiling, A Tractitioneris Guide by Tony Sammes and Brian Jenkinson, Springer International edition. 3. Computer Evidence Collection & Presentation by Christopher L.T. Brown, Firewall Media. 4. Homeland Security, Techniques & Technologies by Jesus Mena, Firewall Media. 5. Software Forensics Collecting Evidence from the Scene of a Digital Crime by Robert M. Slade, TMH 2005 6. Windows Forensics by Chad Steel, Wiley India Edition		
Web References:		
1. https://ocw.cs.pub.ro/courses/_media/isc/lectures/isc_09_forensics.pdf 2. http://www.cse.chalmers.se/edu/course/EDA263/oh07/oh714_forensics.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=GqcdAwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=7CAADAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.edx.org/course/computer-forensics-3 2. https://www.cybrary.it/course/computer-hacking-forensics-analyst/		

DISCRETE STRUCTURES**VII Semester: OPEN ELECTIVE - III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT19	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

1. To learn basic data types and structures (such as numbers, sets, graphs, and trees) used in computer algorithms and systems;
2. To distinguish rigorous definitions and conclusions from merely plausible ones; synthesize elementary proofs, especially proofs by induction.
3. To model and analyze computational processes using analytic and combinatorial methods.
4. To apply principles of discrete probability to calculate probabilities and expectations of simple random processes.
5. To Work in small teams to accomplish all the objectives above.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Use logical notation to define and reason about fundamental mathematical concepts such as sets, relations, functions, and integers.
2. Synthesize induction hypotheses and simple induction proofs.
3. Prove elementary properties of modular arithmetic and explain their applications in Computer Science, for example, in cryptography and hashing algorithms.
4. Apply graph theory models of data structures and state machines to solve problems of connectivity and constraint satisfaction, for example, scheduling.
5. Calculate numbers of possible outcomes of elementary combinatorial processes such as permutations and combinations.

UNIT-I	MATHEMATICAL LOGIC	Classes: 15
Statements and Notations, Connectives, Statement Formulas and Truth Tables, well formed formulas, Tautologies, Equivalence of Formulas, Normal Forms: Disjunctive Normal Forms (DNF), Conjunctive Normal Forms (CNF), Principle Disjunctive Normal Forms (PDNF), Principle Conjunctive Normal Forms (PCNF).		
UNIT-II	PREDICATES	Classes: 15
The Predicate calculus, Free and Bound Variables, Rules of Inference, Consistency Of Premises and Indirect Method of Proof, Automatic Theorem Proving.		
UNIT-III	RELATIONS AND ORDERING	Classes: 15
Relations, Properties of Binary Relations in a Set, Equivalence Relations, Compatibility Relations, and Partial Ordering, Partial Ordered Set – Representation and Associated Terminology.		
FUNCTIONS: Definition and Introduction, Composition of Functions, Inverse Functions, Recursive Functions.		

UNIT-IV	TREES	Classes: 11
Trees, Properties of trees, Representation of trees, Binary trees, Tree traversals, spanning trees, DFS, BFS, Minimal spanning trees, Prim's algorithm and Kruskal's algorithm.		
UNIT-V	GRAPHS	Classes: 10
Representation of Graphs, Planar Graphs, Graph theory and applications. Basic concepts of Isomorphism and sub graphs, Multi graph and Euler circuits, Hamiltonian Graphs, chromatic number.		
Text Books:		
1. J. P. Trembly, R. Manohar (1997), Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, India. 2. Joe L. Mott, Abraham Kandel, Theodore P. Baker (2011), Discrete Mathematics for Computer Scientists and Mathematicians, 2nd edition, Prentice Hall of India Learning Private Limited, New Delhi, India.		
Reference Books:		
1. Kenneth H. Rosen (2007), Discrete Mathematics and its Applications, 6th edition, Tata McGraw Hill, India. 2. C.L. Liu, D.P. Mohapatra (2008), Elements of Discrete Mathematics, 3rd edition, McGraw Hill, India. 3. Ralph P. Grimaldi, B.V. Ramana (2006), Discrete and Combinatorial Mathematics - An Applied Introduction, 5th Edition, Pearson Education, India.		
Web References:		
1. https://web.stanford.edu/class/cs103x/cs103x-notes.pdf 2. https://www.cs.cornell.edu/~rafael/discmath.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=DYKikmfkcE0C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=D8hbO-b6jEcC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.mooc-list.com/course/discrete-structures-saylororg 2. https://online.lewisu.edu/mscs/courses/cpsc-50200		

INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT**VIII Semester: OPEN ELECTIVE - IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT20	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES

1. To learn how to produce apps for Android platform devices
2. To gain a basic understanding of computer architecture and object-oriented programming
3. To develop a working knowledge of Apple's Xcode app development tool
4. To understand mobile design principles
5. To identify need and opportunity in app markets

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Describe those aspects of mobile programming that make it unique from programming for other platforms,
2. Critique mobile applications on their design pros and cons,
3. Utilize rapid prototyping techniques to design and develop sophisticated mobile interfaces,
4. Program mobile applications for the Android operating system that use basic and advanced phone features, and
5. Deploy applications to the Android marketplace for distribution.

UNIT-I INTRODUCTION TO ANDROID**Classes: 10**

The Android Platform, Android SDK, Android Installation, Building you First Android application, Understanding Anatomy of Android Application, Android Manifest file.

UNIT-II ANDROID APPLICATION DESIGN ESSENTIALS**Classes: 08**

Android terminologies, Application Context, Activities.

UNIT-III ANDROID USER INTERFACE DESIGN ESSENTIALS**Classes: 08**

User Interface Screen elements, Designing User Interfaces with Layouts.

UNIT-IV TESTING ANDROID APPLICATIONS**Classes: 08**

Testing Android applications, Publishing Android application, Using Android preferences, managing Application resources in a hierarchy, working with different types of resources.

UNIT-V	USING COMMON ANDROID APIS	Classes: 10
Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers.		
Text Books:		
1. Lauren Darcey and Shane Conder, —Android Wireless Application DevelopmentII, Pearson Education, 2 nd ed. (2011)		
Reference Books:		
1. R1. Reto Meier, —Professional Android 2 Application DevelopmentII, Wiley India Pvt Ltd 2. R2. Mark L Murphy, —Beginning AndroidII, Wiley India Pvt Ltd. 3. R3. Android Application Development All in one for Dummies by Barry Burd, Edition.		
Web References:		
1. https://kmitit4yr1semmadmaterial201516.wordpress.com/ 2. https://www.smartzworld.com/notes/mobile-application-development-notes-pdf-mad-pdf-notes/		
E-Text Books:		
1. https://books.google.co.in/books?id=S7CllnNkg8UC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=zmnhdAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.classcentral.com/course/edx-introduction-to-mobile-application-development-using-android-3758 2. https://www.edx.org/learn/app-development		

DISTRIBUTED DATABASES**VIII Semester: OPEN ELECTIVE - IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT21	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To understand Distributed Database Management System and its Design issues
2. To Explore several algorithms for processing queries and be able to use them
3. To understand the methods to translate complex conceptual data models into logical and Physical database designs
4. To understand query optimization and its algorithms
5. To learn the concepts behind distributed transaction processing

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Demonstrate distributed database concepts and its structures.
2. Produce the transaction management and query processing techniques in DDBMS.
3. Explain the concepts of distributed object database management systems.
4. Relate the importance and application of emerging database technology.
5. Discuss about various databases such as mobile database, web database etc.

Pre-requisites: Basic concept of DBMS and RDBMS.

UNIT-I	INTRODUCTORY CONCEPTS AND DESIGN OF (DDBMS)	Classes: 12
Data Fragmentation; Replication, and allocation techniques for DDBMS; Methods for designing and implementing DDBMS, designing a distributed relational database; Architectures for DDBMS: cluster federated, parallel databases and client server architecture.		
UNIT-II	QUERY PROCESSING & TRANSACTION MANAGEMENT	Classes: 12
Overview Of Query Processing: Query processing problem; Objectives of Query Processing; Complexity of Relational Algebra operations; characterization of Query processors; Layers of Query Processing Introduction To Transaction Management: Definition of Transaction, Properties of Transaction, types of transaction ; Distributed Concurrency Control: Serializability theory; Taxonomy of concurrency control mechanisms; locking bases concurrency control algorithms.		
UNIT-III	DISTRIBUTED OBJECT DATABASE MANAGEMENT SYSTEMS	Classes: 08
Fundamental Object concepts and Object models, Object distribution design, Architectural issues, Object management, Distributed object storage, object query processing.		

UNIT-IV	CURRENT TRENDS & DEVELOPMENTS RELATED TO DISTRIBUTED DATABASE APPLICATIONS TECHNOLOGIES	Classes: 10
Distributed Object/component-based DBMS; Database Interoperability including CORBA; DCOM and Java RMI; Distributed document-based systems; XML and Workflow management.		
UNIT-V	EMERGING RELATED DATABASE TECHNOLOGIES	Classes: 08
Parallel Database Mobile database; Multimedia Database; Spatial Database and Web Databases.		
Text Books:		
1. Distributed Databases - Principles and Systems; Stefano Ceri; Giuseppe Pelagatti; Tata McGraw Hill; 1985. 2. Fundamental of Database Systems; Elmasri & Navathe; Pearson Education; Asia. 3. Database System Concepts; Korth & Sudarshan; TMH. 4. Principles of Distributed Database Systems; M. Tamer Özsu; and Patrick Valduriez Prentice Hall.		
Reference Books:		
1. Data Base Management System; Leon & Leon; Vikas Publications 2. Introduction to Database Systems; Bipin C Desai; Galgotia.		
Web References:		
1. https://edutechlearners.com/distributed-database-systems-notes/ 2. http://www.math-cs.gordon.edu/courses/cps352/lectures-2019/		
E-Text Books:		
1. https://books.google.co.in/books?id=0i9uJJ78LtAC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=EpUAG_HXpEqC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.coursera.org/learn/distributed-database 2. https://www.classcentral.com/course/coursera-distributed-database-systems-11170		

BIG DATA**VIII Semester: OPEN ELECTIVE - IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4IT22	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics and Visualization
3. To demonstrate the usage of various Big Data tools and Data Visualization tools

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Compare various file systems and use an appropriate file system for storing different types of data.
2. Demonstrate the concepts of Hadoop ecosystem for storing and processing of unstructured data.
3. Apply the knowledge of programming to process the stored data using Hadoop tools and generate reports.
4. Connect to web data sources for data gathering, Integrate data sources with hadoop components to process streaming data.
5. Tabulate and examine the results generated using hadoop components.

UNIT-I**INTRODUCTION TO BIG DATA****Classes: 12**

Data and its importance, Big Data- Definition, V's of Big Data, Hadoop Ecosystem

HADOOP ARCHITECTURE

Hadoop Storage : HDFS, Hadoop

Processing : Map Reduce Framework

Hadoop Server Roles : Name Node, Secondary Name Node and Data Node, Job Tracker, Task Tracker

HDFS-HADOOP DISTRIBUTED FILE SYSTEM

Design of HDFS, HDFS Concepts, HDFS Daemons, HDFS High Availability, Block Abstraction, FUSE-File System in User Space. HDFS Command Line Interface (CLI), Concept of File Reading and Writing in HDFS.

UNIT-II**MAPREDUCE PROGRAMMING MODEL****Classes: 12**

Introduction to Map Reduce Programming model to process BigData, key features of MapReduce, MapReduce Job skeleton, Introduction to MapReduce API, Hadoop Data Types, Develop MapReduce Job using Eclipse, build a MapReduce Job export it as a java archive(.jar file).

MAPREDUCE JOB LIFE CYCLE: How Mapreduce Works? Understanding Mapper ,Combiner, Partitioner ,Shuffle & Sort and Reduce phases of MapReduce Application, Developing Map Reduce Jobs based on the requirement using given datasets like weather dataset.

MAPREDUCE API: Understanding new MapReduce API from org.apache.hadoop.mapreduce and its sub packages to develop MapReduce applications ,key difference between old MapReduce API and the new MapReduce API.

UNIT-III	INTRODUCTION TO PIG	Classes: 12
Understanding pig and pig Platform, introduction to Pig Latin Language and Execution engine, running pig in different modes, Pig Grunt Shell and its usage. PIG LATIN LANGUAGE–DATA TYPES IN PIG Pig Latin Basics, Key words ,Pig Data types ,Understanding Pig relation, bag, tuple and writing pig relations or statements using Grunt Shell ,expressions, Data processing operators, using Built in functions. WRITING PIG SCRIPTS USING PIG LATIN: Writing pig scripts and saving them in text editor, running pig scripts from command line.		
UNIT-IV	INTRODUCTION TO HIVE	Classes: 12
Understanding Hive Shell, Running Hive, Understanding Schema on read and Schema on write. HIVE QL DATA TYPES, SEMANTICS: Introduction to Hive QL (Query Language), Language semantics, Hive Data Types. HIVE DDL, DML AND HIVE SCRIPTS: Hive Statements, Understanding and working with Hive Data Definition Languages and Manipulation Language statements, Creating Hive Scripts and running them from hive terminal and commands line.		
UNIT-V	SQOOP, FLUME, OOZIE	Classes: 12
SQOOP: Introduction to Sqoop tool, commands to connect databases and list databases and tables, command to import data from RDBMS into HDFS, Command to export data from HDFS into required tables of RDBMS. FLUME: Introduction to Flume agent, understanding Flume components Source, Channel and Sink, Writing flume configuration file, running flume configuration file to ingest the data into HDFS. OOZIE: Introduction to Oozie, Understanding work flow and how to create Work flow using Work Flow definition language in XML, running a basic Oozie workflow to run a Map Reduce job.		
Text Books:		
1. Hadoop: The Definitive Guide, 4th Edition - O'Reilly Media 2. Big Data and Hadoop- Learn by Example, Mayank Bhushan		
Reference Books:		
1. Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007. 2. Paul Zikopoulos ,Dirk DeRoos , Krishnan Parasuraman , Thomas Deutsch , James Giles , David Corigan , "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.		

Web References:

1. <https://web.stanford.edu/class/cs102/lecturenotes/Overview102.pdf>
2. <https://www.slideshare.net/mohitsainirke/big-data-lecture-notes>
3. <https://www.studocu.com/en/document/national-university-of-ireland-galway/data-science-and-big-data-analytics/lecture-notes/big-data-technologies-lecture-notes/2264642/view>

E-Text Books:

1. https://books.google.co.in/books?id=HpHcGAkFEjC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=a0VHDwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.coursera.org/specializations/big-data>
2. <https://www.edx.org/course/big-data-fundamentals>

**OPEN ELECTIVES
OFFERED BY
MECHANICAL ENGINEERING DEPARTMENT**

ELEMENTS OF MECHANICAL ENGINEERING**V Semester: OPEN ELECTIVE-I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME60	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course deals with fundamental concepts of design , manufacturing and thermal engineering.

COURSE OBJECTIVES:

The course aims to enable the students learn basic concepts of mechanics, materials, thermodynamics, manufacturing and CAD/CAM.

COURSE OUTCOMES:

After completing this course the student must demonstrate the knowledge and ability:

1. To demonstrate basic concepts of mechanics of rigid and deformable bodies.
2. To explain fundamentals of metallurgy and material science.
3. To discuss basic concepts and principles of thermo dynamics.
4. To demonstrate the elements of fabrication processes and equipment.
5. To explain basic concepts of CAD/CAM.

UNIT-I MECHANICS**Classes: 14**

MECHANICS OF RIGID BODIES: Force, Force system, Classification of force systems, triangle law, parallelogram law-derivation and problems, polygon law, resolution of forces in coplanar-concurrent force system-problems; Definition of friction, laws of friction, coefficient of friction, types of friction, angle of friction, angle of repose, definition of centroid and centre of gravity, pappus and gulldinus theorems, problems on centroid of plane areas; Definition of area moment of inertia, parallel axis theorem, perpendicular axis theorem, simple problems on area moment of inertia.

MECHANICS OF DEFORMABLE BODIES: Stress, Strain, Types of stresses and strain, Poisson's ratio, stress-strain curve of mild steel, Hooke's Law, factor of safety, mechanical properties of materials, bulk modulus, shear modulus, bars of uniform and varying sections-simple problems.

UNIT-II METALLURGY & MATERIAL SCIENCE**Classes: 12**

Bonds in Solids, crystallization of metals, Classification of steels, , Necessity of alloying, types of solid solutions, lever rule, phase rule, cooling curve of pure iron, Iron-Iron carbide diagram, Heat treatment- Annealing, quenching, normalizing, Hardening, tempering; Ceramic materials-classification, properties and applications.

UNIT-III	THERMO DYNAMICS	Classes: 12
System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry , First law of Thermodynamics – Corollaries, First law applied to a Closed System , applied to a flow system , Steady Flow Energy Equation and Limitations of the First Law, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements, Carnot cycle and its specialties.		
UNIT-IV	MANUFACTURING	Classes: 11
Casting-Types of casting, advantage of casting and its applications, defects in castings; Patterns - Pattern making, Types, Materials used for patterns, pattern allowances; Welding-classification, Heat affected zone in welding, welding defects-causes and remedies; Basic extrusion process and its characteristics, Forging operations and principles; Basic machining operations-drilling, milling, turning, step turning, threading, grinding and shaping ; Limits, fits and tolerances- Unilateral and bilateral tolerance system, hole and shaft basis system.		
UNIT-V	CAD/CAM	Classes: 11
Fundamentals of CAD/CAM- Benefits of CAD, Computer configuration for CAD applications, Computer peripherals for CAD; Wire frame modeling: Definition, advantages, dis-advantages, wire frame entities-analytic entities and synthetic entities; Surface modeling: Definition, advantages, dis-advantages, surface entities-analytic entities and synthetic entities; Solid Modeling: Definition, constructive solid geometry, advantages, modeling entities.		
Text Books:		
1. Singer's Engineering Mechanics by K.Vijay Kumar Reddy and J.Suresh Kumar, B.S Publications 2. Introduction to physical metallurgy by Avner, TATA McGrawHill Edition. 4. Thermodynamics by P.K Nag, McGrawHill Education. 5. Production Technology by R.K Jain, Khanna Publishers. 6. CAD/CAM Theory & Practice by Ibrahim Zeid & R.Sivasubramanian, McGrawHill Education.		
Reference Books:		
1. Engineering Mechanics By R.K.Bansal, Laxmi Publications 2. Material science and metallurgy by OP Khanna, Dhanpat Ray Publications 3. Engineering Thermodynamics By Pakirappa, V.Naveen Kumar, V.Naresh, Durga Publishing House 4. Production Technology by Dr. P.C Sharma, S.CHAND 5. CAD/CAM By P.Radha Krishnan, S.Subramanyan and V.Raju, New Age International Publishers. 6. Strength of Materials By Dr.Sadhu Singh, Khanna Publishers.		
Web References:		
1. https://www.iitg.ac.in/rkbc/me101/Presentation/L01-03.pdf 2. https://nptel.ac.in/courses/113106032/ 3. https://nptel.ac.in/downloads/112108148/ 4. https://lecturenotes.in/subject/195/cad-and-cam-cac		

E-Text Books:

1. https://books.google.co.in/books?id=8kA8BAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=XYorDAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course:

1. <https://ocw.mit.edu/courses/mechanical-engineering/>

FUNDAMENTALS OF ENGINEERING MATERIALS

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME61	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

This course will enable students to understand basic structure and crystal arrangement of materials, the phase diagrams, advantages of heat treatment, various heat treatment processes, the need and application of composite materials.

COURSE OUTCOMES:

At the end of course students are able to

- 1.Explain basic concepts of crystal structure such as unit cells, crystal systems of metals etc.
- 2.Demonstrate the concept of alloying and formation of different types of phases in alloys.
- 3.Differentiate ferrous and non ferrous alloys.
- 4.Explain various heat treatment processes.
- 5.Classify and explain polymers, ceramics and composites.

UNIT-I	CRYSTAL STRUCTURE	Classes: 10
Unit cells, crystal systems of metals, Imperfection in solids: Point, line, interfacial and volume defects; dislocation strengthening mechanisms and slip systems, determination of grain size, effect of grain size on the properties of alloys.		
UNIT-II	ALLOYS & PHASE DIAGRAMS	Classes: 10
Alloys- substitutional and interstitial solid solutions. Phase diagrams: Interpretation of binary phase diagrams and microstructure development; eutectic, peritectic, peritectoid and monotectic reactions. Iron Iron-carbide phase diagram and microstructure of ledeburite, austenite, ferrite and cementite.		
UNIT-III	FERROUS AND NON FERROUS ALLOYS	Classes: 12
Alloying of steel, properties of stainless steel and tool steels, maraging steels; cast irons-grey, white, malleable and spheroidal cast irons; copper and copper alloys- brass, bronze and cupro-nickel; Aluminium and Aluminium alloys.		
UNIT-IV	HEAT TREATMENT OF STEEL	Classes: 12
Annealing, tempering, normalizing and spheroidising, austempering, martempering, case hardening, carburizing, nitriding, cyaniding, carbo-nitriding, flame and induction hardening, vacuum and plasma hardening.		

UNIT-V	POLYMERS, CERAMICS AND COMPOSITES	Classes: 8
Classification, properties and applications of polymers, ceramics, composites and nano materials.		
Text Books:		
1. V. Raghavan, "Material Science and Engineering", Prentice Hall of India Private Limited, 1999. 2. U. C. Jindal, "Engineering Materials and Metallurgy", Pearson, 2011. 3. Sidney H. Avener (2007,) <i>Introduction to Physical Metallurgy</i>, 2nd edition, Tata McGraw hill education (P) Ltd, New Delhi, India.		
Reference Books:		
1. W. D. Callister, 2006, "Materials Science and Engineering-An Introduction", 6th Edition, Wiley India. 2. Kenneth G. Budinski and Michael K. Budinski, "Engineering Materials", Prentice Hall of India Private Limited, 4th Indian Reprint, 2002. 3. V. D. Kodgire (2006), <i>Material Science and Metallurgy for engineers</i>, 1st Edition, Everest, Pune, India.		
Web References:		
1. https://nptel.ac.in/courses/113106032/ 2. http://web.pdx.edu/~pmoeck/phy381/lecture1.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=XE8BAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=kBM8BAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://ocw.mit.edu/courses/materials-science-and-engineering/ 2. https://www.mooc-list.com/tags/materials-science		

FUNDAMENTALS OF THEORY OF MACHINES

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME62	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Theory of Machines may be defined as that branch of Engineering-science, which deals with the study of relative motion between the various parts of a machine, and forces which act on them. The knowledge of this subject is very essential for an engineer in designing the various parts of a machine.

COURSE OBJECTIVES:

This course aims to enable students to learn basic concepts of kinematics and dynamics of machinery.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Demonstrate various kinematic links, pairs, chains, mechanisms and their inversions.
2. Illustrate and analyze exact and approximate straight line motion mechanisms.
3. Explain concepts of higher pairs.
4. Demonstrate and analyze the effect of gyroscopic couples on stability of motor cycle, motor car and aeroplane.
5. Illustrate and analyze various governors.

UNIT-I	MECHANISMS	Classes: 14
Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematic pairs – Types of constrained motion – kinematic chain – Mechanism-Machine-Structure – Kleins criteria, Degrees of freedom, Grubler's Criterion, Grashof's law – inversion of mechanism – inversions of four bar chain, single and double slider crank chains.		
UNIT-II	STRAIGHT LINE MOTION MECHANISMS	Classes: 12
Exact, approximately copied and generated types – Peaucellier - Hart and Scott Russel – Grasshopper – Watt – Tchebichef's and Robert Mechanisms - Pantograph.		
UNIT-III	GEARS & GEAR TRAINS	Classes: 12
Friction wheels and toothed gears – types – law of gearing, condition for constant velocity ratio for transmission of motion, Form of teeth-cycloid and involute profiles, phenomena of interference.		
Gear trains: Introduction – Types – Simple, compound and reverted gear trains-simple problems.		

UNIT-IV	GYROSCOPES	Classes: 11
Gyroscopes – effect of precession – motion on the stability of moving vehicles such as motorcycle motorcar and aeroplane.		
UNIT-V	GOVERNORS	Classes: 11
Governors: Types of governors - Watt, Porter and Proell governors. Spring loaded governors – Hartnell and Hartung with auxiliary springs. (No derivations)		
Text Books:		
1. Theory of Machines by S.S.Rattan. 2. Theory of Machines by R.S.Khurmi		
Reference Books:		
1. Theory of Machines by Shigley, Mc Graw Hill Publishers 2. Theory of Machines by Thomas Bevan, CBS Publishers 3. Theory of Machines by R.K.Bansal (Lakshmi publications) 4. Mechanism and Machine Theory by JS Rao and RV Duggipati, New Age International		
Web References:		
1. http://nptel.ac.in/courses/112104121/ 2. https://freevideolectures.com/course/2359/kinematics-of-machines 3. https://nptel.ac.in/courses/112104114/ 4. https://freevideolectures.com/course/2364/dynamics-of-machines		
E-Text Books:		
2. https://books.google.co.in/books?id=T_WlquM4EI4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=YI7vAwAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://cosmolearning.org/courses/kinematics-of-machines 2. https://nptel.ac.in/courses/112101096/ 3. https://www.mooc-list.com/course/mechanics-kinematics-and-dynamics-edx		

FUNDAMENTALS OF MECHATRONICS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME63	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

The aim is to introduce students to the fundamental concepts and principles mechatronics. It builds upon the awareness and necessity of interdisciplinary dependency of 21st century. It aim is also to engage students to understand the introduction to mechatronics with emphasis on analog electronics, digital electronics, sensors and transducers, actuators, and microprocessors. Lectures are intended to provide the student with foundational concepts in mechatronics and practical familiarity with common elements making up mechatronic systems.

COURSE OBJECTIVES:

- 1) To understand the basic principles of mechatronics, Differentiating between conventional and mechatronics design process and product.
- 2) Differentiate various sensors, actuators, electromechanical systems.
- 3) To understand and apply interfacing of electromechanical products to microcontrollers and microprocessor.
- 4) To understand the working of various programmable logic controllers (PLCs), Architecture of PLCs.
- 5) Evaluate different mechatronic systems.

COURSE OUTCOMES:

At the end of the course students are able to

- 1) Demonstrate various elements underlying mechatronic systems, electronics, control systems and able differentiate their purpose in the system.
- 2) Analyze and select sensors, actuators, electromechanical components needed for an application.
- 3) To evaluate microprocessor and microcontroller interfacing to mechanical application.
- 4) To choose PLC to a mechanical application.
- 5) To design mechatronic systems for mechanical applications.

UNIT-I	INTRODUCTION	Classes: 12
Introduction to Mechatronics —Mechatronics systems - Mechatronics design process - Mechatronics in Manufacturing —Adaptive and distributed control systems – Modelling and simulation of mechatronics systems.		
UNIT-II	SENSORS AND ACTUATORS	Classes: 12
Sensors and actuators: Overview of sensors and transducers – Microsensors - Signal conditioning – Operational amplifiers – Protection – Filtering - Analog and Digital converters. Electro – pneumatics and Electro – hydraulics - Solenoids – Direct Current motors – Servomotors – Stepper motors - Micro actuators; Drives selection and application.		

UNIT-III	INTERFACING	Classes: 12
Interfacing: Microprocessor based Controllers Architecture of microprocessor and microcontroller – System interfacing for a sensor, keyboard, display and motors - Application cases for temperature control, warning and process control systems		
UNIT-IV	PLCs	Classes: 12
PLCs: Programmable Logic Controllers Architecture of Programmable Logic Controllers – Input/Output modules – programming methods – Timers and counters – Master control – Branching – Data handling – Analog input/output – Selection of PLC and troubleshooting.		
UNIT-V	ARTIFICIAL INTELLIGENCE	Classes: 12
AI: Intelligent Mechatronics and Case Studies Fuzzy logic control and Artificial Neural Networks in mechatronics – Algorithms – Computer – based instrumentation - Real-time Data Acquisition and Control – Software integration – Man Machine interface -Vision system – Mechatronics system case studies.		
Text Books:		
1. Introduction to Mechatronics and Measurement Systems, Tata McGraw Hill		
Reference Books:		
1. Designing Intelligent Machines, Michel B. Hsiao and David G. Alciatore, Open University London		
2. Control Sensors and Actuators, ICW. Desiha, Prentice Hall		
Web References:		
1. https://lecturenotes.in/subject/137/mechatronics-mech		
2. https://sites.google.com/site/profnarendralakal/Home/presentations/mtrx		
E-Text Books:		
1. https://books.google.co.in/books?id=NCEeONKWzX4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
2. https://books.google.co.in/books?id=z8yEUou06cEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://ocw.mit.edu/courses/mechanical-engineering/2-737-mechatronics-fall-2014/		
2. https://nptel.ac.in/courses/112103174/		
3. https://www.igi-global.com/chapter/massive-online-open-course-assisted-mechatronics-learning/137327		

BASICS OF THERMODYNAMICS								
VI Semester: OPEN ELECTIVE -II								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME64	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OVERVIEW: Thermodynamics is the field of physics that deals with the relationship between heat and work in a substance during a thermodynamic process. This course covers fundamental concepts of thermodynamics such as laws of thermodynamics, pure substances, perfect gas laws, mixture of perfect gases, Mollier charts and Psychrometry.								
COURSE OBJECTIVES: To understand the treatment of classical Thermodynamics and to apply the First and Second laws of Thermodynamics for the analysis of thermal equipment								
COURSE OUTCOMES: At the end of the course, the student should be able to 1. Demonstrate basic concepts of thermodynamics. 2. Explain laws of thermodynamics. 3. Explain pure substances and power cycles. 4. Explain perfect gas laws and concepts of mixture of perfect gases. 5. Discuss fundamental concepts of psychrometry.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction: System, Control Volume, Surrounding, Boundaries, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Exact & Inexact Differentials, Cycle – Reversibility – Quasi – static Process, Irreversible Process, Causes of Irreversibility – Energy in State and in Transition, Zeroth Law of Thermodynamics – Concept of Temperature – Principles of Thermometry								
UNIT-II	LAWS OF THERMO DYNAMICS						Classes: 12	
First law of Thermodynamics – Corollaries – First law applied to a Closed System – applied to a flow system – Steady Flow Energy Equation. Limitations of the First Law Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements and their Equivalence / Corollaries, Entropy, Principle of Entropy Increase – Energy Equation, Availability and Irreversibility – Elementary Treatment of the Third Law of Thermodynamics								
UNIT-III	PURE SUBSTANCES & POWER CYCLES						Classes: 12	
Pure Substances, p-V-T- surfaces, T-S and h-s diagrams, Mollier Charts, Phase Transformations – Triple point at critical state properties during change of phase, Dryness Fraction – Clausius – Clapeyron Equation Property tables. Power Cycles: Otto, Diesel, Dual Combustion cycles, Sterling Cycle – Description and representation on P-V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis – comparison of Cycles.								
UNIT-IV	PERFECT GAS LAWS & MIXTURE OF PERFECT GASES						Classes: 12	

Perfect Gas Laws – Equation of State, specific and Universal Gas constants – various Non-flow processes, properties, end states
 Mixtures of perfect Gases – Mole Fraction, Mass fraction Gravimetric and volumetric Analysis – Dalton's Law of partial pressure, Avogadro's Laws of additive volumes – Mole fraction, Volume fraction and partial pressure.

UNIT-V	PSYCHROMETRY
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Classes: 12

Psychrometric Properties – Dry bulb Temperature, Wet Bulb Temperature, Dew point Temperature, Thermodynamic Wet Bulb Temperature, Specific Humidity, Relative Humidity, saturated Air, Vapour pressure, Degree of saturation – Adiabatic Saturation, Carrier's Equation – Psychrometric chart.

Text Books:

1. Engineering Thermodynamics, P.K. Nag, 6th Edition, Mc Graw Hill Education.
2. Thermodynamics an engineering approach, Yunus A. Cengel & Michael A. Boles, 8th Edition, Mc Graw Hill Companies.
3. Fundamentals of Thermodynamics, Richard E. Sonntag Claus Borgnakke, 7th Edition, Wiley.

Reference Books:

1. Fundamentals of engineering thermodynamics, Rathakrishnan. E, 2nd Edition, Prentice hall of India Pvt Ltd., 2006.
2. Thermodynamics, Arora.C.P, Tata Mc Graw Hill, New Delhi.
3. Applied Thermodynamics, Onkar Singh, 3rd Edition, New Age, India

Web References:

1. <https://www.askiitians.com/revision-notes/physics/thermodynamics/>
2. <http://www.ncert.nic.in/ncerts/l/kech106.pdf>

E-Text Books:

3. https://books.google.co.in/books?id=91M72ZOb3n8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
4. https://books.google.co.in/books?id=w8GhW3J8RHIC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.coursera.org/learn/thermodynamics-intro>
2. <https://www.edx.org/course/thermodynamics-2>

FUNDAMENTALS OF SMART MATERIALS

VI Semester: OPEN ELECTIVE -II

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME65	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Smart materials are materials that sense and react to environmental conditions or stimuli (e.g., mechanical, chemical, electrical or magnetic signals). This course is designed to impart knowledge on fundamentals of various smart materials.

COURSE OBJECTIVES:

The objective of this course is to enable the students to learn principles of smart materials such as electronic materials, ionic polymer matrix composites etc.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various smart materials and their principles.
2. Explain different types of electro-active materials.
3. Explain principles of piezoelectricity.
4. Discuss principles of Magnetostriction.
5. Explain about various shape memory alloys.

UNIT-I	INTRODUCTION	Classes: 12
Overview of Smart Materials: Introduction and definition of Smart Materials, principles of smart materials, input – output decision ability, devices based on conductivity changes, devices based on changes in optical response, biological systems smart materials. Devices based on magnetization, artificial structures, surfaces, hetero structures, polycrystalline, amorphous, liquid crystalline materials		
UNIT-II	ELECTRO ACTIVE MATERIALS & PIEZO ELECTRICITY	Classes: 12
Electro-active Materials: Electronic Materials, Electro-active Polymers, Ionic Polymer Matrix Composites (IPMC)		
UNIT-III	PIEZOELECTRICITY	Classes: 12
Piezoelectricity: Principles of Piezoelectricity, Perovskite Piezoceramic Materials, Single Crystals vs Polycrystalline Systems, Piezoelectric Polymers		
UNIT-IV	MAGNETOSTRICTION	Classes: 12
Principles of Magnetostriction, Rare earth Magnetostrictive materials, Giant Magnetostriction and Magnetoresistance Effect, magnetorheological fluids, electrorheological materials and their applications		

UNIT-V	SHAPE MEMORY ALLOYS	CLASSES: 12
Shape Memory Alloys: Shape Memory Effect, Shape Memory Alloys, Shape Memory Polymers, Self-Sensing Piezoelectric Transducers, Energy Harvesting Materials, Self- healing materials.		
Text Books:		
1. B. Culshaw, Smart Structures and Materials, Artech House, 2000 2. P. Gauenzi, Smart Structures: Physical Behaviour, Mathematical Modelling and Applications, Wiley Publishers, 2009 3. W. G. Cady, Piezoelectricity, Dover Publication, 1964 4. M. Addington, Schodek, L. Daniel.: Smart materials and new technologies, Architectural Press, 2005 5. W.D Callister, Materials Science and Engineering an Introduction, Willey 1999		
Reference Books:		
1. M. Schwartz, Encyclopedia of Smart Materials, Volumes 1-2 ,Willey, 2002 2. Smith, C.: Smart material systems, Ralph, SIAM, 2005 3.K Vijay, K.Varadan, J. Vinoy, S.Gopalakrisham, Smart Material Systems and MEMS: Design and Development Methodologies , Willey 2006		
Web References:		
1. https://www.sciencedirect.com/topics/chemistry/smart-material 2. https://www.mide.com/collections/smart-materials		
E-Text Books:		
1. https://books.google.co.in/books?id=KdBr38J-bxoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=CQxwDQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://nptel.ac.in/noc/individual_course.php?id=noc18-me60 2. https://nptel.ac.in/courses/112104251/		

FUNDAMENTALS OF OPERATIONS RESEARCH

VII Semester: OPEN ELECTIVE-III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME66	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE DESCRIPTION:

Operations research (OR) is an analytical method of problem-solving and decision-making that is useful in the management of organizations. In operations research, problems are broken down into basic components and then solved in defined steps by mathematical analysis. This course gives insight of Linear Programming, Transportation, assignment problems, sequencing etc.

COURSE OBJECTIVES:

The objectives of this course are to learn quantitative methods and techniques for effective decisions-making; model formulation and applications that are used in solving business decision problems.

COURSE OUTCOMES:

At the end of course students will be able to

1. Describe types of models and solve linear programming problem.
2. Solve transportation and assignment problems.
3. Analyze sequencing and replacement models and apply them for optimization.
4. Apply gaming theory for optimal decision making.
5. Analyze inventory models to optimize the cost

UNIT-I	INTRODUCTION & ALLOCATION	Classes: 12
Development – Definition– Characteristics and Phases – Types of models – Operations Research models – applications. ALLOCATION: Linear Programming Problem - Formulation – Graphical solution – Simplex method – Artificial variables techniques: Two–phase method, Big-M method; Duality Principle.		
UNIT-II	TRANSPORTATION & ASSIGNMENT PROBLEMS	Classes: 12
TRANSPORTATION PROBLEM: Formulation – Optimal solution, unbalanced transportation problem – Degeneracy. ASSIGNMENT PROBLEM: Formulation – Optimal solution - Variants of Assignment Problem; Traveling Salesman problem.		

UNIT-III	SEQUENCING & REPLACEMENT	Classes: 12
SEQUENCING: Introduction – Flow –Shop sequencing – n jobs through two machines – n jobs through three machines REPLACEMENT: Introduction – Replacement of items that deteriorate with time – when money value is not counted and counted – Replacement of items that fail completely.		
UNIT-IV	THEORY OF GAMES	Classes: 12
THEORY OF GAMES: Introduction –Terminology– Solution of games with saddle points and without saddle points- 2 x 2 games –m x 2 & 2 x n games - graphical method – m x n games - dominance principle.		
UNIT-V	INVENTORY	Classes: 12
INVENTORY: Introduction – Single item, Deterministic models – Types - Purchase inventory models with one price break and multiple price breaks –Stochastic models – demand discrete variable or continuous variable – Single Period model with no setup cost.		
Text Books:		
1. Operation Research by J.K.Sharma, MacMilan. 2. Operations Researchby ACS Kumar, Yesdee		
Reference Books:		
1. Operations Research: Methods and Problems by Maurice Saseini, Arhur Yaspan and Lawrence Friedman 2. Operations Research by A.M.Natarajan, P.Balasubramaniam, A. Tamilarasi, Pearson Education. 3. Operations Research by Wagner, PHI Publications. 4. Introduction to O.Rby Hillier & Libermann, TMH.		
Web References:		
1. http://www.cs.toronto.edu/~stacho/public/IEOR4004-notes1.pdf 2. http://www.mathcity.org/msc/notes/operation_research 3. http://biobharati.com/admin/img/gall/1467763196_BCA-504.pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=6khDDAAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=cwoWJrgCrP8C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 3. https://books.google.co.in/books?id=rj6bBMVzfPsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		

MOOC Course:

1. <http://www.orcomplete.com/internet/enesbilgin/open-courses-on-operations-research>
2. <https://www.coursera.org/learn/wharton-operations>

FUNDAMENTALS OF ENERGY SOURCES AND APPLICATIONS

VII Semester: OPEN ELECTIVE-III

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME67	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course focuses on various renewable energy sources such as solar energy, wind energy, geothermal energy etc.

COURSE OBJECTIVES:

The objectives of this course are to study classifications and comparison of various renewable energy sources, flat plate and concentrating collectors, wind energy, principles of bio mass, geothermal and ocean energy sources and principles of direct energy conversion.

COURSE OUTCOMES:

At the end of the course, the student shall be able to

1. Classify renewable energy sources and explain environmental impact of solar energy, solar radiation, their measurement and instruments.
2. Classify concentrating collectors and explain in detail about various solar gadgets.
3. Explain about wind mill, types of wind mills, its construction, wind energy conversion systems and principles of bio mass.
4. Explain in detail geothermal and ocean energies.
5. Demonstrate the concepts of direct energy conversion.

UNIT-I	INTRODUCTION	Classes: 12
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Introduction – Renewable energy sources – Limitations – Classifications – Comparison – Global and Indian energy scenario. Solar Energy: Introduction – Environmental impact of solar power – extraterrestrial and terrestrial solar radiation – solar radiation on tilted surface – Solar Radiation Measurement and Instruments

UNIT-II	SOLAR ENERGY	Classes: 12
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Flat plate and concentrating collectors – classification of concentrating collectors – orientation and thermal analysis – advanced collectors. Solar Gadgets: Solar furnace, Solar air heater, Solar water heater, Soar cooker, solar still, solar pond, solar pump, solar dryer, solar cell.

UNIT-III	WIND ENERGY & BIO MASS	Classes: 12
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Wind energy utilization: Introduction – Origin and nature of winds – Types of wind mills and its construction – Wind energy conversion systems – performance characteristics.
Bio-Mass: Principles of Bio-Conversion, Anaerobic/aerobic digestion, types of Bio-gas digesters, combustion characteristics of bio-gas, applications and economic aspects.

UNIT-IV	GEO THERMAL ENERGY & OCEAN ENERGY	Classes: 12
Geothermal Energy: Resources, types of wells, methods of harnessing the energy, potential in India. Ocean Energy: OTEC, Principles utilization, setting of OTEC plants. Tidal and wave energy: Potential and conversion techniques, mini-hydel power plants, and their economics.		
UNIT-V	DIRECT ENERGY CONVERSION	Classes: 12
Direct Energy Conversion: Need, limitations, principles of DEC. Thermo-electric generators, seebeck, peltier and Joule-Thomson effects – applications, MHD generator principle, Fuel cells: Classification of fuel cells; Operating principles. Bio-fuels, Urban waste to energy conversion		
Text Books:		
1. Rai G.D, Non-Conventional Energy Sources, Khanna Publishers, New Delhi, 2011. 2. Twidell, J.W. & Weir, A., Renewable Energy Sources, EFN Spon Ltd., UK, 2006		
Reference Books:		
1. Ashok V Desai, <i>Non-Conventional Energy</i> , Wiley Eastern Ltd, New Delhi, 2003. 2. Mittal K M, <i>Non-Conventional Energy Systems</i> , Wheeler Publishing Co. Ltd, New Delhi, 2003. 3. Khan B H, <i>Non-Conventional Energy Resources</i> , Tata McGraw Hill, 2 nd Edn. New Delhi, 2009. 4. Ramesh R & Kumar K U, <i>Renewable Energy Technologies</i> , Narosa Publishing House, New Delhi, 2004 5. Tiwari. G.N., <i>Solar Energy – Fundamentals Design, Modelling & Applications</i> , Narosa Publishing House, New Delhi, 2002.		
Web References:		
1. http://www.afdc.energy.gov/fuels/natural_gas.html 2. https://cleancities.energy.gov/ 3. http://www.firmgreen.com/ 4. https://www.renewableresourcescoalition.org/alternative-energy-sources/ 5. https://www.edx.org/learn/renewable-energy		
E-Text Books:		
1. https://www.sciencedirect.com/book/9780126561531/renewable-energy 2. https://www.pdfdrive.com/renewable-energy-resources-e14705840.html 3. https://www.pdfdrive.com/search?q=renewable&pagecount=&pubyear=&searchin=		
MOOC Course:		
1. https://www.coursera.org/learn/renewable-energy-entrepreneurship 2. https://www.coursera.org/learn/wind-for-renewable-energies 3. https://online-learning.tudelft.nl/courses/sustainable-energy-design-a-renewable-future/		

BASICS OF ROBOTICS**VII Semester: OPEN ELECTIVE-III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME68	OEC	L	T	P	C	CIE	SEE	Total
		3		-	3	30	70	100

COURSE OVERVIEW:

Today robot finds applications in industries, medical and other fields. For example, in eye surgery (replacement of retina), where a cylindrical portion needs to be replaced, the operation is best done by robots. Mobile robots like walking machines, hopping machines are examples of robots, Nuclear and power plants use fish like robots which move inside pipes for purpose of inspection. This course focuses on various types of industrial robots, their kinematic and kinetic aspects, different types of grippers, mechanics of grippers, trajectory planning etc.

COURSE OBJECTIVES:

The objectives of this course are to

1. Study about basic components, architecture, anatomy and specifications of robot.
2. Study about various end effectors and actuators.
3. Study homogeneous transformation, D-H notation and direct kinematics
4. Study Inverse kinematics and Trajectory planning techniques
5. Study feedback components and applications of robots.

COURSE OUTCOMES:

At the end of the course, the student will be able to

1. Demonstrate different types of robots, specifications of robots and different end effectors used in robots.
2. Explain various types of end effectors
3. Evaluate rotation matrices, forward kinematics of RR, RP and 3R Manipulators.
4. Explain inverse kinematics of RR manipulator, RP manipulator and trajectory planning techniques.
5. Explain feedback components used in robots and industrial applications.

UNIT-I	INTRODUCTION	Classes: 12
Introduction: Automation and Robotics, Asimov's laws, Robot Architecture, Components, , Anatomy of robot, Factors to be considered in the selection of robot, present and future applications, Specifications - Degree of freedom, Pay load, Parts per hour, Accuracy, Repeatability, Speed, Work space, Work volume, Work envelope, classification of robots based on configuration and control systems		
UNIT-II	END EFFECTORS & ACTUATORS	Classes: 12
End effectors: Mechanical and Non-mechanical grippers, requirements for the design of grippers, considerations for the selection of grippers, Types of actuation mechanisms.		
Actuators: Pneumatic, Hydraulic actuators, electric & stepper motors, comparison of Actuators		

UNIT-III	MOTION ANALYSIS & DIRECT KINEMATICS	Classes: 12
Motion Analysis: Basic Rotation Matrices, Composite Rotation Matrices. Homogeneous transformations as applicable to rotation and translation – problems. Manipulator Kinematics: D-H notation, D-H method of Assignment of frames, D-H Transformation Matrix, joint coordinates and world coordinates, Forward kinematics of 2R, RP and 3R manipulators		
UNIT-IV	INVERSE KINEMATICS & TRAJECTORY PLANNING	Classes: 12
Inverse kinematics : Inverse kinematics of 2R and RP manipulators. Trajectory Planning: Definition of Trajectory planning, Path, Trajectory, Knot points, Steps involved in trajectory planning, Trajectory planning techniques-Joint space and Cartesian space techniques, Cubic polynomial trajectory		
UNIT-V	FEEDBACK COMPONENTS & APPLICATIONS	Classes: 12
Feedback Components: Position sensors – potentiometers, resolvers, optical encoders, Velocity sensor, Contact Sensors-Touch sensors, Tactile and Range sensors, Force and Torque sensors, Proximity sensor, Inductive sensor. Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading- Processing - spot and continuous arc welding & spray painting - Assembly and Inspection.		
Text Books:		
1. Industrial Robotics by Groover M P, Pearson Edu. 2. Robotics by Fu K S, McGraw Hill. 3. Theory of Applied Robotics (kinematics,Dynamics and Control-Jazar, Springer.		
Reference Books:		
1. Robotics and Control by Mittal R K & Nagrath I J, TMH. 2. Robot Dynamics and Controls by Spony and Vidyasagar, John Wiley 3. Robot Analysis and control by Asada and Slotine, Wiley Inter-Science 4. Introduction to Robotics by John J Craig, Pearson Education		
Web References:		
1. https://www.sciencedaily.com/terms/industrial_robot.htm 2. https://www.robotics.org/robotics/industrial-robot-industry-and-all-it-entails		
E-Text Books:		
1. https://books.google.co.in/books?id=dr9IAI7wucUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=rESVUHwMcvYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course:		
1. https://www.classcentral.com/tag/robotics 2. https://www.mooc-list.com/tags/robotics		

INTRODUCTION TO MATERIAL HANDLING

VIII Semester: OPEN ELECTIVE-IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME69	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

Material handling is the movement, protection, storage and control of materials and products throughout manufacturing, warehousing, distribution, consumption and disposal. As a process, material handling incorporates a wide range of manual, semi-automated and automated equipments and systems that support logistics and make the supply chain work. This course will give an insight material handling equipment such as transport systems and storage systems, various inventory models and forecasting techniques.

COURSE OBJECTIVES:

The aim of this course is to enable the students to learn basic concepts of material handling, consideration in material handling system, materials transport and storage system, various inventory models and forecasting techniques.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate basic concepts and principles of material handling.
2. Discuss about various material transport systems.
3. Explain conventional and automated storage systems.
4. Explain various inventory models.
5. Demonstrate different forecasting techniques.

UNIT-I	INTRODUCTION	Classes: 12
Introduction: Overview of material handling equipment, considerations in material handling system design, The 10 principles of material handling.		
UNIT-II	MATERIAL TRANSPORT SYSTEMS	Classes: 12
Material Transport Systems: Industrial trucks, Automated guided vehicle systems, Monorails and other rail guided vehicles, conveyor systems, cranes and hoists, Analysis of material transport systems.		
UNIT-III	STORAGE SYSTEMS	Classes: 12
Storage Systems: Storage system performance, storage location strategies, conventional storage methods and equipment, automated storage systems and analysis of automated storage systems.		
UNIT-IV	INVENTORY	Classes: 12
Inventory: Introduction, Inventory management, Static Inventory models, ABC Analysis, VED analysis, Just in Time, Aggregate inventory analysis, Inventory models with quantity discounts.		

UNIT-V	FORECASTING	Classes: 12
Forecasting: Forecasting techniques, Make or Buy decision, Acceptance sampling, Materials requirement planning, objectives of master schedule.		
Text Books:		
1. Mikell P. Groover, Automation Production Systems and Computer Integrated Manufacturing, Pearson International Edition. 2. A.K Singh, Materials Management, Laxmi Publications (P) Ltd.		
Reference Books:		
1. Dr.K.C Arora, Vikas V.Shinde, Aspects of Material Handling, Laxmi Publications (P) Ltd. 2. Charles Reese, Material Handling Systems, Taylor & Francis Network		
Web References:		
1. https://www.scribd.com/doc/222647028/Material-Handling-Full-Notes 2. http://users.encs.concordia.ca/~andrea/indu421/Presentation%207%20(MH).pdf		
E-Text Books:		
1. https://books.google.co.in/books?id=AN8dzzlp1wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=d0mhPIDEm0gC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.mooc-list.com/course/materials-and-materials-processing-saylororg 2. https://www.mooc-list.com/tags/material-processing		

NON CONVENTIONAL ENERGY SOURCES

VIII Semester: OPEN ELECTIVE-IV

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME70	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OVERVIEW:

The energy has become an important and one of the basic infrastructures for the economic development of the country. It is imperative for the sustained growth of the economy. This course envisages the new and renewable source of energy, available in nature and to expose the students on sources of energy crisis and the alternates available, also stress up on the application of non-conventional energy technologies.

COURSE OBJECTIVES:

To impart the knowledge of basics of different non conventional types of power generation & power plants in detail so that it helps them in understanding the need and role of Non-Conventional Energy sources particularly when the conventional sources are scarce in nature.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate various energy sources and their availability.
2. Explain concepts and applications of solar radiation
3. Discuss elaboratively the principles of wind energy conversion
4. Explain Biomass conversion techniques and Geothermal Sources and resources
5. Explain about tidal energy and Ocean Thermal Energy.

UNIT- I	ENERGY SOURCES & THEIR AVAILABILITY	Classes: 12
Energy Sources & their Availability - Importance of Non Conventional Energy Resources - Classification of NCES - Solar, Wind, Geothermal, Bio-mass, Nuclear Energy & Fuel Cells, Ocean Energy Sources, Comparison of these Energy Sources, Prospects of Renewable Energy Sources - Criteria for Assessing the Potential of NCES, Statistics on Conventional Energy Sources and supply in Developing Countries		
UNIT- II	INTRODUCTION TO SOLAR RADIATION & APPLICATIONS OF SOLAR ENERGY	Classes: 15
Introduction to Solar radiation - beam and diffuse radiation, solar constant, earth sun angles, attenuation and measurement of solar radiation, local solar time, derived solar angles, sunrise, sunset and day length. Solar Energy Storage – Collectors: Flat plate and Concentrating collectors, collector efficiency, Focusing and non-focusing type Applications of Solar Energy - solar water heater- Solar Cooker-Box type- Solar dryer - solar greenhouse— Summer and winter greenhouse-solar electric power generation- Solar Photo- Voltaic, Solar Cell Principle, Conversion efficiency and power output, Basic Photo Voltaic System for Power Generation.		

UNIT- III	PRINCIPLE OF WIND ENERGY CONVERSION	Classes: 12
Principle of Wind Energy Conversion - Basic components of Wind Energy Conversion Systems : Wind data and energy estimation, Site Selection Considerations - Wind mill Components, Various Types and their Constructional Features - Effect of Density, Frequency Variances, Angle of attack, and Wind Speed - Design Considerations of Horizontal and Vertical Axis Wind Machines - Analysis of Aerodynamic Forces Acting on Wind Mill Blades and Estimation of Power Output.		
UNIT- IV	BIOMASS CONVERSION TECHNIQUES , GEOTHERMAL SOURCES AND RESOURCES	Classes: 15
Biomass conversion techniques - Biogas Generation - Factors affecting biogas Generation-Types of biogas plants - Advantages and disadvantages of biogas plants - site selection, digester design consideration, filling a digester for starting, maintaining biogas production, Fuel properties of bio gas, utilization of biogas - urban waste to energy conversion. Geothermal Sources and resources like hydrothermal, geo-pressured hot dry rock, magma. advantages, disadvantages and application of geothermal energy, prospects of geothermal energy in India - exhaust types of conventional steam turbines.		
UNIT- V	TIDAL ENERGY & OCEAN THERMAL ENERGY CONVERSION	Classes: 12
Tidal Energy - Principle of working, performance and limitations. Wave Energy - Principle of working, performance and limitations. Ocean Thermal Energy Conversion - Availability, theory and working principle, performance and limitations. OTEC power plants, Operational of small cycle experimental facility, Economics of OTEC, Environmental impacts of OTEC.		
Text Books:		
1. G. D. Rai, "Non-Conventional Energy Sources", 4th Edition, Khanna Publishers, 2000 1. B H Khan, " Non-Conventional Energy Resources", 2nd Edition, Tata Mc Graw Hill Education Pvt Ltd, 2011		
Reference Books:		
1. S.Hasan Saeed and D.K.Sharma , "Non-Conventional Energy Resources", 3rd Edition, S.K.Kataria & Sons, 2012 2. Renewable energy sources and conversion technology by N.K. Bansal, M. Kleemann, M. Heliss, Tata McGraw Hill 1990. 3. Ashok V Desai, Non-Conventional Energy, Wiley Eastern Ltd, New Delhi, 2003 4. K M, Non-Conventional Energy Systems, Wheeler Publishing Co. Ltd, New Delhi, 2003 5. S. P. Sukhatme", "Solar Energy Principles and Application", TMH, 2009		
Web References:		
1. https://lecturenotes.in/subject/57/non-conventional-energy-systems-nces 2. https://www.pdfdrive.com/non-conventional-energy-sources-e10086374.html 3. http://www.vssut.ac.in/lecture_notes/lecture1428910296.pdf		
E-Text Books:		
1. http://nptel.ac.in/courses/121106014 2. http://nptel.ac.in/courses/121106014/13 3. http://nptel.ac.in/courses/12110601427		
MOOC Course:		

1. <https://nptel.ac.in/courses/121106014/>
2. <https://nptel.ac.in/syllabus/121106014/>
3. <https://nptel.ac.in/downloads/121106014/>

ASPECTS OF HEAT TRANSFER**VIII Semester: OPEN ELECTIVE-IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4ME22	OEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To understand the mechanisms of heat transfer under steady and transient conditions.
2. To understand the concepts of heat transfer through free and forced convection.
3. To learn the thermal analysis and sizing of heat exchangers.
4. To understand the principles of radiation heat transfer.

COURSE OUTCOMES:

- 1 Demonstrate basic modes of heat transfer and compute temperature distribution in steady state and unsteady state heat conduction.
- 2 Analyze the heat transfer through variable thermal conductivity, system with heat generation and extended surfaces.
- 3 Interpret and analyze the free and forced convection heat transfer.
- 4 Comprehend the concepts of free convection and heat exchangers.
- 5 Explain the principles of heat transfer with phase change.

UNIT-I	INTRODUCTION	Classes: 09
Introduction: Modes and mechanisms of heat transfer, Basic laws of heat transfer, Applications of heat transfer. Fourier rate equation, general heat conduction equations in Cartesian, Cylindrical and Spherical coordinates. Different forms of the field equation, steady, unsteady and periodic heat transfer – Initial and boundary conditions.		
UNIT-II	ONE DIMENSIONAL STEADY STATE HEAT CONDUCTION	Classes: 15
One dimensional steady state heat conduction: Homogeneous slabs, hollow cylinders and spheres, Overall heat transfer coefficient, Electrical analogy, Critical radius of insulation. Variable thermal conductivity and Systems with internal heat generation. Extended surfaces (Fins), Long, Short and insulated tips.		
UNIT-III	CONVECTIVE HEAT TRANSFER	Classes: 12
Convective Heat Transfer: Classification of systems based on causation of flow, condition of flow, configuration of flow and medium of flow – Dimensional analysis as a tool for experimental investigation – Buckingham Theorem and method, application for developing semi – empirical non-dimensional correlation for convection heat transfer – Significance of non-dimensional numbers – Concepts of Continuity, Momentum and Energy Equations.		
UNIT-IV	FREE CONVECTION & HEAT EXCHANGERS	Classes: 12

Free Convection: Development of Hydrodynamic and thermal boundary layer along a vertical plate - Use of empirical relations for Vertical plates and pipes.

Heat Exchangers: Classification of heat exchangers – overall heat transfer Coefficient and fouling factor .

UNIT-V	RADIATION HEAT TRANSFER	Classes: 12
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Radiation Heat Transfer : Emission characteristics and laws of black-body radiation – Irradiation – total and monochromatic quantities – laws of Planck, Wien, Kirchoff, Lambert, Stefan and Boltzmann– heat exchange between two black bodies – concepts of shape factor – Emissivity – heat exchange between grey bodies – radiation shields – electrical analogy for radiation networks.

Text Books:

1. Yunus A. Cengel (2012), Heat Transfer a Practical Approach, 4th edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
2. R. C. Sachdeva (2012), Fundamentals of Engineering, Heat and Mass Transfer, 3rd edition, New Age, New Delhi, India.

Reference Books:

1. Holman (2012), Heat Transfer (SI Units), 10th edition, Tata McGraw hill education (P) Ltd, New Delhi, India.
2. P. S. Ghoshdastidar (2012), Heat Transfer, 2nd edition, Oxford University Press, New Delhi, India.
3. Incropera, Dewitt (2012), Fundamentals of Heat Transfer, 6th edition, John Wiley, UK.

Web References:

1. <http://nptel.ac.in/courses/112104113/>
2. <http://nptel.ac.in/courses/112105123/>
3. <https://www.udemy.com/fundamentals-of-engineering-heat-transfer>

E-Text Books:

1. <https://www.pdfdrive.net/heat-transfer-e18770201.html>
2. <https://www.pdfdrive.net/fundamentals-of-heat-transfer-e18762062.html>
3. <https://www.pdfdrive.net/heat-transfer-e15057935.html>

MOOC Course

1. <https://onlinecourses.nptel.ac.in/>
2. <https://www.coursera.org/learn/heat-transfer-intro>
3. <https://swayam.gov.in/courses/4642-july-2018-conduction-and-convection>

**OPEN ELECTIVES
OFFERED BY
HS DEPARTMENT**

BUSINESS ECONOMICS AND FINANCIAL ANALYSIS**V Semester: OPEN ELECTIVE-I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A4HS10	OEC	3	0	0	3	30	70	100

COURSE OBJECTIVES:

To enable the student to understand and appreciate, with a particular insight, the importance of certain basic issues governing the business operations namely; demand and supply, production function, cost analysis, markets, forms of business organizations, capital budgeting and financial accounting and financial analysis.

COURSE OUTCOMES:

At the end of the course, the student will

1. Describe nature and scope of managerial economics, demand forecasting, elasticity of demand and supply, pricing methods and pricing in different market structures.
2. Analyze how production function is carried out to achieve least cost combination of inputs and cost analysis.
3. Analyze how capital budgeting decisions are carried out.
4. Explain the framework for both manual and computerised accounting process
5. Analyze and interpret the financial statements through ratio analysis.

UNIT-I	INTRODUCTION & DEMAND ANALYSIS	Classes: 12
Definition, Nature and Scope of Managerial Economics. Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of Demand: Definition, Types, Measurement and Significance of Elasticity of Demand. Demand Forecasting, Factors governing demand forecasting, methods of demand forecasting.		
UNIT-II	PRODUCTION FUNCTION & COST ANALYSIS	Classes: 12
Production Function - Isoquants and Isocosts, MRTS, Least Cost Combination of Inputs, Cobb-Douglas Production function, Laws of Returns, Internal and External Economies of Scale. Cost Analysis: Cost concepts. Break-even Analysis (BEA)-Determination of Break-Even Point (simple problems) - Managerial Significance.		
UNIT-III	MARKETS & NEW ECONOMIC ENVIRONMENT	Classes: 12
Types of competition and Markets, Features of Perfect competition, Monopoly and Monopolistic Competition. Price-Output Determination in case of Perfect Competition and Monopoly. Pricing: Objectives and Policies of Pricing. Methods of Pricing. Business: Features and evaluation of different forms of Business Organisation: Sole Proprietorship, Partnership, Joint Stock Company, Public Enterprises and their types, New Economic Environment: Changing Business Environment in Post-liberalization scenario.		
UNIT-IV	CAPITAL BUDGETING	Classes: 12

Capital and its significance, Types of Capital, Estimation of Fixed and Working capital requirements, Methods and sources of raising capital - Trading Forecast, Capital Budget, Cash Budget. Capital Budgeting: features of capital budgeting proposals, Methods of Capital Budgeting: Payback Method, Accounting Rate of return (ARR) and Net Present Value Method (simple problems).		
UNIT-V	INTRODUCTION TO FINANCIAL ACCOUNTING & FINANCIAL ANALYSIS	Classes: 12
Accounting concepts and Conventions - Introduction IFRS - Double - Entry Book Keeping, Journal, Ledger, Trial Balance - Final Accounts (Trading Account, Profit and Loss Account and Balance Sheet with simple adjustments). Financial Analysis: Analysis and Interpretation of Liquidity Ratios, Activity Ratios, and Capital structure Ratios and Profitability ratios. Du Pont Chart.		
Text Books:		
1. Varshney & Maheswari: Managerial Economics, Sultan Chand, 2009. 2. S.A. Siddiqui & A.S. Siddiqui, Managerial Economics and Financial Analysis, New Age international Publishers, Hyderabad 2013. 3. M. Kasi Reddy & Saraswathi, Managerial Economics and Financial Analysis, PHI New Delhi, 2012.		
Reference Books:		
1. Ambrish Gupta, Financial Accounting for Management, Pearson Education, New Delhi, 2012. 2. H. Craig Peterson & W. Cris Lewis, Managerial Economics, Pearson, 2012. 3. Lipsey & Chrystel, Economics, Oxford University Press, 2012. 4. Domnick Salvatore: Managerial Economics In a Global Economy, Thomson, 2012. 5. Narayanaswamy: Financial Accounting - A Managerial Perspective, Pearson, 2012. 6. S.N. Maheswari & S.K. Maheswari, Financial Accounting, Vikas, 2012. 7. Truet and Truet: Managerial Economics: Analysis, Problems and Cases, Wiley, 2012. 8. Dwivedi: Managerial Economics, Vikas, 2012. 9. Shailaja & Usha: MEFA, University Press, 2012. 10. Aryasri: Managerial Economics and Financial Analysis, TMH, 2012. 11. Vijay Kumar & Appa Rao, Managerial Economics & Financial Analysis, Cengage 2011. 12. J.V. Prabhakar Rao & P.V. Rao, Managerial Economics & Financial Analysis, Maruthi Publishers, 2011.		
Web References:		
1. https://16ceea.wordpress.com/business-economics-and-financial-analysis/ 2. https://lecturenotes.in/notes/15627-note-for-managerial-economics-and-financial-analysis-mefa-by-jntu-heroes		
E-Textbooks:		
1. https://books.google.co.in/books?id=QSEjlVMqhYC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=Wd5Fu_ySTBEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.coursera.org/specializations/managerial-economics-business-analysis 2. https://www.udemy.com/the-complete-financial-analyst-course/		

BASICS OF ENTREPRENEURSHIP

V Semester: OPEN ELECTIVE-I

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS11	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The purpose of the course is that the students acquire necessary knowledge and skills required for organizing and carrying out entrepreneurial activities, to develop the ability of analyzing and understanding business situations in which entrepreneurs act and to master the knowledge necessary to plan entrepreneurial activities.

COURSE OUTCOMES:

1. Describe ways to discover a problem worth solving.
2. Identify, track and test their assumptions in the field with customers and pivot if the business model is wrong.
3. Explain how to finance business ideas, various sources of funds to an entrepreneur with their pros and cons and role of a good team in a venture's success
4. Outline various strategies for reaching customers with products and services.
5. Demonstrate business regulations of starting and operating a business

UNIT-I	DISCOVER YOURSELF AND IDENTIFY PROBLEMS WORTH SOLVING	Classes: 10
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Discover Yourself: Find your flow, Effectuation, Case Study: Tristan Walker: The extroverted introvert, Identify your entrepreneurial style,
Identify Problems Worth Solving: What is a business opportunity and how to identify it, Find problems around you that are worth solving, Methods for finding and understanding problems - (Observation, Questioning, DT, Jobs to be done (JTBD), How to run problem interviews to understand the customer's worldview, Introduction to Design Thinking – Process and Examples,
Generate ideas that are potential solutions to the problem identified – DISRUPT, Class Presentation: Present the problem you "love"

UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION	Classes: 10
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CUSTOMER: Identify Your Customer Segments and Early Adopters - The difference between a consumer and a customer (decision maker), Market Types, Segmentation and Targeting, Defining the personas; Understanding Early Adopters and Customer Adoption Patterns, Identify the innovators and early adopters for your start-up; Craft Your Value Proposition - Come up with creative solutions for the identified problems, Deep dive into Gains, Pains and "Jobs-To- Be-Done" (using Value Proposition Canvas, or VPC), Identify the UVP of your solution using the Value Proposition section of the VPC, Outcome-Driven Innovation, Class Presentation: Communicating the Value Proposition- 1 min Customer Pitch .
BUSINESS MODEL: Get Started with Lean Canvas.
VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.

UNIT-III	MONEY AND TEAM	Classes: 12
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MONEY: Revenue Streams - Basics of how companies make money, Understand income, costs, gross and net margins, Identify primary and secondary revenue, streams ; Pricing and Costs - Value, price, and costs; Different pricing, Understand product costs and operations costs; Basics of unit costing strategies; Financing Your New Venture - How to finance business ideas, Various sources of funds available to an entrepreneur and pros and cons of each, What investors expect from you, Practice Pitching to Investors and Corporates

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TEAM: Team Building - Shared Leadership, Role of a good team in a venture's success; What to look for in a

team; How do you ensure there is a good fit? Defining clear roles and responsibilities, How to pitch to candidates to join your startup, Explore collaboration tools and techniques - Brainstorming, Mind mapping, Kanban Board, Slack.

UNIT-IV**MARKETING & SALES****Classes: 10**

MARKETING & SALES: Positioning - Understand the difference between product and brand and the link between them, Define the positioning statement for your product/service and how it should translate into what your customers should see about that brand in the market place.

CHANNELS & STRATEGY: Building Digital Presence and leveraging Social media, Creating your company profile page, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-V**PLANNING & TRACKING, BUSINESS REGULATION****Classes: 08**

Sales Planning : Understanding why customers buy and how buying decisions are made; Listening, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customer needs, not on product features ,Follow-up and closing a sale; Asking for the sale

Planning & Tracking: Importance of project management to launch and track progress, Understanding time management, workflow, and delegation of tasks.

Business Regulation: Basics of business regulations of starting and operating a business; Importance of being compliant and keeping proper documentation, How to find help to get started.

Text Books:

- 1.S. R. Bhowmik & M. Bhowmik, Entrepreneurship, New Age International, 2007.
- 2.Steven Fisher, Ja-nae' Duane, The Startup Equation -A Visual Guidebook for Building Your Startup, Indian Edition, Mc Graw Hill Education India Pvt. Ltd, 2016.
- 3.D F Kuratko and T V Rao —Entrepreneurship-A South-Asian Perspective —Cengage Learning, 2012. (For PPT, Case Solutions Faculty may visit : login.cengage.com)

Reference Books:

1. Vasant Desai —Small Scale industries and entrepreneurship|| Himalaya publishing 2012.
2. Rajeev Roy —Entrepreneurship|| 2e, Oxford, 2012.
3. B.Janakiramand M.Rizwan|| Entrepreneurship Development: Text & Cases,Excel Books,2011.
4. Stuart Read, Effectual Entrepreneurship, Routledge, 2013.
5. Nandan H, Fundamentals of Entrepreneurship, PHI, 2013

Web References:

1. <http://www.onlinevideolecture.com/?course=mba-programs&subject=entrepreneurship>
2. http://nptel.ac.in/courses/122106032/Pdf/7_4.pdf
3. <https://www.scribd.com/doc/21516826/Entrepreneurship-Notes>
4. <http://freevideolectures.com/Course/3514/Economics-/Management-/Entrepreneurship/50>

E-Textbooks:

1. https://books.google.co.in/books?id=kOXbf9chpJMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=Pa1s0A6U91QC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course
1. https://www.openlearning.com/courses/fundamentals-of-entrepreneurship/ 2. https://www.edx.org/course/entrepreneurship-emerging-economies

1. <https://www.openlearning.com/courses/fundamentals-of-entrepreneurship/>
2. <https://www.edx.org/course/entrepreneurship-emerging-economies>

HUMAN VALUES AND PROFESSIONAL ETHICS**V Semester: OPEN ELECTIVE-I**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS12	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

The students will understand the importance of Values and Ethics in their personal lives and professional careers. The students will learn the rights and responsibilities as an employee, team member and a global citizen.

COURSE OUTCOMES:

1. Practice the moral values that ought to guide the Engineering profession.
2. Discover of the set of justified moral principles of obligation, ideals that ought to be endorsed by the engineers and apply them to concrete situations
3. Justify the need for protection of human rights and to know about concept of women empowerment
4. Appreciate the Ethical issues and Know the code of ethics adopted in various professional body's and industries
5. Know the successful functioning of democracy in India

UNIT-I	INTRODUCTION	Classes: 12
Need, basic Guidelines, Content and Process for Value Education: Understanding the need, basic guidelines, content and process for Value Education. Self Exploration - what is it? - its content and process; 'Natural Acceptance' and Experiential Validation - as the mechanism for self exploration. Continuous Happiness and Prosperity - A look at basic Human Aspirations. Right understanding, Relationship and Physical Facilities - the basic requirements for fulfillment of aspirations of every human being with their correct priority. Understanding Happiness and Prosperity correctly - A critical appraisal of the current scenario. Method to fulfill the above human aspirations: understanding and living in harmony at various levels.		
UNIT-II	UNDERSTANDING HARMONY IN THE HUMAN BEING - HARMONY IN MYSELF!	Classes: 12
Understanding human being as a co-existence of the sentient 'I' and the material 'Body'. Understanding the needs of Self ('I') and 'Body' - Sukh and Suvridha. Understanding the Body as an instrument of 'I' (I being the doer, seer and enjoyer). Understanding the harmony of I with the Body: Sanyam and Swasthya; correct appraisal of Physical needs, meaning of Prosperity in detail. Programs to ensure Sanyam and Swasthya.		
UNIT-III	UNDERSTANDING HARMONY IN THE FAMILY AND SOCIETY - HARMONY IN HUMAN - HUMAN RELATIONSHIP	Classes: 12
Understanding harmony in the Family the basic unit of human interaction. Understanding values in human - human relationship; meaning of Nyaya and program for its fulfilment to ensure Ubhay-tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship. Understanding the meaning of Vishwas; Difference between intention and competence. Understanding the meaning of Samman, Difference between respect and differentiation; the other salient values in relationship. Understanding the harmony in the society (society being an extension of family): Samadhan, Samridhi, Abhay, Sah-astiva as comprehensive Human Goals. Visualizing a universal harmonious order in society - Undivided Society (Akhand Samaj), Universal Order (Sarvabhaum Vyawastha) - from family to world family!		
UNIT-IV	UNDERSTANDING HARMONY IN THE NATURE AND EXISTENCE - WHOLE EXISTENCE AS CO-EXISTENCE	Classes: 12
Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature - recyclability and self-regulation in nature. Understanding Existence as Co-existence (Sah-astiva) of mutually interacting units in all-pervasive space. Holistic perception of harmony at all levels of existence.		
UNIT-V	IMPLICATIONS OF THE ABOVE HOLISTIC UNDERSTANDING OF HARMONY ON PROFESSIONAL ETHICS	Classes: 12

Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basic for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics:

- Ability to utilize the professional competence for augmenting universal human order,
- Ability to identify the scope and characteristics of people-friendly and eco-friendly production systems,
- Ability to identify and develop appropriate technologies and management patterns for above production systems.
- Case studies of typical holistic technologies, management models and production systems. Strategy for transition from the present state to Universal Human Order.
- At the level of individual: as socially and ecologically responsible engineers, technologists and managers
- At the level of society: as mutually enriching institutions and organizations.

Text Books:

- R. R. Gaur, R Sangal, G P Bagaria, 2009, A Foundation Course in Human Values and Professional Ethics.
- Prof. K. V. Subba Raju, 2013, Success Secrets for Engineering Students, Smart Student Publications, 3rd Edition.

Reference Books:

- Ivan Illich, 1974, Energy & Equity, The Trinity Press, Worcester, and HarperCollins, USA
- E. F. Schumacher, 1973, Small is Beautiful: a study of economics as if people mattered. Blond & Briggs, Britain.
- A Nagraj, 1998 Jeevan Vidya ek Parichay, Divya Path Sansthan, Amarkantak.
- Sussan George, 1976, How the Other Half Dies, Penguin Press, Reprinted 1986, 1991.
- P. L. Dhar, R. R. Gaur, 1990, Science and Humanism, Commonwealth Publishers.
- A. N. Tripathy, 2003, Human Values, New Age International Publishers.
- Subhas Palekar, 2000, How to practice Natural Farming, Pracheen(Vaidik) Krishi Tantra Shodh, Amravati.
- Donella H. Meadows, Dennis L. Meadows, Jorgen Randers, William W. Behrens III, 1972, Limits to Growth - Club of Rome's report, Universe Books.
- E G Seebauer & Robert L. Berry, 2000, Fundamentals of Ethics for Scientists & Engineers, Oxford University Press.
- M Govindarajan, S Natrajan & V. S Senthil kumar, Engineering Ethics (including Humna Values), Eastern Economy Edition, Prentice Hall of India Ltd.

Web References:

- <https://nptel.ac.in/courses/109104068/30>
- <https://www.srecwarangal.ac.in/cse-downloads/peve-unit-1.pdf>

E-Textbooks:

- <https://soaneemrana.org/onewebmedia/Professional%20Ethics%20and%20Human%20Values%20by%20R.S%20NAAGARAZAN.pdf>
- https://books.google.co.in/books?id=n-KQL3uqeYMC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
- https://books.google.co.in/books?id=6GNDRf0BOyIC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course
1. http://californiauniversity.teachable.com/p/values-and-ethics 2. https://www.matsuniversity.ac.in/professional-ethics.php

ADVANCED ENTREPRENEURSHIP**VI Semester: OPEN ELECTIVE-II**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS13	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand and discover entrepreneurship
2. To build a strong foundation for students to start, build, and grow a viable and sustainable venture
3. To develop an entrepreneurial outlook and mindset, critical skills and knowledge
4. To understand three types of risks: Customer, Business Model, and Product/Technical.

COURSE OUTCOMES:

At the end of the course students are able to

1. Demonstrate basic concepts of advanced entrepreneurship.
2. Identify, track and test their assumptions in the field with customers and pivot if the business model is wrong.
3. Explain how to finance business ideas, various sources of funds to an entrepreneur with their pros and cons and role of a good team in a venture's success.
4. Explain how to finance business ideas, various sources of funds to an entrepreneur with their pros and cons and role of a good team in a venture's success.
5. Demonstrate the legal issues for entrepreneur, importance of proper documentation, patents, intellectual property rights, trademarks and role of business advisors and experts for specific targets in your growth plan.

UNIT-I	ORIENTATION TO GROWTH	Classes: 10
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ORIENTATION TO GROWTH: Getting Ready for Growth-Why growth stage is different compared to start-up phase, Why Product-Market fit is not enough, Case study, To assess readiness for growth, To chart a growth path.

UNIT-II	CUSTOMER, BUSINESS MODEL, VALIDATION	Classes: 10
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CUSTOMERS : Expanding Customer Base-Revisit your business model and develop few variants (more business model types), Identify additional customer segments that your solution can address, Evaluate business models for the new customer segments, Relook at the Problem Statement (can you expand the scope and scalability of your business by repositioning your problem statement?), Explore additional ways to monetize.

BUSINESS MODEL: Get Started with Lean Canvas.

VALIDATION: Develop the Solution, Sizing the Opportunity, Building an MVP.

UNIT-III	TRACTION AND TEAM	Classes: 12
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TRACTION : Scaling - How to gain traction beyond early customers, Defining traction (in quantifiable terms) and identifying the most important metrics to measure traction, Calculate cost of new customer acquisition, Estimate your customer lifetime value(LTV), Identifying waste in your operations and focusing your team on what is important for traction.

Channels and Strategy - The Bulls eye framework, Identify Channels using Bulls Eye Framework, Measuring the effectiveness of selected channels, Budgeting and planning.

UNIT-IV	MONEY & SALES	Classes: 10
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MONEY : Growing Revenues - Stabilizing key revenue streams, Developing additional revenue streams(licensing, franchising), Exploring new channels and partnerships, Sales Planning - Understanding why customers buy and how buying decisions are made; Listening skills, Sales planning, setting targets, Unique Sales Proposition (USP); Art of the sales pitch (focus on customers needs, not on product features, Follow-up

and closing a sale; Asking for the sale.

Strengthening Sales - Follow-up and closing a sale; Asking for the sale, Building a professional sales team, Sales compensation and incentives, Sales planning, setting targets.

Improving Margins - Testing price elasticity, Optimizing costs and operational expenses,

Advanced concepts of unit costing. Financial Modeling - Financial modeling of your venture's growth, Analyzing competitor and peer's financial models.

UNIT-V

SUPPORT

Classes: 08

SUPPORT: Legal - Overview of legal issues and their impact on entrepreneurs, Importance of getting professional help (legal and accounting), Importance of being compliant and keeping proper documentation, Patents and Intellectual property, Trademarks.

Mentors, Advisors, and Experts - The importance of a Mentor and how to find one, Role of business advisors and experts for specific targets in your growth plan

Text Books:

1. Bygrave, W., & Zacharakis, A. (2017) Entrepreneurship, 4th Edition (3rd Edition is ok too) Wiley.
2. Carree, M. A., Thurik, A. R. "The impact of entrepreneurship on economic growth" In: Audretsch, D. B., Acs, Z. J. (eds). Handbook of Entrepreneurship Research. Berlin: Springer Verlag, 2010.

Reference Books:

1. Eric, Reis (2017) The Startup Way: How Entrepreneurial Management Transforms Culture and Drives Growth
2. Thurik, A. R., Audretsch, D. B., Stam, E. "The rise of the entrepreneurial economy and the future of dynamic capitalism" Technovation
3. Audretsch, D. B., Grilo, I., Thurik, A. R. (eds). The Handbook of Research on Entrepreneurship Policy. Cheltenham, UK: Edward Elgar, 2007.

Web References:

1. <https://bbamantra.com/introduction-to-entrepreneurship/>
2. <http://download.nos.org/srsec319new/319EL9.pdf>

E-Textbooks:

1. https://books.google.co.in/books?id=5Rrv37Rlk4C&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=HbN5FyBeqnUC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.coursera.org/specializations/wharton-entrepreneurship>
2. <https://ocw.mit.edu/courses/entrepreneurship/>

INDIAN ETHOS AND BUSINESS ETHICS**VII Semester: OPEN ELECTIVE-III**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS14	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OBJECTIVES:

1. To understand the importance of ethics in business
2. To acquire knowledge and capability to develop ethical practices for effective management
3. To understand the Business Ethics and to provide best practices of business ethics.
4. To learn the values and implement in their careers to become a good managers.
5. To develop various ethical Responsibilities and practice in their professional life
6. To imbibe the ethical issues and to adhere to the ethical codes.

COURSE OUTCOMES:

1. Demonstrate basic concepts Indian Ethos
2. Describe values in business ethics.
3. Discuss various contemporary approaches to Indian Ethos.
4. Analyze various ethical codes in corporate governance
5. Analyze the employees conditions and business ethics.

UNIT-I	INTRODUCTION TO INDIAN ETHOS	Classes: 12
History & Relevance, Principles Practiced by Indian Companies, Role of Indian Ethos in Managerial Practices, Management Lessons from Vedas, Mahabharata, Bible and Quran.		
UNIT-II	UNDERSTANDING VALUES IN BUSINESS	Classes: 12
Kautilya's Arthashastra, Indian Heritage in Business, Management-Production and Consumption. Ethics v/s Ethos , Indian v/s Western Management, Work Ethos and Values for Indian Managers - Relevance of Value Based Management in Global Change- Impact of Values on Stakeholders, Trans-Cultural Human Values, Secular v/s Spiritual Values , Value System in Work Culture, Stress Management-Meditation for mental health, Yoga.		
UNIT-III	CONTEMPORARY APPROACHES TO INDIAN ETHOS	Classes: 12
Contemporary Approaches to Leadership- Joint Hindu Family Business–Leadership Qualities of Karta, Indian Systems of Learning-Gurukul System of Learning, Advantages- Disadvantages of Karma, importance of Karma to Managers-Nishkama Karma-Laws of Karma, Law of Creation- Law of Humility- Law of Growth- Law of Responsibility- Law of Connection-Corporate Karma Leadership.		
UNIT-IV	UNDERSTANDING THE BUSINESS ETHICS	Classes: 12
Understanding the need for ethics, Ethical values, myths and ambiguity, ethical codes, Ethical Principles in Business; Theories of Ethics, Absolutism verses Relativism, Teleological approach, the Deontological approach, Kohlberg's six stages of moral development (CMD).		
UNIT-V	ETHICAL CULTURE IN ORGANIZATION	Classes: 12
Ethical Culture in Organization, Developing codes of Ethics and conduct, Ethical and value based leadership. Role of scriptures in understanding ethics, Indian wisdom & Indian approaches towards business ethics.		

Text Books:

1. M.G. Velasquez, Business Ethics, Prentice Hall India Limited, New Delhi,
2. R.C. Sekhar, Ethical Choices in Business, Response Books, New Delhi, 2007

Reference Books:

1. Chakraborty S.K., —Management Transformation by ValuesII, New Delhi, Sage Publication, 1990.
2. Chakraborty, S.K., Ethics in Management-Vedantic Approach, New Delhi, Oxford India Ltd. 1995.
3. Fernando A.C., Business Ethics: An Indian Perspective, Pearson, 2009.
4. Kautilya's Arthashastra, King, Governance, and Law in Ancient India, Oxford University Press, 2016.
5. Murthy, C.S.R. Business Ethics, Himalaya Publishing House, Mumbai, 2009.
6. Narayana G., —The Responsible Leader: A Journey through Gitall, Ahmedabad, AMA 2000.

Web References:

1. <https://www.slideshare.net/Ravimandal123/indian-ethos-1>
2. <https://www.scribd.com/document/230921031/Indian-Ethos-and-Value-MBA-ZAD-MDU>

E-Textbooks:

1. https://books.google.co.in/books?id=MFmDIM2qvsqC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=NVGI21s5IncC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.edx.org/learn/business-ethics>

MANAGEMENT SCIENCE								
VIII Semester: OPEN ELECTIVE-IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS15	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: This course is intended to familiarise the students with the framework for the managers and leaders available for understanding and making decisions relating to issues related organisational structure, production operations, marketing, Human resource Management, product management and strategy.								
COURSE OUTCOMES: 1. Demonstrate basic concepts of management and organization. 2. Carry out production operations through Work study and Method Study. 3. Plan and control the HR function better 4. Plan, schedule and control projects through PERT and CPM. 5. Evolve a strategy for a business or service organisation								
UNIT-I	INTRODUCTION TO MANAGEMENT AND ORGANIZATION						Classes: 12	
Concepts of Management and organization- nature, importance and Functions of Management, Systems Approach to Management-Taylor's Scientific Management Theory- Fayal's Principles of Management-Maslow's theory of Hierarchy of Human Needs- Douglas McGregor's Theory X and Theory Y - Hertzberg Two Factor Theory of Motivation - Leadership Styles, Social responsibilities of Management, Designing Organizational Structures: Basic concepts related to Organization - Departmentation and Decentralization, Types and Evaluation of mechanisticand organic structures of organization and suitability.								
UNIT-II	OPERATIONS AND MARKETING MANAGEMENT						Classes: 12	
Principles and Types of Plant Layout-Methods of Production(Job, batch and Mass Production), Work Study - Basic procedure involved in Method Study and Work Measurement - Business Process Reengineering(BPR) - Statistical Quality Control: control charts for Variables and Attributes (simple Problems) and Acceptance Sampling, TQM, Six Sigma, Deming's contribution to quality, Objectives of Inventory control, EOQ, ABC Analysis, Purchase Procedure, Stores Management and Store Records - JIT System, Supply Chain Management, Functions of Marketing, Marketing Mix, and Marketing Strategies based on Product Life Cycle, Channels of distribution.								
UNIT-III	HUMAN RESOURCES MANAGEMENT(HRM)						Classes: 12	
Concepts of HRM, HRD and Personnel Management and Industrial Relations (PMIR), HRM vs PMIR, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Placement, Wage and Salary Administration, Promotion, Transfer, Separation, Performance Appraisal, Grievance Handling and Welfare Administration, Job Evaluation and Merit Rating - Capability Maturity Model (CMM) Levels - Performance Management System.								
UNIT-IV	PROJECT MANAGEMENT (PERT/ CPM)						Classes: 12	
Network Analysis, Programme Evaluation and Review Technique (PERT), Critical Path Method (CPM), Identifying critical path, Probability of Completing the project within given time, Project Cost Analysis, Project Crashing (simple problems)								

UNIT-V	STRATEGIC MANAGEMENT AND CONTEMPORARY STRATEGIC ISSUES	Classes: 12
Mission, Goals, Objectives, Policy, Strategy, Programmes, Elements of Corporate Planning Process, Environmental Scanning, Value Chain Analysis, SWOT Analysis, Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. Bench Marking and Balanced Score Card as Contemporary Business Strategies.		
Text Books:		
1.A.R.Aryasri : Management Science, TMH, (Latest edition) 2.Stoner, Freeman, Gilbert, Management, 6th Ed, Pearson Education, New Delhi, 2004.		
Reference Books:		
1. Kotler Philip and Keller Kevin Lane: Marketing Management, Pearson, 2012. 2. Koontz and Weihrich: Essentials of Management, McGraw Hill, 2012. 3. Thomas N. Duening and John M. Ivancevich Management - Principles and Guidelines, Biztantra, 2012. 4. Kanishka Bedi, Production and Operations Management, Oxford University Press, 2012. 5. Samuel C. Certo: Modern Management, 2012. 6. Schermerhorn, Capling, Poole and Wiesner: Management, Wiley, 2012. 7. Parnell: Strategic Management, Cengage, 2012. 8. Lawrence R Jauch, R. Gupta and William F. Glueck: Business Policy and Strategic Management Science, McGraw Hill, 2012.		
Web References:		
1. https://www.smartworld.com/notes/management-science-pdf-notes-ms-notes-pdf 2. https://www.alljntuworld.in/download/management-science-notes/		
E-Textbooks:		
1. https://books.google.co.in/books?id=Pdu9MCkqy5wC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=sTeWF5c8FXcC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://www.classcentral.com/course/independent-principles-of-management-11932 2. https://www.udemy.com/principles-of-management-j/		

INTELLECTUAL PROPERTY RIGHTS								
VIII Semester: OPEN ELECTIVE-IV								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS16	OEC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100
COURSE OBJECTIVES: 1. To understand fundamental aspects of Intellectual property Rights. 2.To disseminate knowledge on patents, patent regime in India and abroad and registration aspects 3.To disseminate knowledge on copyrights and its related rights and registration aspects 4. To aware about current trends in IPR and Govt. steps in fostering IPR 5. To familiarize with the kind of rights, remedies and licensing regime associated with each kind of intellectual property.								
COURSE OUTCOMES: 1. Demonstrate the basic concepts of intellectual property rights. 2. Explain function, acquisition of trademarks and trade mark registration process. 3. Describe laws of copy rights and patents. 4. Explain the concepts of trade secrets and unfair competition. 5. Outline new developments in laws of intellectual property rights.								
UNIT-I	INTRODUCTION						Classes: 12	
Introduction, types of intellectual property, international organizations, agencies and treaties, importance of intellectual property rights.								
UNIT-II	TRADE MARKS						Classes: 12	
Purpose and function of trademarks, acquisition of trade mark rights, protectable matter, selecting and evaluating trade mark, trade mark registration processes.								
UNIT-III	LAW OF COPY RIGHTS AND PATENTS						Classes: 12	
Law of copy rights: Fundamental of copy right law, originality of material, rights of reproduction, rights to perform the work publicly, copy right ownership issues, copy right registration, notice of copy right, international copy right law. Law of patents: Foundation of patent law, patent searching process, ownership rights and transfer.								
UNIT-IV	TRADE SECRETS AND UNFAIR COMPETITION						Classes: 12	
Trade Secrets: Trade secret law, determination of trade secrete status, liability for misappropriations of trade secrets, protection for submission, trade secrete litigation. Unfair competition: Misappropriation right of publicity, False advertising.								
UNIT-V	NEW DEVELOPMENTS IN LAWS OF INTELLECTUAL PROPERTY RIGHTS						Classes: 12	
New developments in trade mark law; copy right law, patent law, intellectual property audits. International overview on intellectual property, international - trade mark law, copy right law, international patent law, and international development in trade secrets law.								
Text Books: 1. Intellectual property right, Deborah, E. Bouchoux, cengage learning. 2.Intellectual property right - Unleashing the knowledge economy, prabuddha ganguli, Tata Mc Graw Hill Publishing Company Ltd.								
Reference Books:								

1. Deborah. E. Bouchoux (2009), Intellectual property, Cengage learning, India.
2. Deborah. E. Bouchoux (2001), Protecting your companies intellectual property, AMACOM, USA.
3. Prabuddaganguli (2003), Intellectual property right, Tata McGraw Hill Publishing company Ltd., India.
4. Robert Hisrich, Michael P. PETER, Dean A. Shepherd (201), Entrepreneurship, Tata McGraw Hill., India.

Web References:

1. <https://slideplayer.com/slide/12065874/>
2. https://www.bits-pilani.ac.in/uploads/Patent_ManualOct_25th_07.pdf
3. <https://www.studocu.com/en/document/the-university-of-adelaide/intellectual-property-law/lecture-notes/lecture-notes-lecture-1-5-revision-notes-intellectual-property-law-rn/191756/view>

E-Textbooks:

1. https://books.google.co.in/books?id=nEc8PtMOuzsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=ydV68o_oMHEC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://www.coursera.org/learn/introduction-intellectual-property>
2. <https://ocw.mit.edu/terms/intellectual-property/>
3. <https://www.edx.org/course/intellectual-property-law-policy-part-1-pennx-iplaw1x-0>

NUMBER THEORY**VIII Semester: OPEN ELECTIVE - IV**

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A4HS17	HSMC	L	T	P	C	CIA	SEE	Total
		3	0	-	3	30	70	100

COURSE OBJECTIVES:

To learn

1. Divisibility test and Euclidian Algorithms.
2. The Arithmetic function and Dirichlet multiplication.
3. The properties of congruences.
4. Application of Lagrange's theorem.
5. Quadratic Residues and Reciprocity Law.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Apply the Fundamental theorem of Arithmetic..
2. Identify the Liouville's function and divisor function.
3. Apply the Euler Fermat theorem in polynomial congruence modulo P
4. Apply Chinese remainder theorem.
5. Find Quadratic residues.

UNIT-I	PRIMES, DIVISIBILITY AND FUNDAMENTAL THEOREM OF ARITHMETIC	Classes: 10
Divisibility, GCD, Prime Numbers, Fundamental theorem of Arithmetic, the series of reciprocal of the Primes, The Euclidean Algorithm.		
UNIT-II	ARITHMETIC FUNCTION & DIRICHLET MULTIPLICATION	Classes: 08
A Arithmetic function and Dirichlet Multiplication, The functions $\phi(n)$, $\mu(n)$ and a relation connecting them, Product formulae for $\phi(n)$, Dirichlet Product, Dirichlet inverse and Mobius inversion formula and Mangoldt function $\Lambda(n)$, multiplication function, multiplication function and Dirichlet multiplication, Inverse of a completely multiplication function, Liouville's function $\lambda(n)$, the divisor function is $\sigma\alpha(n)$		
UNIT-III	CONGRUENCES	Classes: 08
Congruences, Properties of congruences, Residue Classes and complete residue system, linear congruences conversion, reduced residue system and Euler Fermat theorem, polynomial congruence modulo P		
UNIT-IV	LAGRANGE'S AND CHINESE REMAINDER THEOREMS	Classes: 08
Lagrange's theorem, Application of Lagrange's theorem, Chinese remainder theorem and its application, polynomial congruences with prime power module.		
UNIT-V	QUADRATIC RESIDUES	Classes: 08
Quadratic residue and quadratic reciprocity law, Quadratic residues, Legendre's symbol and its properties, evaluation of $(-1/p)$ and $(2/p)$, Gauss Lemma, the quadratic reciprocity law and its applications		

Text Books:

1. "Introduction to analytic Number Theory" by Tom M. Apostol. Chapters 1, 2, 5, 9.
2. Niven, H.S. Zuckerman & H.L. Montgomery, Introduction to the Theory of Numbers, Wiley, 20

Reference Books:

1. Number Theory by Joseph H. Silverman.
2. Theory of Numbers by K. Ramchandra.
3. Elementary Number Theory by James K Strayer.
4. Elementary Number Theory by James Tattusall.

Web References:

1. <http://14.139.172.204/nptel/CSE/Web/111103020-old/NumberTheory-A.Saikia.pdf>
2. <https://www.btechguru.com/courses--nptel--mathematics-video-lecture--maths.html>

E-Text Books:

1. https://books.google.co.in/books?id=G6nz_gNbktcC&printsec=frontcover&source=gbs_ge_summar_y_r&cad=0#v=onepage&q&f=false
2. https://books.google.co.in/books?id=eVwwwZeBf4C&printsec=frontcover&source=gbs_ge_summar_y_r&cad=0#v=onepage&q&f=false

MOOC Course

1. <https://ocw.mit.edu/courses/mathematics/18-781-theory-of-numbers-spring-2012/>
2. <https://ocw.mit.edu/courses/mathematics/18-785-number-theory-i-fall-2017/>
3. <https://www.coursera.org/learn/number-theory-cryptography>

PHYSICS AND TECHNOLOGY OF THIN FILMS**VIII Semester: OPEN ELECTIVE - IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS18	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To understand the bonding, nucleation process in materials.
2. To gain the knowledge on fabrication of thin films.
3. To learn the kinetics of different thin films.
4. To understand the properties of thin films by using characterization methods.
5. To learn the applications of thin films in different fields.

COURSE OUTCOMES:

At the end of the course students are able to

1. Analyze the defects in solids and kinetics.
2. Evaluate the epitaxy of thin films and growth process.
3. Justify how the growth kinetics is more efficient than nucleation kinetics in thin film technology.
4. Recommend appropriate synthesis method and explain the characterization techniques.
5. Analyze which thin films are used in different applications.

UNIT-I	A REVIEW OF MATERIALS SCIENCE	CLASSES: 06
Introduction, Structure Defects in Solids Bonding of Materials, Thermodynamics of Materials, Kinetics, Nucleation and Conclusion.		
UNIT-II	PREPARATION OF THIN-FILMS	CLASSES: 10
Classifications of vacuum ranges – Vacuum pumps - Rotary, Diffusion, Turbo molecular and Ion Pumps – Thin film (epitaxy) – definition & advantages – Types of epitaxy. Different Growth Techniques: Liquid Phase Epitaxy, Molecular Beam Epitaxy, Metal Organic Vapour Phase Epitaxy, Sputtering (RF & DC), Pulsed Laser Deposition. Thickness Measurement: Microbalance technique, Photometry, Interferometry (MBI, FECO)		
UNIT-III	KINETICS OF THIN FILMS	CLASSES: 08
Nucleation Kinetics: types of nucleation – kinetic theory of nucleation – energy formation of a nucleus – critical nucleation parameters; spherical and non-spherical nucleus (cap, disc and cubic shaped) on the substrates. Growth Kinetics: Kinetics of binary (GaAs, InP, etc.), ternary ($\text{Al}_{1-x}\text{Ga}_x\text{As}$, $\text{Ga}_{1-x}\text{In}_x\text{P}$, etc.) and quaternary ($\text{Ga}_{1-x}\text{In}_x\text{As}_{1-y}\text{P}_y$, etc.) semiconductors – derivation of growth rate and composition expressions.		

UNIT-IV	PROPERTIES & CHARACTERIZATION OF THIN FILMS	CLASSES: 10
Properties of Thin Films: Dielectric properties – Important parameters, Measurement of dielectric properties- Effect of annealing and film thickness. Optical properties – Optical constants, determination of optical constants by Brewster angle method, Normal incidence method and graphical method. Mechanical properties – Concept and origin of stress and strain, Lattice misfit, Thermal misfit, Hardness test and Bulge test. Characterization: X-ray diffraction –Photoluminescence –UV-Vis-IR spectrophotometer – Atomic Force Microscope –Scanning Electron Microscope – Hall Effect – Vibrational Sample Magnetometer – Secondary Ion Mass Spectrometry.		
UNIT-V	APPLICATIONS OF THIN FILMS	CLASSES: 08
Optoelectronic devices: LED, LASER and Solar cell – Micro Electromechanical Systems (MEMS) –Fabrication of thin film capacitor – application of ferromagnetic thin films; Data storage, Giant Magneto resistance (GMR).		
Text Books:		
1.A. Goswami, Thin Film Fundamentals, New Age international (P) Ltd. Publishers, New Delhi(1996).		
Reference Books:		
1. K. L. Chopra, Thin Film Phenomena, McGraw- Hill book company New York,(1969). 2. L. Eckertova, Physics of Thin Films, Plenum press, New York(1977). 3. Milton Ohring, Material Science of Thin films, 2 nd Edition, Academic Press(2002).		
Web References:		
1. https://static.ifp.tuwien.ac.at/homepages/Personen/www_samba/DSchichten/pdf/t_p_tf_chapter1.pdf 2. https://static.ifp.tuwien.ac.at/homepages/Personen/duenne_schichten_alt/pdf/t_p_ds_contents_cove_r.pdf		
E-Textbooks:		
1. https://books.google.co.in/books?id=cA-RK8QyLDoC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=qrkCQAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://nptel.ac.in/courses/113104075/40 2. https://www.coursera.org/lecture/high-throughput/vapor-deposition-of-thin-films-introductory-concepts-17l53		

POLYMER CHEMISTRY**VIII Semester: OPEN ELECTIVE - IV**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4HS19	OEC	L	T	P	C	CIA	SEE	Total
		3	0	0	3	30	70	100

COURSE OBJECTIVES:

1. To understand the structure of the polymers, identify and explain types of polymerization.
2. To gain the knowledge to use techniques of polymerization.
3. To understand the properties, processing and applications of natural rubber.
4. To gain the knowledge to design polymers for industrial applications.
5. To learn the structure properties of polymers ,inorganic and its characterization techniques

COURSE OUTCOMES:

At the end of the course students are able to

1. Describe the structures of the polymers, identify and explain types of polymerization and account for the concept of the molecular weight.
2. Explain the various properties of polymers and use techniques of polymerization.
3. Account the properties, processing and applications of natural rubber.
4. Design polymers for industrial application using suitable additives.
5. Correlate structure-properties of polymer composite, inorganic, conducting polymers and its characterization techniques.

UNIT-I	BASICS OF POLYMERS	CLASSES: 10
Introduction: Polymer Definition, Classifications of Polymers: Natural & Synthetic, Biopolymers, Thermoplastic, Thermosets, Elastomers, Fibers etc. Types of Polymerization: Addition & Condensation Polymerization. Molecular Weight : No. Average and Weight Average Molecular Weight of Polymers, Distribution & Poly dispersity Index, Structure of Polymers: Amorphous, Semi crystalline and Crystalline states		
UNIT-II	PROPERTIES & POLYMERIZATION TECHNIQUES	CLASSES: 10
Polymerization Techniques: Mass Polymerization, Bulk Polymerization, Solution Polymerization, Emulsion Polymerization, Suspension Polymerization, Mechanisms with explanation. Structure Property Relationship in Polymers: glass transition, melting and crystallization temperature. Effect of structure on the chemical, mechanical, electrical and optical properties of polymers.		
UNIT-III	PROPERTIES, PROCESSING AND APPLICATIONS OF NATURAL RUBBER	CLASSES: 10
Natural Rubber (NR): Origin – Natural Rubber Latex, Tapping, Processing- Vulcanization, Properties and Applications – Conversion of Latex into Dry Rubber – Properties of Dry Rubber. Synthetic Polyisoprene (IR) Rubbers: Preparation, Properties and Applications, Thermoplastic Elastomers Based on Plastics.		

UNIT-IV	NON RUBBER ADDITIVES	CLASSES: 10
Non Rubber Additives: Part A: Vulcanizing ingredients & other additives: Vulcanizing ingredients & their sequence of mixing: Activators and Accelerators: mechanisms of action Part B: Fillers Carbon black -Its preparation, structure, properties and their effect on rubber properties Silica fillers & coupling agents, Other fillers: Clay, Calcium carbonate, titania etc. Nano-fillers: Reinforcement by filler: Reinforcement, Part C: Processing aids & other additives: Processing aids, plasticizers, process additives, release agents, Prevention of Ozone Attack with the use of waxes & saturated polymer for Ozone Protection.		
UNIT-V	INORGANIC, COMPOSITE AND CONDUCTING POLYMER, CHARACTERIZATION TECHNIQUES OF POLYMERS	CLASSES: 08
Inorganic Polymers: Synthesis, Structure- property Relationship and Applications of Polyphosphates, Polysiloxanes, Polymer Composite- Properties and its Applications. Conducting Polymers- Poly Aniline (PaNi)- Mechanism of Conduction and Applications. Characterization Techniques of Polymers: Thermo gravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC).		
Text Books:		
1. F. W. Billmeyer Jr., Text Book of Polymer Science, Ed. Wiley-Interscience, 1984. 2. V. T. Gowariker, N. V. Viswanathan, and J. Sreedar, Polymer Science, 1988.		
Reference Books:		
1. M. Morton, Rubber Technology, Chapman Hall, 1995. 2. Robert J. Young and Peter A. Lovell, Introduction to Polymers, 3 rd Edition, 2011.		
Web References:		
1. http://web.mit.edu/5.33/www/lec/poly.pdf 2. https://edurev.in/studytube/Lecture-Notes-Introduction-To-Polymer-Chemistry/9ace7013-4014-4d18-8af6-0b8557ccb7a1_p		
E-Textbooks:		
1. https://books.google.co.in/books?id=LivCAAQBAJ&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false 2. https://books.google.co.in/books?id=b6IAVqf_LMsC&printsec=frontcover&source=gbs_ge_summary_r&cad=0#v=onepage&q&f=false		
MOOC Course		
1. https://ocw.mit.edu/courses/chemical-engineering/10-569-synthesis-of-polymers-fall-2006/lecture-notes/ 2. https://www.coursera.org/lecture/material-behavior/1-4-polymers-y1IMY 3. https://www.open.edu/openlearn/science-maths-technology/science/chemistry/introduction-polymers/content-section-0		